

Counting Problems in Graphical Models

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DECLARATION

This thesis is a presentation of my original research work. Whatever contributions of others are involved, every effort is made to indicate this clearly, with due reference to the literature, and acknowledgment of collaborative research and discussions.

The work was done under the guidance of **Piyush Srivastava** at the Tata Institute of Fundamental Research, Mumbai.



Vidya Sagar Sharma

In my capacity as supervisor of the candidate's thesis, I certify that the above statements are true to the best of my knowledge.



Piyush Srivastava

Date: 31-07-2025

Abstract

A graphical model is a probabilistic model that uses a graph to represent conditional independence relations between random variables. In this thesis, we study acyclic directed graphical models, wherein a directed acyclic graph (DAG) is used to represent conditional independence relations and causal relationships between random variables. In this model, a probability distribution P defined over a set of random variables $V = \{X_1, X_2, \dots, X_n\}$ is said to be represented by a DAG D with vertex set V if $P(X_1, X_2, \dots, X_n) = \prod_i P(X_i \mid \text{parent}(X_i))$, where X_j is said to be a parent of X_i in the DAG if $X_j \rightarrow X_i$ is an edge in the DAG, and $\text{parent}(X_i)$ denotes the set of parents of X_i in the DAG. A DAG entails a conditional independence relation $X \perp Y \mid Z$ if all the probability distributions represented by the DAG satisfy the conditional independence relation. There can be more than one DAG that entails the same set of conditional independence relations, and such DAGs are said to be Markov equivalent. Markov equivalent DAGs belong to the same equivalence class, called a Markov equivalence class (MEC), which is graphically represented by the union of the DAGs it contains.

Many interesting combinatorial problems related to MECs exist in the literature. One such problem is determining the size of an MEC given its graphical representation. This problem arises from the fact that a DAG is also used as a causal graph, where a directed edge $X \rightarrow Y$ implies that X is a direct cause of Y . The size of an MEC is then a measure of the uncertainty of the causal model when relying solely on observational data. In applications, additional information about the underlying DAG, beyond that encoded in the MEC, may be available—for example, through access to specific interventions or domain-specific knowledge. Meek referred to this as *background knowledge* and modeled it as a specification of the directions of some of the edges in the underlying DAG. Wienöbst et al. (AAAI 2021) showed that counting such background knowledge-consistent DAGs of an MEC is #P-hard. In this thesis, we present a fixed-parameter tractable (FPT) algorithm for this problem.

Another combinatorial problem related to MECs is counting the number of MECs

that have the same skeleton as a given undirected graph G . As we argue later, MECs with the same skeleton also have a natural statistical interpretation. In this thesis, we present an FPT algorithm that counts MECs with the same skeleton. We also give an FPT algorithm that counts MECs with better time complexity than the previous algorithm when the input graph is chordal. Additionally, we give a polynomial-time algorithm to solve the problem when the input graph is a tree.

List of Publications

Following is a chapter-wise list of publications and preprints based on this thesis.

Chapter 2

1. **Vidya Sagar Sharma.** *Counting Background Knowledge Consistent Markov Equivalent Directed Acyclic Graphs* (2023). In 39th Conference on Uncertainty in Artificial Intelligence (UAI 2023) (pp. 1911-1920).

Chapter 3

1. **Vidya Sagar Sharma.** *Faster Fixed Parameter Tractable Algorithms for Counting Markov Equivalence Classes with Special Skeletons* (preprint, arXiv:2312.17626).

Chapter 4

1. **Vidya Sagar Sharma.** *A Fixed-Parameter Tractable Algorithm for Counting Markov Equivalence Classes with the Same Skeleton* (2024). In 38th AAAI Conference on Artificial Intelligence (AAAI 2024: Oral) (pp. 20532-20539).

Chapter 5

1. **Vidya Sagar Sharma.** *Faster Fixed Parameter Tractable Algorithms for Counting Markov Equivalence Classes with Special Skeletons* (preprint, arXiv:2312.17626).

The following is a list of publications and preprints completed during the PhD that are *not* included in this thesis.

1. D. Kesh and V. S. Sharma. On the Bitprobe Complexity of Two Probe Adaptive Schemes Storing Two Elements. *Discrete Applied Mathematics*, **Volume 319** (Oct 2022), pp. 132–140. Extended abstract in *Proceedings of CALDAM, 2019*. This paper is an extended version of work done as part of my M.Tech. thesis at IIT Guwahati.
2. Vidya Sagar Sharma and Piyush Srivastava. *The PSPACE-hardness of understanding neural circuits* (preprint, arXiv:2006.08266).

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Chapter 1

Introduction

A graphical model is a combinatorial tool for expressing dependencies between random variables. Both directed and undirected versions have been used in the literature to model different kinds of dependency structures. We study *acyclic* directed graphical models, also known as Bayesian networks, which represent conditional independence relations and causal connections between random variables by directed edges [Pearl, 2009]. Such graphical models have been extensively used to model causal relationships across several fields, e.g., material science [Ren et al., 2020], game theory [Kearns et al., 2001], and biology [Friedman, 2004, Finegold and Drton, 2011].

In this model, there can be more than one DAG representing the same conditional independence relations. We call such DAGs *Markov equivalent DAGs*. Markov equivalent DAGs exhibit different causal structures even though they entail the same conditional independence relations between random variables. This makes the study of these equivalence classes important. Many interesting combinatorial problems arise in the study of Markov equivalent DAGs. One of these is to find the number of DAGs that are Markov equivalent and another is to determine the maximum number of DAGs that can have the same skeleton (underlying undirected graph) without being Markov equivalent to each other.

In this thesis, we study variants of these two combinatorial problems in detail. However, before formally describing these problems and the motivation behind them (as is done later in Sections 1.2 and 1.3), it is necessary to set up some basic definitions, notations, and preliminary results. This we now proceed to do. A reader familiar with

these notions can jump directly to Section 1.2 and consult the definitions below as necessary.

1.1 Preliminaries

The following definitions and observations are well-studied terms and results in graph theory and graphical models. We mostly follow the graph theory terminology of Andersson et al. [1997].

Basic Graph Theory Definitions and Notations: A graph G is defined as a pair (V_G, E_G) , where V_G is the set of vertices in G , and $E_G \subseteq V_G \times V_G$ is the set of edges in G . If both (u, v) and (v, u) belong to E_G , then G contains an *undirected edge* between u and v , denoted by $u - v$. If E_G contains (u, v) but not (v, u) , then G contains a *directed edge* from u to v , denoted by $u \rightarrow v$. The graphs considered in this thesis do not have any self-loops; that is, for any node u in a graph G , $(u, u) \notin E_G$. A graph is classified as *directed* if each edge is directed, *undirected* if each edge is undirected, and *partially directed* if it can have both directed and undirected edges. If not specified, we consider a graph to be partially directed. u is said to be a neighbor of v in a graph G if either $(u, v) \in E_G$ or $(v, u) \in E_G$. $N(X, G)$ is the set of neighbors of X in the graph G , i.e., $N(X, G) = \{v : \exists u \in X \text{ and either } (u, v) \in E_G \text{ or } (v, u) \in E_G\}$. The degree of a node is the number of neighbors of the node. The degree of a graph is the maximum degree of its nodes.

Definition 1.1.1 (Clique). For a graph G , a subset $C \subseteq V_G$ is called a *clique* of G if, for all $u, v \in C$, $u - v \in E_G$. The set of maximal cliques of G is denoted by $\Pi(G)$.

Definition 1.1.2 (Path). For a graph G , a sequence of nodes $(v_1, v_2, \dots, v_l)$ is said to be a *path* of length $l - 1$ in G if $v_1, v_2, \dots, v_l$ are distinct, and for all $1 \leq i \leq l - 1$, $(v_i, v_{i+1}) \in E_G$, i.e., either $v_i - v_{i+1} \in E_G$ or $v_i \rightarrow v_{i+1} \in E_G$. For a path $(v_1, v_2, \dots, v_l)$, each (v_i, v_{i+1}) , for all $1 \leq i \leq l - 1$, is called an *edge* of the path. A path is said to be an *undirected path* in G if all the edges of the path are undirected in G , i.e., for all $1 \leq i \leq l - 1$, $v_i - v_{i+1} \in E_G$. Similarly, a path is said to be a *directed path* in G if all the edges of the path are directed, i.e., for all $1 \leq i \leq l - 1$, $v_i \rightarrow v_{i+1} \in E_G$.

We say that a path $P = (v_1, v_2, \dots, v_l)$ is a *path from a to b* if the first node of the path is a and the last node is b , i.e., $v_1 = a$ and $v_l = b$ (the boundary case $l = 1$ is also allowed). P is called a *path from (u, v) to (x, y)* if the first two nodes of P are u and v , and the last two nodes of P are x and y , i.e., $v_1 = u$, $v_2 = v$, $v_{l-1} = x$, and $v_l = y$. Note that the boundary cases $(u, v) = (x, y)$ and $u \neq v = x \neq y$ are both allowed by this definition. P is called a *path from (u, v) to w* if the first two nodes of P are u and v , and the last node of P is w , i.e., $v_1 = u$, $v_2 = v$, and $v_l = w$. Again, note that the degenerate case $v = w$ (i.e., $l = 2$) is allowed by this definition. Similarly, P is called a *path from x to (y, z)* if the first node of P is x and the last two nodes are y and z , i.e., $v_1 = x$, $v_{l-1} = y$, and $v_l = z$. The definition allows the boundary case $x = y$ (i.e., $l = 2$).

Definition 1.1.3 (Reverse of a Path). For an undirected path $P = (u_1, u_2, \dots, u_{l-1}, u_l)$ in a graph G , the *reverse of the path P* is denoted by $\bar{P}$, i.e., $\bar{P} = (u_l, u_{l-1}, \dots, u_2, u_1)$.

Definition 1.1.4 (Chordless Path). For a path P in a graph G , an edge in G between two non-adjacent nodes of P is called a *chord* of P . A path P is called a *chordless path* of a graph G if G does not have any chord of P , i.e., there is no edge in G between any two non-adjacent nodes of P . In other words, if $P = (u_1, u_2, \dots, u_{l-1}, u_l)$ is a chordless path of G , then for $1 \leq i < j \leq l$, if $j \neq i + 1$, it holds that neither $u_i - u_j \in E_G$, nor $u_i \rightarrow u_j \in E_G$, nor $u_j \rightarrow u_i \in E_G$.

Definition 1.1.5 (Descendant and Parent). In a directed graph G , v is said to be a *descendant* of u if there exists a path from u to v . For a directed edge $y \rightarrow x$, y is said to be a *parent* of x .

Definition 1.1.6 (Cycle, Chordless Graph, and DAG). A sequence of nodes $(v_1, v_2, \dots, v_l, v_{l+1} = v_1)$ is said to be a *cycle* of length l in a graph G if $v_1, v_2, \dots, v_l$ are distinct and, for all $1 \leq i \leq l$, $(v_i, v_{i+1}) \in E_G$. For a cycle $(v_1, v_2, \dots, v_l, v_{l+1} = v_1)$, each (v_i, v_{i+1}) , for all $1 \leq i \leq l$, is said to be an edge of the cycle. A cycle $C = (v_1, v_2, \dots, v_l, v_{l+1} = v_1)$ in a graph G is said to be a *directed cycle* in G if *at least one* of the edges of the cycle is directed¹ in G , i.e., there exists $1 \leq i \leq l$ such that $v_i \rightarrow v_{i+1} \in E_G$. C is said to be an *undirected cycle* in G if all the edges of the cycle are undirected in G , i.e., for all $1 \leq i \leq l$,

¹Note that this differs from the notion of a directed path, which is a path in which *all* edges of the path are directed.

$v_i - v_{i+1} \in E_G$. C is said to be a *chordless cycle* in G if the length of the cycle is at least four, i.e., $l \geq 4$, and there is no edge between any two non-adjacent nodes in the cycle, i.e., for $1 \leq i, j \leq l$, if $(v_i, v_j) \in E_G$, then either $|i - j| = 1$ or $\{i, j\} = \{1, l\}$. Henceforth, when we say “chordless cycle”, we will always mean such a cycle of length at least four, though we may also sometimes explicitly include the latter condition for emphasis. A graph is said to be a *chordal graph* if it does not contain a chordless cycle. A graph is said to be a *directed acyclic graph* (DAG) if the graph is directed and does not contain any cycle.

Definition 1.1.7 (Skeleton of a Graph). For a graph G , its *skeleton* is the underlying undirected graph, i.e., we obtain the skeleton of a graph G by replacing each directed edge $u \rightarrow v$ in G with an undirected edge $u - v$.

For example, in Figure 1.2 (on page 9), the skeleton of DAGs D_1 , D_2 , D_3 , and D_4 is G . For a graph G , $\text{skeleton}(G)$ denotes the skeleton of G .

Definition 1.1.8 (v-Structure). For a graph G , an induced subgraph of the form $A \rightarrow B \leftarrow C$ is said to be a v-structure² in G . $\mathcal{V}(G)$ denotes the set of v-structures in G .

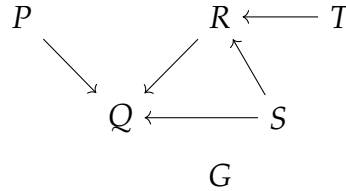


Figure 1.1: v-structures

For example, in Figure 1.1, $P \rightarrow Q \leftarrow R$, $P \rightarrow Q \leftarrow S$, and $S \rightarrow R \leftarrow T$ are v-structures in G . However, $S \rightarrow Q \leftarrow R$ is not a v-structure in G , as it is not an induced subgraph of G . The induced subgraph of G on $\{S, Q, R\}$ also contains the edge $S \rightarrow R$.

Definition 1.1.9 (Union of Graphs). For graphs $G_1, G_2, \dots, G_l$, their *union* forms a graph G , denoted by $G = G_1 \cup G_2 \cup \dots \cup G_l$, where the vertex set of G is the union of the vertex sets of the G_i 's, i.e., $V_G = \bigcup_{i=1}^l V_{G_i}$, and the edge set of G is the union of the edge sets of the G_i 's, i.e., $E_G = \bigcup_{i=1}^l E_{G_i}$.

²In the literature, “v-structures” are often referred to as “immoralities” or “unshielded colliders”.

Definition 1.1.10 (Chain Graph). A graph G is said to be a *chain graph* if there does not exist a directed cycle in G .

Definition 1.1.11 (Topological Ordering of a DAG). For a DAG G , an ordering τ of the vertices in V_G is said to be a topological ordering of G if, for any edge $u \rightarrow v \in E_G$, vertex u comes before vertex v in τ .

There can be more than one topological ordering of a DAG. For example, in Figure 1.1, both (P, T, S, R, Q) and (T, P, S, R, Q) are topological orderings of G .

Definition 1.1.12 (Perfect elimination ordering (PEO)). Let G be an undirected graph. Let τ be an ordering of its vertices. τ is said to be a *perfect elimination ordering* of G if, for all nodes v of G , the subgraph induced in G by the neighbors of v that come before v in τ is a clique, i.e., for $u - v, w - v \in E_G$, if $\tau(u) < \tau(v)$ and $\tau(w) < \tau(v)$, then $u - w \in E_G$.

A graph G has a PEO if and only if G is chordal [Rose et al., 1976].

Definition 1.1.13 (LBFS Ordering). Rose et al. [1976] introduced an algorithm called the LBFS (Lexicographic Breadth First Search) algorithm, which, given an undirected chordal graph as input, outputs a PEO of the graph in linear time. A perfect elimination ordering returned by the LBFS algorithm is referred to as an LBFS ordering. A special feature of an LBFS ordering is that it begins with a maximal clique. A modified version of the LBFS algorithm is provided in Algorithm 3.

Definition 1.1.14 (Tree Decomposition). For an undirected connected graph G , a tuple (T, β) is said to be a *tree decomposition* of G if:

1. T is a tree, and
2. $\beta : V_T \rightarrow 2^{V_G}$ is a function such that (where 2^{V_G} denotes the power set of V_G):
 - (a) For each edge $(x, y) \in E_G$, there exists a node $u \in V_T$ such that $x, y \in \beta(u)$.
 - (b) For all $u, v \in V_T$, and for every node w on the unique path from u to v in T , it holds that $\beta(u) \cap \beta(v) \subseteq \beta(w)$.

If we fix a node $r \in V_T$ as the root of the tree, then (T, β, r) is said to be a *rooted tree decomposition* of G . The *width* of a tree decomposition (T, β) is the maximum value of $|\beta(u)| - 1$ among all nodes $u \in V_T$. The *treewidth* of a graph is the minimum width among all tree decompositions of the graph.

Property 1.1.15 is an intrinsic property of tree decomposition.

Property 1.1.15 (Intersection property of tree decomposition) (Lemma 12.3.1 of [Diestel \[2005\]](#)). Let G be an undirected graph and (T, β) be a tree decomposition of G . For any edge $u - v \in E_T$, $\beta(u) \cap \beta(v)$ is a vertex separator of G .

Definition 1.1.16 (Clique Tree Decomposition). A clique tree decomposition is a specific type of tree decomposition used for chordal graphs. For an undirected connected chordal graph G , a tree decomposition (T, β) of G is said to be a *clique tree decomposition* of G if, for each maximal clique C of G , there exists a unique node $u \in T$ such that $\beta(u) = C$, and for each node $u \in T$, $\beta(u)$ is a maximal clique in G , i.e., $\{\beta(u) : u \in V_T\} = \Pi(G)$ (recall that $\Pi(G)$ denotes the set of all maximal cliques of G). For a chordal graph, a clique tree decomposition always exists, and can be computed in linear time [[Blair and Peyton, 1993](#)].

Definition 1.1.17 (FPT Algorithm, [[Cygan et al., 2015](#)]). A fixed-parameter tractable (FPT) algorithm with input size n and fixed parameter k is an algorithm with $O(f(k) \cdot \text{poly}(n))$ time complexity, where the degree of the polynomial does not depend on k , and $f(k)$ does not depend on n .

1.1.1 Acyclic Directed Graphical Models

An acyclic directed graphical model is a probabilistic model that uses a directed acyclic graph (DAG) to encode conditional independence relations between different subsystems or random variables. In this model, a probability distribution P defined over a set of random variables $V = \{X_1, X_2, \dots, X_n\}$ is represented by a DAG D with vertex set V if $P(X_1, X_2, \dots, X_n) = \prod_i P(X_i \mid \text{parent}(X_i))$, where $\text{parent}(X_i)$ denotes the set of parents of X_i (X_j is said to be a parent of X_i in the DAG if $X_j \rightarrow X_i$ is an edge in the DAG). A DAG entails a conditional independence relation $X \perp Y \mid Z$ if every

probability distribution represented by the DAG obeys the conditional independence relation.

For example, consider a DAG $A \rightarrow B \rightarrow C$. In this DAG, A has no parents, A is the parent of B , and B is the parent of C . A probability distribution P is represented by the DAG if P is defined over the set of random variables $\{A, B, C\}$, and $P(A, B, C) = P(A) \cdot P(B \mid A) \cdot P(C \mid B)$. By the chain rule, we have $P(A, B, C) = P(A) \cdot P(B \mid A) \cdot P(C \mid A, B)$. Therefore, if the DAG represents a probability distribution P , then $P(C \mid A, B) = P(C \mid B)$, implying P obeys the conditional independence relation $C \perp A \mid B$. This further means that the DAG entails the conditional independence relation $C \perp A \mid B$, as all the probability distributions represented by the DAG obey this conditional independence relation.

Above, we use the chain rule to find conditional independence relations entailed by a DAG. Geiger et al. [1990] provide a graphical way to check whether a conditional independence relation is entailed by a DAG.

Definition 1.1.18 (d-Separator [Geiger et al., 1990]). Let G be a DAG. For two distinct nodes u and v of G , and for $Y \subseteq V_G \setminus \{u, v\}$, the set Y *d-separates* u and v if there does not exist a path $Q = (u_1 = u, u_2, \dots, u_l = v)$ from u to v in the skeleton of G such that:

1. For all $1 < i < l$, if $u_{i-1} \rightarrow u_i \in E_G$ and $u_{i+1} \rightarrow u_i \in E_G$, then either u_i belongs to Y , or a descendant of u_i belongs to Y .
2. For all $1 < i < l$, if either $u_{i-1} \rightarrow u_i \notin E_G$ or $u_{i+1} \rightarrow u_i \notin E_G$, then u_i does not belong to Y .

For disjoint subsets X, Y , and Z of nodes in a DAG G , the set Y d-separates X and Z if for all pairs of nodes $(u, v) \in X \times Z$, Y d-separates u and v .

Theorem 1.1.19 (Geiger et al. [1990]). For a DAG G , for disjoint subsets X, Y , and Z of V_G , G entails a conditional independence relation $X \perp Z \mid Y$ if and only if Y d-separates X and Z .

For example, consider a DAG $G := A \rightarrow B \rightarrow C$ (same as DAG D_1 in Figure 1.2). In the DAG, the only path from A to C in the skeleton of G is (A, B, C) . Also, $C \rightarrow B \notin E_G$. This implies $\{B\}$ d-separates $\{A\}$ and $\{C\}$. Therefore, as shown earlier, the DAG entails $\{A\} \perp \{C\} \mid \{B\}$ (for simplicity, we also say $A \perp C \mid B$).

Additionally, using Theorem 1.1.19, we can show that the DAG does not entail any other conditional independence relations.³ Since (A, B, C) is the only path from A to C in $\text{skeleton}(G)$, and $C \rightarrow B \notin E_G$, from Definition 1.1.18, for $X \subseteq V_G \setminus \{A, C\}$, X d-separates A and C if and only if $X = \{B\}$. This implies that for some $X \subseteq V_G \setminus \{A, C\}$, the DAG entails $A \perp C \mid X$ if and only if $X = \{B\}$. Also, since there is an edge between A and B in G , there always exists a path, (A, B) , in the skeleton of G that obeys Items 1 and 2 of Definition 1.1.18. This implies that for all $X \subseteq V_G \setminus \{A, B\}$, X does not d-separate A and B . This implies G does not entail any conditional independence relation of the form $A \perp B \mid X$ for any $X \subseteq V_G \setminus \{A, B\}$. Similarly, G does not entail any conditional independence relation of the form $B \perp C \mid X$ for any $X \subseteq V_G \setminus \{B, C\}$.

The following result by Meek [1995] is a corollary of Theorem 1.1.19. For a DAG G and its nodes u and v , Theorem 1.1.20 provides a necessary and sufficient condition for the existence of a conditional independence relation of the form $u \perp v \mid S$ for some $S \subseteq V_G \setminus \{u, v\}$.

Theorem 1.1.20 (Meek [1995]). *For a DAG G and nodes u and v of G , G entails a conditional independence relation $u \perp v \mid S$ for some $S \subseteq V_G \setminus \{u, v\}$ if and only if there is no edge between u and v in G , i.e., neither $(u, v) \in E_G$ nor $(v, u) \in E_G$.*

For example, consider a DAG $G := A \rightarrow B \rightarrow C$. For all pairs of nodes in G , there is an edge except between nodes A and C . As shown above, the only conditional independence relation entailed by G is $A \perp C \mid B$, which is in accordance with Theorem 1.1.20.

The following definition provides an equivalence relation between DAGs.

Definition 1.1.21 (Markov Equivalent DAGs). Two DAGs are said to be *Markov equivalent* if both entail the same set of conditional independence relations. The equivalence classes determined by this equivalence relation are called *Markov equivalence classes (MECs)*.

For example, consider the DAGs $A \rightarrow B \rightarrow C$ and $A \leftarrow B \rightarrow C$. If a probability distribution P is represented by the first DAG, meaning $P(A, B, C) = P(A) \cdot P(B \mid$

³By conditional independence relation, we mean conditional independence relations of the form $X \perp Z \mid Y$ such that X, Y , and Z are disjoint.

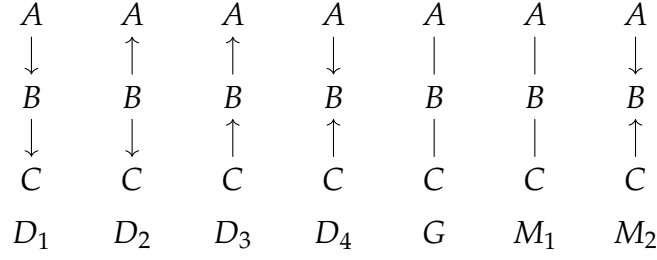


Figure 1.2: Markov equivalent DAGs and graphical representation of MECs

$A) \cdot P(C | B)$, then using Bayes' law, $P(A, B, C) = P(A | B) \cdot P(B) \cdot P(C | B)$, implying P is also represented by the second DAG (in the second DAG, B is the parent of both A and C), and vice versa. This implies that the two DAGs represent the same set of probability distributions. This further implies that the DAGs entail the same conditional independence relations. Therefore, these two DAGs are Markov equivalent.

Above, we use Bayes' law to verify whether two DAGs are Markov equivalent. The following theorem by [Verma and Pearl \[1990\]](#) shows that we can determine whether two DAGs are Markov equivalent simply by comparing their graphical structures.

Theorem 1.1.22 ([Verma and Pearl \[1990\]](#)). *Two DAGs are Markov equivalent if and only if they have the same skeleton and the same set of v-structures.*

The following method is widely used to describe Markov equivalence classes.

Definition 1.1.23 (Graphical Representation of Markov Equivalence Classes). An MEC is graphically represented by the union of the DAGs it contains and entails the same set of conditional independence relations that are entailed by the DAGs it contains.

Using Theorem 1.1.22, just by looking at the graphical structure of D_1, D_2, D_3 , and D_4 (see Figure 1.2), we can determine which of them are Markov equivalent. DAGs D_1, D_2, D_3 , and D_4 are the only DAGs with skeleton G . D_1, D_2 , and D_3 are Markov equivalent as they all possess the same skeleton and have no v-structures. D_4 is not equivalent to D_1, D_2 , and D_3 due to its v-structure $A \rightarrow B \leftarrow C$. Using Theorem 1.1.22, $M_1 = \{D_1, D_2, D_3\}$ is a maximal set of Markov equivalent DAGs, implying M_1 is an MEC and is graphically represented by the union of these DAGs. Similarly, $M_2 = \{D_4\}$ is an MEC and is graphically identical to D_4 .

In general, both an MEC and its graphical representation are referred to as an MEC. Whenever we discuss the graphical features of an MEC, by ‘MEC’ we refer to its graphical representation.

Since a graphical representation of an MEC is the union of the DAGs it contains, Theorem 1.1.22 implies that a directed edge $X \rightarrow Y$ in an MEC means that each DAG in the MEC has the directed edge $X \rightarrow Y$, and an undirected edge $X - Y$ in an MEC means that there exist two DAGs in the MEC: one with the edge $X \rightarrow Y$ and another with the edge $Y \rightarrow X$. Thus, an MEC is a *partially directed graph* that can have both directed and undirected edges. Theorem 1.1.22 also implies that both an MEC and its DAGs have the same skeleton and the same set of v-structures [Verma and Pearl, 1990]. This implies that there cannot be two MECs with the same skeleton and the same v-structures; otherwise, two Markov equivalent DAGs would belong to two different MECs, which is a contradiction.

Clearly, not all graphs represent an MEC. Andersson et al. [1997] provide the following necessary and sufficient conditions for a graph to be an MEC.

Theorem 1.1.24 (Andersson et al. [1997]). *A graph G is an MEC if and only if:*

1. *G is a chain graph, i.e., G does not have a directed cycle.*
2. *Each undirected connected component of G is chordal.*
3. *The configuration $a \rightarrow b - c$ does not occur as an induced subgraph of G .*
4. *Every directed edge $u \rightarrow v \in G$ is strongly protected in G , i.e., $u \rightarrow v$ is part of some induced subgraph of G of one of the forms shown in Figure 1.3.*

In the literature, there are many interesting problems related to the study of MECs. In this thesis, we study two interesting combinatorial problems related to MECs: (a) given the graphical representation of an MEC, count the number of DAGs it contains; and (b) given an undirected graph G , count the number of MECs with G as its skeleton.

In the next two sections, we briefly review these two problems, discuss earlier results, and outline the contributions of this thesis.

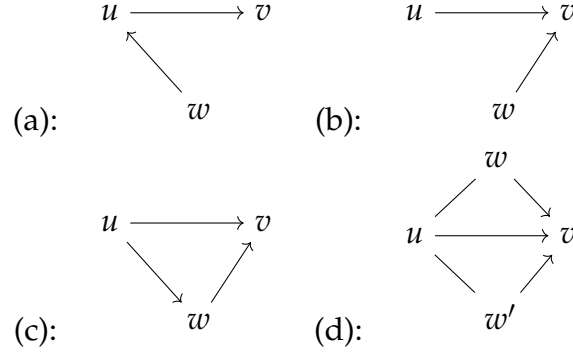


Figure 1.3: Strongly protected edge.

1.2 Counting DAGs of an MEC

As discussed at the beginning of this chapter, we can use acyclic directed graphical models to show causal relationships between random variables, where a directed edge $X \rightarrow Y$ implies that X is a direct cause of Y . Markov equivalent DAGs entail the same conditional independence relations between random variables but may represent different causal relationships. For example, consider the DAGs $A \rightarrow B \rightarrow C$ and $A \leftarrow B \rightarrow C$. These two DAGs are Markov equivalent (using Theorem 1.1.22) and thus entail the same conditional independence relations. However, they represent different causal relationships: the former shows that A is a direct cause of B , and B is a direct cause of C , while the latter suggests that B is the direct cause of both A and C .

Informally, given perfect (non-noisy and “infinite sample”) observational data, one can, in principle, compute the conditional independence relations between random variables. In fact, it may happen that even in the non-noisy case, we observe some “spurious”⁴ conditional independence relations that are obeyed by the observational data but not entailed by the causal DAG that shows the causal relationship between the random variables on which the observational data is defined. [Spirtes et al. \[1990\]](#) show that, under some reasonable assumptions, the occurrence of such spurious in-

⁴Consider three binary random variables, A , B , and C , where A and C are independent of each other, $B = A \oplus C$, and the causal DAG that shows the causal relation between these random variables is $A \rightarrow B \leftarrow C$. For all $x, z \in \{0, 1\}$, suppose $P(A = x) = P(C = z) = 1/2$. Therefore, $P(A, B, C)$ follows this: $P(0, 0, 0) = P(0, 1, 1) = P(1, 0, 1) = P(1, 1, 0) = 1/4$ and $P(0, 0, 1) = P(0, 1, 0) = P(1, 0, 0) = P(1, 1, 1) = 0$. From Theorem 1.1.19, the causal DAG entails only one conditional independence relation: $A \perp C$. However, any observational data that obey P also satisfy $A \perp B$ and $B \perp C$. These are the spurious conditional independence relations that we get from the observational data.

dependencies is a rare event of measure zero and therefore argue that it is natural in causal modeling to assume that the conditional independence relations obeyed by observational data and the conditional independence relations entailed by the causal DAG are the same.

Therefore, we consider the “generic” or “good” setting in which our data does not have these extra spurious conditional independence relations. Yet, there can be multiple DAGs that entail these conditional independence relations, all of which are Markov equivalent and belong to the same MEC. Even in the “good” case considered above, it is not possible to distinguish between these DAGs. Therefore, the size of an MEC measures the amount of uncertainty in the causal model when relying solely on observational data.

Meek [1995] introduced a method to construct the graphical representation of an MEC based on the conditional independence relations it entails. In light of the discussion above, the problem of counting DAGs in an MEC, given the latter’s graphical representation, becomes an important problem.

In practical applications, additional information about the underlying causal DAG may be available beyond what is captured in the Markov equivalence class (MEC). This supplementary information could stem from having access to a specific set of “experimental” (or *interventional*) data or from possessing domain-specific knowledge of the system being studied. Meek [1995] refers to this additional information as *background knowledge* and defines it as a specification of the orientations of certain edges within the underlying DAG. A formal definition of background knowledge for a graph is as follows:

Definition 1.2.1 (Background Knowledge). For a graph G , a background knowledge $\mathcal{K}$ of G is a set of orientations of some undirected edges in G ; i.e., if $u \rightarrow v \in \mathcal{K}$, then $v \rightarrow u \notin \mathcal{K}$, and $u - v$ is an undirected edge in G .

From Definition 1.2.1, a background knowledge of an MEC contains orientations of some undirected edges in the graphical representation of the MEC. When a background knowledge of an MEC is available, rather than merely determining the size of the entire MEC, there is interest in enumerating those DAGs within the MEC that align with the

background knowledge. For example, consider the MEC $M_1 = A - B - C$, as shown in Figure 1.2. As discussed earlier, three DAGs belong to this MEC: $A \rightarrow B \rightarrow C$, $A \leftarrow B \rightarrow C$, and $A \leftarrow B \leftarrow C$. Suppose that, using experimental data, we discover that B is a direct cause of A , which implies that the background knowledge is $\{B \rightarrow A\}$. Consequently, we are no longer interested in counting all DAGs in the MEC. Instead, we want to count the DAGs in the MEC that are consistent with the background knowledge.

In this example, two DAGs are consistent with the given background knowledge: $A \leftarrow B \rightarrow C$ and $A \leftarrow B \leftarrow C$. The DAG $A \rightarrow B \rightarrow C$ is not consistent with the background knowledge because it lacks the edge $B \rightarrow A$. One of the DAGs in the MEC that is consistent with the background knowledge accurately depicts the causal relationship between the random variables. Therefore, background knowledge reduces the uncertainty of the causal model derived from observational data.

Next, we formally state the problem of counting background knowledge-consistent DAGs within an MEC.

Problem 1.1 (Counting Background Knowledge-Consistent DAGs of an MEC). *INPUT: (a) An MEC G , (b) a background knowledge $\mathcal{K}$ of G .*

OUTPUT: The number of DAGs in the MEC that are consistent with $\mathcal{K}$.

As discussed above, the size of an MEC reflects the amount of uncertainty in the causal model when relying solely on observational data. Counting the DAGs in an MEC is equivalent to determining its size. Therefore, the motivation for counting background-knowledge-consistent DAGs of an MEC is to assess how much uncertainty remains in the causal model after incorporating background knowledge. In other words, the main motivation for counting background-knowledge-consistent DAGs is to quantify this residual uncertainty.

Related Work: In the special scenario where no background knowledge exists, an algorithm was proposed by Meek [1995]. This algorithm has since been the focal point of extensive research [Madigan et al., 1996, He et al., 2015, He and Yu, 2016, Bernstein and Tetali, 2017, Ghassami et al., 2019, Talvitie and Koivisto, 2019, Ganian et al., 2020]. He

et al. [2015] evaluated a heuristic based on the partition of essential graphs into chordal components. Ghassami et al. [2019] presented an algorithm that runs in polynomial time for constant degree graphs, though the degree of this polynomial *increases* with the maximum degree of the graph. Talvitie and Koivisto [2019] improved upon this by providing an FPT algorithm for computing the size of an MEC (without background knowledge): for an undirected essential graph on n nodes whose maximum clique is of size c , their algorithm runs in time $\mathcal{O}(c!2^c c^2 n)$. All these results pertain to the special scenario where there is no background knowledge. Finally, Wienöbst et al. [2021, 2023] presented the first polynomial-time algorithm for computing the size of any MEC (again, without background knowledge). However, they demonstrated that, in general, counting background knowledge-consistent DAGs within an MEC is #P-complete (i.e., as hard as counting satisfying assignments to a Boolean formula).

Contribution of this Thesis: As discussed above, Wienöbst et al. [2021] showed that Problem 1.1 is #P-hard. This implies that (under the standard $P \neq NP$ assumption) there *cannot* be an algorithm for solving this problem whose run-time is polynomial in the size of the MEC and background knowledge. In this thesis, we introduce a new parameter, “max-clique knowledge” (Definition 2.1.1). For any MEC G and a background knowledge $\mathcal{K}$ of G , the max-clique knowledge of G for $\mathcal{K}$ can be at most $2|\mathcal{K}|$. However, it can be much smaller than $|\mathcal{K}|$ (see the remark after Definition 2.1.1). We provide a fixed-parameter tractable (FPT) algorithm for the problem that, given an MEC and a background knowledge for the MEC, counts the background knowledge-consistent DAGs of the MEC.

Theorem 1.2.2. *For an MEC G and a background knowledge $\mathcal{K}$ of G , there exists an algorithm that counts $\mathcal{K}$ -consistent DAGs in the MEC in time $\mathcal{O}(k! \cdot k^2 \cdot n^4)$, where k is the max clique knowledge of G for $\mathcal{K}$, and n is the number of vertices in G .*

This is the first non-trivial result that solves Problem 1.1. All previous results have addressed the special scenario where there is no background knowledge. We also provide an empirical exploration of the runtime of our algorithm, demonstrating that it depends on $\mathcal{K}$ only through k and not through $|\mathcal{K}|$ (see Section 2.5).

1.3 Counting MECs with the Same Skeleton

From Theorem 1.1.22, Markov equivalent DAGs have the same skeleton. However, two DAGs with the same skeleton are not necessarily in the same MEC. For example, in Figure 1.2, D_1 and D_4 have the same skeleton but are not Markov equivalent. Similar to the problem of counting DAGs in an MEC (Problem 1.1), where we count DAGs that are Markov equivalent to each other, a natural question arises: given an undirected graph G , how many DAGs with skeleton G can there be so that no two of them are Markov equivalent to each other? For instance, in Figure 1.2, for the undirected graph G , among the DAGs D_1 , D_2 , D_3 , and D_4 with skeleton G , there can be at most two DAGs that are not Markov equivalent to each other, as D_1 , D_2 , and D_3 are Markov equivalent.

Since an MEC has the same skeleton as the DAGs it contains, and DAGs that are not Markov equivalent belong to different MECs (Definition 1.1.21), the question can be reframed as counting the MECs that share the same skeleton. For simplicity, given an undirected graph G , we refer to an MEC M as an MEC of G if the skeleton of M is G . We formally define the problem as follows:

Problem 1.2. INPUT: (a) An undirected connected graph G .

OUTPUT: $|\text{MEC}(G)|$, the number of MECs of G , i.e., the number of MECs with G as their skeleton.

The main motivation for counting the MECs with the same skeleton is that the problem is combinatorially interesting. Another motivation for solving this problem arises from the statistical similarity between MECs that share the same skeleton. We now proceed to describe this notion of similarity.

From Definition 1.1.23, an MEC entails the same set of conditional independence relations as the DAGs it contains. Therefore, from Theorem 1.1.20, an MEC M , defined over a set of random variables V , for any given pair of random variables (P, Q) , entails a conditional independence relation of the form $P \perp Q \mid S$ for some $S \subseteq V \setminus \{P, Q\}$ if and only if there is no edge between P and Q in $\text{skeleton}(M)$.

In other words, for two MECs M_1 and M_2 defined over the set of random variables V , Definition 1.1.23 and Theorem 1.1.20 imply that M_1 and M_2 share the same skeleton

if and only if, for every pair of random variables (P, Q) , M_1 entails $P \perp Q \mid S_1$ for some $S_1 \subseteq V \setminus \{P, Q\}$ if and only if M_2 entails $P \perp Q \mid S_2$ for some $S_2 \subseteq V \setminus \{P, Q\}$ (though S_1 and S_2 may not be the same).

For example, consider the two MECs M_1 and M_2 in Figure 1.2, defined over a set of random variables $V = \{A, B, C\}$. Both MECs have the same skeleton. Since D_1 belongs to M_1 , and D_4 belongs to M_2 , from Definition 1.1.23, M_1 and M_2 entail the same conditional independence relations as those entailed by D_1 and D_4 , respectively. As shown earlier (on page 7), the only conditional independence relation entailed by D_1 is $A \perp C \mid B$. Using the same approach, we show that the only conditional independence relation entailed by D_4 is $A \perp C$.

In the DAG D_4 , between nodes A and B , there is a path (A, B) in $\text{skeleton}(D_4)$ that satisfies Items 1 and 2 of Definition 1.1.18. This implies that for all $X \subseteq V \setminus \{A, B\}$, X does not d-separate A and B . Therefore, from Theorem 1.1.19, for all $X \subseteq V \setminus \{A, B\}$, G does not entail $A \perp B \mid X$. Similarly, for all $X \subseteq V \setminus \{B, C\}$, G does not entail $B \perp C \mid X$. The only path between A and C in $\text{skeleton}(D_4)$ is (A, B, C) , and $A \rightarrow B \leftarrow C$ forms a v-structure in G . From Definition 1.1.18, for $X \subseteq V \setminus \{A, C\}$, X d-separates A and C if and only if $X = \emptyset$. Therefore, from Theorem 1.1.19, for $X \subseteq V \setminus \{A, C\}$, D_4 entails a conditional independence relation $A \perp C \mid X$ if and only if $X = \emptyset$. This implies that the only conditional independence relation entailed by D_4 is $A \perp C$.

The conditional independence relations entailed by the DAGs D_1 and D_4 imply that the set of conditional independence relations entailed by M_1 is $\{A \perp C \mid B\}$, and the set of conditional independence relations entailed by M_2 is $\{A \perp C\}$. This shows the statistical resemblance between the two MECs. Neither entails any conditional independence relations of the form $A \perp B \mid S$ or $B \perp C \mid S$, and both entail a conditional independence relation of the form $A \perp C \mid S$ (for M_1 , $S = \{B\}$, and for M_2 , $S = \emptyset$).

This connection between MECs having the same skeleton motivates the problem of counting MECs with identical skeletons.

Related Work Counting MECs is a well-studied problem. The problem of counting MECs with the same number of nodes has received extensive attention in the literature. [Gillispie and Perlman \[2001\]](#) developed a computer program for computing the number

of MECs with n nodes. [Gillispie and Perlman \[2002\]](#) created a program to enumerate Markov equivalence classes, studying class size distributions and the number of edges for graphs with up to 10 vertices. They also observed that the ratio of the number of DAGs to the number of MECs seems to converge to around 3.7 asymptotically. [Steinsky \[2003\]](#) presented a recursive formula for counting Markov equivalence classes of size 1. [Gillispie \[2006\]](#) provided a recursive algorithm for counting MECs of any size.

[He et al. \[2013\]](#) analyzed the set of MECs with n nodes by constructing a Markov chain on the space of MECs, showing that most edges of MECs are directed. More recently, [Schmid and Sly \[2022\]](#) showed that the expected ratio of the number of DAGs to the number of MECs approaches a positive constant as the number of nodes tends to infinity.

All of these previous results focused on counting MECs based on the number of nodes, not on MECs with a given skeleton. The problem of counting MECs with the same skeleton was introduced by [Radhakrishnan et al. \[2017\]](#), who classified MECs based on the number of v-structures and derived a generating function for counting MECs. They experimentally demonstrated that the generating function varies for graphs with the same number of vertices. In subsequent work, [Radhakrishnan et al. \[2018\]](#) solved the problem for specific graph families such as path graphs, cycle graphs, star graphs, and bi-star graphs. They explored generating functions for these specific graph structures and provided tight lower and upper bounds for the number of MECs in any tree.

Contribution of this Thesis: In this thesis, we present an FPT algorithm for Problem 1.2.

Theorem 1.3.1. *For an undirected graph G , there exists a fixed-parameter tractable algorithm that counts the MECs with skeleton G in $O(n(2^{O(\delta^4 k^4)} + n^2))$ time, where n is the number of nodes in G , and the parameters δ and k are, respectively, the maximum degree of the nodes in G and the treewidth of G .*

Additionally, the thesis presents an FPT algorithm with improved time complexity compared to Theorem 1.3.1 when the input graph is chordal.

Theorem 1.3.2. *For a chordal graph G , there exists a fixed-parameter tractable algorithm that counts the MECs with skeleton G in $O(n(2^{O(\delta^2 k^2)} + n^2))$ time, where n is the number of nodes in G , and the parameters δ and k are, respectively, the maximum degree of the nodes in G and the treewidth of G .*

Furthermore, the thesis provides a polynomial-time solution for Problem 1.2 when the input graph is a tree.

Theorem 1.3.3. *For a tree graph G , the number of MECs with skeleton G can be counted in $O(n\delta)$ time, where n is the number of nodes in G and δ is the maximum degree of a node in G .*

For a tree graph G , Radhakrishnan et al. [2018] provide lower and upper bounds for the number of MECs of G , whereas Theorem 1.3.3 gives the exact count of MECs of G .

1.4 Organization of this Thesis

This thesis contains six chapters. The first chapter (this chapter) introduces and motivates two counting problems in Bayesian networks: the problem of counting background knowledge consistent DAGs and the problem of counting MECs with the same skeleton. It states the results and provides references for closely related lines of work. Chapter 2 addresses the problem of counting background knowledge consistent DAGs of an MEC, while Chapters 3, 4, and 5 focus on counting MECs with the same skeleton for tree graphs, general graphs, and chordal graphs, respectively.

In Chapter 2, we define “max-clique knowledge” and construct an FPT algorithm to count background knowledge consistent DAGs of an MEC. This chapter also includes the proof of Theorem 1.2.2 and an empirical exploration of the runtime of our algorithm, demonstrating that the algorithm depends only on “max-clique knowledge” and not on the size of the background knowledge.

In Chapter 3, we solve Problem 1.2 for tree graphs and construct a polynomial-time algorithm for this problem. The chapter covers the proof of Theorem 1.3.3.

In Chapter 4, we solve Problem 1.2 for general graphs and construct an FPT algorithm for this case. The chapter presents the proof of Theorem 1.3.1.

In Chapter 5, we solve Problem 1.2 for chordal graphs and construct an FPT algorithm for this as well. The chapter includes the proof of Theorem 1.3.2.

Finally, Chapter 6 summarizes the findings from Chapters 2-5 and discusses related open problems in this area.

1.5 Publications from this Thesis

The following is a list of publications that are part of this thesis:

1. **Vidya Sagar Sharma.** *Counting Background Knowledge Consistent DAGs of an MEC.* In the Proceedings of the Thirty-Ninth Conference on Uncertainty in Artificial Intelligence (UAI 2023), [PMLR vol. 216, pp. 1911-1920.](#)
2. **Vidya Sagar Sharma.** *Counting MECs with the Same Skeleton.* In the Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence (AAAI 2024), [vol. 38, no. 18, pp. 20532-20539.](#)

Additionally, the thesis includes results from the following preprint:

1. **Vidya Sagar Sharma.** *Counting MECs with the Same Skeleton for Special Graphs.* Arxiv preprint: [arXiv:2312.17626](#)

Chapter 2

Counting Background Knowledge Consistent DAGs in an MEC

The objective of this chapter is to solve Problem 1.1. For convenience, we restate the problem below.

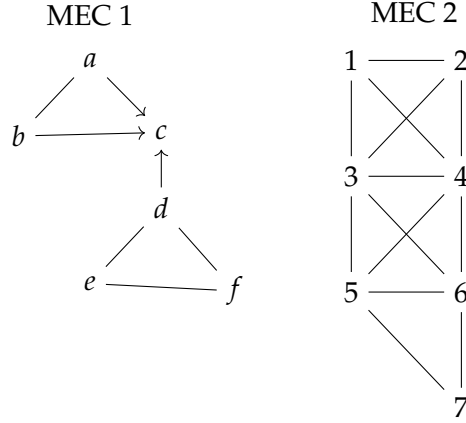
Problem 2.1 (Counting Background Knowledge-Consistent DAGs of an MEC). *INPUT:* (a) An MEC G , (b) A background knowledge $\mathcal{K}$ of G .
OUTPUT: The number of DAGs in the MEC that are consistent with $\mathcal{K}$.

2.1 Overview

In this chapter, we study the problem of counting background knowledge-consistent DAGs of an MEC (Problem 2.1). Recall from Chapter 1 that for a graph G , a background knowledge $\mathcal{K}$ of G (Definition 1.2.1) is a set of orientations of some undirected edges in G . Specifically, if $u \rightarrow v \in \mathcal{K}$, then $v \rightarrow u \notin \mathcal{K}$, and $u - v$ is an undirected edge in G .

As discussed in Chapter 1, Wienöbst et al. [2021] proved that Problem 2.1 is #P-hard. For a special case, where there is no background knowledge, i.e., $\mathcal{K} = \emptyset$, they constructed a polynomial-time algorithm to solve the problem. In this chapter, we construct an FPT algorithm to solve Problem 2.1. Many of the ideas of our algorithm are inspired by the work of Wienöbst et al. [2021].

To solve Problem 2.1, we introduce a new parameter called “*max-clique knowledge*”.



	Background Knowledge	Max-clique Knowledge
Example 1	$\{a \rightarrow b, e \rightarrow d, f \rightarrow d\}$	3
Example 2	$\{a \rightarrow b, e \rightarrow d\}$	2
Example 3	$\{1 \rightarrow 2, 3 \rightarrow 6\}$	2
Example 4	$\{1 \rightarrow 2, 2 \rightarrow 3, 1 \rightarrow 3, 3 \rightarrow 6, 4 \rightarrow 6, 6 \rightarrow 7\}$	3

Figure 2.1: Max-clique knowledge

Definition 2.1.1 (Max-Clique Knowledge). Let G be a graph, and $\mathcal{K}$ be a background knowledge of G . For any clique C in G , its *clique knowledge* for $\mathcal{K}$ is defined as the number of vertices in the clique that are endpoints of edges in $\mathcal{K}[C]$, where $\mathcal{K}[C]$ contains the edges in $\mathcal{K}$ with both endpoints in C . In other words, $\mathcal{K}[C] = \{u \rightarrow v : u \rightarrow v \in \mathcal{K}, \text{ and } u, v \in C\}$. The *max-clique knowledge* of G for $\mathcal{K}$ is the highest value of clique knowledge among all cliques in G .

Figure 2.1 illustrates max-clique knowledge. In MEC 1, there are two maximal cliques: $\{a, b\}$ and $\{d, e, f\}$ (recall from Chapter 1 that for a clique C of a graph G , for all $u, v \in C, u - v \in E_G$). In MEC 2, there are three maximal cliques: $\{1, 2, 3, 4\}$, $\{3, 4, 5, 6\}$, and $\{5, 6, 7\}$. Examples 1 and 2 correspond to MEC 1, and examples 3 and 4 correspond to MEC 2.

In example 1, for the maximal clique $\{d, e, f\}$, d and e are endpoints of the background knowledge edge $e \rightarrow d$, and f is an endpoint of the background knowledge edge $f \rightarrow d$, meaning all nodes in the clique are part of an edge in the background knowledge with both endpoints inside the clique. Therefore, the clique knowledge of

the clique $\{d, e, f\}$ is 3. Similarly, the clique knowledge of the maximal clique $\{a, b\}$ is 2. For the computation of max-clique knowledge, we only consider the maximal cliques because, for cliques C and C' such that $C \subsetneq C'$, the clique knowledge of C must be at most equal to the clique knowledge of C' . This shows that the max-clique knowledge of the background knowledge in example 1 for MEC 1 is 3. The max-clique knowledge for the background knowledge in example 2 for MEC 1 is 2, and for examples 3 and 4 for MEC 2, it is 2 and 3, respectively.

Remark: Note that for any MEC G and a background knowledge $\mathcal{K}$ of G , the max-clique knowledge of G for $\mathcal{K}$ can be at most $2|\mathcal{K}|$ (consider the scenario where there is no common endpoint between any two distinct edges in $\mathcal{K}$, and all edges in $\mathcal{K}$ have endpoints within the same clique). However, it can be much smaller than $|\mathcal{K}|$ (consider the case where, for all maximal cliques C of G , $\mathcal{K}[C]$ contains only a small constant number of edges).

We construct an FPT algorithm for Problem 2.1 based on the parameter *max-clique knowledge*. The following theorem summarizes our result.

Theorem 2.1.2 (Restating Theorem 1.2.2). *For an MEC G and background knowledge $\mathcal{K}$ of G , there exists an algorithm that counts $\mathcal{K}$ -consistent DAGs in the MEC in time $O(k! \cdot k^2 \cdot n^4)$, where k is the max-clique knowledge of G for $\mathcal{K}$, and n is the number of vertices in G .*

The main idea of our result is divided into two phases. In the first phase, we reduce the problem of counting background knowledge-consistent DAGs of an MEC to a graphical problem of counting background knowledge-consistent acyclic moral orientations (AMOs) of a chordal graph. An AMO of a graph is a DAG with the same skeleton and the same v-structures as the graph. In the second phase, we introduce a recursive algorithm to count background knowledge-consistent AMOs of a chordal graph. Here, we follow [Wienöbst et al. \[2021\]](#) to map each AMO of the input chordal graph G to a unique maximal clique of G . For each maximal clique C of G , we then construct an auxiliary graph G^C such that a directed edge in G^C is a directed edge in all the AMOs of the chordal graph that are mapped to C . This implies that if G^C is not background-knowledge consistent, i.e., there exists a directed edge $u \rightarrow v$ in G^C for

which $v \rightarrow u$ is in the background knowledge, then no AMOs of G mapped to C are background-knowledge consistent. Conversely, if G^C is background-knowledge consistent, we use the undirected connected components of G^C to construct our recursive algorithm.

In the algorithm, we also need to compute $\Phi(C, \mathcal{R}, \mathcal{K}[C])$, the number of background knowledge-consistent permutations of C that do not have a prefix in $\mathcal{R}$, where $\mathcal{R} = \{R_1, R_2, \dots, R_l\}$ is a set such that $R_1 \subsetneq R_2 \subsetneq \dots \subsetneq R_l \subsetneq C$, and $\mathcal{K}[C]$ is the set of background-knowledge edges with both endpoints in C . A permutation of C is consistent with $\mathcal{K}[C]$ if, for each $u \rightarrow v \in \mathcal{K}[C]$, u comes before v in the permutation.

The construction of G^C and the computation of $\Phi(C, \mathcal{R}, \mathcal{K}[C])$ constitute the main novelty of this chapter, which we use to construct the recursive algorithm.

Organization of the Chapter: We discuss the first phase of our main idea in Section 2.2, and the second phase in Section 2.3. The mapping of the AMOs of a chordal graph to the maximal cliques of the graph is defined in Section 2.3.1. The construction of G^C , the study of its graphical properties, computation of $\Phi(C, \mathcal{R}, \mathcal{K}[C])$, and the development of the recursive algorithm are detailed in Section 2.3.2. The time complexity analysis of the algorithm discussed in this chapter is presented in Section 2.4.

At the end of this chapter, we present an experimental evaluation of our FPT algorithm (Section 2.5), demonstrating that the algorithm's time complexity depends on max-clique knowledge but not on the size of the background knowledge.

2.2 Reduction of Problem 2.1 to Counting AMOs of a Chordal Graph

In this subsection, we reduce Problem 2.1 to a simpler graphical problem, Problem 2.3. We first establish a graphical relationship between an MEC and the DAGs it contains.

Definition 2.2.1 (Acyclic Moral Orientation (AMO)). A DAG D is said to be an AMO of a graph G if both D and G have the same skeleton and the same set of v-structures.

Lemma 2.2.2 (Verma and Pearl [1990]). *Let G be the graphical representation of an MEC. A DAG D belongs to the MEC if and only if D is an AMO of G .*

From Definition 2.2.1 and Lemma 2.2.2, for an MEC represented by a graph G , and a background knowledge $\mathcal{K}$ of G , the number of $\mathcal{K}$ -consistent DAGs in the MEC is equal to the number of $\mathcal{K}$ -consistent AMOs of G . This reduces Problem 2.1 to the following equivalent problem:

Problem 2.2 (Counting Background Knowledge-Consistent AMOs of an MEC). *INPUT: (a) An MEC G , (b) A background knowledge $\mathcal{K}$ of G .*

OUTPUT: $|\text{AMO}(G, \mathcal{K})|$, the number of AMOs of G that are consistent with $\mathcal{K}$.

Andersson et al. [1997] show that an AMO of an MEC G can be constructed by independently picking an AMO for each undirected connected component of the MEC and combining these with the directed subgraph of G . Since the edges in the background knowledge of an MEC are orientations of undirected edges in the MEC (Definition 1.2.1), from Andersson et al. [1997]’s result, for any MEC G and a background knowledge $\mathcal{K}$ of the MEC, we have:

$$|\text{AMO}(G, \mathcal{K})| = \prod_{H: \text{undirected connected component of } G} |\text{AMO}(H, \mathcal{K}[V_H])| \quad (2.1)$$

where $\mathcal{K}[V_H]$ contains edges in $\mathcal{K}$ with both endpoints in V_H .

From Item 3 of Theorem 1.1.24, the undirected connected components of an MEC are chordal. This reduces Problem 2.2 to the following problem where the input graph G is guaranteed to be an undirected connected chordal graph (UCCG):

Problem 2.3 (Counting Background Knowledge-Consistent AMOs of a UCCG). *INPUT: (a) A UCCG G , (b) A background knowledge $\mathcal{K}$ of G .*

OUTPUT: $|\text{AMO}(G, \mathcal{K})|$, the number of $\mathcal{K}$ -consistent AMOs of G .

In the next section, we construct a fixed-parameter tractable algorithm to solve Problem 2.3.

2.3 FPT Algorithm for Problem 2.3

In this section, we construct an FPT algorithm, Algorithm 3, to solve Problem 2.3 using the parameter “max-clique knowledge” (Definition 2.1.1). Algorithm 3 extends Algorithm 2 from Wienöbst et al. [2021], which addresses a special case of Problem 2.3 when there is no background knowledge. Our algorithm incorporates a background-aware version of their approach.

Specifically, our algorithm utilizes a modified LBFS algorithm, Algorithm 1, which adapts the modified LBFS algorithm from Wienöbst et al. [2021] to accommodate background knowledge. Additionally, we introduce Algorithm 2, a new component that counts background knowledge-consistent permutations of a clique, which is crucial for our final algorithm, Algorithm 3.

We address Problem 2.3 in two main steps:

1. **Partitioning the set of AMOs:** We partition the set of AMOs of the input chordal graph G by mapping each AMO to a unique maximal clique in the graph. This partitioning is based on a structural result of Wienöbst et al. [2021] that provides a correspondence between the AMOs and the maximal cliques of the chordal graph.
2. **Recursive counting for maximal cliques:** For each maximal clique in the chordal graph, we construct a recursive algorithm to count the background knowledge-consistent AMOs that correspond to the clique.

These two steps form the basis of Algorithm 3, which integrates both partitioning and counting techniques to efficiently solve the problem.

We begin by discussing the partitioning of AMOs.

2.3.1 Partition of the Set of AMOs of a Chordal Graph

In this subsection, we partition the set of AMOs of a chordal graph G by mapping each AMO to a unique maximal clique of G . We begin with the following definition.

Definition 2.3.1 (Representation of an AMO; Inspired from Lemma 9 of [Wienöbst et al. \[2023\]](#)). For a UCCG G , an AMO of G is said to be represented by a maximal clique C of G if there exists a topological ordering (Definition 1.1.11) τ of the AMO that starts with C (meaning the first $|C|$ nodes of the topological ordering belong to C). Similarly, we say an AMO of G is represented by a permutation $\pi(C)$ of C if there exists a topological ordering of the AMO that starts with $\pi(C)$ (i.e., the ordering of the first $|C|$ nodes in the topological ordering is the same as $\pi(C)$).

For example, consider a chordal graph $A - B - C$ and an AMO $A \leftarrow B \rightarrow C$ of G . The AMO is represented by the maximal clique $\{A, B\}$ of G since there exists a topological ordering (B, A, C) of the AMO that starts with $\{A, B\}$. The AMO is also represented by the permutation (B, A) of the clique $\{A, B\}$ because the ordering of the first two nodes in the topological ordering is (B, A) .

The following lemma shows that for each maximal clique of a connected chordal graph, there exists an AMO of the graph that is represented by the clique.

Lemma 2.3.2 (Implicit in [Wienöbst et al. \[2021\]](#)). *Let G be a connected chordal graph. For each maximal clique C of G , there exists an AMO of G that is represented by C .*

Proof. For every maximal clique C of G , there exists an LBFS ordering τ of G that starts with C (implicit in the LBFS algorithm of [Wienöbst et al. \[2021\]](#), where, for inputs G and C , τ provides an LBFS ordering of G that starts with C). Let D be a DAG with skeleton G such that, for $u - v \in E_G$, if u comes before v in τ , then $u \rightarrow v \in E_D$. D does not have a v-structure because if $u \rightarrow v$ and $w \rightarrow v \in E_D$, then, from the construction of D , both u and w come before v in τ , i.e., $\tau(u) < \tau(v)$ and $\tau(w) < \tau(v)$. From Definition 1.1.13, $u - w \in E_G$, implying either $u \rightarrow w \in E_D$ (if $\tau(u) < \tau(w)$) or $w \rightarrow u \in E_D$ (if $\tau(w) < \tau(u)$). In both cases, u, v , and w are not part of any v-structure. This implies D is an AMO of G . Additionally, D is represented by C because τ is a topological ordering of D that starts with C . This proves Lemma 2.3.2. $\square$

An AMO of a UCCG G can be represented by more than one maximal clique of G . In the above example, the AMO can also be represented by the maximal clique $\{B, C\}$ of G , as there exists a topological ordering (B, C, A) of the AMO that starts with $\{B, C\}$.

For each AMO of G , we define one of the maximal cliques of G representing the AMO as the canonical representation of the AMO. We use the following definition to achieve this.

Definition 2.3.3 (Canonical Representation of an AMO; Inspired from Lemma 9, Definition 23, Claim 41, and Claim 42 of [Wienöbst et al., 2023]). Let G be an undirected connected chordal graph. Fix a rooted clique tree decomposition $\mathcal{T} = (T, \beta, r)$ of G . For an AMO α of G , a maximal clique C of G canonically represents α , denoted by $C = C_\alpha$, if:

1. C represents α , i.e., there exists a topological ordering τ of α that starts with C , and
2. Let $\beta(v) = C$, and $(u_1 = r, u_2, \dots, u_l = v)$ be the path from r to v in T . Define $\text{FP}(C, \mathcal{T}) = \{S : S = \beta(u_i) \cap \beta(u_{i+1}) \subseteq C \text{ for some } 1 \leq i < l\}$. Then τ does not start with an $S \in \text{FP}(C, \mathcal{T})$.

From now on, whenever we refer to the rooted tree decomposition of G , we mean the tree decomposition $\mathcal{T}$ that is fixed in Definition 2.3.3. The following lemma by Wienöbst et al. [2023] establishes the uniqueness of C_α .

Lemma 2.3.4 (Claims 41 and 42 of Wienöbst et al. [2023]). *For each AMO α of G , there exists a unique maximal clique C_α .*

We denote by $\text{AMO}(G, C, \mathcal{K})$ the set of $\mathcal{K}$ -consistent AMOs α of G for which $C = C_\alpha$. More formally, $\text{AMO}(G, C, \mathcal{K}) = \{\alpha : \alpha \in \text{AMO}(G, \mathcal{K}) \text{ and } C = C_\alpha\}$. The following lemma shows the partition of $\text{AMO}(G, \mathcal{K})$ into the sets $\text{AMO}(G, C, \mathcal{K})$.

Lemma 2.3.5. *Let G be a UCCG, and $\mathcal{K}$ be a background knowledge of G . Then,*

$$|\text{AMO}(G, \mathcal{K})| = \sum_{C \in \Pi(G)} |\text{AMO}(G, C, \mathcal{K})|.$$

Proof. From Lemma 2.3.4, each AMO α in $\text{AMO}(G, \mathcal{K})$ is mapped to a unique maximal clique of G . This implies that each α belongs to exactly one set $\text{AMO}(G, C, \mathcal{K})$, where $C = C_\alpha$. Hence, the set $\text{AMO}(G, \mathcal{K})$ is partitioned into subsets $\text{AMO}(G, C, \mathcal{K})$ for maximal cliques C in G , validating Lemma 2.3.5. $\square$

In the next step, we will construct a recursive algorithm to compute $|\text{AMO}(G, C, \mathcal{K})|$.

2.3.2 Construction of a Recursive Formula to Compute $|\text{AMO}(G, C, \mathcal{K})|$

To compute $|\text{AMO}(G, C, \mathcal{K})|$, we first determine the union of AMOs of G that are represented by C (Definition 2.3.1). The following definition introduces relevant concepts from Wienöbst et al. [2021], which are necessary for constructing the union graph.

Definition 2.3.6 ($G^{\pi(C)}$, G^C , $\mathcal{C}_G(\pi(C))$, and $\mathcal{C}_G(C)$; Definition 1 of Wienöbst et al. [2021]). Let G be a UCCG, C a maximal clique of G , and $\pi(C)$ a permutation of C . Then, G^C (respectively, $G^{\pi(C)}$) denotes the union of all AMOs of G represented by C (respectively, $\pi(C)$). $\mathcal{C}_G(C)$ (respectively, $\mathcal{C}_G(\pi(C))$) denotes the undirected connected components of $G^C[V_G \setminus C]$ (respectively, $G^{\pi(C)}[V_G \setminus C]$).

The structure of G^C provides the set of directed edges in G^C , which allows us to check the $\mathcal{K}$ -consistency of G^C . Since G^C is the union of all AMOs represented by C , a directed edge in G^C appears in all AMOs represented by C . From Item 1 in Definition 2.3.3, if an AMO α of G is canonically represented by C , then it is also represented by C . Therefore, if $u \rightarrow v$ is a directed edge in G^C , it must be a directed edge in all AMOs in $\text{AMO}(G, C, \mathcal{K})$. Consequently, if any directed edge in G^C is not $\mathcal{K}$ -consistent (i.e., for some directed edge $u \rightarrow v$ in G_C , $v \rightarrow u \in \mathcal{K}$), then none of the AMOs in $\text{AMO}(G, C, \mathcal{K})$ are $\mathcal{K}$ -consistent. However, if all the directed edges of G^C are $\mathcal{K}$ -consistent, we reduce the problem to counting $\mathcal{K}$ -consistent AMOs for the undirected connected components of G^C (Lemma 2.3.18).

The key insight required to implement the above discussion is provided by the following definition:

Definition 2.3.7 ($\mathcal{K}$ -Consistency of G^C). Let G be a UCCG and C a maximal clique in G . For a background knowledge $\mathcal{K}$ of G , G^C is said to be $\mathcal{K}$ -consistent if there does not exist a directed edge $u \rightarrow v$ in G^C such that $v \rightarrow u \in \mathcal{K}$.

As discussed, Definition 2.3.7 implies that if G^C is not $\mathcal{K}$ -consistent for a maximal

clique C in G , then there are no $\mathcal{K}$ -consistent AMOs of G in $\text{AMO}(G, C, \mathcal{K})$. That is,

$$|\text{AMO}(G, C, \mathcal{K})| = 0. \quad (2.2)$$

Then, from Lemma 2.3.5,

$$|\text{AMO}(G, \mathcal{K})| = \sum_{\substack{C \in \Pi(C) \\ G^C \text{ is } \mathcal{K}\text{-consistent}}} |\text{AMO}(G, C, \mathcal{K})|. \quad (2.3)$$

We construct Algorithm 1 to check the $\mathcal{K}$ -consistency of G^C for any maximal clique C of G . Wienöbst et al. [2021] present an LBFS algorithm (referred to as “LBFS” because it is a modified version of the LBFS algorithm by Rose et al. [1976]) that, given a chordal graph G and a maximal clique C of G , outputs the undirected connected components (UCCs) of $\mathcal{C}_G(C)$. Their algorithm outputs the UCCs in $\mathcal{C}_G(C)$ in such a way that knowing the UCCs of $\mathcal{C}_G(C)$ allows us to construct G^C . We utilize this fact to construct Algorithm 1, a background-aware version of the LBFS algorithm by Wienöbst et al. [2021].¹

We now describe our background-aware version of the modified LBFS algorithm (see Algorithm 1). We do not change any lines from the LBFS algorithm of Wienöbst et al. [2021]. The modifications we make to their LBFS algorithm are: (a) introducing the “flag” at Line 2, which is used to check the $\mathcal{K}$ -consistency of G^C , (b) lines 11-13, which are used to update the value of the “flag”, and (c) outputting the value of the “flag” along with $\mathcal{C}_G(C)$. The correctness of these modifications (formally stated in Lemma 2.3.9) is based on the following observation, which in turn relies on ideas implicit in the work of Wienöbst et al. [2021].

Lemma 2.3.8 (Based on ideas implicit in the work of Wienöbst et al. [2021]). *Let G be a UCCG, $\mathcal{K}$ be a background knowledge of G , and C be a maximal clique of G . For input G , C , and $\mathcal{K}$, suppose that at some stage of an iteration of the while loop in Algorithm 1, $\mathcal{L} = \{X_1, X_2, \dots, X_l\}$. Then:*

¹If Algorithm 1 is executed with $\mathcal{K} = \emptyset$, it functions identically to the LBFS algorithm constructed by Wienöbst et al. [2021]. If executed with $C = \mathcal{K} = \emptyset$, it functions as the original LBFS algorithm of Rose et al. [1976].

Algorithm 1: LBFS($G, C, \mathcal{K}$) (Background aware LBFS, based on the modified LBFS of [Wienöbst et al. \[2021\]](#))

Input : A UCCG G , a maximal clique C of G , and a background knowledge $\mathcal{K}$ of G .

Output: $(1, \mathcal{C}_G(C))$: if G^C is $\mathcal{K}$ -consistent,
 $(0, \mathcal{C}_G(C))$: otherwise.

- 1 $\mathcal{S} \leftarrow$ sequence of sets initialized with $(C, V \setminus C)$
- 2 $\tau \leftarrow$ empty list, $\mathcal{L} \leftarrow$ empty list, $\text{flag} \leftarrow 1$, $Y \leftarrow$ empty list
- 3 **while** $\mathcal{S}$ is non-empty **do**
- 4 $X \leftarrow$ first non-empty set of $\mathcal{S}$
- 5 $v \leftarrow$ arbitrary vertex from X
- 6 **if** v is neither in a set in $\mathcal{L}$ nor in C **then**
- 7 Append X to the end of the list $\mathcal{L}$.
- 8 Append undirected connected components of $G[X]$ to the end of Y .
- 9 **end**
- 10 Add vertex v to the end of τ .
- 11 **if** $u \rightarrow v \in \mathcal{K}$ for any u which is neither in a set in $\mathcal{L}$ nor in C **then**
- 12 $\text{flag} = 0$;
- 13 **end**
- 14 Replace the set X in the sequence $\mathcal{S}$ by the set $X \setminus \{v\}$.
- 15 $N(v) \leftarrow \{x | x \notin \tau \text{ and } v - x \in E\}$
- 16 Denote the current $\mathcal{S}$ by $(S_1, \dots, S_k)$.
- 17 Replace each S_i by $S_i \cap N(v), S_i \setminus N(v)$.
- 18 Remove all empty sets from $\mathcal{S}$.
- 19 **end**
- 20 **return** (flag, Y)

1. For any $u \in C$ and $v \notin C$, if $(u, v) \in E_G$, then $u \rightarrow v$ is a directed edge in G^C .
2. For $u \in X_i$ and $v \notin C \cup X_1 \cup X_2 \cup \dots \cup X_i$, where $1 \leq i \leq l$, if $(u, v) \in E_G$, then $u \rightarrow v$ is a directed edge in G^C .
3. For $u, v \in X_i$, where $1 \leq i \leq l$, if $(u, v) \in E_G$, then $u - v$ is an undirected edge in G^C .
4. For $u, v \in C$, $u - v$ is an undirected edge in G^C .
5. For any $X_i \in \mathcal{L}$, every undirected connected component of $G[X_i]$ is an element of $\mathcal{C}_G(C)$.

Proof of Lemma 2.3.8. Proof of Item 1: For any edge $u - v \in E_G$, if $u \in C$ and $v \notin C$, then from Definition 1.1.11, for any AMO represented by a topological ordering that starts with C , $u \rightarrow v$ is a directed edge in the AMO. This further implies that for any edge $u - v \in E_G$, if $u \in C$ and $v \notin C$, then $u \rightarrow v$ is a directed edge in G^C , as G^C is the union

of all the AMOs of G that can be represented by an LBFS ordering that starts with C . This proves Item 1.

Proof of Item 2: We prove Item 2 of Lemma 2.3.8 using induction on the size of $\mathcal{L}$.

Base Case: $|\mathcal{L}| = 0$. In this case, Item 2 of Lemma 2.3.8 is vacuously true.

Assume that Item 2 holds when $|\mathcal{L}| = l \geq 0$. We now show that Item 2 holds even when $|\mathcal{L}| = l + 1$.

Let, at some stage of an iteration of the while loop in Algorithm 1, $\mathcal{L} = \{X_1, X_2, \dots, X_l, X_{l+1}\}$. By the induction hypothesis, for an edge $u - v \in E_G$, if $u \in X_i$ and $v \notin C \cup X_1 \cup X_2 \cup \dots \cup X_i$, then $u \rightarrow v$ is in G^C when $i \leq l$. Now, suppose there exists an edge $u - v \in E_G$ such that $u \in X_{l+1}$ and $v \notin C \cup X_1 \cup X_2 \cup \dots \cup X_{l+1}$. Since, at Line 1 of Algorithm 1, u and v are in the same set $V \setminus C$ of $\mathcal{S}$, but now they are in two different sets of $\mathcal{S}$ (with u belonging to the set X_{l+1} of $\mathcal{S}$ and v not in X_{l+1}), there must exist a vertex $x \in C \cup X_1 \cup X_2 \cup \dots \cup X_l$ such that $x - u \in E_G$ and $x - v \notin E_G$, causing u and v to move to two different sets in $\mathcal{S}$ (at Line 17 of Algorithm 1) when x is picked at Line 5 of Algorithm 1. From Item 1 and the induction hypothesis, $x \rightarrow u$ is in G^C . This implies that all the AMOs represented by C have the edge $x \rightarrow u$. Consequently, all the AMOs represented by C must have the edge $u \rightarrow v$, as $v \rightarrow u$ would create a v-structure $x \rightarrow u \leftarrow v$ in the AMOs, which is a contradiction (from Definition 2.2.1, there cannot be a v-structure in an AMO of G because G does not contain a v-structure as it is an undirected graph). This proves Item 2.

Proof of Item 3: As stated earlier, Algorithm 1 is a background-aware version of the LBFS algorithm of Wienöbst et al. [2021]. From the construction of Algorithm 1, the second component of the output does not depend on $\mathcal{K}$, and it is the same as the output of the LBFS algorithm of Wienöbst et al. [2021]. When C is an input of the LBFS algorithm of Wienöbst et al. [2021], it returns the undirected connected components of G^C such that, for two nodes in the same undirected connected component of G^C , if $u - v \in E_G$, then $u - v \in G^C$ (from line 9 of Algorithm 1 of Wienöbst et al. [2021], the undirected connected components of G^C are induced subgraphs of G). This implies that the second component of the output of Algorithm 1 is $\mathcal{C}_G(C)$. At Line 8 of Algorithm 1, $G[X_i]$ is added to Y , which is returned as the second component of the output at Line 20. This implies each $G[X_i]$ is an undirected connected component of G^C . This further

implies that for each $u, v \in X_i$, if $u - v \in E_G$, then $u - v$ is an undirected edge in G^C . This completes the proof of Item 3.

Proof of Item 4: From Lemma 2.3.2, there exists an AMO of G that can be represented by C . Let α be an AMO of G , and τ a topological ordering of α such that τ starts with C . Let $u, v \in C$. Then, $u - v \in E_G$, because C is a clique of G . Without loss of generality, assume that $u \rightarrow v \in \alpha$. We construct another AMO, β , such that β is also represented by C , and $v \rightarrow u \in \beta$. As G^C is the union of AMOs that are represented by C , the existence of α and β implies $u - v \in G^C$.

Since τ is a topological ordering of α , and $u \rightarrow v \in \alpha$, it follows that u comes before v in τ . Let us construct an ordering, τ' , by swapping the positions of u and v in τ . This implies that τ' also starts with C . Now, construct a DAG, β , with skeleton G , such that for $x - y \in E_G$, $x \rightarrow y \in \beta$ if x comes before y in τ' . By construction, β is acyclic, $v \rightarrow u \in \beta$ (as v comes before u in τ' , and $u - v$ is an edge in G), and τ' is a topological ordering of β . We now show that β does not contain any v-structure. Suppose that β contains a v-structure $a \rightarrow b \leftarrow c$. Since τ' is a topological ordering of β , it follows that a and c come before b in τ' . There are two possibilities: either $b \in C$, or $b \notin C$. We show that, in both cases, $a \rightarrow b \leftarrow c$ is not a v-structure in β .

Suppose $b \in C$. Since τ' also starts with C , if b belongs to C , then a and c must also belong to C , as they come before b . This implies $a - c \in E_G$, which means either $a \rightarrow c \in \beta$ (if a comes before c in τ') or $c \rightarrow a \in \beta$ (if c comes before a in τ'). In both cases, $a \rightarrow b \leftarrow c$ is not an induced subgraph of β . Therefore, $a \rightarrow b \leftarrow c$ is not a v-structure in β (from Definition 1.1.8).

Now, suppose $b \notin C$. From the construction, both τ and τ' start with C , and the only difference between τ and τ' is the ordering of u and v . Since u and v are in C and $b \notin C$, and since a and c come before b in τ' , it follows that a and c also come before b in τ . This implies that $a \rightarrow b \leftarrow c$ is a v-structure in α as well, since τ is a topological ordering of α . However, this is a contradiction, as α is an AMO of G . Hence, β does not have any v-structure. This further implies that β is an AMO of G such that $v \rightarrow u \in \beta$. Consequently, $u - v \in G^C$. This proves Item 4.

Proof of Item 5: From Item 1 and Item 2, all the edges with only one endpoint in X_l are directed in G^C . Moreover, from Item 3, all the edges with both endpoints in X_l are

undirected in G^C . This further implies that all the undirected connected components of $G[X_i]$ are undirected connected components of G^C . $\square$

The following lemma encapsulates the correctness of Algorithm 1.

Lemma 2.3.9. *Let G be a UCCG, C a maximal clique of G , and $\mathcal{K}$ be a background knowledge of G . For the input G , C , and $\mathcal{K}$, if G^C is not $\mathcal{K}$ -consistent, Algorithm 1 outputs $(0, \mathcal{C}_G(C))$ at Line 20; otherwise, it returns $(1, \mathcal{C}_G(C))$ at Line 20.*

Proof of Lemma 2.3.9. First, recall that Algorithm 1 is a background-aware version of the modified LBFS algorithm of Wienöbst et al. [2021]. As discussed earlier, we do not change any steps from the LBFS algorithm of Wienöbst et al. [2021]; the only modifications are: (a) the introduction of `flag` at Line 2, which checks the $\mathcal{K}$ -consistency of G^C ; (b) lines 11-13, which update the value of `flag`; and (c) the inclusion of `flag` in the algorithm's output alongside $\mathcal{C}_G(C)$.

The output of Algorithm 1 consists of two components: the first is the value of `flag`, and the second is $\mathcal{C}_G(C)$, the result from the LBFS algorithm by Wienöbst et al. [2021]. Thus, the only thing we need to verify is that `flag` is set to 1 if G^C is $\mathcal{K}$ -consistent, and 0 if G^C is not $\mathcal{K}$ -consistent. This is equivalent to showing that `flag` equals 0 if and only if G^C is not $\mathcal{K}$ -consistent (since `flag` can only be 0 or 1).

Assume G^C is not $\mathcal{K}$ -consistent. Then, from the definition of $\mathcal{K}$ -consistency (Definition 2.3.7), there must exist an edge $v \rightarrow u$ in G^C such that $u \rightarrow v \in \mathcal{K}$. From Lemma 2.3.8, if $v \rightarrow u \in G^C$, then either (a) $v \in C$ and $u \notin C$, or (b) at some iteration, when $\mathcal{L} = \{X_1, X_2, \dots, X_l\}$, $v \in X_l$ and $u \notin C \cup X_1 \cup X_2 \cup \dots \cup X_l$. In both cases, v must be picked before u at Line 5, as v is in a set that precedes the set containing u . When v is picked, the algorithm finds the edge $u \rightarrow v$ and updates `flag` to 0 at Line 11. Once set to 0, `flag` remains 0 for the rest of the algorithm. This shows that if G^C is not $\mathcal{K}$ -consistent, the value of `flag` is 0.

Now suppose the value of `flag` returned by the algorithm is 0. At Line 2, `flag` is initialized to 1. If the returned value of `flag` is 0, then there must have been an edge $u \rightarrow v \in \mathcal{K}$ (found at Line 11) that caused `flag` to be set to 0 at Line 12. Since $u \rightarrow v$ satisfied the condition at Line 11, it must be that at the iteration when v was picked at Line 5, u was neither in C nor in any set of $\mathcal{L}$. Additionally, v must have been in C (if

$\mathcal{L} = \emptyset$) or in X_l (if $\mathcal{L} = \{X_1, X_2, \dots, X_l\}$ with $l \geq 1$). By Lemma 2.3.8, $v \rightarrow u$ must be a directed edge in G^C , making G^C inconsistent with $\mathcal{K}$ since $u \rightarrow v \in \mathcal{K}$ (Definition 2.3.7). This proves that if `flag` is 0, then G^C is not $\mathcal{K}$ -consistent. This completes the proof. $\square$

For any maximal clique C of G such that G^C is $\mathcal{K}$ -consistent, to compute the size of the set of $\mathcal{K}$ -consistent AMOs of G that are canonically represented by C , we further partition the set based on the different permutations of C . For this, we map each AMO in the set with a unique permutation $\pi(C)$ of C . We use the following observation to do this.

Observation 2.3.10. *Let G be a UCCG, and α an AMO of G that is represented by a maximal clique C of G . Then, there exists a unique permutation $\pi(C)$ of C that represents α .*

Proof of Observation 2.3.10. Let $\pi_1(C)$ and $\pi_2(C)$ be two distinct permutations of C . Then, there must exist two vertices $u, v \in C$ such that u comes before v in $\pi_1(C)$, and v comes before u in $\pi_2(C)$. Suppose α is represented by both $\pi_1(C)$ and $\pi_2(C)$. Since u and v are members of the same clique, there exists an edge $u - v \in E_G$. Because the AMO is represented by both $\pi_1(C)$ and $\pi_2(C)$, it would imply that α contains both $u \rightarrow v$ and $v \rightarrow u$, which is a contradiction. This implies that there exists a unique permutation $\pi(C)$ of C that represents α . $\square$

Observation 2.3.10 maps each AMO in the set $\text{AMO}(G, C, \mathcal{K})$ to a unique permutation $\pi(C)$ of C . The following definition formally defines such a permutation as a canonical representation of the AMO.

Definition 2.3.11 (Canonical Representation by $\pi(C)$). Let G be a UCCG, and $\mathcal{T} = (T, \beta, r)$ the rooted clique tree of G on which C_α is defined (Definition 2.3.3), for each AMO α of G . An AMO α is canonically represented by a permutation $\pi(C)$ of a maximal clique C if

1. $\pi(C)$ represents α (Definition 2.3.1), and
2. $C = C_\alpha$, i.e., α is canonically represented by C .

We denote the set of $\mathcal{K}$ -consistent AMOs of G that are canonically represented by $\pi(C)$ by $\text{AMO}(G, \pi(C), \mathcal{K})$. Then, Observation 2.3.10 implies that we can partition the

set of $\mathcal{K}$ -consistent AMOs that are canonically represented by C into $\text{AMO}(G, \pi(C), \mathcal{K})$ s. More formally,

$$|\text{AMO}(G, C, \mathcal{K})| = \sum_{\pi(C)} |\text{AMO}(G, \pi(C), \mathcal{K})| \quad (2.4)$$

To compute $|\text{AMO}(G, \pi(C), \mathcal{K})|$, we first go through the necessary and sufficient conditions for a permutation $\pi(C)$ of a maximal clique C of G to represent an AMO α of G .

Lemma 2.3.12. *Let G be a UCCG, and $\mathcal{T} = (T, \beta, r)$ be the rooted clique tree of G on which C_α is defined (Definition 2.3.3). For an AMO α of G , a permutation $\pi(C)$ of a maximal clique C of G canonically represents α if, and only if,*

1. $\pi(C)$ represents α , i.e., there exists a topological ordering of α that starts with $\pi(C)$, and
2. For all $S \in \text{FP}(C, \mathcal{T})$ (defined at Definition 2.3.3), $\pi(C)$ does not start with S .

Proof of Lemma 2.3.12. Let α be an AMO of G . We first prove $\rightarrow$ of Lemma 2.3.12. Let $\pi(C)$ be a permutation of a maximal clique C of G such that it obeys Items 1 and 2 of Lemma 2.3.12. We show that $\pi(C)$ canonically represents α .

From Item 1 of Lemma 2.3.12, α is represented by $\pi(C)$. This validates Item 1 of Definition 2.3.11. Also, from Item 2 of Lemma 2.3.12, α has a topological ordering τ that starts with $\pi(C)$. Since $\pi(C)$ is a permutation of C , this implies τ also starts with C . Therefore, Item 1 of Definition 2.3.3 is obeyed. Item 2 of Lemma 2.3.12 implies that Item 2 of Definition 2.3.3 is also obeyed. Therefore, from Definition 2.3.3, $C = C_\alpha$. This validates Item 2 of Definition 2.3.11. Therefore, from Definition 2.3.11, $\pi(C)$ canonically represents α .

We now prove $\leftarrow$ of Lemma 2.3.12. Let a permutation $\pi(C)$ of a maximal clique C of G canonically represent α . We show that $\pi(C)$ obeys Items 1 and 2 of Lemma 2.3.12.

From Item 1 of Definition 2.3.11, α is represented by $\pi(C)$. This validates Item 1 of Lemma 2.3.12.

From Item 2 of Definition 2.3.11, $C = C_\alpha$, i.e., C canonically represents α (Definition 2.3.3). From Item 1 of Definition 2.3.3 and Observation 2.3.10, $\pi(C)$ is the unique permutation of C that represents α . And, from Item 2 of Definition 2.3.3, $\pi(C)$ cannot

start with an $S \in \text{FP}(C, \mathcal{T})$. This implies that $\pi(C)$ also obeys Item 2 of Lemma 2.3.12. This completes the proof. $\square$

Based on Lemma 2.3.12, we define $(\mathcal{K}, \mathcal{T})$ -consistency for a permutation $\pi(C)$ to simplify our computation. This definition is one of the main new ingredients that allows us to extend the result of Wienöbst et al. [2021].

Definition 2.3.13 (($\mathcal{K}, \mathcal{T}$)-Consistency of a Permutation of a Maximal Clique). Let G be a UCCG, C a maximal clique in G , $\pi(C)$ a permutation of C , $\mathcal{K}$ be a background knowledge of G , and $\mathcal{T} = (T, \beta, r)$ be the rooted clique tree of G on which C_α is defined (Definition 2.3.3). $\pi(C)$ is said to be $(\mathcal{K}, \mathcal{T})$ -consistent if

1. $\pi(C)$ is $\mathcal{K}$ -consistent, i.e., for any edge $u \rightarrow v \in \mathcal{K}$ such that $u, v \in C$, u occurs before v in $\pi(C)$, and
2. no element of $\text{FP}(C, \mathcal{T})$ (Definition 2.3.3) is a prefix of $\pi(C)$.

Suppose $\pi(C)$ is not $\mathcal{K}$ -consistent, i.e., there exists an edge $u \rightarrow v \in \mathcal{K}$ such that $u, v \in C$, and v comes before u in $\pi(C)$. Then, all the AMOs α that are represented by $\pi(C)$ contain the edge $v \rightarrow u$, which implies that α is not $\mathcal{K}$ -consistent. This means that if $\pi(C)$ itself is not $\mathcal{K}$ -consistent, then all the AMOs represented by $\pi(C)$ are also not $\mathcal{K}$ -consistent, i.e., $|\text{AMO}(G, \pi(C), \mathcal{K})| = 0$.

Suppose $\pi(C)$ has a prefix in $\text{FP}(C, \mathcal{T})$. Then from Lemma 2.3.12, there does not exist an AMO α of G such that α is canonically represented by $\pi(C)$, implying $|\text{AMO}(G, \pi(C), \mathcal{K})| = 0$.

The above discussion yields the following observation.

Observation 2.3.14. *If $\pi(C)$ is not $(\mathcal{K}, \mathcal{T})$ -consistent, then there exist no $\mathcal{K}$ -consistent AMOs of G that can be canonically represented by $\pi(C)$, i.e., $|\text{AMO}(G, \pi(C), \mathcal{K})| = 0$.*

We thus focus on only those permutations $\pi(C)$ of C that are $(\mathcal{K}, \mathcal{T})$ -consistent. The main ingredient towards this end is the following recursive formula.

Lemma 2.3.15. *Let G^C be $\mathcal{K}$ -consistent. Then, for any $(\mathcal{K}, \mathcal{T})$ -consistent permutation $\pi(C)$ of C ,*

$$|AMO(G, \pi(C), \mathcal{K})| = \prod_{H \in \mathcal{C}_G(C)} |AMO(H, \mathcal{K}[H])|. \quad (2.5)$$

For the proof of Lemma 2.3.15, we need the following observation of [Wienöbst et al. \[2021\]](#).

Observation 2.3.16 ([Wienöbst et al. \[2021\]](#), Proposition 1). *For each permutation $\pi(C)$ of a maximal clique C of G , all edges of $G^{\pi(C)}$ coincide with the edges of G^C , excluding the edges connecting the vertices in C . In particular, $\mathcal{C}_G(\pi(C)) = \mathcal{C}_G(C)$.*

Proof of Lemma 2.3.15. Let $\mathcal{C}_G(C) = \{H_1, H_2, \dots, H_l\}$ (the undirected connected components in $\mathcal{C}_G(C)$ are in the same order as we get from Algorithm 1), and let $f : AMO(G, \pi(C), \mathcal{K}) \rightarrow AMO(H_1, \mathcal{K}[V_{H_1}]) \times AMO(H_2, \mathcal{K}[V_{H_2}]) \times \dots \times AMO(H_l, \mathcal{K}[V_{H_l}])$ be a function such that $f(\alpha) = (\alpha_1, \alpha_2, \dots, \alpha_l)$ if, for all $1 \leq i \leq l$, $\alpha[V_{H_i}] = \alpha_i$. To prove Equation (2.5), we show that the function f is bijective, i.e., for each $\alpha \in AMO(G, \pi(C), \mathcal{K})$, $f(\alpha)$ is well-defined, and for each $(\alpha_1, \alpha_2, \dots, \alpha_l) \in AMO(H_1, \mathcal{K}[V_{H_1}]) \times AMO(H_2, \mathcal{K}[V_{H_2}]) \times \dots \times AMO(H_l, \mathcal{K}[V_{H_l}])$, there exists a unique $\alpha \in AMO(G, \pi(C), \mathcal{K})$ such that $f(\alpha) = (\alpha_1, \alpha_2, \dots, \alpha_l)$.

We start with the first part, i.e., for each $\alpha \in AMO(G, \pi(C), \mathcal{K})$, $f(\alpha)$ is well-defined. Let $\alpha \in AMO(G, \pi(C), \mathcal{K})$. For each $1 \leq i \leq l$, $\alpha[V_{H_i}]$ is unique. We show that $\alpha[V_{H_i}] \in AMO(H_i, \mathcal{K}[V_{H_i}])$. Since G is an undirected graph, G does not have a v-structure. Therefore, from Definition 2.2.1, the skeleton of α is G , and α does not have a v-structure. This implies that $\alpha[V_{H_i}]$ is of skeleton $G[V_{H_i}] (= H_i)$, and $\alpha[V_{H_i}]$ also does not have a v-structure. From Definition 2.2.1, this implies that $\alpha[V_{H_i}]$ is an AMO of H_i . We now show that $\alpha[V_{H_i}]$ is $\mathcal{K}[V_{H_i}]$ -consistent.

Suppose $\alpha[V_{H_i}]$ is not $\mathcal{K}[V_{H_i}]$ -consistent. Then, there must exist an edge $u \rightarrow v \in \mathcal{K}[V_{H_i}]$ such that $v \rightarrow u \in \alpha[V_{H_i}]$. However, this would imply that α is not $\mathcal{K}$ -consistent either, which leads to a contradiction, since if $u \rightarrow v \in \mathcal{K}[V_{H_i}]$, then $u \rightarrow v \in \mathcal{K}$, and if $v \rightarrow u \in \alpha[V_{H_i}]$, then $v \rightarrow u \in \alpha$. This contradiction implies that $\alpha[V_{H_i}]$ must be $\mathcal{K}[V_{H_i}]$ -consistent.

This shows that for each $\alpha \in AMO(G, \pi(C), \mathcal{K})$, $f(\alpha)$ is well-defined.

We now prove the second part, i.e., for each $(\alpha_1, \alpha_2, \dots, \alpha_l) \in AMO(H_1, \mathcal{K}[V_{H_1}]) \times AMO(H_2, \mathcal{K}[V_{H_2}]) \times \dots \times AMO(H_l, \mathcal{K}[V_{H_l}])$, there exists a unique $\alpha \in AMO(G, \pi(C), \mathcal{K})$

such that $f(\alpha) = (\alpha_1, \alpha_2, \dots, \alpha_l)$. For each $1 \leq i \leq l$, let $\alpha_i \in \text{AMO}(H_i, \mathcal{K}[V_{H_i}])$. We first construct an $\alpha \in \text{AMO}(G, \pi(C), \mathcal{K})$ such that $f(\alpha) = (\alpha_1, \alpha_2, \dots, \alpha_l)$. Later we show that α is unique.

To construct α , we first construct two auxiliary DAGs: α_{G^C} and $\alpha_{\pi(C)}$. α_{G^C} is the directed subgraph of G^C . From Lemma 2.3.8, directed edges in α_{G^C} are of the following two kinds: (a) for any $u \in C$ and $v \notin C$, if $(u, v) \in E_G$, then $u \rightarrow v \in \alpha_{G^C}$ (by Item 1 of Lemma 2.3.8), and (b) for $u \in H_i$ and $v \in H_j$ such that $i < j$, if $(u, v) \in E_G$, then $u \rightarrow v \in \alpha_{G^C}$ (by Item 2 of Lemma 2.3.8).

We construct $\alpha_{\pi(C)}$ as an AMO of $G[C]$ such that, for $u, v \in C$, if u precedes v in $\pi(C)$, then $u \rightarrow v \in \alpha_{\pi(C)}$.

Let $\alpha = \alpha_{\pi(C)} \cup \alpha_{G^C} \cup \bigcup_i \alpha_i$. From the construction of α , for all $1 \leq i \leq l$, $\alpha[V_{H_i}] = \alpha_i$. This implies, from the definition of the function f , that if $\alpha \in \text{AMO}(G, \pi(C), \mathcal{K})$, then $f(\alpha) = (\alpha_1, \alpha_2, \dots, \alpha_l)$. Therefore, we only need to show that $\alpha \in \text{AMO}(G, \pi(C), \mathcal{K})$.

We first show that α is an AMO of G . This requires showing the following: (i) the skeleton of α is G , (ii) α does not contain a v-structure, and (iii) α is acyclic.

For an edge $u - v \in E_G$, one of the following occurs: (a) $u, v \in C$, (b) $u \in C$ and $v \notin C$, (c) u and v belong to some H_i in $\mathcal{C}_G(C)$, or (d) for some $1 \leq i < j \leq l$, u is in H_i and v is in H_j , where H_i and H_j are in $\mathcal{C}_G(C)$. From the construction of $\alpha_{\pi(C)}$, $u - v \in \text{skeleton}(\alpha_{\pi(C)})$ if and only if $u, v \in C$ and $u - v \in E_G$. From the construction of G^C , $u - v \in \text{skeleton}(\alpha_{G^C})$ if and only if $u - v \in E_G$ and either $u \in C$ and $v \notin C$, or for some $1 \leq i < j \leq l$, u is in H_i and v is in H_j , where H_i and H_j are in $\mathcal{C}_G(C)$. From the construction of α_i , $u - v \in \text{skeleton}(H_i)$ if and only if u and v belong to H_i in $\mathcal{C}_G(C)$. From the construction of α , $\text{skeleton}(\alpha) = \text{skeleton}(\alpha_{\pi(C)}) \cup \text{skeleton}(\alpha_{G^C}) \cup \bigcup_i \text{skeleton}(\alpha_i)$. This implies the skeleton of α is G .

Suppose $u \rightarrow v \leftarrow w$ is a v-structure in α . Since G is the skeleton of α , $u - v, v - w \in E_G$. From the construction of α_{G^C} , for $x \in C$ and $y \notin C$, if $x - y \in E_G$, then $x \rightarrow y \in G^C$, implying $x \rightarrow y \in \alpha$. This further implies that v cannot belong to C ; otherwise, u and w would also belong to C . Since C is a clique in G , the skeleton of α , $u \rightarrow v \leftarrow w$ cannot be a v-structure in α .

Suppose $v \notin C$. Then v must be a node in some H_i in $\mathcal{C}_G(C)$. Again, from the construction of α_{G^C} , for $1 \leq i < j \leq l$, for $x \in V_{H_i}$ and $y \in V_{H_j}$, if $x - y \in E_G$, then $x \rightarrow$

$y \in G_C$, implying $x \rightarrow y \in \alpha$. This implies u and w belong to $V_C \cup V_{H_1} \cup V_{H_2} \cup \dots \cup V_{H_i}$. Since $\alpha[V_{H_i}]$ is an AMO, if $u \rightarrow v \leftarrow w$ is a V-structure in α , then at least one node among u and w belongs to $V_C \cup V_{H_1} \cup V_{H_2} \cup \dots \cup V_{H_{i-1}}$.

Suppose u belongs to H_i and w is not in H_i . This implies u and v are in the same undirected connected component H_i of $\mathcal{C}_G(C)$ and, in Algorithm 1, w has been added to τ before u and v . This further implies $u - w \in E_G$, otherwise, at the iteration of the while loop when w was added to τ , u and v would have been moved to two different groups in Line 17 of Algorithm 1, implying u and w are not in the same connected component of $\mathcal{C}_G(C)$, contradicting our assumption. However, if $u - w \in E_G$, then $u \rightarrow v \leftarrow w$ cannot be a V-structure in α .

Similarly, there cannot be a case where w belongs to H_i and u is not in H_i .

Suppose both u and w are not in H_i . This implies u and w belong to $V_C \cup V_{H_1} \cup V_{H_2} \cup \dots \cup V_{H_{i-1}}$. This implies that u and w must be added to τ in Algorithm 1 before v is added to τ . As discussed earlier, Algorithm 1 is a background-knowledge version of the LBFS algorithm of Wienöbst et al. [2021]. Therefore, τ is an LBFS ordering of G (as stated in Algorithm 1 of Wienöbst et al. [2021]), implying $u - w \in E_G$ (as $u - v, w - v \in E_G$ and both u and w comes before v in the LBFS ordering τ). This implies $u \rightarrow v \leftarrow w$ is not a V-structure in α .

Thus, α does not contain a V-structure.

We now show that α is acyclic. For this, we construct a topological ordering of α . For each $1 \leq i \leq l$, let τ_i be a topological ordering of α_i . Then, $\tau = (\pi(C), \tau_1, \tau_2, \dots, \tau_l)$ is an ordering of the vertices in G . From the construction of $\alpha_{\pi(C)}$, $\pi(C)$ is a topological ordering of $\alpha_{\pi(C)}$. From Items 1 and 2 of Lemma 2.3.8 and the construction of α_{G_C} , τ is a topological ordering of α_{G_C} . Therefore, τ is a topological ordering of α , the union of graphs $\alpha_{\pi(C)}$, α_{G_C} , and the α_i s. This implies α is acyclic.

We now show that $\alpha \in \text{AMO}(G, \pi(C), \mathcal{K})$. Since τ starts with $\pi(C)$, from Definition 2.3.1, α is represented by $\pi(C)$. Since $\pi(C)$ is $(\mathcal{K}, \mathcal{T})$ -consistent, from Item 2 of Definition 2.3.13, $\pi(C)$ does not start with any $S \in \text{FP}(C, \mathcal{T})$. Therefore, from Lemma 2.3.12, $\pi(C)$ canonically represents α . That means $\alpha \in \text{AMO}(G, \pi(C), \mathcal{K})$.

We now prove the uniqueness of α . Suppose there exists an $\alpha' \in \text{AMO}(G, \pi(C), \mathcal{K})$ such that $f(\alpha') = (\alpha_1, \alpha_2, \dots, \alpha_l)$. By the definition of f , for all $1 \leq i \leq l$, $\alpha'[V_{H_i}] = \alpha_i$.

Furthermore, since $\pi(C)$ represents α' , $\alpha_{\pi(C)}$ must be an induced subgraph of α' . Since $\alpha' \in \text{AMO}(G, \pi(C), \mathcal{K})$, by Observation 2.3.16, directed edges in G^C are directed edges in α' . This implies that for all directed edges $u \rightarrow v \in \alpha$, we have $u \rightarrow v \in \alpha'$. Since both α and α' have the same skeleton, this implies $\alpha = \alpha'$. Thus, α is unique.

This proves the second part, i.e., for each $(\alpha_1, \alpha_2, \dots, \alpha_l) \in \text{AMO}(H_1, \mathcal{K}[V_{H_1}]) \times \text{AMO}(H_2, \mathcal{K}[V_{H_2}]) \times \dots \times \text{AMO}(H_l, \mathcal{K}[V_{H_l}])$, there exists a unique $\alpha \in \text{AMO}(G, \pi(C), \mathcal{K})$ such that $f(\alpha) = (\alpha_1, \alpha_2, \dots, \alpha_l)$.

This completes the proof of Lemma 2.3.15. $\square$

Note that the formula obtained in Lemma 2.3.15 depends only on the clique C and *not* on the permutation π of C (as long as π is itself $\mathcal{K}$ -consistent). This immediately implies that $|\text{AMO}(G, C, \mathcal{K})|$ is equal to

$$\text{the number of } (\mathcal{K}, \mathcal{T})\text{-consistent permutations of } C \times \prod_{H \in \mathcal{C}_G(C)} |\text{AMO}(H, \mathcal{K}[H])|. \quad (2.6)$$

This motivates us to count the $(\mathcal{K}, \mathcal{T})$ -consistent permutations of a maximal clique C in G . To do this, we define the following:

Definition 2.3.17. Let S be a set of vertices, $\mathcal{R} = \{R_1, R_2, \dots, R_l\}$ be a set such that $R_1 \subsetneq R_2 \subsetneq \dots \subsetneq R_l \subsetneq S$, and let $\mathcal{K} \subseteq S \times S$ be a set of directed edges. We define $\Phi(S, \mathcal{R}, \mathcal{K})$ as the number of $\mathcal{K}$ -consistent permutations of S that do not have a prefix in $\mathcal{R}$.

Equations (2.3) and (2.4) and Observation 2.3.14, along with Lemma 2.3.15 and the discussion following it, provide us with the following recursion.

Lemma 2.3.18. Let G be a UCCG, $\mathcal{K}$ be a background knowledge of G , and $\mathcal{T} = (T, \beta, r)$ be the rooted clique tree of G on which C_α has been defined (Definition 2.3.3). Then,

$$|\text{AMO}(G, \mathcal{K})| = \sum_{\substack{C \in \Pi(G) \\ G^C \text{ is } \mathcal{K}\text{-consistent}}} \Phi(C, FP(C, \mathcal{T}), \mathcal{K}[C]) \times \prod_{H \in \mathcal{C}_G(C)} |\text{AMO}(H, \mathcal{K}[H])|. \quad (2.7)$$

Proof of Lemma 2.3.18. From Equation (2.3), $|\text{AMO}(G, \mathcal{K})|$ equals the summation of $|\text{AMO}(G, C, \mathcal{K})|$ over all maximal cliques C of G for which G^C is $\mathcal{K}$ -consistent. From

Equation (2.4), for any maximal clique C of G , if G^C is $\mathcal{K}$ -consistent, then $|\text{AMO}(G, C, \mathcal{K})|$ equals the summation of $|\text{AMO}(G, \pi(C), \mathcal{K})|$ over all permutations $\pi(C)$ of C . From Equation (2.6), $|\text{AMO}(G, \pi(C), \mathcal{K})|$ equals the product of the number of $(\mathcal{K}, \mathcal{T})$ -consistent permutations of C and $\prod_{H \in \mathcal{C}_G(C)} |\text{AMO}(H, \mathcal{K}[H])|$. From Definitions 2.3.13 and 2.3.17, the number of $(\mathcal{K}, \mathcal{T})$ -consistent permutations of C equals $\Phi(C, \text{FP}(C, \mathcal{T}), \mathcal{K}[C])$ (the precondition in Definition 2.3.17 for $\text{FP}(C, \mathcal{T})$ is satisfied by Lemma 2.3.19). This implies that $|\text{AMO}(G, \mathcal{K})|$ equals the right side expression of Equation (2.7). This completes the proof of Lemma 2.3.18. $\square$

Lemma 2.3.18 solves our counting problem. The precondition of $\Phi(S, \mathcal{R}, \mathcal{K})$ in Definition 2.3.17, that $\mathcal{R} = \{R_1, R_2, \dots, R_l\}$ satisfies the property $R_1 \subsetneq R_2 \subsetneq \dots \subsetneq R_l \subsetneq S$, is satisfied at the beginning of the recursion, i.e., when $\mathcal{R} = \text{FP}(C, \mathcal{T})$, by Lemma 2.3.19, which follows from the standard clique intersection property of clique trees of chordal graphs.

Lemma 2.3.19 (Wienöbst et al. [2021], Lemma 7). *We can order the elements of $\text{FP}(C, \mathcal{T})$ as $X_1 \subsetneq X_2 \subsetneq \dots \subsetneq X_l \subsetneq C$.*

The following lemma provides a recursive method to compute $\Phi(S, \mathcal{R}, \mathcal{K})$. The precondition of $\Phi(S, \mathcal{R}, \mathcal{K})$ is preserved throughout the recursion described in the lemma.

Lemma 2.3.20. *Let S be a clique, and let $\mathcal{K} \subseteq S \times S$ be a set of directed edges. Let $\mathcal{R} = \{R_1, R_2, \dots, R_l\}$, where $l \geq 1$, be such that $R_1 \subsetneq R_2 \subsetneq \dots \subsetneq R_l \subsetneq S$. Then:*

1. $\Phi(S, \emptyset, \mathcal{K}) = \frac{|S|!}{|V_{\mathcal{K}}|!} \times \Psi(V_{\mathcal{K}}, \mathcal{K})$, where $\Psi(V_{\mathcal{K}}, \mathcal{K})$ is the number of $\mathcal{K}$ -consistent permutations of $V_{\mathcal{K}}$ ($V_{\mathcal{K}}$ is the set of endpoints of edges in $\mathcal{K}$). (Example: Suppose $S = \{1, 2, 3, 4, 5\}$, and $\mathcal{K} = \{1 \rightarrow 2, 2 \rightarrow 3\}$. Then, $V_{\mathcal{K}} = \{1, 2, 3\}$. There exists only one permutation, $(1, 2, 3)$, of $V_{\mathcal{K}}$ that is $\mathcal{K}$ -consistent, i.e., $\Psi(V_{\mathcal{K}}, \mathcal{K}) = 1$. This implies $\Phi(S, \emptyset, \mathcal{K}) = 20$.)
2. If there exists an edge $u \rightarrow v \in \mathcal{K}$ such that $u \in S \setminus R_l$ and $v \in R_l$, then $\Phi(S, \mathcal{R}, \mathcal{K}) = \Phi(S, \mathcal{R} - \{R_l\}, \mathcal{K})$.

Algorithm 2: Valid-Perm($S, \mathcal{R}, \mathcal{K}$)

Input : A clique $S, \mathcal{R} = \{R_1, R_2, \dots, R_l\}$ such that $R_1 \subsetneq R_2 \subsetneq \dots \subsetneq R_l \subsetneq S$,
and background knowledge $\mathcal{K} \subseteq S \times S$.

Output: $\Phi(S, \mathcal{R}, \mathcal{K})$.

```
1 if  $\mathcal{R} = \emptyset$  then
2   | return  $\frac{|S|!}{|V_{\mathcal{K}}|!} \cdot \Psi(V_{\mathcal{K}}, \mathcal{K})$ 
3 end
4  $\text{sum} \leftarrow \text{Valid-Perm}(S, \mathcal{R} - \{R_l\}, \mathcal{K})$ 
5 if  $\{(u, v) : u \rightarrow v \in \mathcal{K}, v \in R_l \text{ and } u \notin R_l\} \neq \emptyset$  then
6   | return  $\text{sum}$ 
7 end
8 return
    $\text{sum} - \text{Valid-Perm}(R_l, \mathcal{R} - \{R_l\}, \mathcal{K}[R_l]) \times \text{Valid-Perm}(S \setminus R_l, \emptyset, \mathcal{K}[S \setminus R_l])$ 
```

3. If there does not exist an edge $u \rightarrow v \in \mathcal{K}$ such that $u \in S \setminus R_l$ and $v \in R_l$, then

$$\Phi(S, R, \mathcal{K}) = \Phi(S, R - \{R_l\}, \mathcal{K}) - \Phi(R_l, R - \{R_l\}, \mathcal{K}[R_l]) \times \Phi(S \setminus R_l, \emptyset, \mathcal{K}[S \setminus R_l]).$$

Proof of Lemma 2.3.20. Proof of Item 1: If $R = \emptyset$, then a permutation $\pi(S)$ of S is $\mathcal{K}$ -consistent if the ordering of nodes in $V_{\mathcal{K}}$ in $\pi(S)$ is $\mathcal{K}$ -consistent. $\Psi(V_{\mathcal{K}}, \mathcal{K})$ gives the number of $\mathcal{K}$ -consistent permutations of $V_{\mathcal{K}}$. For each ordering π_1 of the nodes in $V_{\mathcal{K}}$, the number of permutations $\pi(S)$ of S such that the ordering of the nodes in $V_{\mathcal{K}}$ in $\pi(S)$ follows π_1 is $\frac{|S|!}{|V_{\mathcal{K}}|!}$. This completes the proof of Item 1.

Proof of Item 2: If there exists an edge $u \rightarrow v \in \mathcal{K}$ such that $u \in S \setminus R_l$ and $v \in R_l$, then no $\mathcal{K}$ -consistent permutation of S exists that starts with R_l . Therefore, in that case, $\Phi(S, R, \mathcal{K}) = \Phi(S, R - \{R_l\}, \mathcal{K})$.

Proof of Item 3: If there does not exist an edge $u \rightarrow v \in \mathcal{K}$ such that $u \in S \setminus R_l$ and $v \in R_l$, then one way to compute $\Phi(S, R, \mathcal{K})$ is to first compute the number of $\mathcal{K}$ -consistent permutations of S that do not start with $R_1, R_2, \dots, R_{l-1}$, i.e., $\Phi(S, R - \{R_l\}, \mathcal{K})$. However, $\Phi(S, R - \{R_l\}, \mathcal{K})$ also counts the $\mathcal{K}$ -consistent permutations of S that start with R_l but not with any R_i for $1 \leq i < l$. We subtract such permutations from $\Phi(S, R - \{R_l\}, \mathcal{K})$. To construct a $\mathcal{K}$ -consistent permutation of S that starts with R_l but does not start with any R_i , $1 \leq i < l$, we first need to construct a $\mathcal{K}$ -consistent permutation of R_l that does not start with any R_i , $1 \leq i < l$, and then construct a

$\mathcal{K}$ -consistent permutation of the remaining vertices of S . Since there is no edge in $\mathcal{K}$ from $S \setminus R_l$ to R_l , we can independently choose such a $\mathcal{K}$ -consistent permutation of R_l and a $\mathcal{K}$ -consistent permutation of $S \setminus R_l$. This implies that the number of $\mathcal{K}$ -consistent permutations of S that start with R_l and not with any R_i , $1 \leq i < l$, is $\Phi(R_l, R - \{R_l\}, \mathcal{K}[R_l]) \times \Phi(S \setminus R_l, \emptyset, \mathcal{K}[S \setminus R_l])$. $\square$

Algorithm 2 implements Lemma 2.3.20 to compute $\Phi(S, \mathcal{R}, \mathcal{K})$, and its correctness, given below, is an easy consequence of Lemma 2.3.20.

Observation 2.3.21. For input S , $\mathcal{R} = \{R_1, R_2, \dots, R_l\}$, and $\mathcal{K}$, where $R_1 \subsetneq R_2 \subsetneq \dots \subsetneq R_l \subsetneq S$, and $\mathcal{K} \subseteq S \times S$, Algorithm 2 returns $\Phi(S, \mathcal{R}, \mathcal{K})$.

Proof of Observation 2.3.21. If $R = \emptyset$, then from Item 1 of Lemma 2.3.20, $\Phi(S, R, \mathcal{K}) = \frac{|S|!}{|V_{\mathcal{K}}|!} \times \Psi(V_{\mathcal{K}}, \mathcal{K})$. Line 2 of Algorithm 2 returns the same.

If $R = \{R_1, R_2, \dots, R_l\} \neq \emptyset$, and there exists an edge $u \rightarrow v \in \mathcal{K}$ such that $u \in S \setminus R_l$ and $v \in R_l$, then from Item 2 of Lemma 2.3.20, $\Phi(S, R, \mathcal{K}) = \Phi(S, R - R_l, \mathcal{K})$. Line 6 of Algorithm 2 returns the same.

If $R = \{R_1, R_2, \dots, R_l\} \neq \emptyset$, and there does not exist an edge $u \rightarrow v \in \mathcal{K}$ such that $u \in S \setminus R_l$ and $v \in R_l$, then from Item 3 of Lemma 2.3.20, $\Phi(S, R, \mathcal{K}) = \Phi(S, R - R_l, \mathcal{K}) - (\Phi(R_l, R - \{R_l\}, \mathcal{K}[R_l]) \times \Phi(S \setminus R_l, \emptyset, \mathcal{K}[S \setminus R_l]))$. Line 8 of Algorithm 2 returns the same. $\square$

We now construct Algorithm 3, which computes $|\text{AMO}(G, \mathcal{K})|$. Algorithm 3 implements Lemma 2.3.18 and utilizes memoization to avoid recomputations.

Theorem 2.3.22. For a UCCG G and a background knowledge $\mathcal{K}$ of G , Algorithm 3 returns $|\text{AMO}(G, \mathcal{K})|$.

Proof of Theorem 2.3.22. At Line 4, Algorithm 3 constructs a rooted clique tree $\mathcal{T} = (T, \beta, r)$ of G that is used to define C_α (Definition 2.3.3). Lines 5-8 handle the special case where G is a clique. If $\beta(r) = V_G$ (at line 5), then G is a clique. In this case, for each AMO of G , there exists a unique topological ordering, and each permutation of V_G corresponds to a unique AMO of G . This implies that the number of $\mathcal{K}$ -consistent AMOs of G equals $\Phi(V_G, \emptyset, \mathcal{K})$, which is the number of $\mathcal{K}$ -consistent permutations of V_G . Lines 5-8 implement this.

Algorithm 3: $\text{count}(G, \mathcal{K}, \text{memo})$ (modification of an algorithm of [Wienöbst et al. \[2021\]](#))

Input : A UCCG G , and a background knowledge $\mathcal{K}$ of G .
Output: $|\text{AMO}((G, \mathcal{K}))|$.

```

1 if  $G \in \text{memo}$  then
2   | return  $\text{memo}[G]$ 
3 end
4  $\mathcal{T} = (T, \beta, r) \leftarrow$  a rooted clique tree of  $G$ 
5 if  $\beta(r) = V_G$  then
6   |  $\text{memo}[G] = \Phi(V_G, \emptyset, \mathcal{K})$ 
7   | return  $\text{memo}[G]$ 
8 end
9  $\text{sum} \leftarrow 0$ 
10  $Q \leftarrow$  queue with single element  $r$ 
11 while  $Q$  is not empty do
12   |  $v \leftarrow \text{pop}(Q)$ 
13   |  $\text{push}(Q, \text{children}(v))$ 
14   |  $C \leftarrow \beta(v)$ 
15   |  $(\text{flag}, \mathcal{L}) \leftarrow \text{LBFS}(G, C, \mathcal{K})$ 
16   | if  $\text{flag} = 1$  then
17     |  $\text{prod} \leftarrow 1$ 
18     | foreach  $H \in \mathcal{L}$  do
19       |  $\text{prod} = \text{prod} \times \text{count}(G[H], \mathcal{K}[H], \text{memo})$ 
20     | end
21     |  $\text{sum} = \text{sum} + \text{prod} \times \Phi(C, \text{FP}(C, \mathcal{T}), \mathcal{K}[C])$ 
22   | end
23 end
24  $\text{memo}[G] = \text{sum}$ 
25 return  $\text{sum}$ 

```

For the general case (when G is not a clique), we implement Lemma 2.3.18. From Definition 1.1.16, for each maximal clique C in G , there exists a node v in T such that $C = \beta(v)$. We create a queue Q at Line 10 that initially has only one element, the root node of T . Since T is a tree, each node of T is pushed to the queue exactly once due to the run of lines 12-13 in the while loop.

From the definition of clique tree decomposition (Definition 1.1.16), for each node v in T , $\beta(v)$ is a maximal clique in G , and for each maximal clique C in G , there exists a node v in T such that $\beta(v) = C$. During the run of the while loop (lines 11-23), a node v is popped from the queue (Line 12). Let $C = \beta(v)$ be the maximal clique corresponding to v (Line 14). This implies for each maximal clique C of G , we run lines 15-22.

At Line 15, we call Algorithm 1 (our LBFS-algorithm) for input G , C , and $\mathcal{K}$. If the first component of the output of Algorithm 1 is 0, then G^C is not $\mathcal{K}$ -consistent (from Lemma 2.3.9), implying that $|\{\alpha : \alpha \in \text{AMO}(G, \pi(C), \mathcal{K}) \text{ and } C = C_\alpha\}| = 0$. Hence, we skip lines 17-22 if the first component of $\text{LBFS}(G, C, \mathcal{K})$ is 0. If the first component $\text{LBFS}(G, C, \mathcal{K})$ is 1, then G^C is $\mathcal{K}$ -consistent. In this case, at lines 18-20, we compute $\prod_{H \in \mathcal{C}_G(C)} |\text{AMO}(H, \mathcal{K}[H])|$ by recursively calling Algorithm 3. At Line 21, we compute $|\{\alpha : \alpha \in \text{AMO}(G, \mathcal{K}) \text{ and } C = C_\alpha\}|$ using the discussion following Lemma 2.3.15. At the end of the while loop (Line 23), the variable `sum` has the value $\sum_{C: G^C \text{ is } \mathcal{K}\text{-consistent}} \Phi(C, \text{FP}(C, \mathcal{T}), \mathcal{K}[C]) \times \prod_{H \in \mathcal{C}_G(C)} |\text{AMO}(H, \mathcal{K}[H])|$, i.e., `sum` equals $|\text{AMO}(G, \mathcal{K})|$ (from Lemma 2.3.18). This completes the proof. $\square$

2.4 Time Complexity Analysis

In this section, we analyze the runtime of Algorithms 1 to 3.

Theorem 2.4.1. *For a UCCG G , a maximal clique C of G , and a background knowledge $\mathcal{K}$ of G , Algorithm 1 runs in linear time $O(|V_G| + |E_G|)$.*

Proof of Theorem 2.4.1. The “While” loop on lines 3-19 runs at most $|V_G|$ times. Using two additional arrays (one for checking whether v is in C or not, and another for checking whether v is in $\mathcal{L}$ or not), we can run lines 6-9 in $O(|X|)$ time (for appending X at line 7), and cumulatively in $O(|V_G| + |E_G|)$ time. Checking the existence of the edge $u \rightarrow v \in \mathcal{K}$ at line 11 takes $O(|\mathcal{K}|)$ time cumulatively, for all $v \in V_G$ (assuming an adjacency list representation of $\mathcal{K}$). Finding neighbors of v at line 15 takes $O(E_G)$ time cumulatively, for all $v \in V_G$. Partitioning each set at line 17 also takes $O(E_G)$ time cumulatively. Since the size of $\mathcal{K}$ can be at most E_G , this implies that the overall time complexity of Algorithm 1 is $O(|V_G| + |E_G|)$ (using the same technique as in the standard LBFS implementation by Rose et al. [1976]). $\square$

Similar to Wienöbst et al.’s count function, our count function (Algorithm 3) is also recursively called at most $2|\Pi(G)| - 1$ times, where $\Pi(G)$ is the set of maximal cliques of G . Our approach to compute the background-aware version of Φ (Algorithm 2) is similar to that of Wienöbst et al. [2021]. However, the difference in time complexity

arises from the higher cost of computing $\Phi(S, \emptyset, \mathcal{K})$ at Item 1, which is $O(1)$ for $\mathcal{K} = \emptyset$, as considered by [Wienöbst et al. \[2021\]](#).

Proposition 2.4.2. *Let G be a UCCG, and $\mathcal{K}$ be a background knowledge of G . The number of distinct UCCGs explored by the `count` function (as defined in Algorithm 3) is bounded by $2|\Pi(G)| - 1$.*

Proof of Proposition 2.4.2. Algorithm 3 calls itself at line 19 for each $H \in \mathcal{L}$, when `flag` = 1. The value of `flag` is always 1 when $\mathcal{K} = \emptyset$, and $\mathcal{L} = \mathcal{C}_G(C)$ (from Algorithm 1), which does not depend on $\mathcal{K}$. This further implies that the `count` function makes the maximum number of distinct recursive calls to itself when $\mathcal{K} = \emptyset$. But for $\mathcal{K} = \emptyset$, Algorithm 3 is the same as the Clique-Picking algorithm of [Wienöbst et al. \[2021\]](#), who showed (in Proposition 4 of [Wienöbst et al. \[2021\]](#)) that the number of these distinct recursive calls is at most $2|\Pi(G)| - 1$. This completes the proof. $\square$

Lemma 2.4.3. *For input S , $\mathcal{R} = \{R_1, R_2, \dots, R_l\}$, and $\mathcal{K}$, Algorithm 2 can be implemented using memoization to use $O(k! \cdot k^2 \cdot l^2)$ arithmetic operations, where k is the max-clique knowledge of $\mathcal{K}$ (assuming factorials of integers from 1 to $|V_G|$ are available for free).*

Proof of Lemma 2.4.3. Let us denote $\mathcal{R}_i = \{R_1, R_2, \dots, R_i\}$. For the computation of $\Phi(S, \mathcal{R}_l, \mathcal{K})$, from Items 2 and 3 of Lemma 2.3.20, we need to compute $\Phi(S, \emptyset, \mathcal{K})$, $\Phi(S \setminus R_1, \emptyset, \mathcal{K}[S \setminus R_1])$, $\Phi(S \setminus R_2, \emptyset, \mathcal{K}[S \setminus R_2])$, $\dots$, $\Phi(S \setminus R_l, \emptyset, \mathcal{K}[S \setminus R_l])$, $\Phi(S, \mathcal{R}_1, \mathcal{K})$, $\Phi(S, \mathcal{R}_2, \mathcal{K})$, $\dots$, $\Phi(S, \mathcal{R}_{l-1}, \mathcal{K})$, and for each $1 \leq i \leq l$, $\Phi(R_i, \emptyset, \mathcal{K}[R_i])$, $\Phi(R_i \setminus R_1, \emptyset, \mathcal{K}[R_i \setminus R_1])$, $\dots$, $\Phi(R_i \setminus R_{i-1}, \emptyset, \mathcal{K}[R_i \setminus R_{i-1}])$, and $\Phi(R_i, \mathcal{R}_1, \mathcal{K}[R_i])$, $\Phi(R_i, \mathcal{R}_2, \mathcal{K}[R_i])$, $\dots$, $\Phi(R_i, \mathcal{R}_{i-1}, \mathcal{K}[R_i])$.

Each computation of $\Phi(X, \emptyset, \mathcal{K}' = \mathcal{K}[X])$ takes $O(k! \cdot k^2)$ arithmetic operations. To see this, note that Item 1 of Lemma 2.3.20 gives $\Phi(X, \emptyset, \mathcal{K}') = \frac{|X|!}{|V_{\mathcal{K}'}|} \times \Psi(V_{\mathcal{K}'}, \mathcal{K}')$. From our assumption, the factorials are already precomputed. To compute $\Psi(V_{\mathcal{K}'}, \mathcal{K}')$, we check the $\mathcal{K}'$ -consistency of each permutation of $V_{\mathcal{K}'}$. The number of permutations of $V_{\mathcal{K}'}$ is $O(k!)$, since $|V_{\mathcal{K}'}| \leq k$ (max-clique knowledge), while the verification of whether a permutation of $V_{\mathcal{K}'}$ is $\mathcal{K}'$ -consistent or not takes $O(|\mathcal{K}'|)$ time. Since $|V_{\mathcal{K}'}| \leq k$, we have $|\mathcal{K}'| \leq k^2$. Thus, the computation of $\Psi(V_{\mathcal{K}'}, \mathcal{K}')$ takes $O(k^2 \cdot k!)$ arithmetic operations.

Since the number of required computations of the type $\Phi(X, \emptyset, \mathcal{K}' = \mathcal{K}[X])$ (as already listed above) is $O(l^2)$, their total cost is $O(k! \cdot k^2 \cdot l^2)$. After all the values listed

above of the type $\Phi(X, \emptyset, \mathcal{K}' = \mathcal{K}[X])$ have been computed, we can implement a dynamic programming procedure using Items 2 and 3 of Lemma 2.3.20 (or, alternatively, a memoized version of Algorithm 2) to compute all the required values of type $\Phi(X, Y \neq \emptyset, \mathcal{K}[X])$ (as listed above) using $O(1)$ arithmetic operations each. Since the number of required computations of the type $\Phi(X, Y \neq \emptyset, \mathcal{K}[X])$ (again, as already listed above) is also $O(l^2)$, it follows that the total cost of these computations is $O(l^2)$. Adding the computational costs for both types of computations, we see that the total cost is $O(k! \cdot k^2 \cdot l^2)$ arithmetic operations. $\square$

Theorem 2.4.4 (Final runtime bound of Algorithm 3). *For a UCCG G and a background knowledge $\mathcal{K}$ of G , Algorithm 3 runs in time $O(k!k^2n^4)$, more precisely $O(k!k^2 \cdot |\Pi(G)|^4)$, where n is the number of nodes in G , and k is the max-clique knowledge of $\mathcal{K}$.*

Proof of Theorem 2.4.4. Algorithm 3 is analogous to the Clique-Picking Algorithm of Wienöbst et al. [2021]. We can also say that Algorithm 3 is the background knowledge version of the Clique-Picking Algorithm of Wienöbst et al. [2021]. At line 4, we construct a clique tree of G , which takes $O(|V_G| + |E_G|)$ time. Computing the Φ function at line 6 for a clique C takes $O(k! \cdot k^2 \cdot |\Pi(G)|^2)$ time (from Lemma 2.4.3; the value of l can be at most $|\Pi(G)|$, the number of nodes in the clique tree $\mathcal{T}$). The while loop (at lines 11-23) runs $O(|\Pi(G)|)$ times, as the number of maximal cliques of G is $|\Pi(G)|$. Running the LBFS algorithm (Algorithm 1) at line 15 takes $O(|V| + |E|)$ time (from Theorem 2.4.1). The computation of the function Φ at line 21 takes $O(k! \cdot k^2 \cdot |\Pi(G)|^2)$ time (from Lemma 2.4.3).

Note that it was assumed in Algorithm 2 that factorials of integers from 1 to $|V_G|$ are precomputed. The precomputation of this table can be done using $O(|V_G|)$ arithmetic operations. From Proposition 2.4.2, the number of distinct calls to the count function of Algorithm 3 is $O(|\Pi(G)|)$. Together, the above calculations show that the running time of Algorithm 3 is at most $O(k! \cdot k^2 \cdot |\Pi(G)|^4)$ or $O(k! \cdot k^2 \cdot |V_G|^4)$, as for a chordal graph G , $|\Pi(G)|$ can be at most $|V_G|$. $\square$

Proof of Theorem 2.1.2. We use Equation (2.1) to compute the number of $\mathcal{K}$ -consistent DAGs of an MEC. This implies that the time complexity to compute the number of $\mathcal{K}$ -consistent DAGs of an MEC equals the sum of the time complexities for comput-

ing the number of $\mathcal{K}[V_H]$ -consistent DAGs of the chordal components H of G . Let $H_1, H_2, \dots, H_l$ be the undirected connected components of G . From Theorems 2.3.22 and 2.4.4, the time complexity to compute the number of $\mathcal{K}[V_{H_i}]$ -consistent DAGs of H_i is $O(k! \cdot k^2 \cdot |V_{H_i}|^4)$. Let n be the number of nodes in G . Then, $n \geq \sum_{1 \leq i \leq l} |V_{H_i}|$, and $n^4 \geq \sum_{1 \leq i \leq l} |V_{H_i}|^4$. This implies the time complexity to compute the number of $\mathcal{K}$ -consistent DAGs of the MEC is $O(k! \cdot k^2 \cdot n^4)$. This proves our main result, Theorem 1.2.2. $\square$

2.5 Experimental Evaluation

In this section, we evaluate the performance of Algorithm 3 on a synthetic dataset. For each $n \in \{500, 510, 520, \dots, 1000\}$, we construct 50 random chordal graphs with n nodes, and for each $k \in \{5, 6, \dots, 13\}$, we construct a set of background knowledge edges with k as its max clique knowledge value. We then measure the running time of Algorithm 3 for each of these (graph, background knowledge) pairs, and take the mean running time over all such pairs with the same value of n and k .

Construction of chordal graphs with n vertices: We first construct a *Erdős-Rényi* graph G with n vertices such that each of its edges is picked with probability p , where p is a random value in $[0.1, 0.3)$. After this, we use rejection sampling to get an undirected connected graph, i.e., if G is not connected, we reject G and repeat the above process until we obtain an undirected connected graph. Next, we assign a unique rank to each node of the graph. We then process the vertices in decreasing order of rank. For each vertex x of the graph, if u and v are two neighbors of x such that u and v are not connected, and u and v both have lesser rank than x , we add an edge between u and v in G . This makes G an undirected connected chordal graph, because, by construction, the increasing order of ranks is a perfect elimination ordering.

Construction of background knowledge edges: For each chordal graph G constructed above, and for each $k \in \{5, 6, \dots, 13\}$, we construct a set of background knowledge edges $\mathcal{K}$ such that if there exists a maximal clique C of size greater than or equal to k , then the max-clique knowledge of G for $\mathcal{K}$ is k , otherwise, the max-clique knowledge of G for $\mathcal{K}$ is the size of the maximum clique in G . To do this, we select each

maximal clique of G one by one and choose the edges of the clique such that at most k vertices are involved in the set of selected edges with both endpoints in that clique. We construct a rooted clique tree $\mathcal{T} = (T, \beta, R)$ of G such that $\beta(R)$ is the maximum-sized clique of G . We begin with the maximal clique $\beta(R)$ of G and then perform a depth-first search (DFS) on T to cover all the maximal cliques of G . At the iteration when we are at a maximal clique C , we compute X , the set of vertices of the picked edges having both of their endpoints in C . By the clique intersection property of the clique tree, X is a subset of $C \cap C'$, where C' is the maximal clique of G corresponding to the parent of C in the rooted tree T . Let Y be a node in T such that $\beta(Y) = C$, and Y' the parent of Y in the rooted tree T , then $C' = \beta(Y')$ ($C' = \emptyset$ when $C = \beta(R)$). If $|X|$ equals k (or $|C|$ if $|C| < k$), we move to the next maximal clique. Otherwise, we select edges $u - v$ of C such that neither u nor v belongs to $(C \cap C') \setminus X$. This ensures that the clique knowledge of the maximal cliques that have already been covered does not change. Since we are picking edges with endpoints not in $(C \cap C') \setminus X$, the maximum clique knowledge of C can be at most $|C| - |C \cap C'| + |X|$. If the present clique knowledge of C is $k - 1$, we do not pick an edge $u - v$ such that the clique knowledge of C moves to $k + 1$. We select edges of C until the clique knowledge of C reaches $\min(k, |C| - |C \cap C'| + |X|)$ (the minimum of k and $|C| - |C \cap C'| + |X|$). We do not encounter a situation where $|X|$ exceeds k due to the clique intersection property of the clique tree. For $\beta(R)$, we must have picked edges such that the clique knowledge of $\beta(R)$ equals $\min(k, |\beta(R)|)$. Since we have picked R such that $\beta(R)$ is the maximum size clique of G , the maximum clique knowledge of G for the background knowledge edges that we selected must be k (if the maximum size clique in G is at least k) or the size of maximum clique in G (i.e., $|\beta(R)|$).

We write a Python program for the construction of chordal graphs, background knowledge edges, and the implementation of Algorithm 3. The experiments use the open-source networkx (Hagberg et al. [2008]) package.

Validating the runtime bound To validate the $O(k!k^2n^4)$ runtime bound established in Section 2.4, we draw log-log plots of the mean runtime T against the size n of the graph, for each fixed value of k (Figure 2.2). As predicted by the polynomial (in n) runtime bound in our theoretical result, we obtain, for each value of the parameter k , a

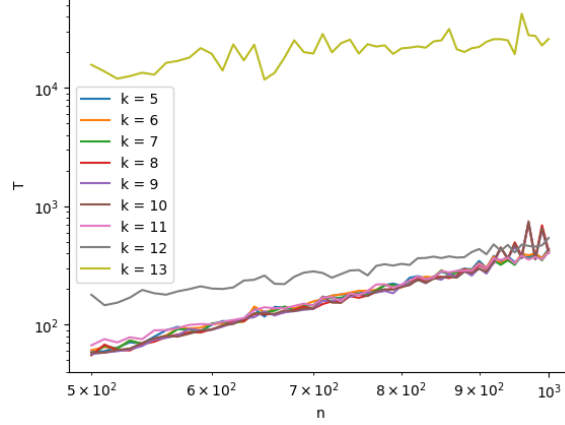


Figure 2.2: log vs. log plot of T vs. n

roughly linear log-log plot.

The intercept of the log-log plot While the plots in Figure 2.2 for $k \in \{5, 6, 7, 8, 9, 10, 11\}$ are quite close to each other, the separation of the plots for $k = 12$ and $k = 13$ is much larger. The reason for this is that the actual time complexity of Algorithm 3 can roughly be bounded as $\log T \leq \log a + \log(k!k^2 + b) + 4 \log n$, where a and b are constants independent of n and k . This observation shows that until about $k \cong 11$, all the plots have intercepts close to each other, as the value of a , b , and n dominates $k!k^2$ for small values of k . The difference starts increasing quickly when k becomes larger than 11.

Effect of changing the size of the background knowledge while keeping k fixed

In this paragraph, we describe the construction of the background knowledge edge sets $\mathcal{K}_1$ and $\mathcal{K}_2$ used in Table 2.1. We first construct $\mathcal{K}_1$ as described above. We then construct $\mathcal{K}_2$ by adding a few edges to $\mathcal{K}_1$ in such a way that the value of k does not change. We perform this experiment for different n and k , where n is the number of nodes of the chordal graph, and k is the maximum number of vertices of any clique of the chordal graph that is part of a background knowledge edge that lies entirely within that clique, as defined earlier. In Table 2.1, T_1 , T_2 represent the running times of Algorithm 3 with background knowledge $\mathcal{K}_1$ and $\mathcal{K}_2$, respectively.

We also observe from the table that the running time decreases slightly by increasing the size of the background knowledge edges. This occurs because k is same in the both case and as the number of background knowledge edges increases, the number of

n	k	$ \mathcal{K}_1 $	$ \mathcal{K}_2 $	T_1	T_2
600	6	58	73	95	92
600	6	52	65	192	190
700	7	47	67	144	144
700	7	49	74	132	128
800	8	56	83	199	195
800	8	55	85	200	195
900	9	46	89	258	256
900	9	39	68	257	252
1000	10	65	116	370	353
1000	10	75	137	346	338
1100	11	55	121	467	455
1100	11	51	104	460	453

Table 2.1: Exploring runtime dependence on the number of background knowledge edges: detailed table

background-consistent permutations of any clique decreases.

Chapter 3

Counting MECs with tree skeleton

The objective of this chapter is to solve Problem 1.2 (the problem of counting MECs with the same skeleton) when the input graph is a tree graph. This chapter serves as a warm-up for the next chapter (Chapter 4), where we construct a recursive algorithm to solve Problem 1.2 for a general graph using a tree decomposition of the graph. We formalize the problem as follows:

Problem 3.1. *INPUT: A tree graph G .*

OUTPUT: $|\text{MEC}(G)|$, the number of MECs of G , i.e., the number of MECs with G as their skeleton.

3.1 Overview

In this chapter, we study the problem of counting MECs with the same skeleton (Problem 1.2) when the input graph is a tree graph, i.e., Problem 3.1. Theorem 3.1.1 summarizes our result.

Theorem 3.1.1 (Restating Theorem 1.3.3). *For a tree graph G , the number of MECs with skeleton G can be counted in $O(nd)$ time, where n is the number of nodes in G , and d is the maximum degree of a node in the graph.*

Here is a technical overview of this chapter.

Technical Overview: To solve Problem 3.1, we fix a node r_1 of the input tree G as its root node. From Item 3 of Theorem 1.1.24, for an MEC of G , if there is a directed edge towards r_1 , then all the edges adjacent to r_1 must be directed in the MEC. We use this property to partition the set of MECs of G into two sets: the set of MECs of G that have a directed edge towards r_1 , and the set of MECs that do not have a directed edge towards r_1 . For the set of MECs of G that do not have a directed edge towards r_1 , we further partition the set based on the number of undirected edges adjacent to r_1 in the MECs.

We construct a recursive algorithm to find the size of the partitions of the set of MECs of G . For this, we arbitrarily pick an edge $r_1 - r_2$ adjacent to r_1 . By cutting the edge, we get two induced subgraphs of G : tree G_1 (containing r_1) and tree G_2 (containing r_2). We define the “projection of an MEC” (Definition 3.7.1) to relate the MECs of G , G_1 , and G_2 . We fix the nodes r_1 and r_2 as the root nodes of G_1 and G_2 , respectively, and partition their MECs similarly to how we partition the MECs of G . We then construct a recursive relation between the number of MECs in the partitions of G and the number of MECs in the partitions of the induced subgraphs. This recursive relation is used to construct an algorithm to solve Problem 3.1. Defining projection to relate the MECs of G , G_1 , and G_2 is the key step in solving the problem.

Furthermore, the notions developed here will be generalized further in the next chapter.

Organization of the chapter: Section 3.1 provides a brief overview of this chapter. In Section 3.2, we present the tree version of Theorem 1.1.24. This provides us with the necessary and sufficient conditions for a graph with a tree skeleton to be an MEC. In Section 3.3, we formally define the partitions of the MECs of a tree graph G . In Section 3.4, we construct two induced subgraphs of G : G_1 and G_2 , and provide a recursive relation (Lemma 3.4.1) between the number of MECs in the partitions of G and the number of MECs in the partitions of the induced subgraphs. In Section 3.5, we use Lemma 3.4.1 to construct a recursive algorithm, Algorithm 4, to count the MECs of a tree graph. In Section 3.6, we compute the time complexity of Algorithm 4. As the proof of Lemma 3.4.1 is lengthy, we place the proof at the end, in Section 3.7.

We begin with the tree version of Theorem 1.1.24.

3.2 Necessary and Sufficient Condition for a Graph to be an MEC of a Tree

The following theorem is the tree version of Theorem 1.1.24.

Theorem 3.2.1. *A graph G with a tree skeleton is an MEC if and only if:*

1. *G does not have an induced subgraph of the form $a \rightarrow b - c$.*
2. *For each directed edge $u \rightarrow v$, there exists a node w such that $u \rightarrow v$ is part of an induced subgraph of G in one of the following forms: either $w \rightarrow u \rightarrow v$ (as shown in Figure 1.3.a) or $u \rightarrow v \leftarrow w$ (as shown in Figure 1.3.b).*

Proof. From Theorem 1.1.24, a graph is an MEC if and only if it satisfies Items 1 to 4 of Theorem 1.1.24. A graph with a tree skeleton cannot have a cycle, meaning it has neither directed cycles nor chordless cycles. Consequently, the graph must be a chain graph. Additionally, each undirected connected component of the graph must be chordal. This implies that for a graph with a tree skeleton, Items 1 and 2 of Theorem 1.1.24 are always satisfied.

Moreover, in a graph with a tree skeleton, there cannot be any induced subgraph of the form shown in Figure 1.3.c or Figure 1.3.d. Therefore, if an edge in the graph is strongly protected, it must be part of an induced subgraph of the form shown in either Figure 1.3.a or Figure 1.3.b. This implies that a graph with a tree skeleton is an MEC if and only if it satisfies Items 1 and 2. This completes the proof of Theorem 3.2.1. $\square$

We now partition the set of MECs of a tree graph, and then use this partitioning to construct a recursive algorithm to count the MECs of a tree graph.

3.3 Partition of the Set of MECs of a Tree Graph

From Item 1 of Theorem 3.2.1, for an MEC of a tree graph G , if there is a directed edge towards a node r in the MEC, then all the edges adjacent to r must be directed in the

MEC. Otherwise, we get an induced subgraph $x \rightarrow r - y$ in the MEC, contradicting Item 1 of Theorem 3.2.1, where $x \rightarrow r$ is a directed edge towards r , and $r - y$ is an undirected edge adjacent to r . We use this property to partition the set of MECs of a tree graph.

Definition 3.3.1 ($\text{MEC}(G, r, 1)$). For a tree graph G , and a node r of G , $\text{MEC}(G, r, 1)$ is defined as the set of MECs of G with an edge directed towards r . We denote the size of $\text{MEC}(G, r, 1)$ by $n_1(G, r)$, i.e., $n_1(G, r) = |\text{MEC}(G, r, 1)|$.

Definition 3.3.2 ($\text{MEC}(G, r, 0)$). For a tree graph G , and a node r of G , $\text{MEC}(G, r, 0)$ is defined as the set of MECs of G with no edge directed towards r . We denote the size of $\text{MEC}(G, r, 0)$ by $n_0(G, r)$, i.e., $n_0(G, r) = |\text{MEC}(G, r, 0)|$.

For any arbitrary node r of a tree graph G , an MEC of G either contains an edge directed towards r or does not contain such an edge. Therefore, Definitions 3.3.1 and 3.3.2 partition the set of MECs of G into $\text{MEC}(G, r, 1)$ and $\text{MEC}(G, r, 0)$. That means,

$$|\text{MEC}(G)| = n_1(G, r) + n_0(G, r) \quad (3.1)$$

From Definition 3.3.2, for all MECs that belong to $\text{MEC}(G, r, 0)$, none of the edges adjacent to r are directed towards r . This implies that all the edges adjacent to r are either directed outward from r or are undirected. We further partition $\text{MEC}(G, r, 0)$ based on the number of undirected edges adjacent to r .

Definition 3.3.3. Let G be a tree graph, r a node of G , and δ the degree of r in G . For $0 \leq i \leq \delta$, $\text{MEC}(G, r, 0, i)$ is defined as the set of MECs of G that have i undirected edges adjacent to r , and the remaining $\delta - i$ edges are directed outward from r . We denote the size of $\text{MEC}(G, r, 0, i)$ by $n_0^i(G, r)$, i.e., $n_0^i(G, r) = |\text{MEC}(G, r, 0, i)|$.

Definition 3.3.3 partitions $\text{MEC}(G, r, 0)$ into $\text{MEC}(G, r, 0, i)$ for all $0 \leq i \leq \delta$. That means,

$$n_0(G, r) = \sum_{i=0}^{\delta} n_0^i(G, r). \quad (3.2)$$

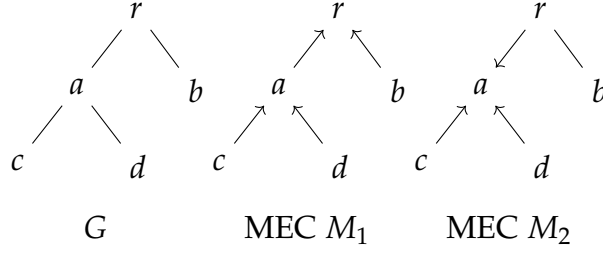


Figure 3.1: Partition of the set of MECs of a tree graph

Figure 3.1 provides examples of the above partitioning scheme. In the figure, G is a tree graph, and r is a node of G . M_1 and M_2 are MECs (both satisfy Items 1 and 2 of Theorem 3.2.1) with skeleton G . M_1 belongs to $\text{MEC}(G, r, 1)$, as there is an edge $a \rightarrow r$ directed towards r . M_2 belongs to $\text{MEC}(G, r, 0)$, as none of the edges adjacent to r in M_2 are directed towards r . M_2 also belongs to $\text{MEC}(G, r, 0, 1)$ as it belongs to $\text{MEC}(G, r, 0)$ and has one undirected edge adjacent to r .

We now use the above-mentioned partitioning of a tree graph to construct a recursive algorithm that counts the MECs of a tree graph.

3.4 Recursive Formula to Count MECs of a Tree Graph

In this section, we provide a recursive formula to count MECs of a tree graph. Let G be a tree graph, r_1 an arbitrary node of G , and δ the degree of r_1 in G . We construct a recursive formula to compute $n_1(G, r_1)$, $n_0^0(G, r_1)$, $n_0^1(G, r_1)$, $\dots$, $n_0^\delta(G, r_1)$. From Equations (3.1) and (3.2),

$$|\text{MEC}(G)| = n_1(G, r_1) + \sum_{i=0}^{\delta} n_0^i(G, r_1). \quad (3.3)$$

Using Equation (3.3), we compute $|\text{MEC}(G)|$.

Base Case: Suppose G is a trivial tree graph, meaning it has only one node, r_1 , and no edges. Then, there exists only one MEC of G , which is G itself (G trivially obeys Items 1 and 2 of Theorem 3.2.1, as there is no edge in G). The MEC G belongs to $\text{MEC}(G, r_1, 0, 0)$, as there is no edge adjacent to r_1 . Therefore, in this case, $n_1(G, r_1) = 0$, and $n_0^0(G, r_1) = 1$.

Inductive Step: Suppose G is not a trivial tree graph. We arbitrarily pick a neighbor r_2 of r_1 in G . By cutting the edge $r_1 - r_2$, we get two induced subgraphs of G : tree G_1 (containing node r_1) and tree G_2 (containing node r_2). The degree of r_1 in G_1 must be $\delta_1 = \delta - 1$, where δ is the degree of r_1 in G . Let the degree of r_2 in G_2 be δ_2 .

Lemma 3.4.1 provides the required recursive formula.

Lemma 3.4.1. *Let G be a tree graph, and $(r_1, r_2) \in E_G$ be an edge of G . By cutting the edge (r_1, r_2) , we get two induced subgraphs of G : G_1 (containing node r_1) and G_2 (containing node r_2). Let δ , δ_1 , and δ_2 be the degrees of r_1 in G , r_1 in G_1 , and r_2 in G_2 , respectively. Then,*

1.

$$\begin{aligned} n_1(G, r_1) &= n_1(G_1, r_1) \times [2 \cdot n_1(G_2, r_2) + \sum_{j=0}^{\delta_2} (j+2) \cdot n_0^j(G_2, r_2)] \\ &\quad + \sum_{i=0}^{\delta_1} n_0^i(G_1, r_1) \times [(i+1) \cdot n_1(G_2, r_2) + i \cdot n_0(G_2, r_2)] \end{aligned}$$

2. $n_0(G, r_1) = n_0(G_1, r_1) \times [n_1(G_2, r_2) + \sum_{j=0}^{\delta_2} j \cdot n_0^j(G_2, r_2)]$

3. For $1 \leq i < \delta$,

$$n_0^i(G, r_1) = n_0^i(G_1, r_1) \times [n_1(G_2, r_2) + \sum_{j=0}^{\delta_2} j \cdot n_0^j(G_2, r_2)] + n_0^{i-1}(G_1, r_1) \times n_0(G_2, r_2)$$

4. $n_0^\delta(G, r_1) = n_0^{\delta-1}(G_1, r_1) \times n_0(G_2, r_2)$

Lemma 3.4.1 shows that, if we know $n_1(G_1, r_1), n_0^0(G_1, r_1), n_0^1(G_1, r_1), \dots, n_0^{\delta_1}(G_1, r_1)$, and $n_1(G_2, r_2), n_0^0(G_2, r_2), n_0^1(G_2, r_2), \dots, n_0^{\delta_2}(G_2, r_2)$, then we can compute $n_1(G, r_1), n_0^0(G, r_1), n_0^1(G, r_1), \dots, n_0^\delta(G, r_1)$.

The proof of Lemma 3.4.1 is deferred to Section 3.7 (page 61).

We now construct an algorithm that implements Lemma 3.4.1.

3.5 Algorithm to Count MECs of a Tree Graph

Let G be a tree graph, r_1 a node in G , and δ the degree of r_1 in G . We implement Lemma 3.4.1 to construct Algorithm 4 that outputs $(n_1(G, r_1), n_0^0(G, r_1), n_0^1(G, r_1), \dots,$

$n_0^\delta(G, r_1)$). We then use Equation (3.3), to count the MECs of G .

Algorithm 4: Count-MEC-Tree(G, r_1)

Input : A tree graph G , and a node $r_1 \in V_G$
Output: $(n_1(G, r_1), n_0^0(G, r_1), n_0^1(G, r_1), \dots, n_0^\delta(G, r_1))$, where δ is the degree of r_1 in G .

- 1 $\delta \leftarrow$ degree of r_1 in G
- 2 **if** $\delta = 0$ **then**
- 3 **return** $(0, 1)$
- 4 **end**
- 5 Arbitrarily pick an edge $(r_1, r_2) \in E_G$.
- 6 $G_1 \leftarrow$ subtree of G containing r_1 after cutting the edge (r_1, r_2) of G .
- 7 $G_2 \leftarrow$ subtree of G containing r_2 after cutting the edge (r_1, r_2) of G .
- 8 $\delta_1 \leftarrow$ degree of r_1 in G_1
- 9 $\delta_2 \leftarrow$ degree of r_2 in G_2
- 10 $(b_1, c_1^0, c_1^1, \dots, c_1^{\delta_1}) \leftarrow$ Count-MEC-Tree(G_1, r_1)
- 11 $(b_2, c_2^0, c_2^1, \dots, c_2^{\delta_2}) \leftarrow$ Count-MEC-Tree(G_2, r_2)
- 12 $X_1 \leftarrow \sum_{i=0}^{\delta_1} c_1^i$
- 13 $X_2 \leftarrow \sum_{j=0}^{\delta_2} c_2^j$
- 14 $Y_1 \leftarrow \sum_{i=0}^{\delta_1} i \cdot c_1^i$
- 15 $Y_2 \leftarrow \sum_{j=0}^{\delta_2} j \cdot c_2^j$
- 16 $b \leftarrow b_1 \times (2 \cdot b_2 + Y_2 + 2 \cdot X_2) + (X_1 + Y_1) \times b_2 + Y_1 \times X_2$
- 17 $c_0 \leftarrow c_1^0 \times (b_2 + Y_2)$
- 18 $i \leftarrow 1$
- 19 **while** $i < \delta$ **do**
- 20 $c_i \leftarrow c_1^i \times (b_2 + Y_2) + c_1^{i-1} \times X_2$
- 21 $i \leftarrow i + 1$
- 22 **end**
- 23 $c_\delta \leftarrow c_1^{\delta-1} \times X_2$
- 24 **return** $(b, c_0, c_1, \dots, c_\delta)$

Lemma 3.5.1 validates Algorithm 4.

Lemma 3.5.1. *Let G be a tree graph, r_1 a node of G , and δ the degree of r_1 in G . For inputs G and r_1 , Algorithm 4 outputs $(n_1(G, r_1), n_0^0(G, r_1), n_0^1(G, r_1), \dots, n_0^\delta(G, r_1))$.*

Proof. Lines 2–4 handle the base case when the degree of the root node in the tree graph G is 0, i.e., when G is a trivial graph. As discussed in Section 3.4 (see the base case paragraph of Section 3.4), in this case, $n_1(G, r_1) = 0$, and $n_0^0(G, r_1) = 1$. Line 3 returns the same.

The remaining lines of Algorithm 4 deal with the case when G is not a trivial graph. In this case, the algorithm implements Lemma 3.4.1. At Line 5, the algorithm arbitrarily picks an edge $r_1 - r_2$. By cutting the edge, we get two induced subgraphs of G : G_1 (containing r_1 ; see Line 6), and G_2 (containing r_2 ; see Line 7). δ_1 and δ_2 are the degree of r_1 and r_2 in G_1 and G_2 , respectively (see Lines 8 and 9). The algorithm recursively computes $n_1(G_1, r_1), n_0^0(G_1, r_1), n_0^1(G_1, r_1), \dots, n_0^{\delta_1}(G_1, r_1)$ (at Line 10), and $n_1(G_2, r_2), n_0^0(G_2, r_2), n_0^1(G_2, r_2), \dots, n_0^{\delta_2}(G_2, r_2)$ (at Line 11). It computes $X_1 = \sum_{i=0}^{\delta_1} c_1^i = n_0(G_1, r_1)$ and $X_2 = \sum_{j=0}^{\delta_2} c_2^j = n_0(G_2, r_2)$ (using Equation (3.2)) at lines 12-13, and $Y_1 = \sum_{i=0}^{\delta_1} i \cdot c_1^i$ and $Y_2 = \sum_{j=0}^{\delta_2} j \cdot c_2^j$ at lines 14-15. It then compute $n_1(G, r_1), n_0^0(G, r_1), n_0^1(G, r_1), \dots, n_0^\delta(G, r_1)$ (at lines 16-23) by implementing Items 1 to 4 of Lemma 3.4.1, and returns them at Line 24. This completes the proof of Lemma 3.5.1. $\square$

3.6 Time Complexity Analysis

In this section, we analyze the runtime of Algorithm 4. In addition to this, we also prove Theorem 3.1.1.

Lemma 3.6.1. *Let G be a tree graph, and r a node of G . For the inputs G and r , we can implement Algorithm 4 in $O(nd)$ time, where n is the number of nodes in G and d is the maximum degree of any node in the graph.*

Proof of Lemma 3.6.1. At Line 5, the algorithm cuts an edge of G to create two induced subgraphs of G and recursively calls itself for these two subgraphs (at Lines 10 and 11). Since the number of edges in G is $n - 1$, the total number of recursive calls to Algorithm 4 is $2(n - 1)$.

Note that we do not need to explicitly construct the subgraphs of G for the recursive calls; each such subgraph must be an induced subgraph of G . If we order the children of each node $x \in V_G$ in the subtree rooted at x and follow this ordering to cut the edge, we can avoid constructing subgraphs at Lines 6 and 7.

The computations of X_1, X_2, Y_1 , and Y_2 (at lines 12-15) each take $O(d)$ time. Similarly, the calculations of the b and c_i values (as in Lines 16, 17, 20 and 23) also take $O(d)$ time.

Therefore, the overall implementation of Algorithm 4 has a time complexity of

$O(nd)$.

□

Proof of Theorem 3.1.1. We can pick a node r_1 of G in $O(1)$ time. Then, we call Algorithm 4 with inputs G and r_1 . From Lemma 3.6.1, the running time of the algorithm is $O(nd)$. The algorithm returns $n_1(G, r_1), n_0^0(G, r_1), n_0^1(G, r_1), \dots, n_0^\delta(G, r_1)$, where δ is the degree of r_1 in G . Then, in $O(d)$ time, we can compute $|\text{MEC}(G)|$ using Equation (3.3). Therefore, we can count the MECs of G in $O(nd)$ time. □

3.7 Proof of Lemma 3.4.1

The proof of Lemma 3.4.1 requires building a relation between the MECs of G , G_1 , and G_2 . We define “projection of an MEC” to build this relation. The ideas developed here (in the simple setting of trees) will serve as the foundation for more general notions in the next chapter (Chapter 4).

Definition 3.7.1 (Projection). Let G be an undirected graph, $X \subseteq V_G$, and $G' = G[X]$, an induced subgraph of G . Let M and M' be MECs of G and G' , respectively. M' is said to be the projection of M on $X = V_{G'}$ if the set of v-structures in M' (Definition 1.1.8) and the set of v-structures in $M[X]$ are the same, i.e., $\mathcal{V}(M') = \mathcal{V}(M[X])$. We denote the projection of M on X by $\mathcal{P}(M, X)$. With slight abuse of notation, we define $\mathcal{P}(M, X_1, X_2) = (\mathcal{P}(M, X_1), \mathcal{P}(M, X_2))$.

Lemmas 3.7.2 and 3.7.3 describe the properties of the projection. Note that the concept of the “projection of an MEC” applies to general graphs. The proofs of these results are given in Chapter 4, where we solve Problem 1.2 (the counting MEC problem) for general graphs.

Lemma 3.7.2 (Lemma 4.4.2 of Chapter 4). *Let M be an MEC of G . For all $X \subseteq V_G$, there exists a unique projection of M on X , i.e., $\mathcal{P}(M, X)$ is unique.*

Lemma 3.7.3 (Lemma 4.4.5 of Chapter 4). *Let M be an MEC, and M' be a projection of M . If $u \rightarrow v \in M'$, then $u \rightarrow v \in M$.*

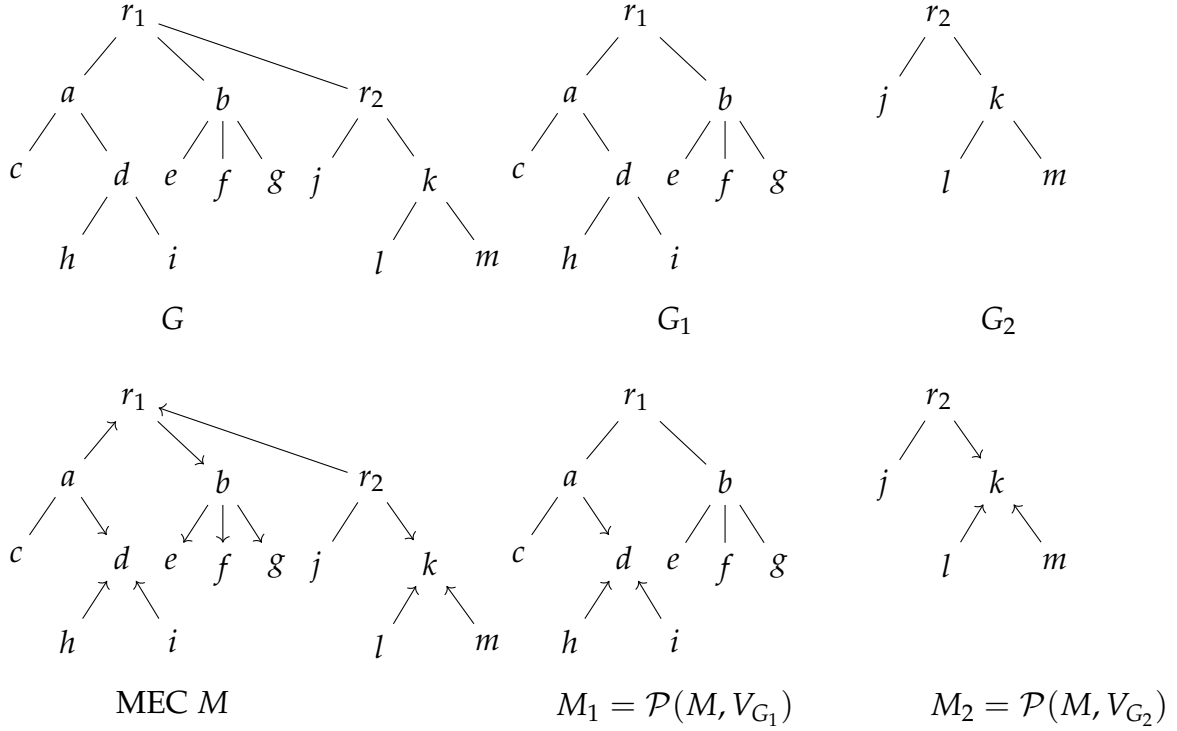


Figure 3.2: Projection of an MEC of a tree graph

Figure 3.2 is provided for a better understanding of the notion of “projection of an MEC.” In the figure, G is a tree, and G_1 and G_2 are two induced subgraphs of G , obtained by cutting the edge $r_1 - r_2$. M , M_1 , and M_2 are MECs of G , G_1 , and G_2 , respectively (all obey Items 1 and 2 of Theorem 3.2.1). M_1 is the projection of M on V_{G_1} since the set of v-structures in $M[V_{G_1}]$ and the set of v-structures in M_1 are the same. Similarly, M_2 is the projection of M on V_{G_2} .

For MECs M_1 and M_2 of G_1 and G_2 , respectively, the following observation provides necessary and sufficient conditions for an MEC M of G such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.

Observation 3.7.4. *Let G be a tree graph, $r_1 - r_2 \in E_G$, and G_1 and G_2 be the two induced subtrees of G obtained by cutting the edge $r_1 - r_2$. Let M , M_1 , and M_2 be MECs of G , G_1 , and G_2 , respectively. Then, $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$ if and only if:*

1. M contains all the v-structures of M_1 and M_2 , i.e., $\mathcal{V}(M_1) \cup \mathcal{V}(M_2) \subseteq \mathcal{V}(M)$; and
2. either all the v-structures of M that are v-structures in neither M_1 nor M_2 contain an

edge $r_1 \rightarrow r_2$, or all the v-structures of M that are v-structures in neither M_1 nor M_2 contain an edge $r_2 \rightarrow r_1$.

Proof. We first prove the $\leftarrow$ direction of Observation 3.7.4. Suppose $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. Since M_1 is a projection of M on V_{G_1} , from Definition 3.7.1, $\mathcal{V}(M_1) = \mathcal{V}(M[V_{G_1}])$. This implies $\mathcal{V}(M_1) \subseteq \mathcal{V}(M)$. Similarly, $\mathcal{V}(M_2) \subseteq \mathcal{V}(M)$. This proves Item 1.

From the structure of M , each edge of M has endpoints such that one of the following holds: both endpoints of the edge are in V_{G_1} , both endpoints of the edge are in V_{G_2} , or the endpoints are r_1 and r_2 . If both edges of the v-structure have endpoints in V_{G_1} , then the v-structure must be in M_1 , since $\mathcal{V}(M[V_{G_1}]) = \mathcal{V}(M_1)$. Similarly, if both edges of the v-structure have endpoints in V_{G_2} , then the v-structure must be in M_2 . From the construction of G_1 and G_2 , there is no vertex common between V_{G_1} and V_{G_2} . Therefore, there cannot be a v-structure in M such that the endpoints of one edge of the v-structure are in V_{G_1} and the endpoints of the other edge are in V_{G_2} . Thus, for a v-structure of M that is neither in M_1 nor in M_2 , the only option that remains is that one edge of the v-structure has endpoints r_1 and r_2 . Since the edges of a v-structure are directed, a v-structure of M that is neither in M_1 nor in M_2 contains either $r_1 \rightarrow r_2$ or $r_2 \rightarrow r_1$. Also, since M cannot have both $r_1 \rightarrow r_2$ and $r_2 \rightarrow r_1$, either all the v-structures of M that are neither in M_1 nor in M_2 contain $r_1 \rightarrow r_2$, or all the v-structures of M that are neither in M_1 nor in M_2 contain $r_2 \rightarrow r_1$. This proves Item 2. This completes the proof of the $\leftarrow$ direction of Observation 3.7.4.

We now prove the $\rightarrow$ direction of Observation 3.7.4. Suppose M satisfies Items 1 and 2 of Observation 3.7.4. We first show that $\mathcal{V}(M[V_{G_1}]) = \mathcal{V}(M_1)$.

Let $u \rightarrow v \leftarrow w$ be a v-structure in $M[V_{G_1}]$. Suppose $u \rightarrow v \leftarrow w$ is not a v-structure in M_1 . Since $V_{G_1} \cap V_{G_2} = \emptyset$, $u \rightarrow v \leftarrow w$ must not be a v-structure in M_2 . As $u \rightarrow v \leftarrow w$ is neither a v-structure in M_1 nor a v-structure in M_2 , from Item 2 of Observation 3.7.4, one of the edges of $u \rightarrow v \leftarrow w$ must be either $r_1 \rightarrow r_2$ or $r_2 \rightarrow r_1$. However, $r_2 \notin V_{G_1}$. This implies that neither $r_1 \rightarrow r_2$ nor $r_2 \rightarrow r_1$ can be an edge of the v-structure $u \rightarrow v \leftarrow w$. This contradicts Item 2 of Observation 3.7.4. Therefore, $u \rightarrow v \leftarrow w$ must be a v-structure in M_1 , implying $\mathcal{V}(M[V_{G_1}]) \subseteq \mathcal{V}(M_1)$.

From Item 1 of Observation 3.7.4, we have $\mathcal{V}(M_1) \subseteq \mathcal{V}(M)$. This implies $\mathcal{V}(M_1) \subseteq$

$\mathcal{V}(M[V_{G_1}])$, because if $u \rightarrow v \leftarrow w$ is a v-structure in M_1 , then $u, v, w \in V_{M_1} = V_{G_1}$. This further implies $\mathcal{V}(M[V_{G_1}]) = \mathcal{V}(M_1)$. Similarly, $\mathcal{V}(M[V_{G_2}]) = \mathcal{V}(M_2)$. Therefore, from Definition 3.7.1, $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.

□

We now begin the proof of Lemma 3.4.1.

Proof of Lemma 3.4.1. From Lemma 3.7.2, for each MEC M of G , there exists a unique pair of MECs $(M_1, M_2) \in \text{MEC}(G_1) \times \text{MEC}(G_2)$ such that $(M_1, M_2) = \mathcal{P}(M, V_{G_1}, V_{G_2})$. For each pair of MECs $(M_1, M_2) \in \text{MEC}(G_1) \times \text{MEC}(G_2)$, we first find the MECs M of G such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. We then classify these MECs M into $\text{MEC}(G, r_1, 1), \text{MEC}(G, r_1, 0, 0), \text{MEC}(G, r_1, 0, 1), \dots, \text{MEC}(G, r_1, 0, \delta)$ (recall δ is the degree of r_1 in G). We further count the MECs in each class. This gives us the required numbers $n_1(G, r_1), n_0^0(G, r_1), n_0^1(G, r_1), \dots, n_0^\delta(G, r_1)$.

For each pair of MECs $(M_1, M_2) \in \text{MEC}(G_1) \times \text{MEC}(G_2)$, we now find MECs M of G such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, i.e., M obeys Items 1 and 2 of Observation 3.7.4. The pair (M_1, M_2) can belong to one of the following categories:

1. $M_1 \in \text{MEC}(G_1, r_1, 1)$ and $M_2 \in \text{MEC}(G_2, r_2, 1)$.
2. $M_1 \in \text{MEC}(G_1, r_1, 1)$ and for some $0 \leq j \leq \delta_2$, $M_2 \in \text{MEC}(G_2, r_2, 0, j)$ (recall that δ_2 is the degree of r_2 in G_2).
3. For some $0 \leq i \leq \delta_1$, $M_1 \in \text{MEC}(G_1, r_1, 0, i)$ and $M_2 \in \text{MEC}(G_2, r_2, 1)$ (recall that δ_1 is the degree of r_1 in G_1).
4. For some $0 \leq i \leq \delta_1$, $M_1 \in \text{MEC}(G_1, r_1, 0, i)$ and for some $0 \leq j \leq \delta_2$, $M_2 \in \text{MEC}(G_2, r_2, 0, j)$.

Lemma 3.7.5 deals with possibility 1. The proof of Lemma 3.7.5 is given on page 70.

Lemma 3.7.5. *Let $M_1 \in \text{MEC}(G_1, r_1, 1)$, and $M_2 \in \text{MEC}(G_2, r_2, 1)$. Then,*

1. *For any MEC M of G , if $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, then one of the following occurs:*

(a)

$$\mathcal{V}(M) = \mathcal{V}(M_1) \cup \mathcal{V}(M_2) \cup \bigcup_{x \rightarrow r_1 \in E_{M_1}} \{x \rightarrow r_1 \leftarrow r_2\} \quad (3.4)$$

(b)

$$\mathcal{V}(M) = \mathcal{V}(M_1) \cup \mathcal{V}(M_2) \cup \bigcup_{y \rightarrow r_2 \in E_{M_2}} \{y \rightarrow r_2 \leftarrow r_1\} \quad (3.5)$$

2. There exists a unique MEC M of G that obeys Equation (3.4). Also, that M belongs to $\text{MEC}(G, r_1, 1)$, and $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.
3. There exists a unique MEC M of G that obeys Equation (3.5). Also, that M belongs to $\text{MEC}(G, r_1, 1)$, and $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.

Lemma 3.7.5 implies the following corollary:

Corollary 3.7.6. For $M_1 \in \text{MEC}(G_1, r_1, 1)$ and $M_2 \in \text{MEC}(G_2, r_2, 1)$, the number of MECs M of G such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$ is two. Further, both the MECs belong to $\text{MEC}(G, r_1, 1)$.

Proof. From Item 1 of Lemma 3.7.5, for any MEC M of G , if $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, then M must obey either Equation (3.4) or Equation (3.5).

From Item 2 of Lemma 3.7.5, there exists a unique MEC M of G that satisfies Equation (3.4) and $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, which belongs to $\text{MEC}(G, r_1, 1)$. Additionally, from Item 3 of Lemma 3.7.5, another unique MEC M follows Equation (3.5) such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, and also belongs to $\text{MEC}(G, r_1, 1)$. The two MECs (one that satisfies Equation (3.4) and the other that satisfies Equation (3.5)) must be different, as they contain distinct sets of v-structures.

Hence, the total count of MECs is two, and both belong to $\text{MEC}(G, r_1, 1)$.

□

Lemma 3.7.7 deals with possibility 2. The proof of Lemma 3.7.7 is given on page 73.

Lemma 3.7.7. Let $M_1 \in \text{MEC}(G_1, r_1, 1)$ and for some $0 \leq j \leq \delta_2$, $M_2 \in \text{MEC}(G_2, r_2, 0, j)$. Then:

1. For any MEC M of G , if $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, then one of the following occurs:

(a)

$$\mathcal{V}(M) = \mathcal{V}(M_1) \cup \mathcal{V}(M_2) \cup \bigcup_{x \rightarrow r_1 \in E_{M_1}} \{x \rightarrow r_1 \leftarrow r_2\} \quad (3.6)$$

(b)

$$\mathcal{V}(M) = \mathcal{V}(M_1) \cup \mathcal{V}(M_2) \quad (3.7)$$

(c) For some undirected edge $y - r_2 \in E_{M_2}$,

$$\mathcal{V}(M) = \mathcal{V}(M_1) \cup \mathcal{V}(M_2) \cup \{r_1 \rightarrow r_2 \leftarrow y\} \quad (3.8)$$

2. There exists a unique MEC M of G that obeys Equation (3.6). Additionally, M belongs to $\text{MEC}(G, r_1, 1)$, and $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.
3. There exists a unique MEC M of G that obeys Equation (3.7). Furthermore, M belongs to $\text{MEC}(G, r_1, 1)$, and $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.
4. For each y such that $y - r_2 \in E_{M_2}$, there exists a unique MEC M of G that obeys Equation (3.8). Also, M belongs to $\text{MEC}(G, r_1, 1)$, and $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.

Lemma 3.7.7 implies the following corollary:

Corollary 3.7.8. For $M_1 \in \text{MEC}(G_1, r_1, 1)$ and $M_2 \in \text{MEC}(G_2, r_2, 0, j)$, the number of MECs M of G such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$ is $j + 2$. More specifically, all these MECs belong to $\text{MEC}(G, r_1, 1)$.

Proof. From Item 1 of Lemma 3.7.7, for any MEC M of G , if $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, then M must obey one of Equation (3.6), Equation (3.7), or Equation (3.8).

From Item 2 of Lemma 3.7.7, there exists a unique MEC satisfying Equation (3.6), which belongs to $\text{MEC}(G, r_1, 1)$. Additionally, from Item 3, another unique MEC, following Equation (3.7), also belongs to $\text{MEC}(G, r_1, 1)$.

Furthermore, given $M_2 \in \text{MEC}(G_2, r_2, 0, j)$, there are j instances of y such that $y - r_2 \in E_{M_2}$. From Item 4, for each $y - r_2 \in E_{M_2}$, there exists a unique MEC satisfying Equation (3.8). Consequently, there are j MECs following Equation (3.8), all of which belong to $\text{MEC}(G, r_1, 1)$ as per Item 4.

Hence, the total count of MECs is $j + 2$, and all of them belong to $\text{MEC}(G, r_1, 1)$. $\square$

Lemma 3.7.9 deals with the third possibility. The proof of Lemma 3.7.9 is given on page 85.

Lemma 3.7.9. *Let $M_1 \in \text{MEC}(G_1, r_1, 0, i)$ and $M_2 \in \text{MEC}(G_2, r_2, 1)$ for some $0 \leq i \leq \delta_1$. Then,*

1. *For any $M \in \text{MEC}(G)$, if $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, then one of the following occurs:*

(a)

$$\mathcal{V}(M) = \mathcal{V}(M_1) \cup \mathcal{V}(M_2) \cup \bigcup_{y \rightarrow r_2 \in E_{M_2}} \{y \rightarrow r_2 \leftarrow r_1\} \quad (3.9)$$

(b)

$$\mathcal{V}(M) = \mathcal{V}(M_1) \cup \mathcal{V}(M_2) \quad (3.10)$$

(c) *For some undirected edge $x - r_1 \in E_{M_1}$,*

$$\mathcal{V}(M) = \mathcal{V}(M_1) \cup \mathcal{V}(M_2) \cup \{x \rightarrow r_1 \leftarrow r_2\} \quad (3.11)$$

2. *There exists a unique MEC M of G that satisfies Equation (3.9). Furthermore, M belongs to $\text{MEC}(G, r_1, 0, i)$, and $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.*
3. *There exists a unique MEC M of G that satisfies Equation (3.10). Moreover, M belongs to $\text{MEC}(G, r_1, 1)$, and $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.*
4. *For each x such that $x - r_1 \in E_{M_1}$, there exists a unique MEC M of G that satisfies Equation (3.11). Additionally, M belongs to $\text{MEC}(G, r_1, 1)$, and $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.*

Lemma 3.7.9 implies the following corollary:

Corollary 3.7.10. *For $M_1 \in \text{MEC}(G_1, r_1, 0, i)$, and $M_2 \in \text{MEC}(G_2, r_2, 1)$, the number of MECs M of G such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$ is $i + 2$. More specifically, $i + 1$ MECs belong to $\text{MEC}(G, r_1, 1)$, and one MEC belongs to $\text{MEC}(G, r_1, 0, i)$.*

Proof. From Item 1 of Lemma 3.7.9, for any MEC M of G , if $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, then M must obey one of Equation (3.9), Equation (3.10), or Equation (3.11).

From Item 2 of Lemma 3.7.9, there exists a unique MEC that satisfies Equation (3.9), which belongs to $\text{MEC}(G, r_1, 0, i)$. As indicated in Item 3, another unique MEC follows Equation (3.10) and belongs to $\text{MEC}(G, r_1, 1)$.

Given $M_1 \in \text{MEC}(G_1, r_1, 0, i)$, there are i instances of x such that $x - r_1 \in E_{M_1}$. From Item 4, for each $x - r_1 \in E_{M_1}$, there exists a unique MEC satisfying Equation (3.11). Consequently, there are i MECs following Equation (3.11), all belonging to $\text{MEC}(G, r_1, 1)$, as per Item 3.

Hence, the total count of MECs is $i + 2$, of which $i + 1$ MECs belong to $\text{MEC}(G, r_1, 1)$, and one MEC belongs to $\text{MEC}(G, r_1, 0, i)$. $\square$

Lemma 3.7.11 deals with the fourth possibility. The proof of Lemma 3.7.11 is given on page 97.

Lemma 3.7.11. *Let for some $0 \leq i \leq \delta_1$, $M_1 \in \text{MEC}(G_1, r_1, 0, i)$, and for some $0 \leq j \leq \delta_2$, $M_2 \in \text{MEC}(G_2, r_2, 0, j)$.*

1. *Let $M \in \text{MEC}(G)$ such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. Then, one of the following occurs:*

(a)

$$\mathcal{V}(M) = \mathcal{V}(M_1) \cup \mathcal{V}(M_2) \quad (3.12)$$

(b) *For some undirected edge $x - r_1 \in E_{M_1}$,*

$$\mathcal{V}(M) = \mathcal{V}(M_1) \cup \mathcal{V}(M_2) \cup \{x \rightarrow r_1 \leftarrow r_2\} \quad (3.13)$$

(c) *For some undirected edge $y - r_2 \in E_{M_2}$,*

$$\mathcal{V}(M) = \mathcal{V}(M_1) \cup \mathcal{V}(M_2) \cup \{y \rightarrow r_2 \leftarrow r_1\} \quad (3.14)$$

2. *There exists a unique MEC M of G that obeys Equation (3.12). Additionally, M belongs to $\text{MEC}(G, r_1, 0, i + 1)$, and $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.*
3. *For each x such that $x - r_1 \in E_{M_1}$, there exists a unique MEC M of G that obeys Equation (3.13). Furthermore, M belongs to $\text{MEC}(G, r_1, 1)$, and $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.*
4. *For each y such that $y - r_2 \in E_{M_2}$, there exists a unique MEC M of G that obeys Equation (3.14). Also, M belongs to $\text{MEC}(G, r_1, 0, i)$, and $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.*

Lemma 3.7.11 implies the following corollary:

Corollary 3.7.12. *For $M_1 \in \text{MEC}(G_1, r_1, 0, i)$ and $M_2 \in \text{MEC}(G_2, r_2, 0, j)$, the number of MECs M of G such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$ is $i + j + 1$. More specifically, i MECs belong to $\text{MEC}(G, r_1, 1)$, j MECs belong to $\text{MEC}(G, r_1, 0, i)$, and one MEC belongs to $\text{MEC}(G, r_1, 0, i + 1)$.*

Proof. From Item 1 of Lemma 3.7.11, for any MEC M of G , if $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, then M must obey one of Equation (3.12), Equation (3.13), or Equation (3.14).

From Item 2 of Lemma 3.7.11, there exists a unique MEC that satisfies Equation (3.12), which belongs to $\text{MEC}(G, r_1, 0, i + 1)$.

Given $M_1 \in \text{MEC}(G_1, r_1, 0, i)$, there are i instances of x such that $x - r_1 \in E_{M_1}$. From Item 3 of Lemma 3.7.11, for each $x - r_1 \in E_{M_1}$, there exists a unique MEC satisfying Equation (3.13). Consequently, there are i MECs following Equation (3.13), all belonging to $\text{MEC}(G, r_1, 1)$ as per Item 3.

Given $M_2 \in \text{MEC}(G_2, r_2, 0, j)$, there are j instances of y such that $y - r_2 \in E_{M_2}$. From Item 4 of Lemma 3.7.11, for each $y - r_2 \in E_{M_2}$, there exists a unique MEC satisfying Equation (3.14). Consequently, there are j MECs following Equation (3.14), all belonging to $\text{MEC}(G, r_1, 0, i)$ as per Item 4.

Hence, the total count of MECs is $i + j + 1$. Of these, i MECs belong to $\text{MEC}(G, r_1, 1)$, j MECs belong to $\text{MEC}(G, r_1, 0, i)$, and one MEC belongs to $\text{MEC}(G, r_1, 0, i + 1)$. $\square$

We now proceed to the proof of Lemma 3.4.1.

From Corollary 3.7.6, corresponding to each tuple (M_1, M_2) such that $M_1 \in \text{MEC}(G_1, r_1, 1)$ and $M_2 \in \text{MEC}(G_2, r_2, 1)$, there are two MECs of G that belong to $\text{MEC}(G, r_1, 1)$. From Corollary 3.7.8, corresponding to each tuple (M_1, M_2) such that $M_1 \in \text{MEC}(G_1, r_1, 1)$ and $M_2 \in \text{MEC}(G_2, r_2, 0, j)$, there are $j + 2$ MECs of G that belong to $\text{MEC}(G, r_1, 1)$.

From Corollary 3.7.10, corresponding to each tuple (M_1, M_2) such that $M_1 \in \text{MEC}(G_1, r_1, 0, i)$ and $M_2 \in \text{MEC}(G_2, r_2, 1)$, there are $i + 1$ MECs of G that belong to $\text{MEC}(G, r_1, 1)$. From Corollary 3.7.12, corresponding to each tuple (M_1, M_2) such that $M_1 \in \text{MEC}(G_1, r_1, 0, i)$ and $M_2 \in \text{MEC}(G_2, r_2, 0, j)$, there are i MECs of G that belong to $\text{MEC}(G, r_1, 1)$. This results in the total number of MECs of G that belong to $\text{MEC}(G, r_1, 1)$ being equal to that in Item 1 of Lemma 3.4.1.

From Corollaries 3.7.6 and 3.7.8, corresponding to each tuple (M_1, M_2) such that $M_1 \in \text{MEC}(G_1, r_1, 1)$ and $M_2 \in \text{MEC}(G_2)$, no MEC of G belongs to $\text{MEC}(G, r_1, 0)$. From Corollary 3.7.10, for all $0 \leq i \leq \delta - 1$, corresponding to each tuple (M_1, M_2) such that $M_1 \in \text{MEC}(G_1, r_1, 0, i)$ and $M_2 \in \text{MEC}(G_2, r_2, 1)$, there is one MEC of G that belongs to $\text{MEC}(G, r_1, 0, i)$. From Corollary 3.7.12, for all $0 \leq i \leq \delta - 1$ and for all $0 \leq j \leq \delta_2$, corresponding to each tuple (M_1, M_2) such that $M_1 \in \text{MEC}(G_1, r_1, 0, i)$ and $M_2 \in \text{MEC}(G_2, r_2, 0, j)$, there are j MECs of G that belong to $\text{MEC}(G, r_1, 0, i)$. From Corollary 3.7.12, for all $1 \leq i \leq \delta$, for each tuple (M_1, M_2) such that $M_1 \in \text{MEC}(G_1, r_1, 0, i - 1)$ and $M_2 \in \text{MEC}(G_2, r_2, 0)$, there is one MEC of G that belongs to $\text{MEC}(G, r_1, 0, i)$. This results in the total number of MECs of G that belong to $\text{MEC}(G, r_1, 0, i)$ being equal to that in Items 2 to 4 of Lemma 3.4.1.

This completes the proof of Lemma 3.4.1. $\square$

3.7.1 Proofs of Auxiliary Lemmas

In this subsection, we provide the proofs of the auxiliary lemmas used in the preceding proofs.

Remark: Some of the proofs presented below could potentially be shortened by applying the rules from [Meek, 1995] and other equivalent definitions of MECs. However, for the sake of consistency, we present all proofs using the criterion of Andersson et al. [1997].

Proof of Lemma 3.7.5. We first prove Item 1 of Lemma 3.7.5.

Proof of Item 1 of Lemma 3.7.5. Let M be an MEC of G such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. From Observation 3.7.4, M has the v-structures of M_1 and M_2 , and additionally, it can only have the v-structures that involve an edge with endpoints r_1 and r_2 . From Definition 3.3.1, for an MEC in $\text{MEC}(G_1, r_1, 1)$, there exists a directed edge towards r_1 in the MEC. Since $M_1 \in \text{MEC}(G_1, r_1, 1)$, there exists a node $x \in V_{G_1}$ such that $x \rightarrow r_1 \in M_1$. Since M_1 is an MEC, from Item 1 of Theorem 3.2.1, all the edges adjacent to r_1 in M_1 are directed. This implies that we can partition the set of neighbors of r_1 in G_1 into X and X' , where $X = \{x : x \rightarrow r_1 \in M_1\}$ and $X' = \{x : x \leftarrow r_1 \in M_1\}$, with X being

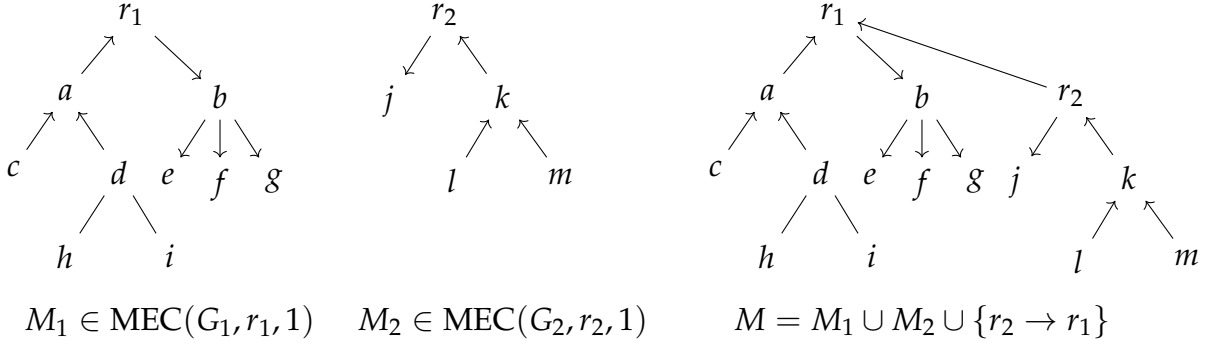


Figure 3.3: Auxiliary picture for the proof of Item 2 of Lemma 3.7.5.

non-empty. Similarly, since $M_2 \in \text{MEC}(G_2, r_2, 1)$, we can partition the set of neighbors of r_2 in G_2 into Y and Y' , where $Y = \{y : y \rightarrow r_2 \in M_2\}$ and $Y' = \{y : y \leftarrow r_2 \in M_2\}$, with Y being non-empty. From Lemma 3.7.3, M has edges $x \rightarrow r_1$ for $x \in X$, $r_1 \rightarrow x$ for $x \in X'$, $y \rightarrow r_2$ for $y \in Y$, and $r_2 \rightarrow y$ for $y \in Y'$. From Item 1 of Theorem 3.2.1, either $r_2 \rightarrow r_1 \in E_M$ or $r_1 \rightarrow r_2 \in E_M$. Otherwise, M has an induced subgraph of the form $x \rightarrow r_1 - r_2$ for $x \in X$, contradicting Item 1 of Theorem 3.2.1. If $r_2 \rightarrow r_1 \in E_M$, then, in addition to the v-structures of M_1 and M_2 , M has the v-structures of the form $x \rightarrow r_1 \leftarrow r_2$ for $x \in X$, i.e., M obeys $\mathcal{V}(M)$ as stated in Equation (3.4). Similarly, if $r_1 \rightarrow r_2 \in E_M$, then M obeys Equation (3.5). This completes the proof of Item 1. $\square$

We now prove Item 2 of Lemma 3.7.5.

Proof of Item 2 of Lemma 3.7.5. As discussed in the introduction, there cannot exist two distinct MECs with the same skeleton and the same set of v-structures. This implies that there can be at most one MEC of G containing the v-structures in Equation (3.4). The uniqueness of the MEC presented in Item 2 of Lemma 3.7.5 follows from this fact. Therefore, to prove Item 2 of Lemma 3.7.5, we only need to show that there exists an MEC M of G that satisfies Item 2 of Lemma 3.7.5.

We construct an MEC $M = M_1 \cup M_2 \cup \{r_2 \rightarrow r_1\}$ (i.e., M is the union of graphs M_1 , M_2 , and the graph $r_1 \rightarrow r_2$, where the graph $r_1 \rightarrow r_2$ contains two nodes r_1 and r_2 with one edge $r_1 \rightarrow r_2$) of G that satisfies Item 2 of Lemma 3.7.5.

Figure 3.3 is provided as an example. In the figure, $M_1 \in \text{MEC}(G_1, r_1, 1)$, $M_2 \in \text{MEC}(G_2, r_2, 1)$, and $M = M_1 \cup M_2 \cup \{r_2 \rightarrow r_1\}$, where G , G_1 , and G_2 are from Figure 3.2.

To verify that M satisfies Item 2 of Lemma 3.7.5, we need to show the following: (a) M is an MEC of G , (b) M obeys Equation (3.4), (c) $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, and (d) $M \in \text{MEC}(G, r_1, 1)$. Note that M , constructed in Figure 3.3, has all these four properties.

We first show that M is an MEC of G . From the construction of M , the skeleton of M is G . Therefore, it suffices to show that M obeys Items 1 and 2 of Theorem 3.2.1.

We start by showing that M obeys Item 1 of Theorem 3.2.1. Suppose M contains an induced subgraph of the form $a \rightarrow b - c$. There are three possible cases based on the construction of M : (a) $a, b, c \in V_{G_1}$, (b) $a, b, c \in V_{G_2}$, or (c) $a = r_2$, $b = r_1$, and $c \in V_{G_1}$. From the construction of M , the induced subgraph of M on V_{G_1} is M_1 . This implies that if $a, b, c \in V_{G_1}$, then $a \rightarrow b - c \in M_1$. But since M_1 is an MEC of the tree G_1 , this would contradict Item 1 of Theorem 3.2.1. Thus, case (a) cannot occur. A similar argument applies to case (b). If, for some $c \in V_{G_1}$, $r_1 - c \in M$, then $r_1 - c \in M_1$ (as M_1 is an induced subgraph of M on V_{G_1}). Since $M_1 \in \text{MEC}(G_1, r_1, 1)$, there must exist a node x such that $x \rightarrow r_1 \in M_1$, implying M_1 contains an induced subgraph of the form $x \rightarrow r_1 - c$, which cannot occur due to Item 1 of Theorem 3.2.1. This implies that case (c) also cannot occur. Thus, M does not have any induced subgraph of the form $a \rightarrow b - c$. This shows that M satisfies Item 1 of Theorem 3.2.1.

We now show that each directed edge $u \rightarrow v$ in M satisfies Item 2 of Theorem 3.2.1. Let $u \rightarrow v$ be a directed edge in M . From the construction of M , one of the following holds: $u \rightarrow v \in M_1$, $u \rightarrow v \in M_2$, or $u = r_2$ and $v = r_1$. Since M_1 is an MEC of the tree graph G_1 , from Item 2 of Theorem 3.2.1, any directed edge $u \rightarrow v$ in M_1 must be part of an induced subgraph of M_1 of the form either $w \rightarrow u \rightarrow v$ or $u \rightarrow v \leftarrow w$. From the construction of M , M_1 is an induced subgraph of M , which implies that any induced subgraph of M_1 is also an induced subgraph of M . Therefore, if $u \rightarrow v \in M_1$, then the directed edge $u \rightarrow v$ in M satisfies Item 2 of Theorem 3.2.1. A similar argument applies for $u \rightarrow v \in M_2$. For the case $u = r_2$ and $v = r_1$, since $M_1 \in \text{MEC}(G_1, r_1, 1)$, there exists a node $x \in V_{G_1}$ such that $x \rightarrow r_1 \in M_1$. From the construction of M , $x \rightarrow r_1 \in M$. This implies that $x \rightarrow r_1 \leftarrow r_2$ is an induced subgraph of M , implying $r_2 \rightarrow r_1$ also satisfies Item 2 of Theorem 3.2.1. This shows that all directed edges in M satisfy Item 2 of Theorem 3.2.1.

Next, we show that M obeys Equation (3.4). From the construction of M , the induced subgraph of M on V_{G_1} is M_1 . Therefore, $\mathcal{V}(M[V_{G_1}]) = \mathcal{V}(M_1)$. Similarly, $\mathcal{V}(M[V_{G_2}]) = \mathcal{V}(M_2)$. This implies that M contains the v-structures of M_1 and M_2 .

Additionally, from the construction of M , besides these v-structures, any v-structure contained in M must involve one edge $r_2 \rightarrow r_1$. And, if $x \rightarrow r_1 \in M$, then either $x = r_2$ or $x \rightarrow r_1 \in M_1$. Hence, the v-structures of M with one edge $r_2 \rightarrow r_1$ are of the form $x \rightarrow r_1 \leftarrow r_2$ for $x \rightarrow r_1 \in M_1$. This shows that the v-structures in M obey Equation (3.4).

We now prove that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. Since M satisfies Items 1 and 2 of Observation 3.7.4, it follows from Observation 3.7.4 that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.

We now prove the remaining part of Item 2, that is, $M \in \text{MEC}(G, r_1, 1)$. The following observation provides the proof.

Observation 3.7.13. *Let M and M_1 be MECs of G and G_1 , respectively, such that $\mathcal{P}(M, V_{G_1}) = M_1$, and $M_1 \in \text{MEC}(G_1, r_1, 1)$. Then, $M \in \text{MEC}(G, r_1, 1)$.*

Proof. Since $M_1 \in \text{MEC}(G_1, r_1, 1)$, there exists an x such that $x \rightarrow r_1 \in M_1$. Since M_1 is a projection of M , by Lemma 3.7.3, $x \rightarrow r_1 \in M$. Therefore, $M \in \text{MEC}(G, r_1, 1)$. $\square$

Since $M_1 \in \text{MEC}(G_1, r_1, 1)$ and $\mathcal{P}(M, V_{G_1}) = M_1$, from Observation 3.7.13, we have $M \in \text{MEC}(G, r_1, 1)$. This completes the proof of Item 2 of Lemma 3.7.5. $\square$

The proof of Item 3 of Lemma 3.7.5 is similar to Item 2 of Lemma 3.7.5. We simply swap the roles of M_1 and r_1 with M_2 and r_2 , respectively. The required MEC that satisfies Item 3 of Lemma 3.7.5 is $M = M_1 \cup M_2 \cup \{r_1 \rightarrow r_2\}$.

This completes the proof of Lemma 3.7.5. $\square$

The proofs of Lemmas 3.7.7, 3.7.9 and 3.7.11 are similar in nature to the proof of Lemma 3.7.5. For completeness, the proofs are presented below.

Proof of Lemma 3.7.7. We first prove Item 1 of Lemma 3.7.7.

Proof of Item 1 of Lemma 3.7.7. Let M be an MEC of G such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. From Observation 3.7.4, M has the v-structures of M_1 and M_2 . In addition to those, it can only have v-structures that involve an edge with endpoints r_1 and r_2 .

Since $M_1 \in \text{MEC}(G_1, r_1, 1)$, there exists an $x \in V_{G_1}$ such that $x \rightarrow r_1$ in M_1 . Following Item 1 of Theorem 3.2.1, there cannot be an induced subgraph $x \rightarrow r_1 - y$ in M_1 . Therefore, each edge adjacent to r_1 in M_1 is directed. Consequently, we can partition the neighbors of r_1 in G_1 into X and X' , where $X = \{x : x \rightarrow r_1 \in M_1\}$ and $X' = \{x : r_1 \rightarrow x \in M_1\}$, with X being non-empty.

From Lemma 3.7.3, for $x \in X$, $x \rightarrow r_1 \in M$, and for $x \in X'$, $r_1 \rightarrow x \in M$. Further, from Item 1 of Theorem 3.2.1, either $r_2 \rightarrow r_1 \in M$ or $r_1 \rightarrow r_2 \in M$. Otherwise, for some $x \in X$, M contains an induced subgraph $x \rightarrow r_1 - r_2$, contradicting Item 1 of Theorem 3.2.1.

If $r_2 \rightarrow r_1 \in M$, then the v-structures in M involving an edge with endpoints r_1 and r_2 are of the form $x \rightarrow r_1 \leftarrow r_2$ for each $x \in X$.

Suppose $r_1 \rightarrow r_2 \in M$. Since $M_2 \in \text{MEC}(G_2, r_2, 0, j)$, from Definition 3.3.3, there cannot be an edge $y \rightarrow r_2$ in M_2 . Thus, we can partition the neighbors of r_2 in G_2 into Y and Y' , where $Y = \{y : r_2 - y \in M_2\}$ (a set of j elements), and $Y' = \{y : r_2 \rightarrow y \in M_2\}$ (a set of $\delta_2 - j$ elements). From Lemma 3.7.3, for $y \in Y'$, $r_2 \rightarrow y \in M$. Since $r_1 \rightarrow r_2 \in M$, from Item 1 of Theorem 3.2.1, for $y \in Y$, either $y \rightarrow r_2 \in M$ or $r_2 \rightarrow y \in M$, otherwise, $r_1 \rightarrow r_2 - y$ would be an induced subgraph of M , contradicting Item 1 of Theorem 3.2.1.

If for two distinct $y_1, y_2 \in Y$, $y_1 \rightarrow r_2$ and $y_2 \rightarrow r_2 \in M$, then we get a v-structure $y_1 \rightarrow r_2 \leftarrow y_2$ in M where the nodes are in V_{G_2} , but the v-structure is not in M_2 , implying $\mathcal{V}(M[V_{G_2}]) \neq \mathcal{V}(M_2)$, which further implies that $\mathcal{P}(M, V_{G_2}) \neq M_2$, a contradiction. This implies that there exists at most one $y \in Y$ such that $y \rightarrow r_2 \in M$.

If there does not exist any $y \in Y$ such that $y \rightarrow r_2 \in M$, then no v-structure in M involves an edge with endpoints r_1 and r_2 . In this case, M only contains the v-structures of M_1 and M_2 , and obeys Equation (3.7).

However, if there exists a $y \in Y$ such that $y \rightarrow r_2 \in M$, then $r_1 \rightarrow r_2 \leftarrow y$ is the only v-structure that involves an edge with endpoints r_1 and r_2 in M . In this case, M obeys Equation (3.8).

The above discussion shows that in each possibility, M obeys Item 1 of Lemma 3.7.7. This completes the proof of Item 1 of Lemma 3.7.7. $\square$

As discussed in the introduction, there cannot be two distinct MECs with the same skeleton and the same set of v-structures. The uniqueness of the MECs present in

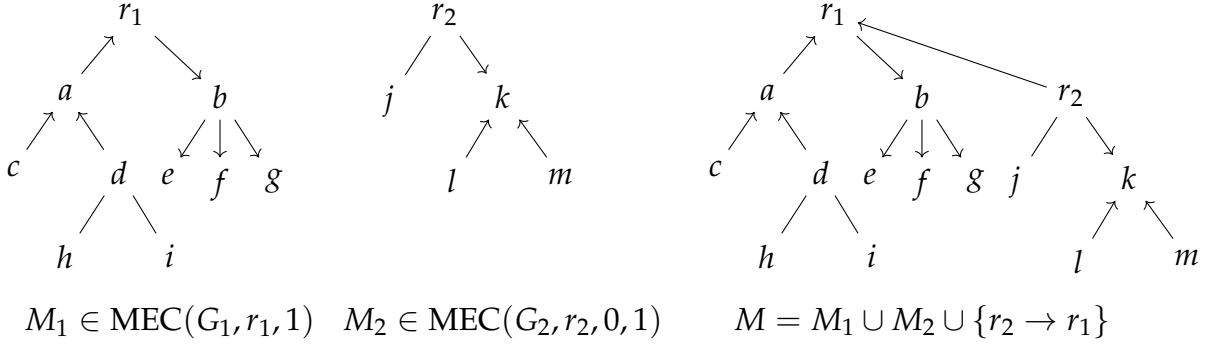


Figure 3.4: Auxiliary picture for the proof of Item 2 of Lemma 3.7.7.

Items 2 to 4 of Lemma 3.7.7 follows from this fact. To complete the proofs of these items, we now prove the existence of each of these MECs.

Proof of Item 2 of Lemma 3.7.7. The proof follows a similar approach to that of verifying Item 2 in Lemma 3.7.5. For completeness, we provide the proof. We construct an MEC $M = M_1 \cup M_2 \cup \{r_2 \rightarrow r_1\}$ of G that satisfies Item 2 of Lemma 3.7.7. As already discussed, there can be almost one such MEC, so this will complete the proof of Item 2 of Lemma 3.7.7.

Figure 3.4 is provided as an example. In the figure, $M_1 \in \text{MEC}(G_1, r_1, 1)$, $M_2 \in \text{MEC}(G_2, r_2, 0, 1)$, and $M = M_1 \cup M_2 \cup \{r_2 \rightarrow r_1\}$, where G , G_1 , and G_2 are from Figure 3.2.

To verify that M satisfies Item 2 of Lemma 3.7.7, we need to show the following: (a) M is an MEC of G , (b) M satisfies Equation (3.6), (c) $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, and (d) $M \in \text{MEC}(G, r_1, 1)$. Note that M , constructed in Figure 3.4, has all these four properties.

Firstly, we establish that M is an MEC of G . The construction of M implies that its skeleton is identical to G . Hence, we only need to confirm that M is an MEC, i.e., it satisfies Items 1 and 2 of Theorem 3.2.1.

Suppose M contains an induced subgraph of the form $a \rightarrow b - c$. There are three possible scenarios from the construction of M : (a) $a, b, c \in V_{G_1}$, (b) $a, b, c \in V_{G_2}$, or (c) $a = r_2$, $b = r_1$, and $c \in V_{G_1}$. If $a, b, c \in V_{G_1}$, then $a \rightarrow b - c \in M_1$, as M_1 is the induced subgraph of M on V_{G_1} . Since M_1 is an MEC, this contradicts Item 1 of Theorem 3.2.1. Therefore, this scenario is not possible. Similarly, $a, b, c \in V_{G_2}$ is also not possible.

Considering $a = r_2$, $b = r_1$, and $c \in V_{G_1}$, from the construction of M , $r_1 - c \in M_1$. As $M_1 \in \text{MEC}(G_1, r_1, 1)$, there must exist a node x such that $x \rightarrow r_1 \in M_1$. This implies that M_1 contains an induced subgraph of the form $x \rightarrow r_1 - c \in M_1$. However, this contradicts Item 1 of Theorem 3.2.1, as M_1 is an MEC of the tree graph G_1 . Consequently, M cannot contain any induced subgraph of the form $a \rightarrow b - c$. Thus, M satisfies Item 1 of Theorem 3.2.1.

We now show that each directed edge $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1. Let $u \rightarrow v$ be a directed edge in M . Based on the construction of M , one of the following holds: $u \rightarrow v \in M_1$, $u \rightarrow v \in M_2$, or $u = r_2$ and $v = r_1$.

Considering M_1 as an MEC of the tree graph G_1 , if $u \rightarrow v \in M_1$, then from Item 2 of Theorem 3.2.1, $u \rightarrow v$ is part of an induced subgraph in M_1 of the form $w \rightarrow u \rightarrow v$ or $w \rightarrow v \leftarrow u$. From the construction of M , M_1 is an induced subgraph of M . This implies that an induced subgraph of M_1 is also an induced subgraph of M . This further implies that $u \rightarrow v$ in M also obeys Item 2 of Theorem 3.2.1. Similarly, if $u \rightarrow v \in M_2$, then $u \rightarrow v$ in M satisfies Item 2 of Theorem 3.2.1 for M .

Now, let us consider the case when $u = r_2$ and $v = r_1$. Since $M_1 \in \text{MEC}(G_1, r_1, 1)$, from Definition 3.3.1, there exists a node $x \in V_{G_1}$ such that $x \rightarrow r_1 \in M_1$. From the construction of M , $x \rightarrow r_1 \leftarrow r_2$ is an induced subgraph of M . Therefore, in this case, $u \rightarrow v$ in M also obeys Item 2 of Theorem 3.2.1.

Thus, this analysis shows that M satisfies Item 2 of Theorem 3.2.1. The aforementioned discussions imply that M complies with Items 1 and 2 of Theorem 3.2.1. Consequently, this verifies that M is an MEC.

Next, we show that the v-structures of M obey Equation (3.6). From the construction of M , M_1 and M_2 are the induced subgraphs of M on V_{G_1} and V_{G_2} , respectively. Thus, $\mathcal{V}(M[V_{G_1}]) = \mathcal{V}(M_1)$ and $\mathcal{V}(M[V_{G_2}]) = \mathcal{V}(M_2)$. Since $r_2 \rightarrow r_1 \in M$, apart from the v-structures of M_1 and M_2 , M possesses v-structures involving the edge $r_2 \rightarrow r_1$. Given $M_1 \in \text{MEC}(G_1, r_1, 1)$, there exists a non-empty set $X = \{x : x \rightarrow r_1 \in M\}$. For each $x \in X$, $x \rightarrow r_1 \leftarrow r_2$ is a v-structure in M . This shows that the v-structures in M satisfy Equation (3.6).

Next, we verify $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. As per the construction of M , $M[V_{G_1}] = M_1$ and $M[V_{G_2}] = M_2$. Consequently, $\mathcal{V}(M[V_{G_1}]) = \mathcal{V}(M_1)$ and $\mathcal{V}(M[V_{G_2}]) = \mathcal{V}(M_2)$.

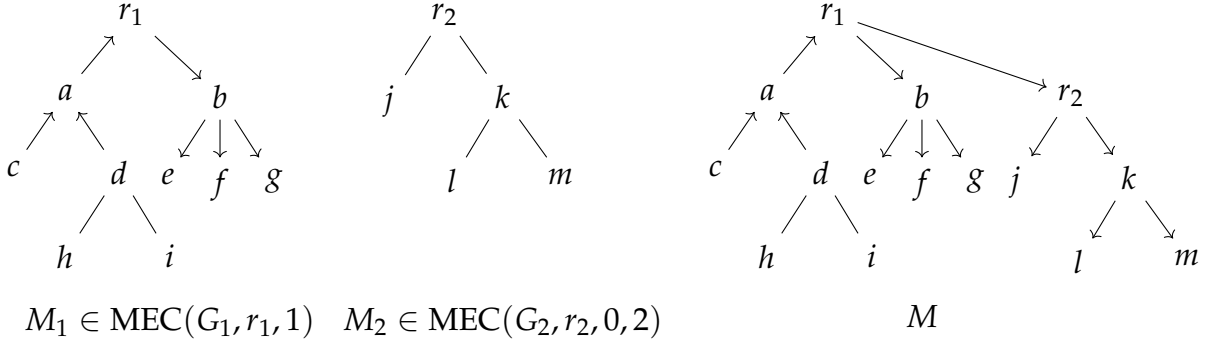


Figure 3.5: Auxiliary picture for the proof of Item 3 of Lemma 3.7.7.

Therefore, from Definition 3.7.1, $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.

From Observation 3.7.13, $M \in \text{MEC}(G, r_1, 1)$. This completes the proof of Item 2 of Lemma 3.7.7. $\square$

Proof of Item 3 of Lemma 3.7.7. We create an MEC M of G using the following steps:

1. Initialize $M = M_1 \cup M_2 \cup \{r_1 \rightarrow r_2\}$.
2. Update M by replacing $u - v$ in M with $u \rightarrow v$ if $u - v \in M_2$ and there exists an undirected path in M_2 from r_2 to (u, v) (i.e., the first node of the path is r_2 , and the last two nodes are u and v).

Step 2 implies that we repeat the second step as long as there exists an edge $u - v$ in M such that $u - v \in M_2$ and there exists an undirected path in M_2 from r_2 to (u, v) . Figure 3.5 is provided as an example. In the figure, $M_1 \in \text{MEC}(G_1, r_1, 1)$, $M_2 \in \text{MEC}(G_2, r_2, 0, 2)$, and M is constructed using the above two steps, where G , G_1 , and G_2 are from Figure 3.2. After the initialization step (the first step), M has an undirected edge $k - l$. We replace it with $k \rightarrow l$ at the second step, since M_2 has an undirected path $P = (r_2, k, l)$ from r_2 to (k, l) (the first node of the path is r_2 , and the last two nodes of the path are k and l). Similarly, the other edges of M_2 get directed in M (as shown in the figure).

We now show that M , constructed using the aforementioned steps, satisfies Item 3 of Lemma 3.7.7. To prove this, we need to establish the following: (a) M is an MEC of G ; (b) M satisfies Equation (3.7); (c) $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$; and (d) $M \in$

$\text{MEC}(G, r_1, 1)$. Note that M , constructed in Figure 3.5 using the above steps, has all these four properties.

We first show that M is an MEC of G . From the construction of M , the skeleton of M is G . Therefore, it suffices to show that M obeys Items 1 and 2 of Theorem 3.2.1.

Suppose M contains an induced subgraph of the form $a \rightarrow b - c$. From the construction of M , there are three possibilities: (a) $a, b, c \in V_{G_1}$, (b) $a, b, c \in V_{G_2}$, or (c) $a = r_1$, $b = r_2$, and $c \in V_{G_2}$. One by one, we will show that none of the possibilities occurs.

From the construction of M , M_1 is an induced subgraph of M (note that step 2 of the construction of M does not make any changes to an edge with endpoints in V_{G_1}). This implies that an induced subgraph of M with vertices in V_{G_1} is also an induced subgraph of M_1 . Since M_1 is an MEC of the tree graph G_1 , from Item 1 of Theorem 3.2.1, there cannot be an induced subgraph $a \rightarrow b - c \in M_1$. This implies that the first possibility cannot occur.

Suppose M contains an induced subgraph of the form $a \rightarrow b - c$ such that $a, b, c \in V_{G_2}$. From the construction of M , $b - c \in M_2$ (note that step 2 of the construction of M does not change a directed edge in $M_1 \cup M_2 \cup \{r_1 \rightarrow r_2\}$ into an undirected edge), and either $a \rightarrow b \in M_2$ or $a - b \in M_2$. If $a \rightarrow b \in M_2$, then $a \rightarrow b - c$ is an induced subgraph in the MEC M_2 of the tree graph G_2 , contradicting Item 1 of Theorem 3.2.1. If $a - b \in M_2$, then it must have been directed at step 2 of the construction of M . This implies that there exists a path $P = (v_1 = r_2, v_2, \dots, v_{l-1} = a, v_l = b)$ in M_2 , from r_2 to (a, b) . This further implies that there exists a path $P' = (v_1 = r_2, v_2, \dots, v_{l-1} = a, v_l = b, v_{l+1} = c)$ in M_2 , from r_2 to (b, c) . However, then we have $b \rightarrow c \in M$ at step 2 of the construction of M . This shows that the second possibility also cannot occur.

Suppose $r_1 \rightarrow r_2 - c$ is an induced subgraph in M such that $c \in V_{G_2}$. From the construction of M , this implies $r_2 - c \in M_2$. However, the edge $r_2 - c$ forms a path $P = (r_2, c)$ from r_2 to (r_2, c) (note that the first node of the path is r_2 , and the last two nodes of the path are r_2 and c). Thus, we have $r_2 \rightarrow c$ at step 2 of the construction of M . This implies that the third possibility cannot occur either.

This shows that M cannot have an induced subgraph of the form $a \rightarrow b - c$. Therefore, M obeys Item 1 of Theorem 3.2.1.

We now show that M obeys Item 2 of Theorem 3.2.1. Suppose $u \rightarrow v$ is a directed

edge in M . Then one of the following occurs: $u, v \in V_{G_1}$, $u, v \in V_{G_2}$, or $u = r_1$ and $v = r_2$. We will show that in each of these possibilities, $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1.

Suppose $u, v \in V_{G_1}$. From the construction of M , $u \rightarrow v \in M_1$. Since M_1 is an MEC of the tree graph G_1 , from Item 2 of Theorem 3.2.1, $u \rightarrow v$ is part of an induced subgraph of M_1 of the form either $u \rightarrow v \leftarrow w$ or $w \rightarrow u \rightarrow v$. From the construction of M , an induced subgraph of M_1 is also an induced subgraph of M . Therefore, $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1.

Suppose $u, v \in V_{G_2}$. From the construction of M , there are two cases: either $u \rightarrow v \in M_2$ or $u - v \in M_2$.

Suppose $u \rightarrow v \in M_2$. Since M_2 is an MEC of the tree graph G_2 , from Item 2 of Theorem 3.2.1, $u \rightarrow v$ is part of an induced subgraph of M_2 of the form either $u \rightarrow v \leftarrow w$ or $w \rightarrow u \rightarrow v$. Since, from the construction of M , a directed induced subgraph of M_2 is also an induced subgraph of M , it follows that in the case when $u \rightarrow v \in M_2$, $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1.

Suppose $u - v \in M_2$. Then it must have been directed in M at step 2 of the construction of M . This implies that there exists an undirected path from r_2 to (u, v) in M_2 . Let the path be $P = (x_0 = r_2, x_1, \dots, x_{l-1} = u, x_l = v)$ for some $l \geq 1$. From the construction of M , for all $0 \leq i < l$, $x_i \rightarrow x_{i+1} \in M$ (directed at step 2 of the construction of M) because $P_i = (x_0 = r_2, \dots, x_i, x_{i+1})$ is an undirected path in M_2 from r_2 to (x_i, x_{i+1}) . If $l > 1$, then $u \rightarrow v$ is part of an induced subgraph $x_{l-2} \rightarrow x_{l-1} \rightarrow x_l$ in M . And, if $l = 1$, then $r_2 = u$, and $u \rightarrow v$ is part of an induced subgraph $r_1 \rightarrow r_2 \rightarrow v$ in M . This shows that if $u, v \in V_{G_2}$, then $u \rightarrow v$ obeys Item 2 of Theorem 3.2.1.

We now show that $r_1 \rightarrow r_2$ in M also satisfies Item 2 of Theorem 3.2.1. Since $M_1 \in \text{MEC}(G_1, r_1, 1)$, there must exist an edge $x \rightarrow r_1 \in M_1$. From the construction of M , we have $x \rightarrow r_1 \in M$. Thus, $r_1 \rightarrow r_2$ is part of an induced subgraph $x \rightarrow r_1 \rightarrow r_2$ in M (similar to Figure 1.3.a). Therefore, we have shown that every edge of M satisfies Item 2 of Theorem 3.2.1.

The above discussion shows that M is an MEC, as it obeys Item 1 and Item 2 of Theorem 3.2.1.

We now show that M obeys Equation (3.7). From the construction of M , the directed

edges of M_1 and M_2 are also directed edges of M . This implies that the v-structures of M_1 and M_2 are v-structures of M , i.e., $\mathcal{V}(M_1) \cup \mathcal{V}(M_2) \subseteq \mathcal{V}(M)$. For the completeness of the proof, we show that if $u \rightarrow v \leftarrow w$ is a v-structure in M , then it is either a v-structure in M_1 or a v-structure in M_2 , i.e., $\mathcal{V}(M) \subseteq \mathcal{V}(M_1) \cup \mathcal{V}(M_2)$. This implies that $\mathcal{V}(M) = \mathcal{V}(M_1) \cup \mathcal{V}(M_2)$, and thus M obeys Equation (3.7).

Suppose there exists a v-structure $u \rightarrow v \leftarrow w$ in M . We will show that either $u \rightarrow v \leftarrow w$ is a v-structure in M_1 , or it is a v-structure in M_2 . From the construction of M , the following possibilities arise: (a) $u, v, w \in V_{G_1}$, (b) $u, v, w \in V_{G_2}$, or (c) $u = r_1$, $v = r_2$, and $w \in V_{G_2}$ is a neighbor of r_2 (or equivalently, $w = r_1$, $v = r_2$, and $u \in V_{G_2}$ is a neighbor of r_2 ; both are equivalent as both of them imply the same v-structure).

From the construction of M , M_1 is an induced subgraph of M . This implies that if $u, v, w \in V_{G_1}$, then $u \rightarrow v \leftarrow w$ is a v-structure in M_1 .

Next, suppose $u, v, w \in V_{G_2}$. From the construction of M , if $a \rightarrow b \in M$, then either $a \rightarrow b \in M_2$ or $a - b \in M_2$. This implies that either $u - v \in M_2$ or $u \rightarrow v \in M_2$, and either $v - w \in M_2$ or $v \leftarrow w \in M_2$. From Item 1 of Theorem 3.2.1, either $u \rightarrow v \leftarrow w \in M_2$ or $u - v - w \in M_2$ (recall that M_2 is an MEC of the tree graph G_2).

If $u \rightarrow v \leftarrow w \in M_2$, then this v-structure is indeed a v-structure in M_2 . Suppose instead that $u - v - w \in M_2$. If $u \rightarrow v \in M$, then it must have been directed at step 2 of the construction of M , i.e., there exists an undirected path from r_2 to (u, v) . Since $u - v - w \in M_2$, the existence of the path from r_2 to (u, v) in M_2 implies the existence of the path from r_2 to (v, w) in M_2 . Also, since M_2 is an MEC of a tree graph, there cannot be two different paths from r_2 to w . This implies that at step 2 of the construction of M , if we have $u \rightarrow v \in M$, then we must also have $v \rightarrow w \in M$. However, this leads to a contradiction, as we have assumed that $u \rightarrow v \leftarrow w \in M$. Therefore, we conclude that it is impossible to have both $u - v - w \in M_2$ and $u \rightarrow v \leftarrow w \in M$. This further implies that if $u, v, w \in V_{G_2}$, and $u \rightarrow v \leftarrow w$ is a v-structure in M then $u \rightarrow v \leftarrow w$ is a v-structure in M_2 as well.

We now move to the final possibility. Suppose $u = r_1$, $v = r_2$, and $w \in V_{G_2}$ is a neighbor of r_2 . We show that this possibility cannot occur; that is, $r_1 \rightarrow r_2 \leftarrow w$ is not a v-structure in M . Since $M_2 \in \text{MEC}(G_2, r_2, 0)$, from Definition 3.3.2, either $r_2 \rightarrow w \in M_2$ or $r_2 - w \in M_2$. From the construction of M (see Items 1 and 2), if $r_2 \rightarrow w \in M_2$, then

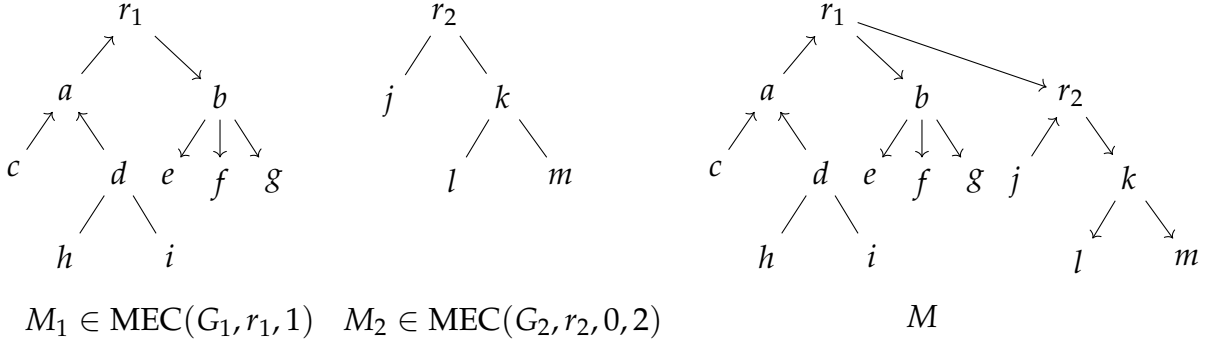


Figure 3.6: Auxiliary picture for the proof of Item 4 of Lemma 3.7.7.

$r_2 \rightarrow w \in M$. This implies that $r_1 \rightarrow r_2 \leftarrow w$ is not a v-structure in M .

Now, suppose $r_2 - w \in M_2$. Then, from step 2 of the construction of M , it follows that $r_2 \rightarrow w \in M$, which again implies that $r_1 \rightarrow r_2 \leftarrow w$ is not a v-structure in M . Thus, this possibility cannot occur.

Thus, we have shown that in all possible scenarios, if $u \rightarrow v \leftarrow w$ is a v-structure in M , then it is a v-structure either in M_1 or in M_2 . This completes the proof that M obeys Equation (3.7). The above discussion implies that $\mathcal{V}(M[V_{G_1}]) = \mathcal{V}(M_1)$ and $\mathcal{V}(M[V_{G_2}]) = \mathcal{V}(M_2)$. Therefore, from Definition 3.7.1, we have $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. From Observation 3.7.13, it follows that $M \in \text{MEC}(G, r_1, 1)$. This completes the proof of Item 3 of Lemma 3.7.7. $\square$

Proof of Item 4 of Lemma 3.7.7. We construct an MEC of G using the following steps:

1. Initialize $M = M_1 \cup M_2 \cup \{r_1 \rightarrow r_2\}$.
2. Pick an edge $y - r_2 \in M_2$. Replace the edge $y - r_2$ in M with $y \rightarrow r_2$.
3. Update M by replacing $u - v$ in M with $u \rightarrow v$ if $u - v \in M_2$ and there exists an undirected path in M_2 from r_2 to (u, v) in M_2 such that the path does not contain y .

Figure 3.6 is provided as an example. In the figure, $M_1 \in \text{MEC}(G_1, r_1, 1)$, $M_2 \in \text{MEC}(G_2, r_2, 0, 2)$, and M is constructed using the steps mentioned above, where G , G_1 , and G_2 are from Figure 3.2. In the second step, we pick the edge $j - r_2$ of M_2 and replace the edge $j - r_2$ in M with $j \rightarrow r_2$ (here j plays the role of y). In the third step, the

undirected edges of M_2 that satisfy the conditions of the third step get directed in M . For instance, the undirected edge $k - l$ of M_2 is directed in M . We replace the edge $k - l$ in M (after the second step) with $k \rightarrow l$ in the third step, since M_2 has an undirected path $P = (r_2, k, l)$ from r_2 to (k, l) , and the path does not contain j .

We show that M constructed using the above steps will obey Item 4 of Lemma 3.7.7. To prove this, we have to show the following: (a) M is an MEC of G , (b) M obeys Equation (3.8), (c) $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, and (d) $M \in \text{MEC}(G, r_1, 1)$. Note that M , constructed in Figure 3.6 using the above steps, has all these four properties.

We first show that M is an MEC of G . From the construction of M , the skeleton of M is G . Therefore, it suffices to show that M obeys Items 1 and 2 of Theorem 3.2.1.

We first show that M obeys Item 1 of Theorem 3.2.1. Suppose M contains an induced subgraph of the form $a \rightarrow b - c$. From the construction of M , there are three possibilities: (a) $a, b, c \in V_{G_1}$, (b) $a, b, c \in V_{G_2}$, or (c) $a = r_1$, $b = r_2$, and $c \in V_{G_2}$.

We will show that none of these possibilities occurs.

Suppose $a, b, c \in V_{G_1}$. From the construction of M , an induced subgraph of M with vertices in V_{G_1} is also an induced subgraph of M_1 (note that steps 2 and 3 in the construction of M do not change any edge with endpoints in V_{G_1} from $M_1 \cup M_2 \cup \{r_1 \rightarrow r_2\}$). Since M_1 is an MEC of the tree graph G_1 , from Item 1 of Theorem 3.2.1, there cannot be an induced subgraph $a \rightarrow b - c \in M_1$. This implies that the first possibility cannot occur.

Suppose $a, b, c \in V_{G_2}$. From the construction of M , $b - c \in M_2$ (note that steps 2 and 3 do not change a directed edge in $M_1 \cup M_2 \cup \{r_1 \rightarrow r_2\}$ into an undirected edge), and either $a \rightarrow b \in M_2$ or $a - b \in M_2$. If $a \rightarrow b \in M_2$, then $a \rightarrow b - c$ is an induced subgraph in the MEC M_2 of the tree graph G_2 , contradicting Item 1 of Theorem 3.2.1. If $a - b \in M_2$, then it must have been directed either at step 2 (first case) or at step 3 (second case). If it is directed at step 2, then $a = y$, $b = r_2$, and $c \neq y$ is a neighbor of r_2 . This implies that $r_2 - c$ is an undirected path in M_2 from r_2 to (r_2, c) . However, at step 3, $r_2 - c$ has been converted into $r_2 \rightarrow c$. This implies that the first case cannot occur. Now consider the second case. Suppose $a - b$ is directed at step 3. This implies that there exists an undirected path $P = (u_1 = r_2, u_1, \dots, u_{l-1} = a, u_l = b)$, from r_2 to (a, b) in M_2 , and the path does not contain y . This further implies that there exists an

undirected path $P' = (u_1 = r_2, u_1, \dots, u_{l-1} = a, u_l = b, u_{l+1} = c)$, from r_2 to (b, c) in M_2 , and the path does not contain y . But then, we have $b \rightarrow c \in M$ at step 3. This shows that the second possibility cannot occur.

Suppose $a = r_1$, $b = r_2$, and $c \in V_{G_2}$. From the construction of M , this implies that $r_2 - c \in M_2$. There are two cases: either $c = y$, or $c \neq y$. If $c = y$, then at step 2, we have $c \rightarrow b \in M$. Therefore, this case cannot occur. Suppose $c \neq y$. However, the edge $r_2 - c$ is a path from r_2 to (r_2, c) in M , and the path does not contain y . Then, it must have been directed at step 3. This implies that the third possibility also cannot occur. Thus, we have shown that M cannot have an induced subgraph of the form $a \rightarrow b - c$. This implies that M obeys Item 1 of Theorem 3.2.1.

We now show that M obeys Item 2 of Theorem 3.2.1. Suppose $u \rightarrow v$ is a directed edge in M ; then one of the following occurs: $u, v \in V_{G_1}$, $u, v \in V_{G_2}$, or $u = r_1$ and $v = r_2$. We show that in each of the possibilities, $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1.

Suppose $u, v \in V_{G_1}$. From the construction of M , M_1 is an induced subgraph of M (note that steps 2 and 3 do not change an edge in $M_1 \cup M_2 \cup \{r_1 \cup r_2\}$ if the endpoints of the edge is in V_{G_1}). This implies that $u \rightarrow v \in M_1$. Since M_1 is an MEC of the tree graph G_1 , from Item 2 of Theorem 3.2.1, $u \rightarrow v$ is part of an induced subgraph of M_1 of the form either $u \rightarrow v \leftarrow w$ or $w \rightarrow u \rightarrow v$. Since M_1 is an induced subgraph of M , an induced subgraph of M_1 is an induced subgraph of M . Therefore, $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1.

Suppose $u, v \in V_{G_2}$. From the construction of M , there are two cases: either $u \rightarrow v \in M_2$ or $u - v \in M_2$. Suppose $u \rightarrow v \in M_2$. Since M_2 is an MEC of the tree graph G_2 , from Item 2 of Theorem 3.2.1, $u \rightarrow v$ is part of an induced subgraph of M_2 of the form either $u \rightarrow v \leftarrow w$ or $w \rightarrow u \rightarrow v$. From the construction of M , a directed induced subgraph of M_2 is an induced subgraph of M (because from the construction, a directed edge of M_2 is a directed edge of M , and the skeletons of M_2 and $M[V_{G_2}]$ are the same). Therefore, in this case also, $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1.

Suppose $u - v \in M_2$. Then, it must have been directed in M either at step 2 or at step 3. Suppose $u - v \in M_2$ and it gets directed in M at step 2. Then, $u = y$ and $v = r_2$. In that case, $u \rightarrow v$ is part of an induced subgraph $r_1 \rightarrow r_2 \leftarrow y$, obeying Item 2 of Theorem 3.2.1. Suppose $u - v$ gets directed in M at step 3. This implies that there exists

an undirected path from r_2 to (u, v) in M_2 , and the path does not contain y . Let the path be $P = (x_0 = r_2, x_1, \dots, x_{l-1} = u, x_l = v)$ for some $l \geq 1$. From the construction of M , for all $0 \leq i < l$, $x_i \rightarrow x_{i+1} \in M$ because $P_i = (x_0 = r_2, \dots, x_i, x_{i+1})$ is an undirected path in M_2 from r_2 to (x_i, x_{i+1}) , and P_i does not contain y either. If $l > 1$, then $u \rightarrow v$ is part of an induced subgraph $x_{l-2} \rightarrow x_{l-1} \rightarrow x_l$ in M . And, if $l = 1$, then $r_2 = u$, and $u \rightarrow v$ is part of an induced subgraph $r_1 \rightarrow r_2 \rightarrow v$ in M . Thus, we have shown that if $u, v \in V_{G_2}$, then $u \rightarrow v$ obeys Item 2 of Theorem 3.2.1.

We now show that $r_1 \rightarrow r_2$ in M also obeys Item 2 of Theorem 3.2.1. The edge $r_1 \rightarrow r_2$ is part of an induced subgraph $r_1 \rightarrow r_2 \leftarrow y$ in M . Thus, we have shown that each edge of M obeys Item 2 of Theorem 3.2.1.

The above discussion shows that M obeys Items 1 and 2 of Theorem 3.2.1. This implies that M is an MEC of G .

We now show that M obeys Equation (3.8). From the construction of M , the directed edges of M_1 and M_2 are also directed edges of M . Additionally, $r_1 \rightarrow r_2 \leftarrow y$ is a v-structure in M . This implies that $\mathcal{V}(M_1) \cup \mathcal{V}(M_2) \cup \{r_1 \rightarrow r_2 \leftarrow y\} \subseteq \mathcal{V}(M)$. For completeness, we show that if $u \rightarrow v \leftarrow w$ is a v-structure in M , then one of the following holds: $u \rightarrow v \leftarrow w$ is a v-structure of M_1 , $u \rightarrow v \leftarrow w$ is a v-structure of M_2 , or $u = r_1, v = r_2$, and $w = y$, i.e., $\mathcal{V}(M) \subseteq \mathcal{V}(M_1) \cup \mathcal{V}(M_2) \cup \{r_1 \rightarrow r_2 \leftarrow y\}$. This further implies that $\mathcal{V}(M) = \mathcal{V}(M_1) \cup \mathcal{V}(M_2) \cup \{r_1 \rightarrow r_2 \leftarrow y\}$, which means that M obeys Equation (3.8).

Suppose there exists a v-structure $u \rightarrow v \leftarrow w$ in M . From the construction of M , the following possibilities arise: (a) $u, v, w \in V_{G_1}$, (b) $u = r_1, v = r_2$, and w is a neighbor of r_2 in V_{G_2} , or (c) $u, v, w \in V_{G_2}$.

Suppose $u, v, w \in V_{G_1}$. From the construction of M , M_1 is an induced subgraph of M . This implies that if $u, v, w \in V_{G_1}$, then $u \rightarrow v \leftarrow w$ is a v-structure in M_1 .

Suppose $u = r_1, v = r_2$, and w is a neighbor of r_2 in V_{G_2} . Since $M_2 \in \text{MEC}(G_2, r_2, 0)$, either $r_2 \rightarrow w \in M_2$ or $r_2 - w \in M_2$. Suppose $r_2 \rightarrow w \in M_2$. From the construction of M (steps 1-3), if $r_2 \rightarrow w \in M_2$, then $r_2 \rightarrow w \in M$. However, we have $r_2 \leftarrow w \in M$. This implies that $r_2 \rightarrow w \notin M_2$. Suppose $r_2 - w \in M_2$. From steps 2 and 3, if we have $w \rightarrow r_2 \in M$, then $w = y$. Otherwise, we would have $r_2 \rightarrow w \in M$ at step 3 since $r_2 - w$ is an undirected path from r_2 to (r_2, w) , and this path does not contain y . Thus,

we conclude that $u = r_1$, $v = r_2$, and $w = y$.

Suppose $u, v, w \in V_{G_2}$. From the construction of M , if $a \rightarrow b \in M$, then either $a \rightarrow b \in M_2$ or $a - b \in M_2$. This implies that either $u - v \in M_2$ or $u \rightarrow v \in M_2$, and either $v - w \in M_2$ or $v \leftarrow w \in M_2$. Since M_2 is an MEC of the tree graph G_2 , from Item 1 of Theorem 3.2.1, either $u \rightarrow v \leftarrow w \in M_2$ or $u - v - w \in M_2$.

If $u \rightarrow v \leftarrow w \in M_2$, then this is a v-structure in M_2 . Suppose $u - v - w \in M_2$. Then, if $u \rightarrow v \in M$, it must have been directed either at step 2 or at step 3. Suppose $u \rightarrow v$ gets directed in M at step 2. Then $u = y$ and $v = r_2$. However, in that case, $v - w$ forms an undirected path from r_2 to (v, w) (and the path does not contain y). Moreover, we have $v \rightarrow w \in M$ at step 3, leading to a contradiction, as the v-structure implies that $w \rightarrow v \in M$. This means that this case cannot occur.

Suppose now that $u \rightarrow v$ gets directed in M at step 3. Then, there exists an undirected path in M_2 , from r_2 to (u, v) , and the path does not contain y . Since $u - v - w \in M_2$, the existence of an undirected path from r_2 to (u, v) in M_2 implies the existence of an undirected path from r_2 to (v, w) in M_2 , and this new path also does not contain y . Additionally, since M_2 is an MEC of the tree graph G_2 , there cannot be two distinct paths from r_2 to w . This implies that at step 3, if we have $u \rightarrow v \in M$, then we also have $v \rightarrow w \in M$. However, this leads to a contradiction, as we initially assumed that $u \rightarrow v \leftarrow w \in M$.

Thus, we conclude that there cannot be $u - v - w \in M_2$ and $u \rightarrow v \leftarrow w \in M$. Therefore, we have shown that in all possible scenarios, if $u \rightarrow v \leftarrow w$ is a v-structure in M , then it is a v-structure either in M_1 or in M_2 , or $u = r_1$, $v = r_2$, and $w = y$.

This completes the proof that M obeys Equation (3.8).

The above discussion implies that $\mathcal{V}(M[V_{G_1}]) = \mathcal{V}(M_1)$ and $\mathcal{V}(M[V_{G_2}]) = \mathcal{V}(M_2)$. Therefore, from Definition 3.7.1, we have $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.

From Observation 3.7.13, it follows that $M \in \text{MEC}(G, r_1, 1)$. This completes the proof of Item 4 in Lemma 3.7.7. □

This completes the proof of Lemma 3.7.7. □

Proof of Lemma 3.7.9. We first prove Item 1 of Lemma 3.7.9.

Proof of Item 1 of Lemma 3.7.9. Let M be an MEC of G such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. From Observation 3.7.4, M contains the v-structures of both M_1 and M_2 . Additionally, it may have v-structures involving an edge with endpoints r_1 and r_2 .

Since $M_2 \in \text{MEC}(G_2, r_2, 1)$, there exists a $y \in V_{G_2}$ such that $y \rightarrow r_2$ in M_2 . From Item 1 of Theorem 3.2.1, there cannot be an induced subgraph $y \rightarrow r_2 - y'$ in M_2 . Therefore, each edge adjacent to r_2 in M_2 must be directed. This allows us to partition the neighbors of r_2 in G_2 into two sets: Y and Y' , where $Y = \{y : y \rightarrow r_2 \in M_2\}$ and $Y' = \{y : r_2 \rightarrow y \in M_2\}$, with Y being non-empty.

From Lemma 3.7.3, for each $y \in Y$, $y \rightarrow r_2 \in M$, and for each $y \in Y'$, $r_2 \rightarrow y \in M$. From Item 1 of Theorem 3.2.1, either $r_1 \rightarrow r_2 \in M$ or $r_2 \rightarrow r_1 \in M$. If neither condition holds, then for some $y \in Y$, M would contain an induced subgraph $y \rightarrow r_2 - r_1$, contradicting Item 1 of Theorem 3.2.1.

Assume $r_1 \rightarrow r_2 \in M$. Then, the v-structures in M involving r_1 and r_2 are of the form $y \rightarrow r_2 \leftarrow r_1$ for each $y \in Y$, thus satisfying Equation (3.9).

Now, suppose $r_2 \rightarrow r_1 \in M$. Since $M_1 \in \text{MEC}(G_1, r_1, 0, i)$, there cannot be an edge $x \rightarrow r_1$ in M_1 . Thus, we partition the neighbors of r_1 in G_1 into two sets: X and X' , where $X = \{x : x \leftarrow r_1 \in M_1\}$, consisting of i elements (from Definition 3.3.3), and $X' = \{x : r_1 \rightarrow x \in M_1\}$, consisting of $\delta_1 - i$ elements. For $x \in X'$, from Lemma 3.7.3, $r_1 \rightarrow x \in M$.

Since $r_2 \rightarrow r_1 \in M$, from Item 1 of Theorem 3.2.1, for each $x \in X$, either $x \rightarrow r_1 \in M$ or $r_1 \rightarrow x \in M$. If neither condition holds, then an induced subgraph $r_2 \rightarrow r_1 - x$ would appear in M , contradicting Item 1 of Theorem 3.2.1.

If for distinct $x_1, x_2 \in X$, we have $x_1 \rightarrow r_1, x_2 \rightarrow r_1 \in M$, then we obtain a v-structure $x_1 \rightarrow r_1 \leftarrow x_2$ in M . The nodes of this v-structure are in V_{G_1} , but the v-structure is not present in M_1 , implying $\mathcal{V}(M[V_{G_1}]) \neq \mathcal{V}(M_1)$, leading to the conclusion that $\mathcal{P}(M, V_{G_1}) \neq M_1$, a contradiction. Thus, at most one $x \in X$ has $x \rightarrow r_1 \in M$.

If no such x exists, then M has no v-structure involving an edge with endpoints r_1 and r_2 . Therefore, M contains only the v-structures of M_1 and M_2 , satisfying Equation (3.10).

If an $x \in X$ exists with $x \rightarrow r_1 \in M$, then $r_2 \rightarrow r_1 \leftarrow x$ is the only v-structure involving r_1 and r_2 in M , which satisfies Equation (3.11).

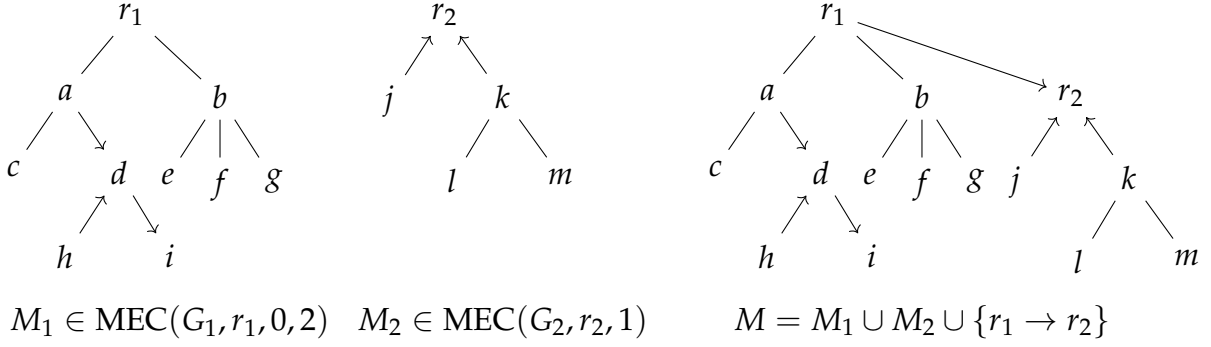


Figure 3.7: Auxiliary picture for the proof of Item 2 of Lemma 3.7.9.

This completes the proof of Item 1 of Lemma 3.7.9. □

As discussed in the introduction, there cannot be two distinct MECs with the same skeleton and the same set of v-structures. The uniqueness of the MECs presented in Items 2 to 4 of Lemma 3.7.9 follows from this fact. We now prove Items 2 to 4 of Lemma 3.7.9.

Proof of Item 2 of Lemma 3.7.9. We construct an MEC $M = M_1 \cup M_2 \cup \{r_1 \rightarrow r_2\}$ of G that obeys Item 2 of Lemma 3.7.9.

Figure 3.7 is provided as an example. In the figure, $M_1 \in \text{MEC}(G_1, r_1, 0, 2)$, $M_2 \in \text{MEC}(G_2, r_2, 1)$, and $M = M_1 \cup M_2 \cup \{r_1 \rightarrow r_2\}$, where G , G_1 , and G_2 are from Figure 3.2.

To prove that M obeys Item 2, we have to show the following: (a) M is an MEC of G , (b) M obeys Equation (3.9), (c) $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, and (d) $M \in \text{MEC}(G, r_1, 0, i)$. Note that M , constructed in Figure 3.7, has all these four properties.

We first show that M is an MEC of G . From the construction of M , it is clear that the skeleton of M is G . Thus, the only thing we have to show is that M is an MEC, i.e., it obeys Items 1 and 2 of Theorem 3.2.1.

Suppose M contains an induced subgraph of the form $a \rightarrow b - c$. From the construction of M , there are 3 possibilities: (a) $a, b, c \in V_{G_1}$, (b) $a, b, c \in V_{G_2}$, or (c) $a = r_1$, $b = r_2$, and $c \in V_{G_2}$. We show that none of these possibilities occurs.

Suppose $a, b, c \in V_{G_1}$. From the construction of M , M_1 is an induced subgraph of M . This implies that $a \rightarrow b - c \in M_1$. This contradicts Item 1 of Theorem 3.2.1, as M_1 is

an MEC of the tree graph G_1 . This implies that this possibility cannot occur. Similarly, there cannot be $a, b, c \in V_{G_2}$, as from the construction of M , M_2 is also an induced subgraph of M .

Suppose $a = r_1$, $b = r_2$, and $c \in V_{G_2}$. Then, from the construction of M , $r_2 - c \in M_2$. Since $M_2 \in \text{MEC}(G_2, r_2, 1)$, there must exist a node y such that $y \rightarrow r_2 \in M_2$. This implies that M_2 contains an induced subgraph of the form $y \rightarrow r_2 - c \in M_2$. Since M_2 is an MEC of the tree graph G_2 , this contradicts Item 1 of Theorem 3.2.1. This implies that this possibility also cannot occur. This further implies that M cannot have any induced subgraph of the form $a \rightarrow b - c$.

We now show that each directed edge $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1. Let $u \rightarrow v$ be a directed edge in M . From the construction of M , one of the following holds: $u \rightarrow v \in M_1$, $u \rightarrow v \in M_2$, or $u = r_1$ and $v = r_2$.

Since M_1 is an MEC of the tree graph G_1 , if $u \rightarrow v \in M_1$, then from Item 2 of Theorem 3.2.1, $u \rightarrow v$ is part of an induced subgraph of M_1 of the form either $w \rightarrow u \rightarrow v$ or $w \rightarrow v \leftarrow u$. From the construction of M , the induced subgraph of M_1 is an induced subgraph of M . Thus, when $u \rightarrow v \in M_1$, $u \rightarrow v$ obeys Item 2 of Theorem 3.2.1 for M . Similarly, if $u \rightarrow v \in M_2$, then $u \rightarrow v$ obeys Item 2 of Theorem 3.2.1 for M .

We now go through the final possibility when $u = r_1$ and $v = r_2$. Since $M_2 \in \text{MEC}(G_2, r_2, 1)$, there exists a node $y \in V_{G_2}$ such that $y \rightarrow r_2 \in M_2$. From the construction of M , $y \rightarrow r_2 \leftarrow r_1$ is an induced subgraph of M . This implies that in this case as well, $u \rightarrow v$ obeys Item 2 of Theorem 3.2.1.

The above discussion implies that M obeys Items 1 and 2 of Theorem 3.2.1. This further implies that M is an MEC.

We now show that the v-structures of M obey Equation (3.9). From the construction of M , M_1 and M_2 are the induced subgraphs of M on V_{G_1} and V_{G_2} , respectively. Therefore, $\mathcal{V}(M[V_{G_1}]) = \mathcal{V}(M_1)$, and $\mathcal{V}(M[V_{G_2}]) = \mathcal{V}(M_2)$. Since $r_1 \rightarrow r_2 \in M$, other than the v-structures of M_1 and M_2 , M can have v-structures with one edge $r_1 \rightarrow r_2$. Since $M_2 \in \text{MEC}(G_2, r_2, 1)$, there exists a non-empty set $Y = \{y : y \rightarrow r_2 \in M_2\}$. For each $y \in Y$, $y \rightarrow r_2 \leftarrow r_1$ is a v-structure in M . This shows that the v-structures in M obey Equation (3.9).

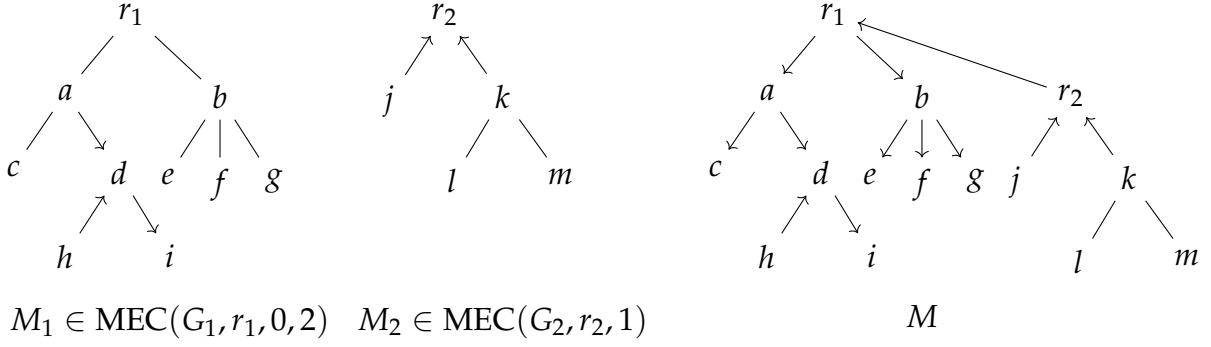


Figure 3.8: Auxiliary picture for the proof of Item 3 of Lemma 3.7.9.

We now show that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. From the construction of M , $M[V_{G_1}] = M_1$, and $M[V_{G_2}] = M_2$. This implies that $\mathcal{V}(M[V_{G_1}]) = \mathcal{V}(M_1)$, and $\mathcal{V}(M[V_{G_2}]) = \mathcal{V}(M_2)$. From Definition 3.7.1, this further implies that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.

Finally, we show that that $M \in \text{MEC}(G, r_1, 0, i)$. From the construction of M , M_1 is an induced subgraph of M . In M_1 , i edges adjacent to r_1 are undirected, and the remaining $\delta_1 - i$ adjacent edges of r_1 in M_1 are directed outward from r_1 . The only edge that is adjacent to r_1 in M but not adjacent to r_1 in M_1 is $r_1 \rightarrow r_2$, which is directed outward from r_1 . Therefore, no edge adjacent to r_1 in M is directed towards r_1 , and i undirected edges adjacent to r_1 in M are undirected. This implies that $M \in \text{MEC}(G, r_1, 0, i)$ (from Definition 3.3.3). This completes the proof of Item 2 of Lemma 3.7.9. $\square$

Proof of Item 3 of Lemma 3.7.9. We construct an MEC M of G using the following steps:

1. Initialize $M = M_1 \cup M_2 \cup \{r_2 \rightarrow r_1\}$.
2. Update M by replacing $u - v$ in M with $u \rightarrow v$ if $u - v \in M_1$ and there exists an undirected path in M_1 from r_1 to (u, v) in M_1 .

Figure 3.8 is provided as an example. In the figure, $M_1 \in \text{MEC}(G_1, r_1, 0, 2)$, $M_2 \in \text{MEC}(G_2, r_2, 1)$, and M is constructed using the above two steps, where G , G_1 , and G_2 are from Figure 3.2. We direct an undirected edge of M (after the first step) if it satisfies the condition given at step 2. For instance, after the initialization step (step 1 of the construction of M), M has an undirected edge $a - c$. We replace it with $a \rightarrow c$ at the second step, since M_1 has an undirected path $P = (r_1, a, c)$ from r_1 to (a, c) .

We show that M constructed using the above steps obeys Item 3 of Lemma 3.7.9. To prove this, we have to show the following: (a) M is an MEC of G , (b) M obeys Equation (3.10), (c) $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, and (d) $M \in \text{MEC}(G, r_1, 1)$. Note that M , constructed in Figure 3.8 using the above steps, has all these four properties.

We first show that M is an MEC of G . From the construction of M , the skeleton of M is G . Therefore, it suffices to show that M obeys Items 1 and 2 of Theorem 3.2.1.

We start by showing that M obeys Item 1 of Theorem 3.2.1. Suppose M contains an induced subgraph of the form $a \rightarrow b - c$. From the construction of M , there are three possibilities: (a) $a, b, c \in V_{G_1}$, (b) $a, b, c \in V_{G_2}$, or (c) $a = r_2$, $b = r_1$, and $c \in V_{G_1}$. One by one, we show that none of these possibilities occurs.

Suppose M contains an induced subgraph of the form $a \rightarrow b - c$ such that $a, b, c \in V_{G_1}$. From the construction of M , $b - c \in M_1$ (note that step 2 does not change a directed edge in $M_1 \cup M_2 \cup \{r_2 \rightarrow r_1\}$ into an undirected edge), and either $a \rightarrow b \in M_1$ or $a - b \in M_1$. If $a \rightarrow b \in M_1$, then $a \rightarrow b - c$ is an induced subgraph in the MEC M_1 , contradicting Item 1 of Theorem 3.2.1. If $a - b \in M_1$, it must have been directed at step 2. This implies that there exists an undirected path from r_2 to (a, b) in M_1 , which further implies that there exists an undirected path from r_2 to (b, c) in M_1 . But then, we have $b \rightarrow c \in M$ at step 2. This shows that the first possibility cannot occur.

From the construction of M , an induced subgraph of M with vertices in V_{G_2} is also an induced subgraph of M_2 (note that step 2 does not make any changes to an edge with endpoints in V_{G_2}). Since M_2 is an MEC of the tree graph G_2 , from Item 1 of Theorem 3.2.1, there cannot be an induced subgraph of M_2 of the form $a \rightarrow b - c$. This implies that the second possibility cannot occur.

Suppose $r_2 \rightarrow r_1 - c$ in M such that $c \in V_{G_1}$. From the construction of M , this implies that $r_1 - c \in M_1$. However, the edge $r_1 - c$ is a path from r_1 to (r_1, c) , and it must have been directed at step 2. This implies that the third possibility also cannot occur. This shows that M cannot have an induced subgraph of the form $a \rightarrow b - c$, implying that M obeys Item 1 of Theorem 3.2.1.

We now show that M obeys Item 2 of Theorem 3.2.1. Suppose $u \rightarrow v$ is a directed edge in M , then one of the following occurs: $u, v \in V_{G_1}$, $u, v \in V_{G_2}$, or $u = r_1$ and $v = r_2$. We show that in each of these cases, $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1.

Suppose $u, v \in V_{G_1}$. From the construction of M , a directed edge in M_1 is also a directed edge in M , and $\text{skeleton}(M[V_{G_1}]) = \text{skeleton}(M_1)$. This implies that if $u \rightarrow v \in M$, then either $u \rightarrow v \in M_1$, or $u - v \in M_1$. Suppose $u \rightarrow v \in M_1$. Since M_1 is an MEC of the tree graph G_1 , by Item 2 of Theorem 3.2.1, $u \rightarrow v$ is part of an induced subgraph of M_1 of the form either $u \rightarrow v \leftarrow w$, or $w \rightarrow u \rightarrow v$. From the construction of M , a directed induced subgraph of M_1 is also an induced subgraph of M , as a directed edge in M_1 remains directed in M . Therefore, $u \rightarrow v$ is strongly protected in M .

If $u - v \in M_1$, then it must have been directed in M at step 2. This implies that there exists an undirected path from r_1 to (u, v) in M_1 . Let the path be $P = (x_0 = r_1, x_1, \dots, x_{l-1} = u, x_l = v)$ for some $l \geq 1$. From the construction of M , for all $0 \leq i < l$, $x_i \rightarrow x_{i+1} \in M$ (directed at step 2 of the construction of M), because $P_i = (x_0 = r_1, \dots, x_i, x_{i+1})$ is an undirected path in M_1 from r_1 to (x_i, x_{i+1}) . If $l > 1$, then $u \rightarrow v$ is part of an induced subgraph $x_{l-2} \rightarrow x_{l-1} \rightarrow x_l$ in M . And, if $l = 1$, then $r_1 = u$, and $u \rightarrow v$ is part of an induced subgraph $r_2 \rightarrow r_1 \rightarrow v$ in M . Thus, we have shown that if $u, v \in V_{G_1}$, then $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1.

Now, suppose $u, v \in V_{G_2}$. From the construction of M , M_2 is an induced subgraph of M (after step 1, M_2 is an induced subgraph of M , and step 2 does not change any edge with endpoints in V_{M_2}). This implies that $u \rightarrow v \in M_2$. Since M_2 is an MEC of the tree graph G_2 , by Item 2 of Theorem 3.2.1, $u \rightarrow v$ is part of an induced subgraph of M_2 of the form either $u \rightarrow v \leftarrow w$, or $w \rightarrow u \rightarrow v$. Since, from the construction of M , an induced subgraph of M_2 is also an induced subgraph of M , $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1.

Finally, we show that $r_2 \rightarrow r_1$ in M also obeys Item 2 of Theorem 3.2.1. Since $M_2 \in \text{MEC}(G_2, r_2, 1)$, there must exist $y \rightarrow r_2 \in M_2$. From the construction of M , $y \rightarrow r_2 \in M$. Then, $r_2 \rightarrow r_1$ is part of an induced subgraph of the form $y \rightarrow r_2 \rightarrow r_1$. Thus, we have shown that each edge of M obeys Item 2 of Theorem 3.2.1.

The above discussion shows that M is an MEC, as it obeys Items 1 and 2 of Theorem 3.2.1.

We now show that M obeys Equation (3.10). From the construction of M , the directed edges of M_1 and M_2 are directed edges of M . This implies that the v-structures of M_1 and M_2 are also the v-structures of M , i.e., $\mathcal{V}(M_1) \cup \mathcal{V}(M_2) \subseteq \mathcal{V}(M)$. For

completeness, we now show that if $u \rightarrow v \leftarrow w$ is a v-structure in M , then it must either be a v-structure of M_1 or a v-structure of M_2 , i.e., $\mathcal{V}(M) \subseteq \mathcal{V}(M_1) \cup \mathcal{V}(M_2)$. This implies that $\mathcal{V}(M) = \mathcal{V}(M_1) \cup \mathcal{V}(M_2)$, showing that M obeys Equation (3.10).

Suppose there exists a v-structure $u \rightarrow v \leftarrow w$ in M . We now show that either $u \rightarrow v \leftarrow w$ is a v-structure in M_1 , or it is a v-structure in M_2 . From the construction of M , there are the following possibilities: (a) $u, v, w \in V_{G_1}$, (b) $u = r_2, v = r_1$, and w is a neighbor of r_1 in V_{G_1} , or (c) $u, v, w \in V_{G_2}$.

Suppose $u, v, w \in V_{G_1}$. From the construction of M , $\text{skeleton}(M[V_{G_1}]) = \text{skeleton}(M_1)$, and if $a \rightarrow b \in M_1$, then $a \rightarrow b \in M$. This implies that either $u \rightarrow v \in M_1$ or $u \rightarrow v \in M_1$, and either $v \rightarrow w \in M_1$ or $v \leftarrow w \in M_1$. From Item 1 of Theorem 3.2.1, either $u \rightarrow v \leftarrow w \in M_1$ or $u \rightarrow v \rightarrow w \in M_1$ (since M_1 is an MEC).

If $u \rightarrow v \leftarrow w \in M_1$, then this v-structure is also a v-structure in M_1 .

Now, suppose $u \rightarrow v \rightarrow w \in M_1$. In this case, if $u \rightarrow v \in M$, it must have been directed at step 2, indicating that there is an undirected path from r_1 to (u, v) in M_1 . Since $u \rightarrow v \rightarrow w \in M_1$, the existence of an undirected path from r_1 to (u, v) in M_1 implies that there is an undirected path from r_1 to (v, w) in M_1 . Given that M_1 is an MEC of the tree graph G_1 , there cannot be two different paths from r_1 to w . Thus, at step 2, if $u \rightarrow v \in M$, we must also have $v \rightarrow w \in M$. However, this leads to a contradiction, as we initially assumed $u \rightarrow v \leftarrow w \in M$. Therefore, both $u \rightarrow v \rightarrow w \in M_1$ and $u \rightarrow v \leftarrow w \in M$ cannot hold simultaneously.

Now suppose $u = r_2, v = r_1$, and w is a neighbor of r_1 in V_{G_1} . Since $M_1 \in \text{MEC}(G_1, r_1, 0)$, either $r_1 \rightarrow w \in M_1$ or $r_1 \leftarrow w \in M_1$. From the construction of M (Items 1 and 2), if $r_1 \rightarrow w \in M_1$, then $r_1 \rightarrow w \in M$. But we have $r_1 \leftarrow w \in M$. This implies that $r_1 \leftarrow w \in M_1$. However, from step 2, $r_1 \rightarrow w \in M$ (the existence of the undirected edge $r_1 \leftarrow w$ in M_1 implies the existence of an undirected path from r_1 to (r_1, w) in M_1). Therefore, this case cannot occur.

Finally, consider the case when $u, v, w \in V_{G_2}$. From the construction of M , M_2 is an induced subgraph of M . This implies that if $u \rightarrow v \leftarrow w$ is a v-structure in M , then $u \rightarrow v \leftarrow w$ is a v-structure in M_2 . Thus, we have shown that in all possible cases, if $u \rightarrow v \leftarrow w$ is a v-structure in M , it is a v-structure either in M_1 or in M_2 .

This completes the proof that M obeys Equation (3.10).

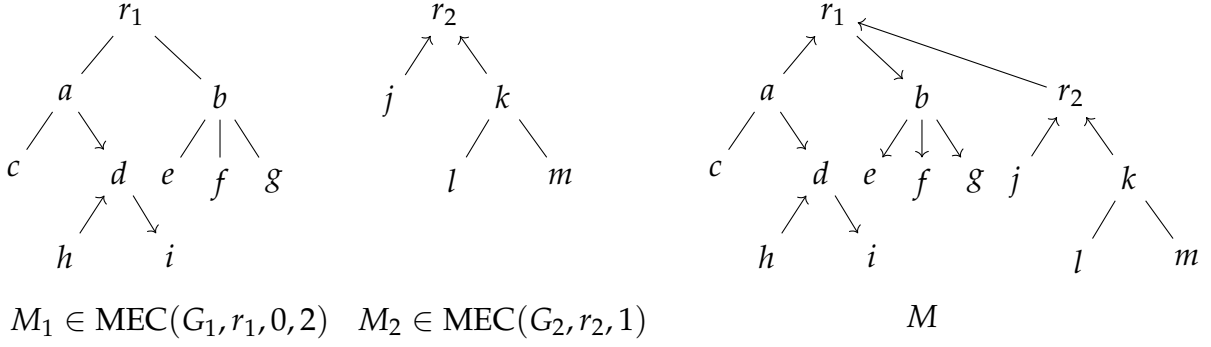


Figure 3.9: Auxiliary picture for the proof of Item 4 of Lemma 3.7.9.

The above discussion implies that $\mathcal{V}(M[V_{G_1}]) = \mathcal{V}(M_1)$ and $\mathcal{V}(M[V_{G_2}]) = \mathcal{V}(M_2)$. Therefore, from Definition 3.7.1, $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.

Since there is a directed edge $r_2 \rightarrow r_1$ towards r_1 in M , we have $M \in \text{MEC}(G, r_1, 1)$. This completes the proof of Item 3 of Lemma 3.7.9. □

Proof of Item 4 of Lemma 3.7.9. We construct an MEC M of G using the following steps:

1. Initialize $M = M_1 \cup M_2 \cup \{r_2 \rightarrow r_1\}$.
2. Pick an edge $x - r_1 \in M_1$. Replace the edge $x - r_1$ in M with $x \rightarrow r_1$.
3. Update M by replacing $u - v$ in M with $u \rightarrow v$ if $u - v \in M_1$ and there exists an undirected path in M_1 from r_1 to (u, v) that does not contain x .

Figure 3.9 is provided as an example. In the figure, $M_1 \in \text{MEC}(G_1, r_1, 0, 2)$, $M_2 \in \text{MEC}(G_2, r_2, 1)$, and M is constructed using the steps mentioned above, where G , G_1 , and G_2 are from Figure 3.2. In the second step, we pick the edge $a - r_1$ of M_1 and replace the edge $a - r_1$ in M with $a \rightarrow r_1$ (here a plays the role of x). In the third step, the undirected edges of M_1 that satisfy the conditions of the third step get directed in M . For instance, the undirected edge $b - e$ of M_1 is directed in M . We replace the edge $b - e$ in M (after the second step) with $b \rightarrow e$ in the third step, since M_1 has an undirected path $P = (r_1, b, e)$ from r_1 to (b, e) , and the path does not contain a .

We show that M , constructed using the above steps, satisfies Item 4 of Lemma 3.7.9. To prove this, we have to show the following: (a) M is an MEC of G , (b) M obeys

Equation (3.11), (c) $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, and (d) $M \in \text{MEC}(G, r_1, 1)$. Note that M , constructed in Figure 3.9 using the above steps, has all these four properties.

We first show that M is an MEC of G . From the construction of M , the skeleton of M is G . Therefore, it suffices to show that M obeys Items 1 and 2 of Theorem 3.2.1.

We begin by showing that M satisfies Item 1 of Theorem 3.2.1. Suppose M contains an induced subgraph of the form $a \rightarrow b - c$. From the construction of M , there are three possibilities: (a) $a, b, c \in V_{G_1}$, (b) $a, b, c \in V_{G_2}$, or (c) $a = r_2, b = r_1$, and $c \in V_{G_1}$. We show that none of these possibilities can occur.

Suppose $a, b, c \in V_{G_1}$. From the construction of M , $\text{skeleton}(M[V_{G_1}]) = \text{skeleton}(M_1)$, and if $u \rightarrow v \in M_1$ then $u \rightarrow v \in M$. This implies that $b - c \in M_1$, and either $a \rightarrow b \in M_1$ or $a - b \in M_1$. If $a \rightarrow b \in M_1$, then $a \rightarrow b - c$ is an induced subgraph of the MEC M_1 of the tree graph G_1 , contradicting Item 1 of Theorem 3.2.1. If $a - b \in M_1$, then it must have been directed in M either at step 2 (first case) or at step 3 (second case). If it was directed at step 2, then $a = x, b = r_1$, and $c \neq x$ is a neighbor of r_1 . This implies that $r_1 - c \in M_1$. But, then, at step 3, we have $r_1 \rightarrow c \in M$ (the undirected edge $r_1 - c$ in M_1 implies the existence of an undirected path from r_1 to (r_1, c) in M_1 that does not contain x), contradicting our assumption that $b - c \in M$. This implies that the first case cannot occur. Moving to the second case, if $a - b$ was directed at step 3, then there exists an undirected path from r_1 to (a, b) in M_1 (and the path does not contain x). Since $b - c \in M_1$, this further implies that there exists an undirected path from r_1 to (b, c) in M_1 (and this path also does not contain x), and hence $b \rightarrow c \in M$ at step 3, which contradicts the assumption that $b - c \in M$. This implies that the second case also cannot occur. Thus, the first possibility cannot occur.

Now, suppose $a, b, c \in V_{G_2}$. From the construction of M , M_2 is an induced subgraph of M . Since M_2 is an MEC of the tree graph G_2 , by Item 1 of Theorem 3.2.1, there cannot be an induced subgraph of the form $a \rightarrow b - c$ in M_2 , implying $a \rightarrow b - c$ cannot be in M as well (because M_2 is an induced subgraph of M). This shows that the second possibility also cannot occur.

Finally, suppose $a = r_2, b = r_1$, and $c \in V_{G_1}$. From the construction of M , this implies that $r_1 - c \in M_1$. There are two cases to consider: either $c = x$, or $c \neq x$. If $c = x$, then at step 2, we have $c \rightarrow b \in M$. Therefore, this case cannot occur. If $c \neq x$,

the edge $r_1 - c \in M_1$ forms an undirected path in M_1 from r_1 to (r_1, c) . Thus, the edge must have been directed at step 3, implying $r_1 - c \notin M$. This implies that the third possibility cannot occur either.

Thus, we conclude that M cannot contain an induced subgraph of the form $a \rightarrow b - c$. Therefore, M obeys Item 1 of Theorem 3.2.1.

We now show that M satisfies Item 2 of Theorem 3.2.1. Suppose $u \rightarrow v$ is a directed edge in M . Then, one of the following occurs: $u, v \in V_{G_1}$, $u, v \in V_{G_2}$, or $u = r_2$ and $v = r_1$. We show that in each of these cases, $u \rightarrow v$ in M satisfies Item 2 of Theorem 3.2.1.

Suppose $u, v \in V_{G_1}$. Based on the construction of M , there are two possibilities: either $u \rightarrow v \in M_1$ or $u - v \in M_1$.

Suppose $u \rightarrow v \in M_1$. Since M_1 is an MEC of the tree graph G_1 , by Item 2 of Theorem 3.2.1, $u \rightarrow v$ is part of an induced subgraph of M_1 that is either of the form $u \rightarrow v \leftarrow w$, or $w \rightarrow u \rightarrow v$. From the construction of M , a directed induced subgraph of M_1 is also an induced subgraph in M . Therefore, $u \rightarrow v$ is strongly protected in M .

Suppose $u - v \in M_1$. In that case, $u - v$ must have been directed in M either at step 2 or at step 3. If $u - v \in M_1$ and it is directed in M at step 2, then $u = x$ and $v = r_2$. In this situation, $u \rightarrow v$ is part of the induced subgraph $r_2 \rightarrow r_1 \leftarrow x$, satisfying Item 2 of Theorem 3.2.1. Alternatively, if $u - v \in M_1$ and it is directed in M at step 3, there exists an undirected path from r_1 to (u, v) in M_1 , and this path does not contain x . Let the path be $P = (x_0 = r_1, x_1, \dots, x_{l-1} = u, x_l = v)$ for some $l \geq 1$. From the construction of M , for all $0 \leq i < l$, $x_i \rightarrow x_{i+1} \in M$ because $P_i = (x_0 = r_1, \dots, x_i, x_{i+1})$ is an undirected path in M_1 from r_1 to (x_i, x_{i+1}) , and P_i also does not contain x . If $l > 1$, then $u \rightarrow v$ is part of the induced subgraph $x_{l-2} \rightarrow x_{l-1} \rightarrow x_l$ in M . If $l = 1$, then $r_1 = u$, and $u \rightarrow v$ is part of the induced subgraph $r_2 \rightarrow r_1 \rightarrow v$ in M . Thus, we have shown that if $u, v \in V_{G_1}$, then $u \rightarrow v$ satisfies Item 2 of Theorem 3.2.1.

Suppose $u, v \in V_{G_2}$. From the construction of M , M_2 is an induced subgraph of M . This implies that $u \rightarrow v \in M_2$. Since M_2 is an MEC of the tree graph G_2 , by Item 2 of Theorem 3.2.1, $u \rightarrow v$ is part of an induced subgraph of M_2 of the form either $u \rightarrow v \leftarrow w$ or $w \rightarrow u \rightarrow v$. From the construction of M , an induced subgraph of M_2 is an induced subgraph of M . Therefore, $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1.

We now show that $r_2 \rightarrow r_1$ in M also obeys Item 2 of Theorem 3.2.1. $r_2 \rightarrow r_1$ is part

of an induced subgraph $r_2 \rightarrow r_1 \leftarrow x$ in M . Thus, we have shown that each edge of M obeys Item 2 of Theorem 3.2.1.

The above discussion shows that M is an MEC as it obeys Items 1 and 2 of Theorem 3.2.1.

We now show that M obeys Equation (3.11). From the construction of M , the directed edges of M_1 and M_2 are also directed edges of M , $\text{skeleton}(M[V_{G_1}]) = \text{skeleton}(M_1)$, and $\text{skeleton}(M[V_{G_2}]) = \text{skeleton}(M_2)$. Additionally, $r_2 \rightarrow r_1 \leftarrow x$ is a v-structure in M . This implies that $\mathcal{V}(M_1) \cup \mathcal{V}(M_2) \cup \{r_2 \rightarrow r_1 \leftarrow x\} \subseteq \mathcal{V}(M)$. For completeness, we show that if $u \rightarrow v \leftarrow w$ is a v-structure in M , then one of the following occurs: $u \rightarrow v \leftarrow w$ is a v-structure in M_1 , $u \rightarrow v \leftarrow w$ is a v-structure in M_2 , or $u = r_2$, $v = r_1$, and $w = x$, i.e., $\mathcal{V}(M) \subseteq \mathcal{V}(M_1) \cup \mathcal{V}(M_2) \cup \{r_2 \rightarrow r_1 \leftarrow x\}$. This further implies that $\mathcal{V}(M) = \mathcal{V}(M_1) \cup \mathcal{V}(M_2) \cup \{r_2 \rightarrow r_1 \leftarrow x\}$, showing that M obeys Equation (3.11).

Suppose there exists a v-structure $u \rightarrow v \leftarrow w$ in M . From the construction of M , the following possibilities arise: (a) $u, v, w \in V_{G_2}$, (b) $u = r_2$, $v = r_1$, and w is a neighbor of r_1 in V_{G_1} , or (c) $u, v, w \in V_{G_1}$.

Suppose $u, v, w \in V_{G_2}$. From the construction of M , M_2 is an induced subgraph of M . This implies that $u \rightarrow v \leftarrow w$ is a v-structure in M_2 (recall that M_2 is an MEC of G_2 , which means $V_{M_2} = V_{G_2}$).

Suppose $u = r_2$, $v = r_1$, and w is a neighbor of r_1 in V_{G_1} . Since $M_1 \in \text{MEC}(G_1, r_1, 0)$, either $r_1 \rightarrow w \in M_1$ or $r_1 - w \in M_1$. Suppose $r_1 \rightarrow w \in M_1$. From the construction of M (steps 1-3), if $r_1 \rightarrow w \in M_1$, then $r_1 \rightarrow w \in M$. However, we have $r_1 \leftarrow w \in M$, which implies $r_1 \rightarrow w \notin M_1$. Suppose $r_1 - w \in M_1$. From steps 2 and 3, if $w \rightarrow r_1 \in M$, then $w = x$. Otherwise, we would have $r_1 \rightarrow w \in M$ at step 3, as $r_1 - w$ forms an undirected path from r_1 to (r_1, w) that does not contain x . Thus, we conclude that $w = x$.

Finally, suppose $u, v, w \in V_{G_1}$. From the construction of M , $\text{skeleton}(M[V_{G_1}]) = \text{skeleton}(M_1) = G_1$, and if $a \rightarrow b \in M_1$, then $a \rightarrow b \in M$. This implies that either $u - v \in M_1$ or $u \rightarrow v \in M_1$, and either $v - w \in M_1$ or $v \leftarrow w \in M_1$. Since M_1 is an MEC of the tree graph G_1 , from Item 1 of Theorem 3.2.1, either $u \rightarrow v \leftarrow w \in M_1$ or $u - v - w \in M_1$.

If $u \rightarrow v \leftarrow w \in M_1$, then it is a v-structure in M_1 . Suppose $u - v - w \in M_1$. Then

the directed edge $u \rightarrow v$ in M must have been directed either at step 2 or at step 3. Suppose $u \rightarrow v$ gets directed in M at step 2. Then $u = x$ and $v = r_1$. However, in that case, $v - w$ forms an undirected path from r_1 to (v, w) (that does not contain x). Moreover, we have $v \rightarrow w \in M$ at step 3, leading to a contradiction, as the v-structure implies that $w \rightarrow v \in M$. Hence, this case cannot occur.

Suppose $u \rightarrow v$ gets directed in M at step 3. Then there exists an undirected path in M_1 from r_1 to (u, v) , and the path does not contain x . Since $u - v - w \in M_1$, this implies the existence of an undirected path from r_1 to (v, w) in M_1 , which also does not contain x . Since M_1 is an MEC of the tree graph G_1 , there cannot be two distinct paths from r_1 to w . This implies that if $u \rightarrow v \in M$, then $v \rightarrow w \in M$, leading to a contradiction, as we initially assumed $u \rightarrow v \leftarrow w \in M$.

Thus, there cannot be $u - v - w \in M_1$ and $u \rightarrow v \leftarrow w \in M$. Therefore, we have shown that if $u \rightarrow v \leftarrow w$ is a v-structure in M , then it must be a v-structure in M_1 or M_2 , or $u = r_2, v = r_1$, and $w = x$.

This completes the proof that M obeys Equation (3.11).

The above discussion implies that $\mathcal{V}(M[V_{G_1}]) = \mathcal{V}(M_1)$ and $\mathcal{V}(M[V_{G_2}]) = \mathcal{V}(M_2)$. Therefore, by Definition 3.7.1, $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.

Since there is a directed edge $r_2 \rightarrow r_1$ in M , we have $M \in \text{MEC}(G, r_1, 1)$. This completes the proof of Item 4 of Lemma 3.7.9. □

This completes the proof of Lemma 3.7.9. □

Proof of Lemma 3.7.11. We first prove Item 1 of Lemma 3.7.11.

Proof of Item 1 of Lemma 3.7.11. Let M be an MEC of G such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. From Observation 3.7.4, M contains the v-structures of M_1 and M_2 . Other than that, it can only contain v-structures that involve an edge with endpoints r_1 and r_2 . There are three possibilities: $r_1 - r_2 \in M$, $r_2 \rightarrow r_1 \in M$, or $r_1 \rightarrow r_2 \in M$.

Suppose $r_1 - r_2 \in M$. Then, there cannot be any v-structure involving an edge with endpoints r_1 and r_2 , because a v-structure contains directed edges, and $r_1 - r_2$ is undirected. This implies that if $r_1 - r_2 \in M$, then all the v-structures in M are v-structures in either M_1 or M_2 , i.e., M satisfies Equation (3.12).

Suppose $r_2 \rightarrow r_1 \in M$. Since $M_1 \in \text{MEC}(G_1, r_1, 0, i)$, we can partition the set of neighbors of r_1 in G_1 into X and X' such that $X = \{x : x \rightarrow r_1 \in M_1\}$ is a set of i nodes, and $X' = \{x : r_1 \rightarrow x \in M_1\}$ is a set of $\delta_1 - i$ nodes. From Lemma 3.7.3, for $x \in X'$, $r_1 \rightarrow x \in M$. Since $r_2 \rightarrow r_1 \in M$, from Item 1 of Theorem 3.2.1, for $x \in X$, either $r_1 \rightarrow x \in M$ or $x \rightarrow r_1 \in M$, otherwise, $r_2 \rightarrow r_1 - x$ is an induced subgraph in M , contradicting Item 1 of Theorem 3.2.1.

If for two distinct $x_1, x_2 \in X$, $x_1 \rightarrow r_1, x_2 \rightarrow r_1 \in M$, then we get a v-structure $x_1 \rightarrow r_1 \leftarrow x_2$ in M such that the nodes of the v-structure are in V_{G_1} , but the v-structure is not in M_1 , implying $\mathcal{V}(M[V_{G_1}]) \neq \mathcal{V}(M_1)$, which further implies that $\mathcal{P}(M, V_{G_1}) \neq M_1$, a contradiction. This implies that there exists at most one $x \in X$ such that $x \rightarrow r_1 \in M$. If there is no $x \in X$ such that $x \rightarrow r_1 \in M$, then no v-structure in M involves an edge with endpoints r_1 and r_2 . In this case, M contains only the v-structures of M_1 and M_2 , as in Equation (3.12). If there exists an $x \in X$ such that $x \rightarrow r_1 \in M$, then other than the v-structures of M_1 and M_2 , M also contains a v-structure $r_2 \rightarrow r_1 \leftarrow x$. In this case, M obeys Equation (3.13).

Suppose now that $r_1 \rightarrow r_2 \in M$. Since $M_2 \in \text{MEC}(G_2, r_2, 0, j)$, we can partition the set of neighbors of r_2 in G_2 into Y and Y' such that $Y = \{y : y \rightarrow r_2 \in M_2\}$ is a set of j nodes, and $Y' = \{y : r_2 \rightarrow y \in M_2\}$ is a set of $\delta_2 - j$ nodes. From Lemma 3.7.3, for $y \in Y'$, $r_2 \rightarrow y \in M$. Since $r_1 \rightarrow r_2 \in M$, from Item 1 of Theorem 3.2.1, for $y \in Y$, either $r_2 \rightarrow y \in M$ or $y \rightarrow r_2 \in M$, otherwise, $r_1 \rightarrow r_2 - y$ is an induced subgraph in M , contradicting Item 1 of Theorem 3.2.1.

If for two distinct $y_1, y_2 \in Y$, $y_1 \rightarrow r_2, y_2 \rightarrow r_2 \in M$, then we get a v-structure $y_1 \rightarrow r_2 \leftarrow y_2$ in M such that the nodes of the v-structure are in V_{G_2} , but the v-structure is not in M_2 , implying $\mathcal{V}(M[V_{G_2}]) \neq \mathcal{V}(M_2)$, which further implies that $\mathcal{P}(M, V_{G_2}) \neq M_2$, a contradiction. This implies that there exists at most one $y \in Y$ such that $y \rightarrow r_2 \in M$. If there is no $y \in Y$ such that $y \rightarrow r_2 \in M$, then no v-structure in M involves an edge with endpoints r_1 and r_2 . In this case, M contains only the v-structures of M_1 and M_2 , as in Equation (3.12). If there exists a $y \in Y$ such that $y \rightarrow r_2 \in M$, then other than the v-structures of M_1 and M_2 , M also contains a v-structure $r_1 \rightarrow r_2 \leftarrow y$. In this case, M obeys Equation (3.14).

This implies that in all the possibilities, M follows one of Equation (3.12), Equ-

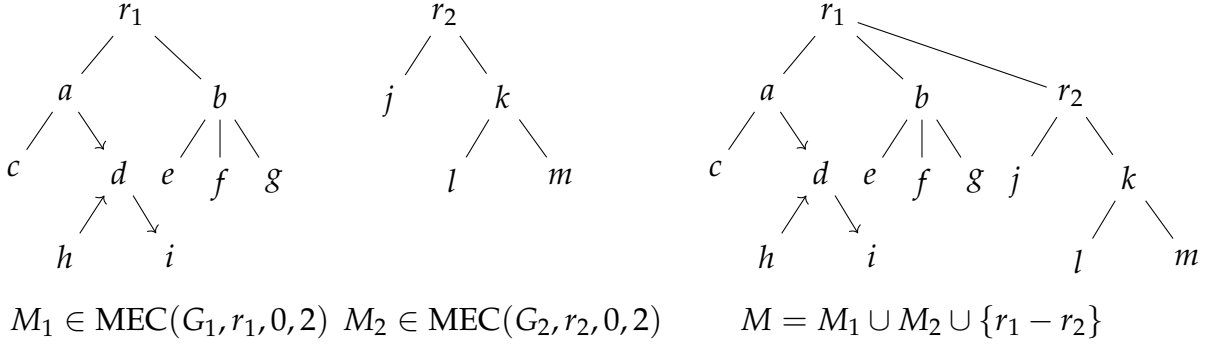


Figure 3.10: Auxiliary picture for the proof of Item 2 of Lemma 3.7.11.

tion (3.13), or Equation (3.14). This completes the proof of Item 1. $\square$

As discussed in the introduction, there cannot be two distinct MECs with the same skeleton and the same set of v-structures. The uniqueness of the MECs in Items 2 to 4 of Lemma 3.7.11 follows from this fact. To prove Item 2 of Lemma 3.7.11, we show that an MEC exists that satisfies it. We apply the same approach to prove Items 3 and 4 of Lemma 3.7.11.

Proof of Item 2 of Lemma 3.7.11. We construct an MEC $M = M_1 \cup M_2 \cup \{r_1 - r_2\}$ of G that obeys Item 2 of Lemma 3.7.11.

Figure 3.10 is provided as an example. In the figure, $M_1 \in \text{MEC}(G_1, r_1, 0, 2)$, $M_2 \in \text{MEC}(G_2, r_2, 0, 2)$, and $M = M_1 \cup M_2 \cup \{r_1 - r_2\}$, where G , G_1 , and G_2 are from Figure 3.2.

To prove that M obeys Item 2, we need to show the following: (a) M is an MEC of G , (b) M obeys Equation (3.12), (c) $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, and (d) $M \in \text{MEC}(G, r_1, 0, i + 1)$. Note that M , constructed in Figure 3.10, has all these four properties.

We first show that M is an MEC of G . From the construction of M , it is clear that the skeleton of M is G . Thus, the only thing we need to show is that M is an MEC, i.e., it obeys Items 1 and 2 of Theorem 3.2.1.

We start by proving Item 1 of Theorem 3.2.1. Suppose M contains an induced subgraph of the form $a \rightarrow b - c$. Since $M_1 \in \text{MEC}(G_1, r_1, 0, i)$, there cannot be an edge of the form $x \rightarrow r_1$ in M_1 . Similarly, since $M_2 \in \text{MEC}(G_2, r_2, 0, j)$, there cannot be an

edge of the form $y \rightarrow r_2$ in M_2 . Since $M = M_1 \cup M_2 \cup \{r_1 - r_2\}$, there cannot be an edge of the form $x \rightarrow r_1$ and $y \rightarrow r_2$ in M . This implies that either (a) $a, b, c \in V_{G_1}$ or (b) $a, b, c \in V_{G_2}$. If $a, b, c \in V_{G_1}$, then from the construction of M , $a \rightarrow b - c \in M_1$. Since M_1 is an MEC of the tree graph G_1 , this contradicts Item 1 of Theorem 3.2.1. This implies that this case cannot occur. Similarly, there cannot be $a, b, c \in V_{G_2}$. Thus, M cannot have any induced subgraph of the form $a \rightarrow b - c$.

Next, we show that each directed edge $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1. Let $u \rightarrow v$ be a directed edge in M . From the construction of M , either $u \rightarrow v \in M_1$ or $u \rightarrow v \in M_2$. Since M_1 is an MEC of the tree graph G_1 , if $u \rightarrow v \in M_1$, then from Item 2 of Theorem 3.2.1, $u \rightarrow v$ is part of an induced subgraph of M_1 of the form either $w \rightarrow u \rightarrow v$ or $w \rightarrow v \leftarrow u$. From the construction of M , the induced subgraph of M_1 is also an induced subgraph of M . Thus, if $u \rightarrow v \in M_1$, then $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1 for M . Similarly, if $u \rightarrow v \in M_2$, then $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1. This shows that each directed edge $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1.

The above discussion implies that M obeys Items 1 and 2 of Theorem 3.2.1. This further implies that M is an MEC.

We now show that the v-structures in M obey Equation (3.12). From the construction of M , M_1 and M_2 are the induced subgraphs of M on V_{G_1} and V_{G_2} . Therefore, $\mathcal{V}(M[V_{G_1}]) = \mathcal{V}(M_1)$ and $\mathcal{V}(M[V_{G_2}]) = \mathcal{V}(M_2)$. Since $r_1 - r_2 \in M$, aside from the v-structures of M_1 and M_2 , M cannot have any additional v-structures. This shows that the v-structures in M obey Equation (3.12).

Next, we show that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. From the construction of M , $M[V_{G_1}] = M_1$ and $M[V_{G_2}] = M_2$. This implies that $\mathcal{V}(M[V_{G_1}]) = \mathcal{V}(M_1)$ and $\mathcal{V}(M[V_{G_2}]) = \mathcal{V}(M_2)$. From Definition 3.7.1, this further implies that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.

We now show that $M \in \text{MEC}(G, r_1, 0, i + 1)$. From the construction of M , M_1 is an induced subgraph of M . In M_1 , i edges adjacent to r_1 are undirected, and the remaining $\delta_1 - i$ adjacent edges of r_1 in M_1 are directed outward from r_1 . Since M_1 is an induced subgraph of M , the i adjacent undirected edges of M_1 are undirected in M as well. Additionally, M contains an undirected edge $r_1 - r_2$, adjacent to r_1 . This implies that the total number of undirected edges adjacent to r_1 in M is $i + 1$. Therefore, no edge

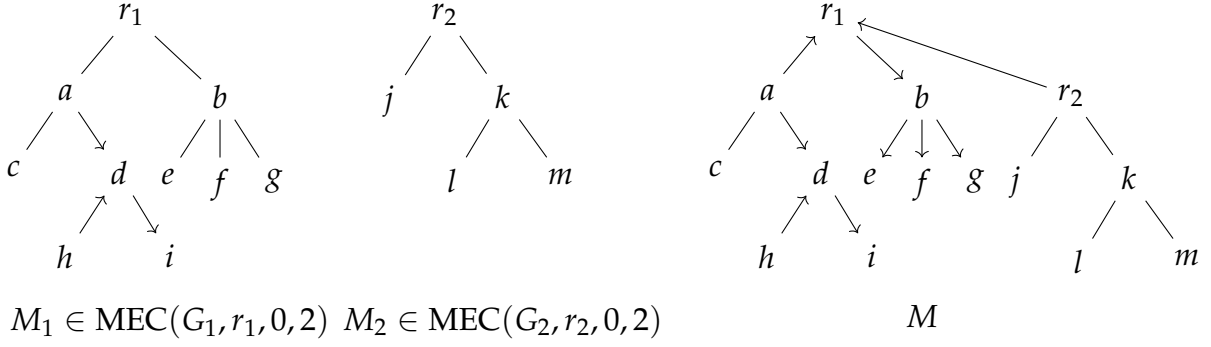


Figure 3.11: Auxiliary picture for the proof of Item 3 of Lemma 3.7.11.

adjacent to r_1 in M is directed towards r_1 , and $i + 1$ undirected edges adjacent to r_1 in M are undirected. This implies that $M \in \text{MEC}(G, r_1, 0, i + 1)$. This completes the proof of Item 2 of Lemma 3.7.11. □

Proof of Item 3 of Lemma 3.7.11. We construct an MEC of G using the following steps:

1. Initialize $M = M_1 \cup M_2 \cup \{r_2 \rightarrow r_1\}$.
2. Pick an edge $x - r_1 \in M_1$. Replace the edge $x - r_1$ in M with $x \rightarrow r_1$.
3. Update M by replacing $u - v$ in M with $u \rightarrow v$ if $u - v \in M_1$ and there exists an undirected path in M_1 from r_1 to (u, v) such that the path does not contain x .

Figure 3.11 is provided as an example. In the figure, $M_1 \in \text{MEC}(G_1, r_1, 0, 2)$, $M_2 \in \text{MEC}(G_2, r_2, 0, 2)$, and M is constructed using the steps mentioned above, where G , G_1 , and G_2 are from Figure 3.2. In the second step, we pick the edge $a - r_1$ of M_1 and replace the edge $a - r_1$ in M (after the first step) with $a \rightarrow r_1$ (here a plays the role of x). In the third step, the undirected edges of M_1 that satisfy the conditions of the third step get directed in M . For instance, the undirected edge $b - e$ of M_1 is directed in M . We replace the edge $b - e$ in M (after the second step) with $b \rightarrow e$ in the third step, since M_1 has an undirected path $P = (r_1, b, e)$ from r_1 to (b, e) , and the path does not contain a .

We show that M , constructed using the above steps, obeys Item 3 of Lemma 3.7.11. To prove this, we need to show the following: (a) M is an MEC of G , (b) M obeys Equation (3.13), (c) $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, and (d) $M \in \text{MEC}(G, r_1, 1)$. Note that M , constructed in Figure 3.11 using the above steps, has all these four properties.

We first show that M is an MEC of G . From the construction of M , the skeleton of M is G . Therefore, it suffices to show that M obeys Items 1 and 2 of Theorem 3.2.1.

We begin by showing that M obeys Item 1 of Theorem 3.2.1. Suppose M contains an induced subgraph $a \rightarrow b - c$. From the construction of M , there are three possibilities: (a) $a, b, c \in V_{G_1}$, (b) $a, b, c \in V_{G_2}$, or (c) $a = r_2$, $b = r_1$, and $c \in V_{G_1}$. We next show that none of these possibilities occurs.

First, suppose $a, b, c \in V_{G_1}$. From the construction of M , $\text{skeleton}(M[V_{G_1}]) = \text{skeleton}(M_1) = G_1$, and for $u \rightarrow v \in M_1$, $u \rightarrow v \in M$. This implies that $b - c \in M_1$, and either $a \rightarrow b \in M_1$ or $a - b \in M_1$.

If $a \rightarrow b \in M_1$, then $a \rightarrow b - c$ is an induced subgraph in the MEC M_1 , contradicting Item 1 of Theorem 3.2.1.

If $a - b \in M_1$, it must have been directed in M either at step 2 (first case) or at step 3 (second case). If it was directed at step 2, then $a = x$, $b = r_1$, and $c \neq x$ is a neighbor of r_1 . This implies that $r_1 - c$ is an undirected path from r_1 to (r_1, c) in M_1 . And, at step 3, we have $(b = r_1) \rightarrow c \in M$, which contradicts our assumption that $b - c \in M$. Therefore, the first case cannot occur.

Now, suppose $a - b$ was directed at step 3. This implies that there exists an undirected path from r_1 to (a, b) in M_1 . Consequently, there must also be an undirected path from r_1 to (b, c) in M_1 . Therefore, we would have $b \rightarrow c \in M$ at step 3. This means that there cannot be an induced subgraph $a \rightarrow b - c$ in M with $a, b, c \in V_{G_1}$.

Next, suppose $a, b, c \in V_{G_2}$. From the construction of M , M_2 is an induced subgraph of M . Since M_2 is an MEC of the tree graph G_2 , by Item 1 of Theorem 3.2.1, there cannot be an induced subgraph $a \rightarrow b - c \in M_2$. Thus, there cannot be an induced subgraph $a \rightarrow b - c$ in M with $a, b, c \in V_{G_2}$.

Finally, suppose $a = r_2$, $b = r_1$, and $c \in V_{G_1}$. From the construction of M , this implies that $r_1 - c \in M_1$. There are two cases: either $c = x$ or $c \neq x$.

If $c = x$, then at step 2, we have $c \rightarrow b \in M$. Therefore, this case cannot occur.

If $c \neq x$, then $r_1 - c$ is an undirected path from r_1 to (r_1, c) . And, at step 3, we have $(b = r_1) \rightarrow c \in M$, contradicting our assumption that $b - c \in M$. This implies that this possibility also cannot occur.

This shows that M cannot have an induced subgraph of the form $a \rightarrow b - c$. Conse-

quently, M obeys Item 1 of Theorem 3.2.1.

We now show that M obeys Item 2 of Theorem 3.2.1. Suppose $u \rightarrow v$ is a directed edge in M . Then one of the following holds: $u, v \in V_{G_1}$, $u, v \in V_{G_2}$, or $u = r_2$ and $v = r_1$. We show that in each of these cases, $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1.

Suppose $u, v \in V_{G_1}$. From the construction of M , $\text{skeleton}(M[V_{G_1}]) = \text{skeleton}(M_1)$, and for $a \rightarrow b \in M_1$, $a \rightarrow b \in M$. Therefore, since $u \rightarrow v \in M$, either $u \rightarrow v \in M_1$ or $u - v \in M_1$. Suppose $u \rightarrow v \in M_1$. Since M_1 is an MEC of the tree graph G_1 , from Item 2 of Theorem 3.2.1, $u \rightarrow v$ is part of an induced subgraph of M_1 of the form either $u \rightarrow v \leftarrow w$ or $w \rightarrow u \rightarrow v$. From the construction of M , a directed induced subgraph of M_1 is also an induced subgraph of M . This implies that $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1.

If $u - v \in M_1$, then it must have been directed in M either at step 2 or at step 3. Suppose $u - v \in M_1$ and it gets directed in M at step 2. Then, $u = x$ and $v = r_1$. In that case, $u \rightarrow v$ is part of an induced subgraph $r_2 \rightarrow r_1 \leftarrow x$, satisfying Item 2 of Theorem 3.2.1.

Suppose $u - v \in M_1$ and it gets directed in M at step 3. This implies that there exists an undirected path from r_1 to (u, v) in M_1 , and the path does not contain x . Let the path be $P = (x_0 = r_1, x_1, \dots, x_{l-1} = u, x_l = v)$ for some $l \geq 1$. For all $0 \leq i < l$, at step 3 of the construction of M , we have $x_i \rightarrow x_{i+1} \in M$ because $P_i = (x_0 = r_1, \dots, x_i, x_{i+1})$ is an undirected path in M_1 from r_1 to (x_i, x_{i+1}) , and P_i does not contain x either. If $l > 1$, then $u \rightarrow v$ is part of an induced subgraph $x_{l-2} \rightarrow x_{l-1} \rightarrow x_l$ in M , satisfying Item 2 of Theorem 3.2.1. And, if $l = 1$, then $r_1 = u$, and $u \rightarrow v$ is part of an induced subgraph $r_2 \rightarrow r_1 \rightarrow v$ in M , satisfying Item 2 of Theorem 3.2.1.

Thus, we have shown that if $u, v \in V_{G_1}$, then $u \rightarrow v$ satisfies Item 2 of Theorem 3.2.1.

Suppose $u, v \in V_{G_2}$. From the construction of M , M_2 is an induced subgraph of M . This implies that $u \rightarrow v \in M_2$. Since M_2 is an MEC of the tree graph G_2 , from Item 2 of Theorem 3.2.1, $u \rightarrow v$ is part of an induced subgraph of M_2 of the form either $u \rightarrow v \leftarrow w$ or $w \rightarrow u \rightarrow v$. Since from the construction of M , an induced subgraph of M_2 is an induced subgraph of M . Therefore, $u \rightarrow v$ in M obeys Item 2 of Theorem 3.2.1.

We now show that $r_2 \rightarrow r_1$ in M also obeys Item 2 of Theorem 3.2.1. $r_2 \rightarrow r_1$ is part of an induced subgraph $r_2 \rightarrow r_1 \leftarrow x$ in M . Thus, we have shown that each edge of M

obeys Item 2 of Theorem 3.2.1.

The above discussion shows that M is an MEC as it obeys Items 1 and 2 of Theorem 3.2.1.

We now show that M obeys Equation (3.13). From the construction of M , the directed edges of M_1 and M_2 are also directed edges of M . Additionally, $r_2 \rightarrow r_1 \leftarrow x$ is a V-structure in M . This implies that $\mathcal{V}(M_1) \cup \mathcal{V}(M_2) \cup \{r_2 \rightarrow r_1 \leftarrow x\} \subseteq \mathcal{V}(M)$. For the completeness of the proof, we show that if $u \rightarrow v \leftarrow w$ is a V-structure in M , then one of the following holds: $u \rightarrow v \leftarrow w$ is a V-structure of M_1 , $u \rightarrow v \leftarrow w$ is a V-structure of M_2 , or $u = r_2, v = r_1$, and $w = x$, i.e., $\mathcal{V}(M) \subseteq \mathcal{V}(M_1) \cup \mathcal{V}(M_2) \cup \{r_2 \rightarrow r_1 \leftarrow x\}$. This further implies that $\mathcal{V}(M) = \mathcal{V}(M_1) \cup \mathcal{V}(M_2) \cup \{r_2 \rightarrow r_1 \leftarrow x\}$, i.e., M obeys Equation (3.13).

Suppose there exists a V-structure $u \rightarrow v \leftarrow w$ in M . From the construction of M , the following possibilities arise: (a) $u, v, w \in V_{G_1}$, (b) $u = r_2, v = r_1$, and w is a neighbor of r_1 in V_{G_1} , or (c) $u, v, w \in V_{G_2}$.

Suppose $u, v, w \in V_{G_1}$. From the construction of M , we have $\text{skeleton}(M[V_{G_1}]) = \text{skeleton}(M_1) = G_1$, and for $a \rightarrow b \in M_1$, we also have $a \rightarrow b \in M$. This implies that either $u - v \in M_1$ or $u \rightarrow v \in M_1$, and either $v - w \in M_1$ or $v \leftarrow w \in M_1$. Since M_1 is an MEC of the tree graph G_1 , from Item 1 of Theorem 3.2.1, we conclude that either $u \rightarrow v \leftarrow w \in M_1$ or $u - v - w \in M_1$.

If $u \rightarrow v \leftarrow w \in M_1$, then the V-structure is a V-structure of M_1 .

Suppose $u - v - w \in M_1$. Then, if $u \rightarrow v \in M$, it must have been directed either at step 2 or at step 3. Suppose $u \rightarrow v \in M$ at step 2. Then, $u = x$ and $v = r_1$. Additionally, $v - w \in M_1$ is an undirected path from r_1 to (v, w) , and the path does not contain x . Therefore, we have $v \rightarrow w \in M$ at step 3, which contradicts our assumption that $u \rightarrow v \leftarrow w \in M$. This implies that this case cannot occur.

Suppose $u \rightarrow v \in M$ at step 3. Then, there exists an undirected path from r_1 to (u, v) in M_1 , and the path does not contain x . Since $u - v - w \in M_1$, the existence of an undirected path from r_1 to (u, v) in M_1 implies the existence of an undirected path from r_1 to (v, w) in M_1 , and the new path also does not contain x . Additionally, since M_1 is an MEC of the tree graph G_1 , there cannot be two different paths from r_1 to w . This implies that at step 2, if we have $u \rightarrow v \in M$, then we also have $v \rightarrow w \in M$. But

this is a contradiction, as we have assumed that $u \rightarrow v \leftarrow w \in M$. Therefore, there cannot be $u - v - w \in M_1$ and $u \rightarrow v \leftarrow w \in M$.

Suppose $u = r_2$, $v = r_1$, and w is a neighbor of r_1 in V_{G_1} . Since $M_1 \in \text{MEC}(G_1, r_1, 0)$, either $r_1 \rightarrow w \in M_1$ or $r_1 - w \in M_1$. From the construction of M , if $r_1 \rightarrow w \in M_1$, then $r_1 \rightarrow w \in M$. But we have $r_1 \leftarrow w \in M$, which implies that $r_1 - w \in M_1$. From steps 2 and 3, if we have $w \rightarrow r_1 \in M$, then $w = x$; otherwise, we have $r_1 \rightarrow w \in M$ at step 3, as $r_1 - w$ is an undirected path from r_1 to (r_1, w) , and the path does not contain x . This implies that $w = x$.

Finally, suppose $u, v, w \in V_{G_2}$. From the construction of M , M_2 is an induced subgraph of M . This implies that if $u, v, w \in V_{G_2}$, then $u \rightarrow v \leftarrow w$ is a V-structure in M_2 .

Thus, we have shown that in all the possible scenarios, if $u \rightarrow v \leftarrow w$ is a V-structure in M , then one of the following occurs: $u \rightarrow v \leftarrow w$ is a V-structure in M_1 , $u \rightarrow v \leftarrow w$ is a V-structure in M_2 , or $u = r_2$, $v = r_1$, and $w = x$. This completes the proof that M obeys Equation (3.13).

The above discussion implies that $\mathcal{V}(M[V_{G_1}]) = \mathcal{V}(M_1)$ and $\mathcal{V}(M[V_{G_2}]) = \mathcal{V}(M_2)$. Therefore, from Definition 3.7.1, $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.

Since there is a directed edge $r_2 \rightarrow r_1$ towards r_1 in M , we have $M \in \text{MEC}(G, r_1, 1)$. This completes the proof of Item 3 of Lemma 3.7.11. □

Proof of Item 4 of Lemma 3.7.11. We construct an MEC of G using the following steps:

1. Initialize $M = M_1 \cup M_2 \cup \{r_1 \rightarrow r_2\}$.
2. Pick an edge $y - r_2 \in M_2$. Replace the edge $y - r_2$ in M with $y \rightarrow r_2$.
3. Update M by replacing $u - v$ in M with $u \rightarrow v$, if $u - v \in M_2$ and there exists an undirected path in M_2 from r_2 to (u, v) such that the path does not contain y .

Figure 3.12 is provided as an example. In the figure, $M_1 \in \text{MEC}(G_1, r_1, 0, 2)$, $M_2 \in \text{MEC}(G_2, r_2, 0, 2)$, and M is constructed using the steps mentioned above, where G , G_1 , and G_2 are from Figure 3.2. In the second step, we pick the edge $k - r_2$ of M_2 and replace the edge $k - r_2$ in M (after the first step) with $k \rightarrow r_2$ (here k plays the role of y).

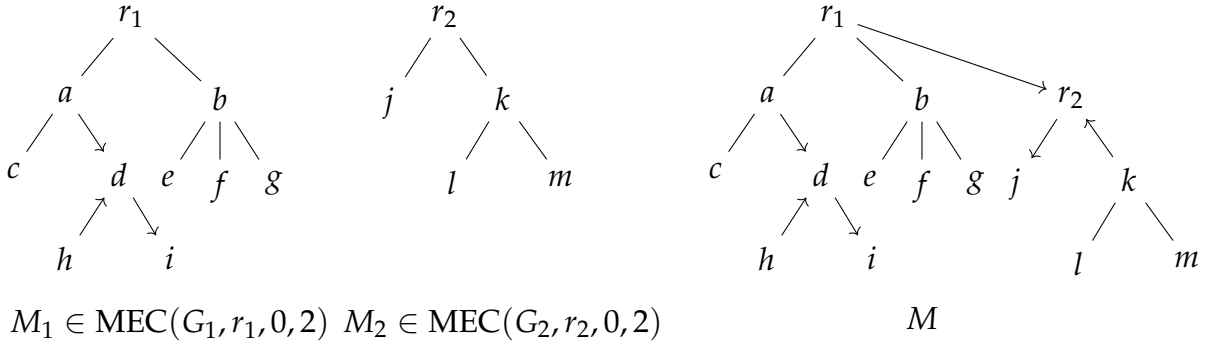


Figure 3.12: Auxiliary picture for the proof of Item 4 of Lemma 3.7.11.

In the third step, the undirected edges of M_2 that satisfy the conditions of the third step get directed in M . For instance, the undirected edge $r_2 - j$ of M_2 is directed in M by following the third step. We replace the edge $r_2 - j$ in M (after the second step) with $r_2 \rightarrow j$ in the third step, since M_2 has an undirected path $P = (r_2, j)$ from r_2 to (r_2, j) (the first node of the path is r_2 , and the last two nodes of the path are r_2 and j), and the path does not contain k .

We show that M constructed using the above steps will obey Item 4 of Lemma 3.7.11. To prove this, we need to show the following: (a) M is an MEC of G , (b) M obeys Equation (3.14), (c) $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$, and (d) $M \in \text{MEC}(G, r_1, 0, i)$. Note that M , constructed in Figure 3.12 using the above steps, satisfies all these four properties.

Since the construction of M is similar to the construction of M in the proof of Item 3 of Lemma 3.7.11 (with the roles of M_1 and M_2 swapped, the roles of r_1 and r_2 swapped, and y in place of x), we can follow a similar argument to show that M satisfies (a), (b), and (c). We now show that $M \in \text{MEC}(G, r_1, 0, i)$.

From the construction of M , M_1 is an induced subgraph of M . Since $M_1 \in \text{MEC}(G_1, r_1, 0, i)$, i edges adjacent to r_1 are undirected, and the remaining $\delta_1 - i$ adjacent edges of r_1 in M_1 are directed outward from r_1 . The only edge that is adjacent to r_1 in M but not adjacent to r_1 in M_1 is $r_1 \rightarrow r_2$, which is directed outward from r_1 . Therefore, no edge adjacent to r_1 in M is directed toward r_1 , and i edges adjacent to r_1 in M are undirected. This implies that $M \in \text{MEC}(G, r_1, 0, i)$.

This completes the proof of Item 4 of Lemma 3.7.11.

□

This completes the proof of Lemma 3.7.11.

□

Chapter 4

Counting MECs with the same skeleton

The objective of this chapter is to solve Problem 1.2. For convenience, we restate the problem below.

Problem 4.1. INPUT: (a) An undirected connected graph G .

OUTPUT: $|\text{MEC}(G)|$, the number of MECs of G , i.e., the number of MECs with G as their skeleton.

4.1 Overview

Theorem 4.1.1 summarizes our result.

Theorem 4.1.1 (Restating Theorem 1.3.1). *For an undirected graph G , there exists a fixed-parameter tractable algorithm that can count the MECs with skeleton G in $O(n(2^{O(d^4 k^4)} + n^2))$ time, where n is the number of nodes in G , and the parameters d and k are, respectively, the maximum degree and the treewidth of G .*

A trivial brute-force approach to solve Problem 4.1 for an input graph G is to go through each possible graph H with skeleton G and check whether H obeys Items 1 to 4 of Theorem 1.1.24 (the necessary and sufficient conditions for a graph to be an MEC). For each edge $u - v$ in G , H has three possibilities: $u \rightarrow v \in E_H$, $v \rightarrow u \in E_H$, or $u - v \in E_H$. This implies that the number of graphs H having skeleton G is $3^{|E_G|}$, and therefore, the runtime of this approach is exponential in the size of G .

In this chapter, we construct an FPT algorithm to solve Problem 4.1. The idea for the algorithm comes from the previous chapter, where we construct a polynomial algorithm to solve Problem 4.1 for a special case when the input graph is a tree graph. To solve Problem 4.1 for general graphs, we construct a rooted tree decomposition (T, β, r_1) of the input graph G . We define “shadow of an MEC” to provide a unique succinct identity to each MEC. We use this identity to partition the MECs of the input graph G . Similar to the tree case (Chapter 3), we cut an edge adjacent to r_1 in T to construct two induced subgraphs of G : G_1 and G_2 . The main idea of our algorithm is to find a relation between the shadows of the MECs of G , G_1 , and G_2 . We use this relation to answer the number of MECs of G having a specific shadow when for all possible shadows S_1 and S_2 , we have the information about the number of MECs of G_1 and G_2 with shadows S_1 and S_2 , respectively. We use this recursive relation to construct our algorithm.

Here is a detailed technical overview of this chapter.

Technical Overview: As described above, the main technical contribution is the notion of a “shadow of an MEC” (Definition 4.3.1), that provides a unique and succinct identity to an MEC. For an MEC M of an undirected graph G , the shadow of an MEC M on $X \subseteq V_G$ is a triple (O, P_1, P_2) , where O is the induced subgraph of M on X , $P_1 : E_O \times E_O \rightarrow \{0, 1\}$ is a function that maps $((u, v), (x, y)) \in E_O \times E_O$ with 1 if and only if there exists a triangle-free path (a path of a graph in which for any three consecutive nodes a, b , and c of the path, there is no edge in the graph with endpoints a and c ; Definition 4.2.4) from (u, v) to (x, y) in M of length three or more, and $P_2 : E_O \times V_O \rightarrow \{0, 1\}$ is a function that maps $((u, v), w) \in E_O \times V_O$ with 1 if and only if there exists a triangle-free path in M from (u, v) to w of length three or more. The definition of “shadow” implies that for an MEC M of an undirected graph G and $X \subseteq V_G$, there exists a unique shadow of M on X . We use these shadows to partition the MECs of G .

We find that the induced subgraph of an MEC is a *partial MEC* [Van Der Zander and Liskiewicz, 2016], a graph that obeys Items 1 to 3 of Theorem 1.1.24 (Observation 4.2.3). With a slight abuse of notation, we define a triplet (O, P_1, P_2) as a shadow of an undirected graph G if O is a partial MEC with skeleton G , and $P_1 : E_O \times E_O \rightarrow \{0, 1\}$ and

$P_2 : E_O \times V_O \rightarrow \{0, 1\}$ are two functions. Our definitions imply that for an MEC M of an undirected graph G and $X \subseteq V_G$, the shadow of M on X is a shadow of $G[X]$ (Observation 4.3.5). We show that for any undirected graph G , for any $X \subseteq V_G$, summation of $|\text{MEC}(G, O, P_1, P_2)|$, the size of the set of MECs of G having shadow (O, P_1, P_2) , over all shadows (O, P_1, P_2) of $G[X]$, equals the number of MECs of G (Lemma 4.3.7). This reduces the problem of counting MECs of an undirected graph G into counting MECs of G with shadow (O, P_1, P_2) for each possible shadow (O, P_1, P_2) of $G[X]$ for some $X \subseteq V_G$. It is the idea of “shadow”, a nontrivial concise representation of an MEC, which is the key for the construction of the fixed-parameter tractable algorithm (Algorithm 9).

We provide a recursive algorithm (Algorithm 10) to solve the reduced problem. For this, we find a relation between the MECs of an undirected graph G and the MECs of an induced subgraph G' of G . We note that for an MEC M of G , $M[V_{G'}]$ may not be an MEC of G' . But, we found that there exists a unique MEC M' of G that has the same set of v -structure as $M[V_{G'}]$ (Lemma 4.4.2). We call the MEC M' as the *projection* of M on $V_{G'}$ (Definition 4.4.1).

For the recursion, we divide the input graph G using its tree decomposition. Suppose (T, β, r_1) is a tree decomposition of G . We arbitrarily pick a neighbor r_2 of r_1 in T . We cut the edge $r_1 - r_2$ of T . By this, we get two induced subtrees T_1 (containing r_1) and T_2 (containing r_2) of T , and their corresponding induced subgraphs of G , G_1 and G_2 , that are represented by T_1 and T_2 , respectively. From the tree decomposition properties, we have (a) G is the union of G_1 and G_2 , and (b) $I = V_{G_1} \cap V_{G_2}$ is a vertex separator of G . We start with finding a graphical relation between an MEC M of G and its projections M_1 and M_2 on V_{G_1} and V_{G_2} respectively. Let $\beta(r_1) = S_1$, $\beta(r_2) = S_2$, (O, P_1, P_2) be the shadow of M on $S_1 \cup S_2 \cup N(S_1 \cup S_2, G)$ ($N(X, G)$ is the set of neighbors of the vertices in X in G), (O_1, P_{11}, P_{12}) be the shadow of M_1 on $S_1 \cup N(S_1, G_1)$, and (O_2, P_{21}, P_{22}) be the shadow of M_2 on $S_2 \cup N(S_2, G_2)$. We find a graphical resemblance between the shadows of M , M_1 and M_2 . We find a necessary condition obeyed by the shadows (Lemma 4.5.7). We later show that the conditions are also sufficient (Lemma 4.5.8). That means, if we have shadows (O, P_1, P_2) of $G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)]$, (O_1, P_{11}, P_{12}) of $G_1[S_1 \cup N(S_1, G_1)]$, and (O_2, P_{21}, P_{22}) of $G_2[S_2 \cup N(S_2, G_2)]$ that obey the necessary

and sufficient condition then for each pair of MECs (M_1, M_2) such that M_1 is an MEC of G_1 with shadow (O_1, P_{11}, P_{12}) and M_2 is an MEC of G_2 with shadow (O_2, P_{21}, P_{22}) there exists a unique MEC M of G with shadow (O, P_1, P_2) . This relation shows that if we have the knowledge of $|\text{MEC}(G_1, O_1, P_{11}, P_{12})|$ for each shadow (O_1, P_{11}, P_{12}) of $G_1[S_1 \cup N(S_1, G_1)]$ and $|\text{MEC}(G_2, O_2, P_{21}, P_{22})|$ for each shadow (O_2, P_{21}, P_{22}) of $G_2[S_2 \cup N(S_2, G_2)]$, we can recursively calculate the number of MECs of G with shadow (O, P_1, P_2) for each shadow (O, P_1, P_2) of $G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)]$ (Lemma 4.5.13).

Lemma 4.5.13 provides an approach to compute $|\text{MEC}(G, O, P_1, P_2)|$ for each shadow (O, P_1, P_2) of $G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)]$. However, for the recursion, we need to compute $|\text{MEC}(G, O, P_1, P_2)|$ for each shadow (O, P_1, P_2) of $G[S_1 \cup N(S_1, G)]$. Since a shadow of $G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)]$ contains more information than a shadow of $G[S_1 \cup N(S_1, G)]$, we can trim the former to fulfill our purpose. We define the “projection of a shadow” (Definition 4.5.14) for this. The definition implies that the projection of a shadow is unique.

We show that if a shadow (O', P'_1, P'_2) of $G[S_1 \cup N(S_1, G)]$ is the projection of a shadow (O, P_1, P_2) of $G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)]$, then if M is an MEC of G with shadow (O, P_1, P_2) , M also has (O', P'_1, P'_2) as its shadow. For each shadow (O', P'_1, P'_2) of $G[S_1 \cup N(S_1, G)]$, this leads us to count MECs of G with shadow (O', P'_1, P'_2) when, for each shadow (O, P_1, P_2) of $G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)]$, we know $|\text{MEC}(G, O, P_1, P_2)|$ (Lemma 4.5.15).

As discussed earlier, for the recursion, we cut an edge of the tree decomposition of G to construct induced subgraphs G_1 and G_2 . Since the number of edges in the tree decomposition of G is linear in the size of G , the number of times we recursively call our algorithm is linear in the size of the input graph. The number of possible shadows is exponential in the treewidth and the degree of the input graph. The computation of the number of MECs having a specific shadow also takes exponential time in terms of the treewidth and the degree of the input graph. We implement the above discussion to construct an algorithm (Algorithm 9) to count the MECs of an undirected graph. The overall time complexity (Theorem 4.8.5) of the algorithm is $O(n(2^{O(k^4\delta^4)} + n^2))$, where the parameters k and δ are the treewidth and the degree of the input graph, respectively, and n is the number of nodes of the input graph.

Organization of the chapter: In Section 4.2, we define the concepts of “partial MEC” and “triangle-free path” along with their related results, which are foundational for addressing Problem 4.1. In Section 4.3, we introduce the “shadow of an MEC” (Definition 4.3.1) and the “shadow of an undirected graph” (Definition 4.3.4), which are utilized to partition the MECs of an undirected graph (Lemma 4.3.7). Section 4.4 defines the “projection of an MEC” (Definition 4.4.1) and presents associated results.

In Section 4.5, we describe our approach for counting the MECs of an undirected graph. Section 4.6 provides the proofs omitted from Sections 4.4 and 4.5. In Section 4.7, we extend the discussion in Section 4.5 to develop a recursive algorithm for counting the MECs of an undirected graph. Finally, Section 4.8 analyzes the time complexity of the algorithms presented in this chapter.

4.2 Partial MECs and Triangle-Free Paths

In this section, we define “partial MECs” and “triangle-free paths”. These two notions are crucial for this chapter. The first of these, the notion of a partial MEC, has appeared earlier in the literature, e.g., in the work of [Van Der Zander and Liskiewicz \[2016\]](#).

Definition 4.2.1 (Partial MEC). A graph M is said to be a *partial MEC* if it satisfies the following conditions:

1. M is a chain graph.
2. Undirected connected components of M are chordal.
3. There does not exist a subgraph of the form $u \rightarrow v - w \in M$.

i.e., a partial MEC obeys Items 1 to 3 of Theorem 1.1.24. It’s worth noting that every MEC is a partial MEC, and a partial MEC that obeys Item 4 of Theorem 1.1.24 is an MEC.

Notation 4.2.2. For an undirected graph G , we say that an MEC M is an MEC of G if $\text{skeleton}(M) = G$, and we denote by $\text{MEC}(G)$ the set of MECs of G . Similarly, we say that a partial MEC M is a partial MEC of G if $\text{skeleton}(M) = G$, and we denote by $\text{PMEC}(G)$ the set of partial MECs of G .

The following observation shows that an induced subgraph of an MEC is a partial MEC.

Observation 4.2.3. *For an MEC M of G , for $X \subseteq V_G$, $M[X]$ is a partial MEC of $G[X]$.*

Proof. Since M is an MEC, as stated in Theorem 1.1.24, M is a chain graph with chordal undirected connected component and does not possess any induced subgraph of the form $u \rightarrow v - w$. It's worth noting that an induced subgraph of a chain graph with chordal undirected connected component is also a chain graph with chordal undirected connected component. Furthermore, if M has no induced subgraph of the form $u \rightarrow v - w$, then its induced subgraphs must also lack such a structure. From Definition 4.2.1, this implies that an induced subgraph of an MEC is a partial MEC. From the construction, the skeleton of $M[X]$ is $G[X]$, implying $M[X]$ is a partial MEC of $G[X]$. $\square$

We now define “triangle-free path”.

Definition 4.2.4 (Triangle-free path). For a graph G , a path $P = (u_1, u_2, \dots, u_{l-1}, u_l)$ is said to be a *triangle-free path* of G if for any node u_i in P , there is no edge between the two adjacent nodes of u_i in P . More formally, for any $1 \leq i \leq l - 2$, neither $u_i - u_{i+2} \in E_G$, nor $u_i \rightarrow u_{i+2} \in E_G$, nor $u_{i+2} \rightarrow u_i \in E_G$, i.e., $u_i - u_{i+2} \notin \text{skeleton}(G)$.

The following observation shows the transitive nature of triangle-free paths in a chain graph with chordal undirected components.

Lemma 4.2.5 (Concatenation of Triangle-Free Paths). *Consider a chain graph G with chordal undirected components. Let $u, v, x, y, w \in V_G$ be (possibly non-distinct) vertices of G . Suppose $P_1 = (a_1 = u, a_2 = v, \dots, a_{l-1} = x, a_l = y)$ and $P_2 = (b_1 = x, b_2 = y, \dots, b_{m-1}, b_m = w)$ are triangle-free paths in G from (u, v) to (x, y) and from (x, y) to w , respectively. Then, their concatenation $P := (a_1 = u, a_2 = v, \dots, a_{l-1} = x, a_l = y, b_3, b_4, \dots, b_{m-1}, b_m = w)$ is a triangle-free path in G from (u, v) to w .*

Proof. From the definition of P , we observe that (i) for any two consecutive vertices p and q in P , either the edge $p \rightarrow q$ or the edge $p - q$ exists in G because p and q must be consecutive in either P_1 or P_2 , both of which are paths in G , and (ii) for any three

consecutive vertices p, q , and r in P , the vertices p, q , and r are distinct, and p and r are not adjacent in G because p, q , and r must be consecutive in one of P_1 or P_2 , both of which are triangle-free paths.

Thus, to show that P is a triangle-free path in G it only remains to show that all vertices in P are distinct. Suppose, if possible, that this is not the case. Then, there must exist a segment of P of the form $C := (c_1 = p, c_2, \dots, c_{k-1}, c_k = p)$ in which the vertices $c_1, c_2, \dots, c_{k-1}$ are distinct. From the observations above about P , we see that C must be a cycle in G . Since G is a chain graph, C cannot have any directed edges. Therefore, C must be entirely contained within an undirected component H of G . Furthermore, from our second observation, C must have at least four nodes. If C does not have a chord in H then C is a chordless cycle in G , which contradicts the assumption that the undirected connected components of G are chordal. Suppose C has a chord (c_i, c_j) . We choose such (c_i, c_j) with $i < j$ and $j - i$ being minimal. In this case, we have another cycle $C' := (c_i, c_{i+1}, \dots, c_j, c_{j+1} = c_i)$. We have selected c_i and c_j such that there is no edge between any two non-adjacent nodes of C' (due to the minimality of $j - i$). According to our second observation, C' must have at least four nodes. This makes C' a chordless cycle, contradicting the assumption that the undirected components of G are chordal.

We conclude that all vertices in P are distinct. Combined with the above two observations about P , this shows that P is a triangle-free path in G . $\square$

Lemma 4.2.5 applies to MECs because, as shown in Theorem 1.1.24, an MEC is a chain graph with undirected connected components.

The following lemma shows the unidirectional nature of triangle-free paths in an MEC. We use this later for the verification of an MEC.

Lemma 4.2.6. *Let M be an MEC of an undirected graph G . For $(u, v), (x, y) \in E_G$ such that $(u, v) \neq (x, y)$, if there exists a triangle-free path from (u, v) to (x, y) in M then there cannot exist a triangle-free path from (x, y) to (u, v) .*

Proof. Suppose, if possible, that there exist triangle-free paths P_1 from (u, v) to (x, y) and P_2 from (x, y) to u (P_2 is a prefix of the hypothesized triangle-free path from (x, y) to (u, v)). Then by Lemma 4.2.5, the concatenation Q of P_1 and P_2 (in the sense of

Lemma 4.2.5) is a triangle-free path from (u, v) to u . But this is a contradiction, since by the definition of path, Definition 1.1.2, no vertex can appear twice in a path. This implies there cannot exist a triangle-free path from (x, y) to u . Thus, there cannot exist a triangle-free path from (x, y) to (u, v) . $\square$

4.3 Shadow of an MEC

In this section, we introduce “shadow of an MEC”. As discussed in the introduction, we use it to partition the MECs of a graph.

Definition 4.3.1 (Shadow of an MEC). Let M be an MEC of an undirected graph G , $Y \subseteq V_G$, $O \in \text{PMEC}(G[Y])$ (i.e., O be a partial MEC with skeleton $G[Y]$), and $P_1 : E_O \times E_O \rightarrow \{0, 1\}$ and $P_2 : E_O \times V_O \rightarrow \{0, 1\}$ be two functions. We define (O, P_1, P_2) as the *shadow* of the MEC M on Y , denoted by $(O, P_1, P_2) = \text{shadow}(M, Y)$, if the following occur:

1. $M[Y] = O$, i.e., O is an induced subgraph of M on Y .
2. For $(u, v), (x, y) \in E_O$, $P_1((u, v), (x, y)) = 1$ if $(u, v) \neq (x, y)$ and there exists a triangle-free path from (u, v) to (x, y) in M , else $P_1((u, v), (x, y)) = 0$.
3. For $(u, v) \in E_O$ and $w \in V_O$, $P_2((u, v), w) = 1$, if $v \neq w$ and there exists a triangle-free path from (u, v) to w in M , else $P_2((u, v), w) = 0$.

Definition 4.3.1 shows that the shadow of an MEC M on $Y \subseteq V_M$ is unique because the induced subgraph of M on Y (the first term of the shadow) is unique, and the other two terms of the shadow, P_1 and P_2 , can be computed uniquely using Items 2 and 3 of Definition 4.3.1.

Remark 4.3.2 (Purpose of triangle-free paths information in shadow). Each MEC has a unique skeleton and a set of v-structures. In our case, we assume the skeleton of the MEC is known. However, we cannot construct an MEC for every possible set of v-structures. There exist sets of v-structures for which no MEC exists with the given skeleton. For example, there cannot exist an MEC that has C_4 (a cycle with four nodes)

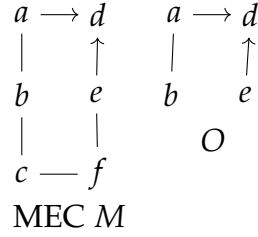


Figure 4.1: Auxiliary figure to show that P_1 does not determine P_2 .

as its skeleton and no v-structures, as Item 2 of Theorem 1.1.24 is not satisfied. To determine whether an MEC exists for a given skeleton and a given set of v-structures, we use information about triangle-free paths. This property of triangle-free paths is used in Lemma 4.5.8.

Remark 4.3.3 (P_1 does not determine P_2). Note that P_2 is not necessarily determined by P_1 . This is because there can exist $(u, v) \in E_O$ and $w \in Y$ such that there exists a triangle-free path from (u, v) to w in M , i.e., $P_2((u, v), w) = 1$, but for which there does not exist any edge $(x, w) \in E_O$ such that $P_1((u, v), (x, w)) = 1$. This can happen when every triangle-free path $P = (u_1 = u, u_2 = v, \dots, u_{l-1}, u_l = w)$ in M from (u, v) to w has $u_{l-1} \notin Y$. For example, consider the MEC M in Figure 4.1. Let $Y = \{a, b, d, e\}$, and $\text{shadow}(M, Y) = (O, P_1, P_2)$ (O is as shown in the figure). For $(a, b) \in E_O$ and $e \in V_O$, there exists a triangle-free path, (a, b, c, f, e) , from (a, b) to e in the MEC. But, there is no $x \in Y$ such that there exists a triangle-free path from (a, b) to (x, e) in the MEC.

With a slight abuse of notation, we define the following:

Definition 4.3.4 (Shadow of an Undirected Graph). For an undirected graph G , we say a triplet (O, P_1, P_2) is a *shadow* of G , denoted by $(O, P_1, P_2) \in \text{shadow}(G)$, if $O \in \text{PMEC}(G)$, and $P_1 : E_O \times E_O \rightarrow \{0, 1\}$ and $P_2 : E_O \times V_O \rightarrow \{0, 1\}$ are two functions.

The following observation is intuitive from Definition 4.3.1.

Observation 4.3.5. Let M be an MEC of an undirected graph G . For $Y \subseteq V_G$, if (O, P_1, P_2) is the shadow of the MEC M on Y , then (O, P_1, P_2) is a shadow of the undirected graph $G[Y]$.

Proof. Suppose (O, P_1, P_2) is the shadow of M on Y . According to Definition 4.3.1, $O = M[Y]$, and $P_1 : E_O \times E_O \rightarrow \{0, 1\}$ and $P_2 : E_O \times V_O \rightarrow \{0, 1\}$ are two functions.

As $\text{skeleton}(M) = G$, the skeleton of $O = M[Y]$ must be $G[Y]$. Thus, to establish that (O, P_1, P_2) is a shadow of $G[Y]$, we only need to demonstrate that O is a partial MEC of $G[Y]$. From Observation 4.2.3, O is a partial MEC. This concludes the proof of Observation 4.3.5. $\square$

We denote the set of MECs of an undirected graph G having shadow (O, P_1, P_2) by $\text{MEC}(G, O, P_1, P_2)$. Notation 4.3.6 formally states this.

Notation 4.3.6 ($\text{MEC}(G, O, P_1, P_2)$). *Let G be an undirected graph, $Y \subseteq V_G$, and $(O, P_1, P_2) \in \text{shadow}(G[Y])$. We denote by $\text{MEC}(G, O, P_1, P_2)$ the set of MECs M of G for which (O, P_1, P_2) is the shadow of M on Y . More formally, $\text{MEC}(G, O, P_1, P_2) := \{M : M \in \text{MEC}(G) \text{ and } (O, P_1, P_2) = \text{shadow}(M, V_O)\}$.*

Lemma 4.3.7 partitions the set of MECs of G based on its shadow.

Lemma 4.3.7. *Let G be an undirected graph. Then, for any $Y \subseteq V_G$,*

$$|\text{MEC}(G)| = \sum_{(O, P_1, P_2) \in \text{shadow}(G[Y])} |\text{MEC}(G, O, P_1, P_2)| \quad (4.1)$$

Proof. From the definition of shadow (Definition 4.3.1), for each $M \in \text{MEC}(G)$, there exists a unique shadow (O, P_1, P_2) of M on Y . From Observation 4.3.5, $(O, P_1, P_2) \in \text{shadow}(G[Y])$. This proves Lemma 4.3.7. $\square$

Since we have partitioned the set of MECs based on their shadows, we do not need to account for the v-structures that an MEC contains.

4.4 Projection of an MEC

To construct a recursive algorithm to count the MECs of a graph, we define “*projection of an MEC*”, to relate an MEC of a graph G and an MEC of an induced subgraph of G . The definition is identical to the one given in Chapter 3, but we restate it here for the convenience of the reader.

Definition 4.4.1 (Projection of an MEC). Let G be an undirected graph, $S \subseteq V_G$, $G' = G[S]$ an induced subgraph of G , M an MEC of G , and M' an MEC of G' . We

call M' the *projection* of M on S , denoted by $\mathcal{P}(M, S) = M'$, if the set of v-structures (Definition 1.1.8) in M' equals the set of v-structures in $M[S]$, i.e., $\mathcal{V}(M') = \mathcal{V}(M[S])$. With a slight abuse of notation, for two subsets $S_1, S_2 \subseteq V_G$, we call (M_1, M_2) the projection of M on (S_1, S_2) , denoted by $\mathcal{P}(M, S_1, S_2) = (M_1, M_2)$, if $M_1 = \mathcal{P}(M, S_1)$ and $M_2 = \mathcal{P}(M, S_2)$.

The following results illustrate the properties of a projection of an MEC. In Lemma 4.4.2 and Corollaries 4.4.3 and 4.4.4, we record some easy consequences of the notion of a projection. A more interesting structural similarity between an MEC M and its projection is recorded in Lemma 4.4.5.

Lemma 4.4.2. *Let G be an undirected graph, and M an MEC of G . For any $S \subseteq V_G$, there exists a unique MEC M' of the graph $G[S]$ such that $\mathcal{P}(M, S) = M'$.*

Proof. As a reminder from the introduction, every DAG corresponds to a unique MEC, and the skeleton and the set of v-structures of a DAG are identical to the skeleton and the set of v-structures of its corresponding MEC. We first show the existence of such an MEC M' . Let D be a DAG such that $D \in M$. The set of v-structures contained in $D[S]$ equals $\mathcal{V}(M[S])$ (since both D and M have the same set of v-structures). Let M' be the MEC of $G[S]$ such that $D[S] \in M'$. Then, the set of v-structures of M' must be the same as the set of v-structures of $D[S]$, implying $\mathcal{V}(M') = \mathcal{V}(M[S])$. From Definition 4.4.1, this further implies $\mathcal{P}(M, S) = M'$. The uniqueness of M' comes from the fact that if two MECs of a graph have the same set of v-structure, then they must be the same (as from Theorem 1.1.22, DAGs contained by such MECs are Markov equivalent). $\square$

Lemma 4.4.2 further implies the following corollaries:

Corollary 4.4.3. *Let G be an undirected graph, and M be an MEC of G . For $S_1, S_2 \subseteq V_G$, there exists a unique tuple $(M_1, M_2) \in \text{MEC}(G[S_1]) \times \text{MEC}(G[S_2])$ such that $\mathcal{P}(M, S_1, S_2) = (M_1, M_2)$.*

Proof. From Definition 4.4.1, $M_1 = \mathcal{P}(M, S_1)$ and $M_2 = \mathcal{P}(M, S_2)$. From Lemma 4.4.2, $\mathcal{P}(M, S_1)$ and $\mathcal{P}(M, S_2)$ both are unique. This proves the uniqueness of (M_1, M_2) . $\square$

Corollary 4.4.4. *Let G be an undirected graph, $S \subseteq V_G$, $M \in \text{MEC}(G)$, and $M' \in \text{MEC}(G[S])$. If $\mathcal{P}(M, S) = M'$ then for any $S' \subseteq S$, $\mathcal{V}(M[S']) = \mathcal{V}(M'[S'])$, and $\mathcal{P}(M, S') = \mathcal{P}(M', S')$.*

Proof. Since $S' \subseteq S$, for $u, v, w \in S'$, if a v-structure $u \rightarrow v \leftarrow w \in \mathcal{P}(M', S')$ then from Definition 4.4.1, the v-structures also belongs to M' , M and $\mathcal{P}(M, S')$. Similarly, for $u, v, w \in S'$, if a v-structure $u \rightarrow v \leftarrow w \in \mathcal{P}(M, S')$ then from Definition 4.4.1, the v-structures also belongs to M' , and $\mathcal{P}(M', S')$. This implies that $\mathcal{V}(M[S']) = \mathcal{V}(M'[S'])$. This further implies that both $\mathcal{P}(M, S')$ and $\mathcal{P}(M', S')$ have the same set of v-structures (from Definition 4.4.1). And, from the construction, both $\mathcal{P}(M, S')$ and $\mathcal{P}(M', S')$ have the same skeleton, $G[S']$, implying $\mathcal{P}(M, S') = \mathcal{P}(M', S')$. $\square$

The following important lemma shows some structural similarity between an MEC M and its projection M' .

Lemma 4.4.5. *Let G be a graph, $S \subseteq V_G$, $M \in \text{MEC}(G)$, and $\mathcal{P}(M, S) = M'$. If $u \rightarrow v \in M'$, then $u \rightarrow v \in M$.*

The proof of Lemma 4.4.5 is given in Section 4.6.2 (page 145). Recall that Lemmas 4.4.2 and 4.4.5 were used in Chapter 3.

Lemma 4.4.5 further implies the following corollaries:

Corollary 4.4.6. *Let G be a graph, $S \subseteq V_G$, $M \in \text{MEC}(G)$, and $\mathcal{P}(M, S) = M'$. If $u \rightarrow v \in M$, and $u, v \in S$ then either $u \rightarrow v \in M'$, or $u - v \in M'$.*

Proof. Suppose $u, v \in S$ and $u \rightarrow v \in M$. From the construction of projection, $\text{skeleton}(M') = \text{skeleton}(M[S])$. This implies if $u \rightarrow v \in M$ then either $u \rightarrow v \in M'$ or $v \rightarrow u \in M'$ or $u - v \in M'$. But, if $v \rightarrow u \in M'$ then from Lemma 4.4.5, $v \rightarrow u \in M$, a contradiction. This implies either $u \rightarrow v \in M'$, or $u - v \in M'$. $\square$

Corollary 4.4.7. *Let G be a graph, $S \subseteq V_G$, $M \in \text{MEC}(G)$, and $\mathcal{P}(M, S) = M'$. If $u - v \in M$, and $u, v \in S$ then $u - v \in M'$.*

Proof. Suppose $u, v \in S$ and $u - v \in M$. From the construction of projection, $\text{skeleton}(M') = \text{skeleton}(M[S])$. This implies if $u - v \in M$ then either $u \rightarrow v \in M'$ or $v \rightarrow u \in M'$ or $u - v \in M'$. From Lemma 4.4.5, if there is a directed edge between u and v in M' then M also has a directed edge between u and v , a contradiction. This implies $u - v \in M'$. $\square$

Corollary 4.4.8. *Let M be an MEC and M' a projection of M . Let $P = (u_1, u_2, \dots, u_l)$ be a triangle-free path in M such that all nodes of P are in M' , i.e., for each $1 \leq i \leq l$, $u_i \in V_{M'}$. Then P is also a triangle-free path in M' .*

Proof. Suppose $P = (u_1, u_2, \dots, u_l)$ is a triangle-free path in M such that all nodes of P are in M' . From Corollaries 4.4.6 and 4.4.7, P is a path in M' . If P is not a triangle-free path in M' , there exist u_{i-1}, u_i, u_{i+1} such that there is an edge between u_{i-1} and u_{i+1} in M' (either $u_{i-1} - u_{i+1} \in M'$, or $u_{i-1} \rightarrow u_{i+1} \in M'$, or $u_{i+1} \rightarrow u_{i-1} \in M'$). Since M' is a projection of M , it follows that if $u - v \in \text{skeleton}(M')$, then $u - v \in \text{skeleton}(M[V_M])$. Thus, there is an edge between u_{i-1} and u_{i+1} in M , implying P is not a triangle-free path in M , a contradiction. Therefore, P is a triangle-free path in M' . $\square$

The following corollary is a partial converse of Corollary 4.4.8.

Corollary 4.4.9. *Let M be an MEC and M' a projection of M . Let $P = (u_1, u_2, \dots, u_l)$ be a triangle-free path in M' . If $u_2 \rightarrow u_1 \notin M$, then P is also a triangle-free path in M .*

Proof. Suppose $u_2 \rightarrow u_1 \notin M$. From the construction, $\text{skeleton}(M') = \text{skeleton}(M[V_{M'}])$. Since P is a triangle-free path in M' , if P is a path in M , then it is a triangle-free path in M . We show that P is a path in M .

Suppose P is not a path in M . Then there exists an edge $u_i \leftarrow u_{i+1} \in M$. Pick the least i such that $u_i \leftarrow u_{i+1} \in M$. Clearly, $i \geq 2$ (from the assumption $u_2 \rightarrow u_1 \notin M$). Since P is a triangle-free path in M' , there does not exist an edge between u_{i-1} and u_{i+1} . This implies that either $u_{i-1} - u_i \leftarrow u_{i+1}$ or $u_{i-1} \rightarrow u_i \leftarrow u_{i+1}$ is an induced subgraph of M .

From Item 3 of Theorem 1.1.24, $u_{i-1} - u_i \leftarrow u_{i+1}$ cannot be an induced subgraph of M . From Definition 4.4.1, $\mathcal{V}(M') = \mathcal{V}(M[V_{M'}])$. This implies that if the v-structure $u_{i-1} \rightarrow u_i \leftarrow u_{i+1} \in M$, then $u_{i-1} \rightarrow u_i \leftarrow u_{i+1} \in M'$, as $u_{i-1}, u_i, u_{i+1} \in V_{M'}$. However, this would imply P is not a path in M' (since $u_i \leftarrow u_{i+1} \in M'$), a contradiction.

Thus, P is a path in M . $\square$

The following corollary is an extension of Corollary 4.4.9.

Corollary 4.4.10. *Let M be an MEC and M' a projection of M . If $P = (u_1 = x, u_2 = y, u_3, \dots, u_{l-1} = u, u_l = v)$ is a triangle-free path from (x, y) to (u, v) in M' , and $x \rightarrow y \in M$, then for all $1 \leq i < l$, $u_i \rightarrow u_{i+1} \in M$.*

Proof. From Corollary 4.4.9, P is a triangle-free path in M . This implies that for all $1 \leq i < l$, either $u_i - u_{i+1} \in M$ or $u_i \rightarrow u_{i+1} \in M$; otherwise, if for some i , $u_i \leftarrow u_{i+1} \in M$, then P is not a path in M .

Suppose for some i , $u_i - u_{i+1} \in M$. Pick the least i such that $u_i - u_{i+1} \in M$. Note that $i > 1$ since $u_1 \rightarrow u_2 \in M$ is given. However, this results in $u_{i-1} \rightarrow u_i - u_{i+1}$ being an induced subgraph of M , contradicting Item 3 of Theorem 1.1.24.

Thus, no such i exists, and for all $1 \leq i < l$, $u_i \rightarrow u_{i+1} \in M$. $\square$

We now describe our approach to count the MECs of an undirected graph.

4.5 Counting MECs of a Graph

We present a recursive algorithm to solve Problem 4.1. For this recursion, we divide the input graph G using its tree decomposition. We construct a tree decomposition (T, β) of G . Let $V_T = \{X_1, X_2, \dots, X_s\}$. We select X_1 as the root node of T . For each node X_i in T , we denote the subtree of T rooted at X_i by T_i , and G_i as the induced subgraph of G that is represented by T_i , i.e., $G_i = G[\bigcup_{X_j \in V_{T_i}} \beta(X_j)]$. This implies $T = T_1$, $G = G_1$, and our goal is to count the MECs of G_1 .

In the subtree T_i , let the children of X_i be $(X_{i1}, X_{i2}, \dots, X_{il})$ (i.e., for T_i , we consider X_{i1} as the first child of X_i , X_{i2} as the second child of X_i , and so on; the ordering of the children is arbitrary). For any $0 \leq j \leq l$, we denote by T_i^j the induced subtree of T_i that contains the node X_i and the nodes of $T_{i1}, T_{i2}, \dots, T_{ij}$. That means T_i^0 contains only the node X_i , and $T_i^l = T_i$. We denote by G_i^j the induced subgraph of G that is represented by T_i^j , i.e., $G_i^j = G[\bigcup_{X_j \in V_{T_i^j}} \beta(X_j)]$. If the number of children of X_1 is p , then $T_1^p = T_1 = T$, and $G_1^p = G_1 = G$, and our goal is to count the MECs of G_1^p . We provide an example illustrating the notation introduced here in Figure 4.2.

In Figure 4.2, G is an undirected graph, and (T, β) is a tree decomposition of G where $\beta(X_1) = \{1, 2, 3\}$, $\beta(X_2) = \{2, 4, 5\}$, $\beta(X_3) = \{2, 3, 6\}$, $\beta(X_4) = \{3, 7, 8\}$,

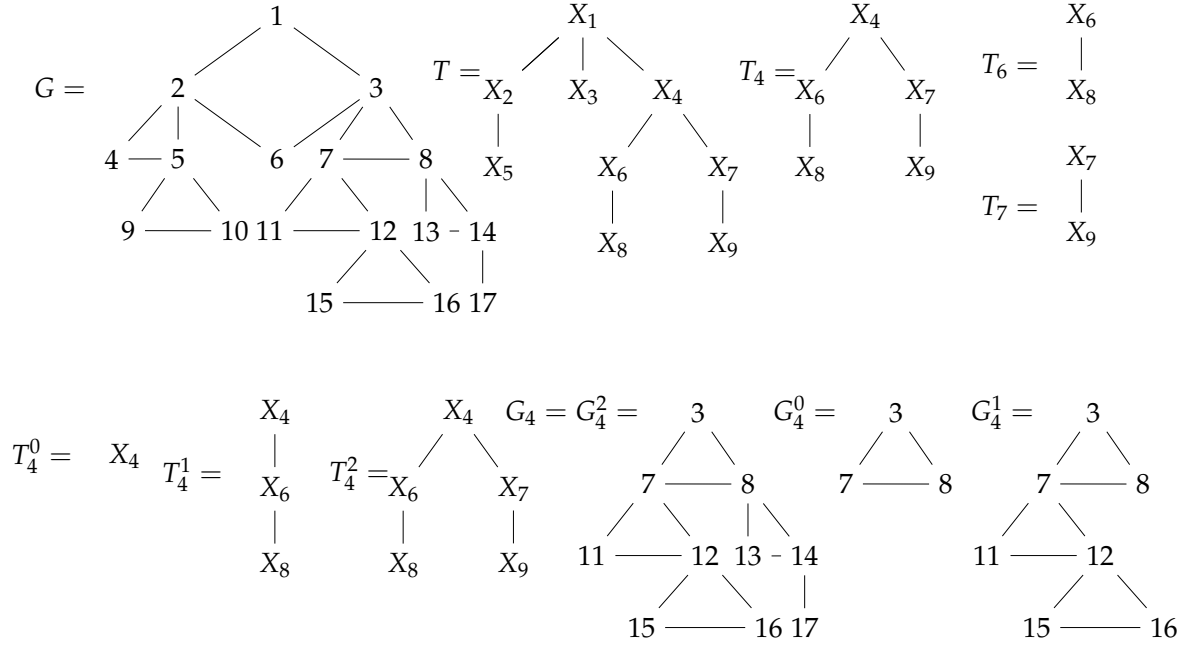


Figure 4.2: Partition of G using its tree decomposition

$\beta(X_5) = \{5, 9, 10\}$, $\beta(X_6) = \{7, 11, 12\}$, $\beta(X_7) = \{8, 13, 14\}$, $\beta(X_8) = \{12, 15, 16\}$, and $\beta(X_9) = \{14, 17\}$. T_4 , T_6 , and T_7 are induced subtrees of T rooted at X_4 , X_6 , and X_7 , respectively. We assume X_6 and X_7 are the first and second children of X_4 . T_4^0 is an induced subgraph of T containing only the node X_4 . T_4^1 is an induced subtree of T containing node X_4 and the nodes of T_6 , and T_4^2 is an induced subtree of T_4 containing node X_4 and the nodes of T_6 and T_7 . G_4 is the induced subgraph of G represented by T_4 . G_4^0 is the induced subgraph of G represented by T_4^0 . G_4^1 is the induced subgraph of G represented by T_4^1 . G_4^2 is the induced subgraph of G represented by T_4^2 . Since $T_4 = T_4^2$, it follows that $G_4 = G_4^2$.

Let $X_{i_1}, X_{i_2}, \dots, X_{i_l}$ be the children of X_i in the tree T_i . For $1 \leq j \leq l$, Observation 4.5.1 establishes a relationship between G_i^j , G_i^{j-1} , and G_{i_j} .

Observation 4.5.1. *Let G be an undirected graph, and (T, β) be a tree decomposition of G with $V_T = \{X_1, X_2, \dots, X_s\}$, rooted at X_1 . For $1 \leq i \leq s$, let $X_{i_1}, X_{i_2}, \dots, X_{i_l}$ be the children of X_i in the tree T_i . Then, for all $j > 0$:*

1. G_i^{j-1} and G_{i_j} are induced subgraphs of G_i^j .
2. $G_i^j = G_i^{j-1} \cup G_{i_j}$.

3. $\beta(X_i) \subseteq V_{G_i^{j-1}}$.
4. $\beta(X_{i_j}) \subseteq V_{G_{i_j}}$.
5. $I_i^j = \beta(X_i) \cap \beta(X_{i_j})$ is a vertex separator of G_i^j that separates $V_{G_i^{j-1}} \setminus I_i^j$ and $V_{G_{i_j}} \setminus I_i^j$.
6. $V_{G_i^{j-1}} \cap V_{G_{i_j}} = \beta(X_i) \cap \beta(X_{i_j}) = I_i^j$

Proof. From the construction of G_i^j, G_i^{j-1} and G_{i_j} , Items 1 to 4 are true. Let $\beta_i^j : V_{T_i^j} \rightarrow V_{G_i^j}$ be a function such that for $X_k \in V_{T_i^j}$, $\beta_i^j(X_k) = \beta(X_k)$. Then, (T_i^j, β_i^j) is a tree decomposition of G_i^j , as it obeys Items 1 and 2 of Definition 1.1.14.¹ Since $X_i - X_{i_j}^j$ is an edge of T_i^j , from Property 1.1.15, $I_i^j = \beta_i^j(X_i) \cap \beta_i^j(X_{i_j}) = \beta(X_i) \cap \beta(X_{i_j})$ serves as a vertex separator of G_i^j that separates $V_{G_i^{j-1}} \setminus I_i^j$ and $V_{G_{i_j}} \setminus I_i^j$, validating Item 5. Item 6 follows by Item 2b of Definition 1.1.14 for the tree decomposition (T_i^j, β_i^j) of G_i^j . $\square$

Recall from the above discussion that we want to count the MECs of G_1^p , where p is the number of children of the root node X_1 in the tree decomposition of G . Observation 4.5.1 shows that if we cut the edge $X_i - X_{i_j}$ in the tree T_i^j then we get two subtrees T_i^{j-1} and T_{i_j} of T_i . This further gives us two induced subgraphs G_i^{j-1} and G_{i_j} of G_i^j that have the properties given in Items 1 to 6 of Observation 4.5.1. We establish a relation between the MECs of G_i^j, G_i^{j-1} , and G_{i_j} that if we have the knowledge about (a) $|\text{MEC}(G_i^{j-1}, O_1, P_{11}, P_{12})|$ for each shadow (O_1, P_{11}, P_{12}) of $G_i^{j-1}[\beta(X_i) \cup N(\beta(X_i), G_i^{j-1})]$, and (b) $|\text{MEC}(G_{i_j}, O_2, P_{21}, P_{22})|$ for each shadow (O_2, P_{21}, P_{22}) of $G_{i_j}[\beta(X_{i_j}) \cup N(\beta(X_{i_j}), G_{i_j})]$ then we can compute $|\text{MEC}(G_i^j, O, P_1, P_2)|$ for each shadow (O, P_1, P_2) of $G_i^j[\beta(X_i) \cup N(\beta(X_i), G_i^j)]$. This gives us a recursive method to compute $|\text{MEC}(G_1^p, O, P_1, P_2)|$ for each shadow (O, P_1, P_2) of G_1^p on $\beta(X_1) \cup N(\beta(X_1), G_1^p)$. Then, with the assistance of Lemma 4.3.7, we can effectively count the MECs of $G = G_1^p$.

We first show the relation between $\text{MEC}(G_i^j)$, $\text{MEC}(G_i^{j-1})$, and $\text{MEC}(G_{i_j})$, using which we construct a recursive algorithm to count the MECs of G . To understand this relation, we abstract out the following basic setup underlying it. The technical results below will all be under this abstract setup.

¹Items 1 and 2b of Definition 1.1.14 hold because (T, β) is a tree decomposition of G . The correctness of Item 2a follows from the fact that for any node u in G , the induced subgraph of T on $\{x : x \in V_T \text{ and } u \in \beta(x)\}$ is connected, and any path in T from a node in T_i^j to a node not in it must pass through X_i .

Setup for technical results Let H , H_1 , and H_2 be three undirected graphs, and S_1 and S_2 be subsets of V_{H_1} and V_{H_2} , respectively, such that

1. H_1 and H_2 are induced subgraphs of H such that $H = H_1 \cup H_2$,
2. $I = V_{H_1} \cap V_{H_2}$ is a vertex separator of H that separates $V_{H_1} \setminus I$ and $V_{H_2} \setminus I$, and
3. both S_1 and S_2 contain the nodes of I , i.e., $S_1 \cap S_2 = I$.

Here, H resembles G_i^j , H_1 resembles G_i^{j-1} , H_2 resembles G_{ij} , S_1 resembles $\beta(X_i)$, S_2 resembles $\beta(X_{ij})$, and I resembles I_i^j . We show a relation between the shadows of the MECs of H , H_1 , and H_2 .

Lemma 4.5.2 (along with its reformulation in Lemma 4.5.7) and Lemma 4.5.8 are the main results of this section, and serve as the foundations of the above counting principle. Along the way, we will also develop auxiliary notions such as *derived path function* (Definition 4.5.3), *valid derived path function* (Definition 4.5.5), and *extension of a shadow* (Definition 4.5.6).

Lemma 4.5.2. Let M , M_1 , and M_2 be MECs of H , H_1 , and H_2 , respectively, $(O, P_1, P_2) = \text{shadow}(M, S_1 \cup S_2 \cup N(S_1 \cup S_2, H))$, $(O_1, P_{11}, P_{12}) = \text{shadow}(M_1, S_1 \cup N(S_1, H_1))$, and $(O_2, P_{21}, P_{22}) = \text{shadow}(M_2, S_2 \cup N(S_2, H_2))$. If $\mathcal{P}(M, V_{H_1}, V_{H_2}) = (M_1, M_2)$, then:

1. For each $a \in \{1, 2\}$, for $u, v \in V_{O_a}$, if $u \rightarrow v \in O_a$, then $u \rightarrow v \in O$.
2. For each $a \in \{1, 2\}$, the set of v-structures in O_a and the set of v-structure in $O[V_{O_a}]$ are equal, i.e., $\mathcal{V}(O_a) = \mathcal{V}(O[V_{O_a}])$.
3. For each $a \in \{1, 2\}$, for $u - v \in O_a$, $u \rightarrow v \in O$ if and only if at least one of the following occurs:
 - (a) $u \rightarrow v$ is strongly protected in O (see Item 4 of Theorem 1.1.24 for the definition of strongly protected).
 - (b) There exists $x - y \in O_a$ such that $x \rightarrow y \in O$, and $P_{a1}((x, y), (u, v)) = 1$ (i.e., there exists a triangle-free path from (x, y) to (u, v) in M_a with $(x, y) \neq (u, v)$).
 - (c) There exists $x - y \in O_a$ such that $x \rightarrow y \in O$, and $P_{a2}((x, y), v) = P_{a2}((v, u), x) = 1$ (i.e., $y \neq v$, $u \neq x$, and there exist triangle-free paths in M_a from (x, y) to v , and from (v, u) to x).

4. For $((u, v), (x, y)) \in E_O \times E_O$, $P_1((u, v), (x, y)) = 1$ if and only if $(u, v) \neq (x, y)$ and at least one of the following occurs:

(a) There exists a triangle-free path from (u, v) to (x, y) in O .

(b) For some $a \in \{1, 2\}$, $P_{a1}((u, v), (x, y)) = 1$.

(c) There exists $(z_1, z_2) \in E_O$ such that $P_1((u, v), (z_1, z_2)) = P_1((z_1, z_2), (x, y)) = 1$.

5. For $((u, v), w) \in E_O \times V_O$, $P_2((u, v), w) = 1$ if and only if $v \neq w$ and at least one of the following occurs:

(a) There exists a triangle-free path from (u, v) to w in O .

(b) For some $a \in \{1, 2\}$, $P_{a2}((u, v), w) = 1$.

(c) There exists $(z_1, z_2) \in E_O$ such that $P_1((u, v), (z_1, z_2)) = P_2((z_1, z_2), w) = 1$.

The proof of Lemma 4.5.2 is given in Section 4.6.3 (page 147).

4.5.1 Derived Path Function and Extensions of Shadows

We note that P_1 and P_2 in Lemma 4.5.2 are not independent. Knowing $O, P_{11}, P_{12}, P_{21}$, and P_{22} , we can compute P_1 and P_2 using Items 4 and 5 of Lemma 4.5.2 (this is shown in Lemma 4.5.4). This motivates us to define (P_1, P_2) as a derived path function of $(O, P_{11}, P_{12}, P_{21}, P_{22})$. Definition 4.5.3 formally captures this idea. H, H_1 , and H_2 in Definition 4.5.3 are defined in the paragraph before Lemma 4.5.2.

Definition 4.5.3 (Derived Path Function). Let $(O, P_1, P_2) \in \text{shadow}(H[S_1 \cup S_2 \cup N(S_1 \cup S_2, H)])$, and for $a \in \{1, 2\}$, $(O_a, P_{a1}, P_{a2}) \in \text{shadow}(H_a[S_a \cup N(S_a, H_a)])$. We say that (P_1, P_2) is the *derived path function* of $(O, P_{11}, P_{12}, P_{21}, P_{22})$, denoted by $(P_1, P_2) = \text{DPF}(O, P_{11}, P_{12}, P_{21}, P_{22})$, if P_1 and P_2 are computed through the following steps:

1. Step 1 (Initialization):

(a) $\forall ((u, v), (x, y)) \in E_O \times E_O$, if $(u, v) \neq (x, y)$ and there exists a triangle-free path from (u, v) to (x, y) in O then $P_1((u, v), (x, y)) = 1$ else $P_1((u, v), (x, y)) = 0$.

- (b) $\forall((u, v), w) \in E_O \times V_O$, if $v \neq w$ and there exists a triangle-free path from (u, v) to w in O then $P_2((u, v), w) = 1$ else $P_2((u, v), w) = 0$.

2. Step 2 (Update 1):

- (a) $\forall((u, v), (x, y)) \in E_O \times E_O$ such that $(u, v) \neq (x, y)$ and $P_1((u, v), (x, y)) = 0$, if for some $a \in \{1, 2\}$, $P_{a1}((u, v), (x, y)) = 1$ then update P_1 with $P_1((u, v), (x, y)) = 1$.
- (b) $\forall((u, v), w) \in E_O \times V_O$ such that $v \neq w$ and $P_2((u, v), w) = 0$, if for some $a \in \{1, 2\}$, $P_{a2}((u, v), w) = 1$ then update P_2 with $P_2((u, v), w) = 1$.

3. Step 3 (Update 2: adding transitivity):

Repeat the following until no further change in (P_1, P_2) :

- (a) $\forall((u, v), (x, y)) \in E_O \times E_O$ such that $(u, v) \neq (x, y)$, and $P_1((u, v), (x, y)) = 0$, if there exists $(z_1, z_2) \in E_O$ such that $P_1((u, v), (z_1, z_2)) = P_1((z_1, z_2), (x, y)) = 1$ then update P_1 with $P_1((u, v), (x, y)) = 1$.
- (b) $\forall((u, v), w) \in E_O \times V_O$, such that $v \neq w$ and $P_2((u, v), w) = 0$, if there exists $(z_1, z_2) \in E_O$ such that $P_1((u, v), (z_1, z_2)) = P_2((z_1, z_2), w) = 1$ then update P_2 with $P_2((u, v), w) = 1$.

We emphasize again that P_2 is not in general determined by P_1 ; see Remark 4.3.3 following Definition 4.3.1.

The following lemma shows that $(P_1, P_2) = \text{DPF}(O, P_{11}, P_{12}, P_{21}, P_{22})$ in Lemma 4.5.2.

Lemma 4.5.4. *Let M, M_1, M_2 be the MECs of H, H_1, H_2 , respectively, (O, P_1, P_2) be the shadow of M on $S_1 \cup S_2 \cup N(S_1 \cup S_2, H)$, (O_1, P_{11}, P_{12}) be the shadow of M_1 on $S_1 \cup N(S_1, H_1)$, and (O_2, P_{21}, P_{22}) be the shadow of M_2 on $S_2 \cup N(S_2, H_2)$. If $\mathcal{P}(M, V_{H_1}, V_{H_2}) = (M_1, M_2)$ then $(P_1, P_2) = \text{DPF}(O, P_{11}, P_{12}, P_{21}, P_{22})$, i.e., (P_1, P_2) is the derived path function of $(O, P_{11}, P_{12}, P_{21}, P_{22})$.*

The proof of Lemma 4.5.4 is given in Section 4.6.3 (page 165).

We define the following, to summarize the necessary conditions provided by Lemma 4.5.2.

Definition 4.5.5 (Valid derived path function). (P_1, P_2) is said to be a *valid derived path function* of $(O, P_{11}, P_{12}, P_{21}, P_{22})$ if:

1. $(P_1, P_2) = \text{DPF}(O, P_{11}, P_{12}, P_{21}, P_{22})$, and
2. For all $(u, v), (x, y) \in E_O$, if $P_1((u, v), (x, y)) = 1$ then $P_1((x, y), (u, v)) = 0$.

Definition 4.5.6 (Extension). Let $(O_1, P_{11}, P_{12}) \in \text{shadow}(H_1[S_1 \cup N(S_1, H_1)])$, $(O_2, P_{21}, P_{22}) \in \text{shadow}(H_2[S_2 \cup N(S_2, H_2)])$, and $(O, P_1, P_2) \in \text{shadow}(H[S_1 \cup S_2 \cup N(S_1 \cup S_2, H)])$. We define (O, P_1, P_2) to be an *extension* of (O_1, P_{11}, P_{12}) and (O_2, P_{21}, P_{22}) , denoted by $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$, if

1. For each $a \in \{1, 2\}$, for $u, v \in S_a \cup N(S_a, H_a)$, if $u \rightarrow v \in O_a$ then $u \rightarrow v \in O$.
2. For each $a \in \{1, 2\}$, $\mathcal{V}(O_a) = \mathcal{V}(O[S_a \cup N(S_a, H_a)])$.
3. For each $a \in \{1, 2\}$, for $u - v \in O_a$, $u \rightarrow v \in O$ if and only if at least one of the following occurs:
 - (a) $u \rightarrow v$ is strongly protected in O .
 - (b) $\exists x - y \in O_a$ such that $x \rightarrow y \in O$, and $P_{a1}((x, y), (u, v)) = 1$.
 - (c) $\exists x - y \in O_a$ such that $x \rightarrow y \in O$, $P_{a2}(x, y, v) = 1$, and $P_{a2}(v, u, x) = 1$.
4. (P_1, P_2) is a valid derived path function of $(O, P_{11}, P_{12}, P_{21}, P_{22})$.

Since P_1 in Lemma 4.5.2 obeys Item 2 of Definition 4.5.5 (from Lemma 4.2.6), therefore, Lemma 4.5.4 and Definitions 4.5.3, 4.5.5 and 4.5.6 allow us to summarize Lemma 4.5.2 as follows:

Lemma 4.5.7. Let M, M_1, M_2 be MECs of H, H_1, H_2 , respectively, (O, P_1, P_2) be the shadow of M on $S_1 \cup S_2 \cup N(S_1 \cup S_2, H)$, (O_1, P_{11}, P_{12}) be the shadow of M_1 on $S_1 \cup N(S_1, H_1)$, and (O_2, P_{21}, P_{22}) be the shadow of M_2 on $S_2 \cup N(S_2, H_2)$. If $\mathcal{P}(M, V_{H_1}, V_{H_2}) = (M_1, M_2)$ then $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$.

The following lemma (Lemma 4.5.8) shows that the necessary condition obeyed by the shadow of an MEC of H , provided in Lemma 4.5.7, is also sufficient. We use this result along with Lemma 4.5.7 to construct an algorithm that counts MECs of H .

Lemma 4.5.8. Let M_1 be an MEC of H_1 , and M_2 be an MEC of H_2 , $(O_1, P_{11}, P_{12}) = \text{shadow}(M_1, S_1 \cup N(S_1, H_1))$, $(O_2, P_{21}, P_{22}) = \text{shadow}(M_2, S_2 \cup N(S_2, H_2))$, and $(O, P_1, P_2) \in \text{shadow}(H[S_1 \cup S_2 \cup N(S_1 \cup S_2, H)])$ ². If $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$ then there exists a unique MEC M of H such that $\text{shadow}(M, V_O) = (O, P_1, P_2)$, and $\mathcal{P}(M, V_{H_1}, V_{H_2}) = (M_1, M_2)$.

The proof of Lemma 4.5.8 is given in Section 4.6.4 (page 170).

4.5.2 Finding Size of $\text{MEC}(H, O, P_1, P_2)$

In this subsection, we construct a recursive formula to compute $|\text{MEC}(H, O, P_1, P_2)|$ for each $(O, P_1, P_2) \in \text{shadow}(H[S_1 \cup S_2 \cup N(S_1 \cup S_2, H)])$. We use Definition 4.5.6 to define a one-to-one function f_{O, P_1, P_2} , which we then use for the computation of $|\text{MEC}(H, O, P_1, P_2)|$.

Definition 4.5.9. Let $(O, P_1, P_2) \in \text{shadow}(H[S_1 \cup S_2 \cup N(S_1 \cup S_2, H)])$, $T_1 = \text{MEC}(H, O, P_1, P_2)$, and $T_2 = \{(M_1, M_2) : \exists (O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22}) \text{ such that } (O_1, P_{11}, P_{12}) \in \text{shadow}(H_1[S_1 \cup N(S_1, H_1)]), (O_2, P_{21}, P_{22}) \in \text{shadow}(H_2[S_2 \cup N(S_2, H_2)]), (O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22}), M_1 \in \text{MEC}(H_1, O_1, P_{11}, P_{12}), \text{ and } M_2 \in \text{MEC}(H_2, O_2, P_{21}, P_{22})\}$. The function $f_{O, P_1, P_2} : T_1 \rightarrow T_2$ is defined as $f_{O, P_1, P_2}(M) = (M_1, M_2)$ if $\mathcal{P}(M, V_{H_1}, V_{H_2}) = (M_1, M_2)$.

Lemma 4.5.10. f_{O, P_1, P_2} is a bijection between T_1 and T_2 .

Proof. Claims 4.5.11 and 4.5.12 show that f_{O, P_1, P_2} is bijective.

Claim 4.5.11. For each $M \in \text{MEC}(H, O, P_1, P_2)$, $f_{O, P_1, P_2}(M) = (M_1, M_2)$ belongs to T_2 .

Proof. Let $M \in \text{MEC}(H, O, P_1, P_2)$, i.e., M is an MEC of H whose shadow on $V_O = S_1 \cup S_2 \cup N(S_1 \cup S_2, H)$ is (O, P_1, P_2) . Let $\mathcal{P}(M, V_{H_1}, V_{H_2}) = (M_1, M_2)$. From Corollary 4.4.3, there exists a unique (M_1, M_2) such that $\mathcal{P}(M, V_{H_1}, V_{H_2}) = (M_1, M_2)$. Thus, we only need to show that $(M_1, M_2) \in T_2$.

Let $(O_1, P_{11}, P_{12}) = \text{shadow}(M_1, S_1 \cup N(S_1, H_1))$ and $(O_2, P_{21}, P_{22}) = \text{shadow}(M_2, S_2 \cup N(S_2, H_2))$. This implies that $M_1 \in \text{MEC}(H_1, O_1, P_{11}, P_{12})$ and $M_2 \in \text{MEC}(H_2, O_2, P_{21}, P_{22})$.

²As defined in Definition 4.3.4.

From Observation 4.3.5, $(O_1, P_{11}, P_{12}) \in \text{shadow}(H_1[S_1 \cup N(S_1, H_1)])$ and $(O_2, P_{21}, P_{22}) \in \text{shadow}(H_2[S_2 \cup N(S_2, H_2)])$.

Additionally, from Lemma 4.5.7, $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$. This implies that $(M_1, M_2) \in T_2$. This concludes the proof. $\square$

Claim 4.5.12. *Let $(O_1, P_{11}, P_{12}) \in \text{shadow}(H_1[S_1 \cup N(S_1, H_1)])$ and $(O_2, P_{21}, P_{22}) \in \text{shadow}(H_2[S_2 \cup N(S_2, H_2)])$ such that (O, P_1, P_2) is an extension of $(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$. For each $M_1 \in \text{MEC}(H_1, O_1, P_{11}, P_{12})$ and $M_2 \in \text{MEC}(H_2, O_2, P_{21}, P_{22})$, there exists a unique $M \in \text{MEC}(H, O, P_1, P_2)$ such that $f_{O, P_1, P_2}(M) = (M_1, M_2)$.*

Proof. Lemma 4.5.8 proves this. $\square$

Claims 4.5.11 and 4.5.12 prove Lemma 4.5.10. $\square$

Lemma 4.5.10 implies the following:

Lemma 4.5.13. *For any $(O, P_1, P_2) \in \text{shadow}(H[S_1 \cup S_2 \cup N(S_1 \cup S_2, H)])$,*

$$|\text{MEC}(H, O, P_1, P_2)| = \sum_{\substack{(O_1, P_{11}, P_{12}) \in \text{shadow}(H_1[S_1 \cup N(S_1, H_1)]) \\ (O_2, P_{21}, P_{22}) \in \text{shadow}(H_2[S_2 \cup N(S_2, H_2)]) \\ \text{such that } (O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})}} |\text{MEC}(H_1, O_1, P_{11}, P_{12})| \times |\text{MEC}(H_2, O_2, P_{21}, P_{22})|.$$

Proof. From Lemma 4.5.10, $|T_1| = |T_2|$. Since $T_1 = \text{MEC}(H, O, P_1, P_2)$, therefore, $|T_1| = |\text{MEC}(H, O, P_1, P_2)|$. Since T_2 contains tuples (M_1, M_2) such that there exist $(O_1, P_{11}, P_{12}) \in \text{shadow}(H_1[S_1 \cup N(S_1, H_1)])$ and $(O_2, P_{21}, P_{22}) \in \text{shadow}(H_2[S_2 \cup N(S_2, H_2)])$ such that $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$, $M_1 \in \text{MEC}(H_1, O_1, P_{11}, P_{12})$ and $M_2 \in \text{MEC}(H_2, O_2, P_{21}, P_{22})$. This implies

$$T_2 = \bigcup_{\substack{(O_1, P_{11}, P_{12}) \in \text{shadow}(H_1[S_1 \cup N(S_1, H_1)]) \\ (O_2, P_{21}, P_{22}) \in \text{shadow}(H_2[S_2 \cup N(S_2, H_2)]) \\ \text{such that } (O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})}} \text{MEC}(H_1, O_1, P_{11}, P_{12}) \times \text{MEC}(H_2, O_2, P_{21}, P_{22}). \quad (4.2)$$

We show that the sets $\text{MEC}(H_1, O_1, P_{11}, P_{12}) \times \text{MEC}(H_2, O_2, P_{21}, P_{22})$, in Equation (4.2), are disjoint. Suppose the sets $\text{MEC}(H_1, O_1, P_{11}, P_{12}) \times \text{MEC}(H_2, O_2, P_{21}, P_{22})$ are not disjoint. Then there exists $(M_1, M_2) \in T_2$ such that, for two distinct pairs $(O_1, P_{11}, P_{12}, O_2, P_{21},$

$P_{22})$ and $(O'_1, P'_{11}, P'_{12}, O'_2, P'_{21}, P'_{22})$, we have $(M_1, M_2) \in \text{MEC}(H_1, O_1, P_{11}, P_{12}) \times \text{MEC}(H_2, O_2, P_{21}, P_{22})$ as well as $(M_1, M_2) \in \text{MEC}(H_1, O'_1, P'_{11}, P'_{12}) \times \text{MEC}(H_2, O'_2, P'_{21}, P'_{22})$, where (O_1, P_{11}, P_{12}) and (O'_1, P'_{11}, P'_{12}) belong to $\text{shadow}(H_1[S_1 \cup N(S_1, H_1)])$, and (O_2, P_{21}, P_{22}) and (O'_2, P'_{21}, P'_{22}) belong to $\text{shadow}(H_2[S_2 \cup N(S_2, H_2)])$. This implies that $M_1 \in \text{MEC}(H_1, O_1, P_{11}, P_{12})$ as well as $M_1 \in \text{MEC}(H_1, O'_1, P'_{11}, P'_{12})$. However, from Definition 4.3.1, the shadow of M on $S_1 \cup N(S_1, H_1)$ is unique, i.e., $(O_1, P_{11}, P_{12}) = (O'_1, P'_{11}, P'_{12})$. Similarly, $(O_2, P_{21}, P_{22}) = (O'_2, P'_{21}, P'_{22})$. This shows that $(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$ and $(O'_1, P'_{11}, P'_{12}, O'_2, P'_{21}, P'_{22})$ are equal. This implies that the sets $\text{MEC}(H_1, O_1, P_{11}, P_{12}) \times \text{MEC}(H_2, O_2, P_{21}, P_{22})$ are disjoint. This further implies

$$|T_2| = \sum_{\substack{(O_1, P_{11}, P_{12}) \in \text{shadow}(H_1[S_1 \cup N(S_1, H_1)]) \\ (O_2, P_{21}, P_{22}) \in \text{shadow}(H_2[S_2 \cup N(S_2, H_2)]) \\ \text{such that } (O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})}} |\text{MEC}(H_1, O_1, P_{11}, P_{12})| \times |\text{MEC}(H_2, O_2, P_{21}, P_{22})|. \quad (4.3)$$

This proves Lemma 4.5.13. $\square$

Lemma 4.5.13 gives us a method to compute $|\text{MEC}(H, O, P_1, P_2)|$ for each $(O, P_1, P_2) \in \text{shadow}(H[S_1 \cup S_2 \cup N(S_1 \cup S_2, H)])$. But, for the recursion, we need $|\text{MEC}(H, O', P'_1, P'_2)|$ for each $(O', P'_1, P'_2) \in \text{shadow}(H[S_1 \cup N(S_1, H)])$. As (O, P_1, P_2) contains more information than (O', P'_1, P'_2) , we define “*projection of a shadow*” to get $|\text{MEC}(H, O', P'_1, P'_2)|$.

Definition 4.5.14 (Projection of a shadow). Let H be an undirected graph. Let S and S' be two vertex subsets of H such that $S' \subseteq S \subseteq V_H$. Let $(O, P_1, P_2) \in \text{shadow}(G[S])$ and $(O', P'_1, P'_2) \in \text{shadow}(G[S'])$. We say (O', P'_1, P'_2) is a projection of (O, P_1, P_2) on S' , denoted by $\mathcal{P}(O, P_1, P_2, S') = (O', P'_1, P'_2)$, if

1. $O[S'] = O'$,
2. for $((u, v), (x, y)) \in E_{O'} \times E_{O'}$, $P'_1((u, v), (x, y)) = P_1((u, v), (x, y))$, and
3. for $((u, v), w) \in E_{O'} \times V_{O'}$, $P'_2((u, v), w) = P_2((u, v), w)$.

We now compute $|\text{MEC}(H, O', P'_1, P'_2)|$.

Lemma 4.5.15. *Let H be an undirected graph, and S and S' be two vertex subsets of H such that $S' \subseteq S \subseteq V_H$. Let $(O', P'_1, P'_2) \in \text{shadow}(H[S'])$. Then,*

$$|\text{MEC}(H, O', P'_1, P'_2)| = \sum_{\substack{(O, P_1, P_2) \in \text{shadow}(H[S]) \\ \text{such that } (O', P'_1, P'_2) = \mathcal{P}(O, P_1, P_2, S')}} |\text{MEC}(H, O, P_1, P_2)| \quad (4.4)$$

Proof. We first show the following:

$$\text{MEC}(H, O', P'_1, P'_2) = \bigcup_{\substack{(O, P_1, P_2) \in \text{shadow}(H[S]) \\ \text{such that } (O', P'_1, P'_2) = \mathcal{P}(O, P_1, P_2, S')}} \text{MEC}(H, O, P_1, P_2) \quad (4.5)$$

Claims 4.5.16 and 4.5.17 show the correctness of Equation (4.5).

Claim 4.5.16. *For each $M \in \text{MEC}(H, O', P'_1, P'_2)$, there exists a unique $(O, P_1, P_2) \in \text{shadow}(H[S])$ such that $\mathcal{P}(O, P_1, P_2, S') = (O', P'_1, P'_2)$, and $M \in \text{MEC}(H, O, P_1, P_2)$.*

Proof. Let $M \in \text{MEC}(H, O', P'_1, P'_2)$. From Definition 4.3.1, there exists a unique triplet (O, P_1, P_2) such that $(O, P_1, P_2) \in \text{shadow}(H[S])$ and $M \in \text{MEC}(H, O, P_1, P_2)$. We only need to verify that $\mathcal{P}(O, P_1, P_2, S') = (O', P'_1, P'_2)$.

Since both (O, P_1, P_2) and (O', P'_1, P'_2) are shadows of M , from Definition 4.3.1, $O = M[S]$, and $O' = M[S']$. As $S' \subseteq S$, it follows that $O[S'] = O'$. Moreover, $E_{O'} \times E_{O'} \subseteq E_O \times E_O$, and for all $((u, v), (x, y)) \in E_{O'} \times E_{O'}$, $P'_1((u, v), (x, y)) = P_1((u, v), (x, y))$ (from Item 2 of Definition 4.3.1, both return 1 if $(u, v) \neq (x, y)$ and there exists a triangle-free path from (u, v) to (x, y) in M ; otherwise, they return 0). Similarly, for all $((u, v), w) \in E_{O'} \times V_{O'}$, $P'_2((u, v), w) = P_2((u, v), w)$. This further implies that $\mathcal{P}(O, P_1, P_2, S') = (O', P'_1, P'_2)$ (from Definition 4.5.14). This proves the claim. $\square$

Claim 4.5.17. *Suppose there exists a triplet (O, P_1, P_2) such that $(O, P_1, P_2) \in \text{shadow}(H[S])$ and $\mathcal{P}(O, P_1, P_2, S') = (O', P'_1, P'_2)$. Then, for each $M \in \text{MEC}(H, O, P_1, P_2)$, we have $M \in \text{MEC}(H, O', P'_1, P'_2)$.*

Proof. Let $M \in \text{MEC}(H, O, P_1, P_2)$. From Definition 4.3.1, $M[S] = O$. Since $S' \subseteq S$, $M[S'] = O[S']$. Since $\mathcal{P}(O, P_1, P_2, S') = (O', P'_1, P'_2)$, from Item 1 of Definition 4.5.14, $O[S'] = O'$. This implies that $M[S'] = O'$, and $E_{O'} \times E_{O'} \subseteq E_O \times E_O$. From Item 2 of Definition 4.5.14, for all $((u, v), (x, y)) \in E_{O'} \times E_{O'}$, $P'_1((u, v), (x, y)) = P_1((u, v), (x, y))$.

This implies for each $((u, v), (x, y)) \in E_{O'} \times E_{O'}$, $P'_1((u, v), (x, y)) = 1$ if and only if $(u, v) \neq (x, y)$ and there exists a triangle-free path from (u, v) to (x, y) in M . Similarly, for all $((u, v), w) \in E_{O'} \times V_{O'}$, $P'_1((u, v), w) = 1$ if and only if $v \neq w$ and there exists a triangle-free path from (u, v) to w in M . The above discussion implies that (O', P'_1, P'_2) is the shadow of M on S' (from Definition 4.3.1). This implies that $M \in \text{MEC}(H, O', P'_1, P'_2)$. $\square$

Claim 4.5.16 shows that every M in the L.H.S. of Equation (4.5) is in the R.H.S., and Claim 4.5.17 shows the converse. Therefore, the L.H.S. and R.H.S. of Equation (4.5) are equal.

Finally, since the shadow of an MEC of H on S is unique (from Definition 4.3.1), an MEC M of H belongs to a unique $\text{MEC}(H, O, P_1, P_2)$. This implies that the sets in the union on the R.H.S. of Equation (4.5) are disjoint, which establishes Equation (4.4). This completes the proof of Lemma 4.5.15. $\square$

Combining Lemmas 4.5.13 and 4.5.15, we get the following lemma:

Lemma 4.5.18. *Let H be an undirected graph, and H_1 and H_2 be two induced subgraphs of H such that $H = H_1 \cup H_2$, and $I = V_{H_1} \cap V_{H_2}$ is a vertex separator of H that separates $V_{H_1} \setminus I$ and $V_{H_2} \setminus I$. Let S_1 and S_2 be subsets of V_{H_1} and V_{H_2} , respectively, such that $S_1 \cap S_2 = I$, and $S' = S_1 \cup N(S_1, H)$. Then, for all $(O', P'_1, P'_2) \in \text{shadow}(H[S_1 \cup N(S_1, H)])$,*

$$|\text{MEC}(H, O', P'_1, P'_2)| = \sum_{\substack{(O_1, P_{11}, P_{12}) \in \text{shadow}(H_1[S_1 \cup N(S_1, H_1)]) \\ (O_2, P_{21}, P_{22}) \in \text{shadow}(H_2[S_2 \cup N(S_2, H_2)]) \\ \text{such that } \exists (O, P_1, P_2) \in \text{shadow}(H[S_1 \cup S_2 \cup N(S_1 \cup S_2, H)]) \\ \text{such that } (O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22}) \text{ and} \\ (O', P'_1, P'_2) = \mathcal{P}(O, P_1, P_2, S')}} |\text{MEC}(H_1, O_1, P_{11}, P_{12})| \times |\text{MEC}(H_2, O_2, P_{21}, P_{22})|.$$

Proof. Consider $S = S_1 \cup S_2 \cup N(S_1 \cup S_2, H)$. Since $S' \subseteq S$, combining Lemmas 4.5.13 and 4.5.15, we get Lemma 4.5.18. $\square$

Using Lemma 4.5.18, we can compute $|\text{MEC}(H, O', P'_1, P'_2)|$ for all shadows $(O', P'_1, P'_2) \in \text{shadow}(H[S_1 \cup N(S_1, H)])$, given that $|\text{MEC}(H_1, O_1, P_{11}, P_{12})|$ is known for each $(O_1, P_{11}, P_{12}) \in \text{shadow}(H_1[S_1 \cup N(S_1, H_1)])$ and $|\text{MEC}(H_2, O_2, P_{21}, P_{22})|$ is known for

each $(O_2, P_{21}, P_{22}) \in \text{shadow}(H_2[S_2 \cup N(S_2, H_2)])$. As discussed in Section 4.5 (the paragraph above Lemma 4.5.2), H resembles G_i^j , H_1 resembles G_i^{j-1} , H_2 resembles G_{ij} , S_1 resembles $\beta(X_i)$, and S_2 resembles $\beta(X_{ij})$. Lemma 4.3.7, along with Lemma 4.5.18, gives us a recursive method to compute the MECs of G .

In Section 4.7 (page 233), we implement Lemmas 4.3.7 and 4.5.18 to construct a recursive algorithm (Algorithm 9) to count the MECs of G . Before that, we first present the proofs omitted from Sections 4.4 and 4.5.

4.6 Proofs of Main Lemmas

In this section, we present the missing proofs of the results in Sections 4.4 and 4.5. Before that, we provide some definitions and preliminary results, which help us to simplify the proofs. We begin with preliminary results on chain graphs and chordal graphs.

Proposition 4.6.1. *Let G be a chain graph, C an undirected connected component of G , and $u, v \in C$. Then $u \rightarrow v \notin G$.*

Proof. Since both u and v belong to the same undirected connected component of G , there must exist an undirected path connecting them. If $u \rightarrow v \in G$, then combining this directed edge with the undirected path would form a directed cycle in G , contradicting the fact that G is a chain graph. Hence, we conclude that $u \rightarrow v \notin G$. This completes the proof. $\square$

Proposition 4.6.2. *Let G be a graph with a chordal skeleton. If $C = (u_1, u_2, \dots, u_l, u_1)$ is a directed cycle in G , then there exists a directed cycle $C' = (a, b, c, a)$ in G of length three such that a, b, c are in the cycle C .*

Proof. Consider the smallest length directed cycle C' in G such that the vertices in C' are vertices in C . Such a cycle exists because C exists. We aim to show that C' must be of length three.

Suppose, for the sake of contradiction, that C' has more than three vertices, so it can be expressed as $C' = (v_1, v_2, \dots, v_m, v_1)$. Since the skeleton of G is chordal, there must

exist a chord in C' . Let us assume that this chord is between vertices v_i and v_j , where $i < j$, and v_i and v_j are not adjacent in C .

There are three possible cases: $v_i - v_j \in E_G$, $v_i \rightarrow v_j \in E_G$, or $v_i \leftarrow v_j \in E_G$. In each of these cases, either $(v_1, v_2, \dots, v_i, v_j, v_{j+1}, v_m, v_1)$ or $(v_i, v_{i+1}, \dots, v_j, v_i)$ forms a directed cycle in G of length less than m , which contradicts the minimality of C' . Therefore, we conclude that C' must be of length three.

This completes the proof of Proposition 4.6.2. $\square$

Proposition 4.6.3. *For every chain graph G , there always exists a topological ordering τ of the vertices of G such that for $u \rightarrow v \in E_G$, $\tau(u) < \tau(v)$.*

Proof. We can construct a topological ordering τ by first creating a topological ordering τ_1 of the vertices of the directed subgraph of G and then placing the nodes in τ_1 into τ in the same order as they appear in τ_1 . Next, we add the remaining nodes of G in any arbitrary order to complete τ . $\square$

The following observations on chordal graphs come from Rose et al. [1976].

Observation 4.6.4. *Let τ be an LBFS ordering of a chordal graph G . For $x, y, z \in V_G$, if both x and y are adjacent to z (i.e., $x - z, y - z \in E_G$) and both are of lesser rank than z in τ (i.e., $\tau(x) < \tau(z)$ and $\tau(y) < \tau(z)$) then x and y are also adjacent in G (i.e., $x - y \in E_G$).*

Proof. An LBFS ordering of a chordal graph G is known to be a perfect elimination ordering of G [Rose et al., 1976]. Given that τ represents an LBFS ordering of a chordal graph G , we can deduce that the subgraph induced by the neighbors of z that appear before z in τ forms a clique. Consequently, if x and y are neighbors of z and satisfy $\tau(x) < \tau(z)$ and $\tau(y) < \tau(z)$ (i.e., both appear before z in τ), then x and y must belong to the same clique, implying that $(x, y) \in E_G$. $\square$

Observation 4.6.5. *Let G be an undirected chordal graph, and τ be an LBFS ordering of the vertices of G . For vertices $u, v \in V_G$ if $P = (p_1 = u, p_2, \dots, p_l = v)$ is a chordless path from u to v in G then there exists $1 \leq i \leq l$ such that that $\tau(p_1) > \tau(p_2) > \dots > \tau(p_i)$, and $\tau(p_i) < \tau(p_{i+1}) \dots < \tau(p_l)$ (note that when $i = 1$ or $i = l$ then one or the other of the above series of inequalities is empty). In other words, there are no “internal local maxima” of τ in P :*

τ is either strictly increasing, or strictly decreasing, or has a unique global minimum in the interior of P .

Proof. There cannot exist vertices p_{j-1}, p_j, p_{j+1} such that $\tau(p_{j-1}) < \tau(p_j)$, and $\tau(p_{j+1}) < \tau(p_j)$. Otherwise, from Observation 4.6.4, $p_{j-1} - p_{j+1} \in E_G$, which further implies that P is not a chordless path. $\square$

We introduce the following definitions and observations that play a crucial role in simplifying our subsequent discussion.

Definition 4.6.6 (Synchronous Graphs). Two graphs G and H are said to be *synchronous graphs* if there do not exist vertices $x, y \in V_G \cap V_H$ such that $x \rightarrow y \in E_G$ and $y \rightarrow x \in E_H$.

Definition 4.6.7 (Markov Union of Synchronous Graphs). Let $G_1, G_2, \dots, G_l$ be pairwise synchronous graphs. The *Markov union* of $G_1, G_2, \dots, G_l$ is a graph G , denoted by $G = \mathcal{U}_M(G_1, G_2, \dots, G_l)$, such that:

1. $\text{skeleton}(G)$ is the skeleton of the union of $G_1, G_2, \dots, G_l$ (see Definition 1.1.9 for the definition of the union of graphs).
2. For any edge $u - v \in \text{skeleton}(G)$, $u \rightarrow v \in G$ if $u \rightarrow v \in G_i$ for any G_i .

More formally, the vertex set of G is $V_G = \bigcup_{i=1}^l V_{G_i}$, the set of directed edges of G is $D_G = \{u \rightarrow v : u \rightarrow v \in E_{G_i} \text{ for some } 1 \leq i \leq l\}$, and the set of undirected edges of G is $U_G = \{u - v : u - v \in E_{G_i} \text{ for some } 1 \leq i \leq l, \text{ and there does not exist a } j \text{ such that } u \rightarrow v \in E_{G_j} \text{ or } v \rightarrow u \in E_{G_j}\}$. In other words, if $u \rightarrow v$ is a directed edge in any graph G_i , then $u \rightarrow v$ is also a directed edge in the Markov union graph. An undirected edge $u - v$ in G_i is an undirected edge in the Markov union graph if neither $u \rightarrow v$ nor $v \rightarrow u$ is part of any graph G_j in the union. Note that, because the graphs are pairwise synchronous, if one graph G_i contains a directed edge $u \rightarrow v$, then no other graph G_j can contain the directed edge $v \rightarrow u$.

We now go through some preliminary results on triangle-free paths.

4.6.1 Properties of Triangle Free Paths

The following results illustrate the properties of triangle-free paths.

Observation 4.6.8. *For a graph G , every chordless path $P = (u_1, u_2, \dots, u_l)$ of G is a triangle-free path of G .*

Proof. Since P is a chordless path, there cannot be an edge between two non-adjacent nodes of P . This implies there cannot be an edge between u_i and u_{i+2} for any $1 \leq i \leq l - 2$. This further implies that P is a triangle-free path of G . $\square$

For our purposes, the utility of the notion of a triangle-free path comes from the following partial converse.

Proposition 4.6.9. *Let P be a triangle-free path of an undirected chordal graph G . Then, P is also a chordless path of G .*

Proof. Suppose, for the sake of contradiction, that there exists a triangle-free path $P = (u_0, u_1, \dots, u_l)$ of G such that P is not a chordless path of G . Then, there must exist a chord $u_i - u_j$ between two non-adjacent nodes u_i and u_j of P .

Without loss of generality, assume $i < j$, and pick such u_i and u_j such that $j - i$ is minimized. Since P is a triangle-free path of G , we have $j - i > 2$, and $u_k - u_{k+2} \notin E_G$ for all $i \leq k \leq j - 2$. Moreover, due to our choice of i and j , we also have $u_{k_1} - u_{k_2} \notin E_G$ for all $i \leq k_1 < k_2 \leq j$ with $2 < k_2 - k_1 < j - i$.

These conditions imply that $C = (u_i, u_{i+1}, \dots, u_j, u_i)$ forms a chordless cycle of length at least four in G , which contradicts the fact that G is a chordal graph. Therefore, no such chord $u_i - u_j$ can exist, and we conclude that P must be a chordless path. $\square$

The following corollary of this partial converse is also useful.

Corollary 4.6.10. *Let $P = (v_1 = u_1, v_2, v_3, \dots, v_l)$ be a triangle-free path of an undirected chordal graph G , and suppose there exists a node $u_2 \in V_G$ such that $u_1 - u_2 \in E_G$ and u_2 is also a node in P . Then, it must be the case that u_2 is the second vertex in P , i.e., $v_2 = u_2$.*

Proof. By Proposition 4.6.9, P is a chordless path in G , and hence no v_i for $i > 2$ can be adjacent to $v_1 = u_1$ (otherwise, $v_1 - v_i$ would be a chord of P). Since u_2 is adjacent to u_1 and is contained in P , this only leaves open the possibility $v_2 = u_2$. $\square$

The following observation extends the above to MECs.

Observation 4.6.11. *Let G be a chain graph with chordal undirected components, and P an undirected triangle-free path in G . Then, P is an undirected chordless path in G .*

Proof. Since P is undirected, it must be contained wholly in some undirected chordal component H of G . Thus, Proposition 4.6.9 implies that P must in fact be a chordless path in H (and therefore also in G). $\square$

The following observation is an easy consequence of the fact that the reverse of an undirected path is also a path.

Observation 4.6.12. *For a graph G , if $P = (u_1 = u, u_2 = v, \dots, u_{l-1} = x, u_l = y)$ is an undirected triangle free path from (u, v) to (x, y) in G then $\bar{P} = (u_l = y, u_{l-1} = x, \dots, u_2 = v, u_1 = u)$, the reverse of the path P , is also a triangle-free path from (y, x) to (v, u) .*

Proof. Suppose that $\bar{P}$ is not a triangle-free path, then there must exist nodes u_i and u_{i+2} such that there is an edge between u_i and u_{i+2} . But, then, P is also not a triangle-free path, which is a contradiction. $\square$

The following observation comes from Item 3 of Theorem 1.1.24.

Observation 4.6.13. *Let $P = (u_1 = x, u_2 = y, \dots, u_{l-1} = u, u_l = v)$ be a triangle-free path, from (x, y) to (u, v) , in an MEC M . If $u - v \in M$, then P is an undirected chordless path in M . In particular, $x - y \in M$.*

Proof. Since P is a path of M , for each $1 \leq i \leq l - 1$, either $u_i - u_{i+1} \in E_M$ or $u_i \rightarrow u_{i+1} \in E_M$. Suppose P is not an undirected path of G . Then, pick the highest i such that $u_i \rightarrow u_{i+1} \in E_M$. Note that i must be less than $l - 1$, as $u_{l-1} - u_l \in E_M$. However, in this case, M has an induced subgraph $u_i \rightarrow u_{i+1} - u_{i+2}$, which contradicts Item 3 of Theorem 1.1.24. Therefore, all the edges of P must be undirected. This further implies that P is an undirected triangle-free path in M . From Observation 4.6.11, P is a chordless path in M (since, by Theorem 1.1.24, M is a chain graph with chordal undirected components). $\square$

Observation 4.6.8, Proposition 4.6.9, and Lemma 4.2.5 imply the following:

Lemma 4.6.14. *Let G be an undirected chordal graph. Suppose $P_1 = (a_1 = u, a_2 = v, \dots, a_{l-1} = x, a_l = y)$ and $P_2 = (b_1 = x, b_2 = y, \dots, b_{m-1}, b_m = w)$ are chordless paths in G from (u, v) to (x, y) and from (x, y) to w , respectively. Then, their concatenation $P := (a_1 = u, a_2 = v, \dots, a_{l-1} = x, a_l = y, b_3, b_4, \dots, b_{m-1}, b_m = w)$ is a chordless path in G from (u, v) to w .*

Proof. From Observation 4.6.8, P_1 and P_2 are triangle-free paths in G . From Lemma 4.2.5, the path P , concatenation of P_1 and P_2 , is a triangle-free path in G . And, from Proposition 4.6.9, P is a chordless path in G . $\square$

We now introduce a triangle-free path variant of a source node.

Definition 4.6.15 (Canonical Source Node). Let G be a graph. A node $s \in V_G$ is said to be a *canonical source node* of G if there does not exist a directed edge $u \rightarrow v \in G$ such that there is a triangle-free path from (u, v) to s in G .

In particular, note that the existence of an edge of the form $u \rightarrow s$ also prohibits s from being a canonical source node, due to the presence of the (degenerate) triangle-free path (u, s) from (u, s) to s .

Definition 4.6.16 (Certificate for a non-canonical source node). Let G be a graph. For any non-canonical source node $z \in V_G$, we say x is a *certificate* of z if there exists a $y \in V_G$ such that $x \rightarrow y \in G$ and there exists a triangle-free path from (x, y) to z . In other words, x certifies that z is not a canonical source node in G .

The following observation shows that self-certification is not possible.

Observation 4.6.17. *Let G be a graph, and $u \in V_G$ such that u is not a canonical source node. Then u cannot be a certificate of itself.*

Proof. Suppose that, if possible, u is a certificate of itself. Then from Definition 4.6.16, there exists a $v \in V_G$ such that $u \rightarrow v \in G$ and there exists a triangle-free path $Q = (u_1 = u, u_2 = v, \dots, u_l = u)$ from (u, v) to u . But, then from the definition of a path (Definition 1.1.2), Q is not a path. Hence, Q is not a triangle-free path. Therefore, u cannot be a certificate of itself. $\square$

Algorithm 5: TFP(G)

Input : A chain graph G with chordal undirected connected components

Output: (P_1, P_2) such that

$P_1 : E_G \times E_G \rightarrow \{0, 1\}$ and $P_2 : E_G \times V_G \rightarrow \{0, 1\}$ such that

(a) for $(u, v), (x, y) \in E_G$, $P_1((u, v), (x, y)) = 1$ if $(u, v) \neq (x, y)$ and there exists a triangle-free path from (u, v) to (x, y) in G else $P_1((u, v), (x, y)) = 0$, and

(b) for $((u, v), w) \in E_G \times V_G \rightarrow \{0, 1\}$, $P_2((u, v), w) = 1$ if $v \neq w$ and there exists a triangle-free path from (u, v) to w in G else $P_2((u, v), w) = 0$.

```
1 Construct a function  $P_1 : E_G \times E_G \rightarrow \{0, 1\}$ .
2 Construct a function  $P_2 : E_G \times V_G \rightarrow \{0, 1\}$ .
3 foreach  $((u, v), (x, y)) \in E_G \times E_G$  do
4   |  $P_1((u, v), (x, y)) \leftarrow 0$ 
5 end
6 foreach  $((u, v), w) \in E_G \times V_G$  do
7   |  $P_2((u, v), w) \leftarrow 0$ 
8 end
9 foreach  $2 \leq i < |V_G|$  do
10  |  $Q_i \leftarrow$  an empty list
11 end
12 foreach triple  $(u, v, w)$  do
13   | if  $(u, v, w)$  is a triangle-free path in  $G$  then
14     |  $P_1((u, v), (v, w)) = 1$ 
15     |  $P_2((u, v), w) = 1$ 
16     | Add  $((u, v), (v, w))$  to  $Q_2$ .
17   | end
18 end
19  $l \leftarrow 3$ 
20 while  $l < |V_G|$  do
21   | foreach  $((u, v), (z, x))$  in  $Q_{l-1}$  do
22     | foreach  $x - y \in E_G$  do
23       | if  $(z, x, y)$  is a triangle-free path in  $G$  and  $P_1((u, v), (x, y)) = 0$  then
24         | Add  $((u, v), (x, y))$  to  $Q_l$ .
25       | end
26     | end
27   | end
28   | foreach  $((u, v), (x, y))$  in  $Q_l$  do
29     |  $P_1((u, v), (x, y)) = 1$ 
30     |  $P_2((u, v), y) = 1$ 
31   | end
32   |  $l = l + 1$ 
33 end
34 return  $(P_1, P_2)$ 
```

We now present an algorithm (Algorithm 5) that, given a chain graph G with chordal undirected connected components, outputs a pair of functions (P_1, P_2) , where P_1 determines the existence of triangle-free paths of length three or more between two edges of G , and P_2 determines the existence of triangle-free paths of length three or more between an edge and a node of G . Lemma 4.6.18 establishes the correctness of the algorithm.

Lemma 4.6.18. *Let G be a chain graph with chordal undirected connected components. For input G , Algorithm 5 outputs a pair of functions (P_1, P_2) such that:*

1. $P_1 : E_G \times E_G \rightarrow \{0, 1\}$. For edges $(u, v), (x, y) \in E_G$, $P_1((u, v), (x, y)) = 1$ if $(u, v) \neq (x, y)$ and there exists a triangle-free path from (u, v) to (x, y) in G ; otherwise, $P_1((u, v), (x, y)) = 0$.
2. $P_2 : E_G \times V_G \rightarrow \{0, 1\}$. For $((u, v), w) \in E_G \times V_G$, $P_2((u, v), w) = 1$ if $v \neq w$ and there exists a triangle-free path from (u, v) to w ; otherwise, $P_2((u, v), w) = 0$.

Proof. We first prove the following claim:

Claim 4.6.19. *For $2 \leq k < |V_G|$,*

1. *For $(u, v), (x, y) \in E_G$, $((u, v), (x, y))$ is in the list Q_k if and only if there exists a triangle-free path from (u, v) to (x, y) in G , and the shortest triangle-free path from (u, v) to (x, y) in G has length k .*
2. *At the start of the while loop (lines 20-33) when $l = k + 1$, for $(u, v), (x, y) \in E_G$, $P_1((u, v), (x, y)) = 1$ if and only if $(u, v) \neq (x, y)$ and there exists a triangle-free path from (u, v) to (x, y) in G of length at most k .*
3. *At the start of the while loop (lines 20-33) when $l = k + 1$, for $((u, v), w) \in E_G \times V_G$, $P_2((u, v), w) = 1$ if and only if $v \neq w$ and there exists a triangle-free path from (u, v) to w in G of length at most k .*

Proof. We proceed by induction on k .

Base Case: Let $k = 2$. Algorithm 5 initializes Q_2 as an empty list (Line 10), and updates it at Line 16. For $(u, v), (x, y) \in E_G$, if the shortest length triangle-free path from (u, v) to (x, y) is two, then $(u, v = x, y)$ is the shortest triangle-free path from (u, v) to (x, y) . Then, at Line 16, $((u, v), (x, y))$ has been added to Q_2 . Suppose for some $(u, v), (x, y) \in E_G$, $((u, v), (x, y))$ is in the list Q_2 , then it must have been added to Q_2 at Line 16, implying $Q = (u, v = x, y)$ is a triangle-free path from (u, v) to (x, y) (lines 13-17). There cannot be any triangle-free path from (u, v) to $(v = x, y)$ of length less than two; otherwise, we have $u = v = x = y$, and Q is not a path, a contradiction. This shows that for $(u, v), (x, y) \in E_G$, $((u, v), (x, y))$ is in the list Q_2 if and only if there exists a triangle-free path from (u, v) to (x, y) in G , and the shortest triangle-free path from (u, v) to (x, y) in G has length two. This implies Item 1 of Claim 4.6.19 holds true for $k = 2$.

At lines 3-5, we initialize P_1 such that for all $(u, v), (x, y) \in E_G$, $P_1((u, v), (x, y)) = 0$. At lines 6-8, we initialize P_2 such that for all $((u, v), w) \in E_G \times V_G$, $P_2((u, v), w) = 0$. We update P_1 and P_2 at lines 14-15 such that for all triangle-free path (u, v, w) of length two, $P_1((u, v), (v, w)) = 1$, and $P_2((u, v), w) = 1$. Also, if $(u, v) = (x, y)$ or $v = w$, then there could not be a triangle-free path of length two either from (u, v) to (x, y) or from (u, v) to w . This implies Items 2 and 3 of Claim 4.6.19 hold true for $k = 2$.

Induction Hypothesis: Let us assume that Items 1 to 3 of Claim 4.6.19 hold true for $k = k'$ for some $2 \leq k' \leq |V_G| - 2$. Under this assumption, the following statements hold:

1. For all $((u, v), (x, y)) \in E_G \times E_G$, $Q_{k'}$ contains $((u, v), (x, y))$ if and only if there exists a triangle-free path from (u, v) to (x, y) and the shortest triangle-free path from (u, v) to (x, y) has length k' .
2. At the start of the iteration of the while loop (lines 20-33) when $l = k' + 1$, for all $((u, v), (x, y)) \in E_G \times E_G$, $P_1((u, v), (x, y)) = 1$ if and only if $(u, v) \neq (x, y)$ and there exists a triangle-free path from (u, v) to (x, y) of length at most k' .
3. At the start of the iteration of the while loop when $l = k' + 1$, for all $((u, v), w) \in E_G \times V_G$, $P_2((u, v), w) = 1$ if and only if $v \neq w$ and there exists a triangle-free path

from (u, v) to w of length at most k' .

We now show that Items 1 to 3 of Claim 4.6.19 hold true even for $k = k' + 1$.

Inductive Step: $Q_{k'+1}$ is updated only at lines 21-27. This shows that for $(u, v), (x, y) \in E_G$, $((u, v), (x, y))$ is added to $Q_{k'+1}$ if $P_1((u, v), (x, y)) = 0$ and there exists a z such that $((u, v), (z, x))$ is in $Q_{k'}$ and (z, x, y) is a triangle-free path in G . From our assumption, $P_1((u, v), (x, y)) = 0$ implies that there is no triangle-free path from (u, v) to (x, y) of length k' , and $((u, v), (z, x))$ being in the list $Q_{k'}$ implies that there exists a triangle-free path from (u, v) to (z, x) of length k' . From Lemma 4.2.5, the existence of the triangle-free path from (z, x) to (x, y) further implies that there exists a triangle-free path of length $k' + 1$ from (u, v) to (x, y) , and the path is of the shortest length. This confirms that if $((u, v), (x, y))$ is in $Q_{k'+1}$, then there exists a triangle-free path from (u, v) to (x, y) and the shortest triangle-free path from (u, v) to (x, y) has length $k' + 1$. This proves the “only if” part of Item 1 of Claim 4.6.19.

We now prove the “if” part of Item 1 of Claim 4.6.19. Suppose there exists a triangle-free path $Q = (u_1 = u, u_2 = v, \dots, u_{k'}, u_{k'+1} = x, u_{k'+2} = y)$ from (u, v) to (x, y) of length $k' + 1$, and there is no path from (u, v) to (x, y) of length smaller than $k' + 1$. From our assumption, this implies $P_1((u, v), (x, y)) = 0$. Also, $(u_{k'}, x, y)$, the subpath of Q containing the last three nodes of Q , must be a triangle-free path in G . The existence of the path Q implies that $Q' = (u_1 = u, u_2 = v, \dots, u_{k'}, u_{k'+1} = x)$ (the subpath of Q from u_1 to $u_{k'+1}$) is a k' -length triangle-free path from (u, v) to $(u_{k'}, x)$. We claim that Q' is a triangle-free path of the shortest possible length. Suppose our claim is not true. Then, there must exist a triangle-free path from (u, v) to $(u_{k'}, x)$ of length less than k' . From Lemma 4.2.5, concatenating this path with the triangle-free path $(u_{k'}, x, y)$ gives us a triangle-free path from (u, v) to (x, y) of length less than $k' + 1$, contradicting our assumption that there exists no triangle-free path from (u, v) to (x, y) of length smaller than $k' + 1$. This implies Q' is a triangle-free path of the shortest possible length. Since Q' is of length k' , from our assumption, $((u, v), (u_{k'}, x))$ must be in $Q_{k'}$. This implies $((u, v), (x, y))$ has been added to $Q_{k'+1}$ at Line 24. This proves the “if” part of Item 1 of Claim 4.6.19. This shows that Item 1 of Claim 4.6.19 holds true for $k = k' + 1$.

We now show that Item 2 of Claim 4.6.19 hold true for $k = k' + 1$. For all $((u, v), (x, y)) \in$

$E_G \times E_G$, if $(u, v) \neq (x, y)$ and there exists a triangle-free path from (u, v) to (x, y) of length at most k' , then from our assumption we already have $P_1((u, v), (x, y)) = 1$. As proved earlier, Item 1 of Claim 4.6.19 holds true for $k = k' + 1$. Therefore, for $((u, v), (x, y)) \in E_G \times E_G$, if the shortest triangle-free path from (u, v) to (x, y) is $k' + 1$, then $((u, v), (x, y))$ must be in $Q_{k'+1}$, and at Line 29, we have $P_1((u, v), (x, y)) = 1$. This implies that at the end of the while loop iteration when $l = k' + 1$, for all $((u, v), (x, y)) \in E_G \times E_G$, if $(u, v) \neq (x, y)$ and there exists a triangle-free path from (u, v) to (x, y) of length at most $k' + 1$ then $P_1((u, v), (x, y)) = 1$.

Suppose at the end of the iteration of the while loop when $l = k' + 1$, $P_1((u, v), (x, y)) = 1$. There are two possibilities: either even at the start of the iteration, we have $P_1((u, v), (x, y)) = 1$ or in the current iteration (when $l = k' + 1$), we have updated P_1 at Line 29 with $P_1((u, v), (x, y)) = 1$. If at the start of the iteration, we have $P_1((u, v), (x, y)) = 1$ then from our assumption, $(u, v) \neq (x, y)$ and there exists a triangle-free path from (u, v) to (x, y) of length at most k' . And, if at the current iteration, we update P_1 with $P_1((u, v), (x, y)) = 1$ then there must be $((u, v), (x, y))$ in $Q_{k'+1}$, implying there exists a triangle-free path of length $k' + 1$ from (u, v) to (x, y) . Since $k' + 1 > 1$, $(u, v) \neq (x, y)$. This implies that at the end of the iteration of the while loop when $l = k' + 1$, if $P_1((u, v), (x, y)) = 1$ then $(u, v) \neq (x, y)$ and there exists a triangle-free path of length $k' + 1$ from (u, v) to (x, y) . This shows that at the start of the while loop iteration when $l = k' + 2$, for all $((u, v), (x, y)) \in E_G$, $P_1((u, v), (x, y)) = 1$ if and only if $(u, v) \neq (x, y)$ and there exists a triangle-free path from (u, v) to (x, y) of length at most $k' + 1$. This shows that Item 2 of Claim 4.6.19 holds true for $k = k' + 1$.

We now show that Item 3 of Claim 4.6.19 hold true for $k = k' + 1$. For all $((u, v), w) \in E_G \times V_G$, if $v \neq w$ and there exists a triangle-free path from (u, v) to w of length at most k' , then from our assumption we already have $P_2((u, v), w) = 1$. As proved earlier, Item 1 of Claim 4.6.19 holds true for $k = k' + 1$. Therefore, for $((u, v), w) \in E_G \times V_G$, if the shortest triangle-free path from (u, v) to w is $k' + 1$, then there exist $x \in V_G$ such that the shortest triangle-free path is from (u, v) to (x, w) , and $((u, v), (x, y))$ must be in $Q_{k'+1}$, and at Line 30, we have $P_2((u, v), w) = 1$. This implies that at the end of the while loop iteration when $l = k' + 1$, for all $((u, v), w) \in E_G \times V_G$, if $v \neq w$ and there exists a triangle-free path from (u, v) to w of length at most $k' + 1$ then $P_2((u, v), w) = 1$.

Suppose at the end of the iteration of the while loop when $l = k' + 1$, $P_2((u, v), w) = 1$. There are two possibilities: either even at the start of the iteration, we have $P_2((u, v), w) = 1$ or in the current iteration (when $l = k' + 1$), we have updated P_2 at Line 30 with $P_2((u, v), w) = 1$. If at the start of the iteration, we have $P_2((u, v), w) = 1$ then from our assumption, $v \neq w$ and there exists a triangle-free path from (u, v) to w of length at most k' . And, if at the current iteration, we update P_2 with $P_2((u, v), w) = 1$ then there must exist some $x \in V_G$ such that $((u, v), (x, w))$ in $Q_{k'+1}$, implying there exists a triangle-free path of length $k' + 1$ from (u, v) to (x, w) . Since $k' + 1 > 1$, $v \neq w$. This implies that at the end of the iteration of the while loop when $l = k' + 1$, if $P_2((u, v), w) = 1$ then $v \neq w$ and there exists a triangle-free path of length $k' + 1$ from (u, v) to w . This shows that at the start of the while loop iteration when $l = k' + 2$, for all $((u, v), w) \in E_G \times V_G$, $P_2((u, v), w) = 1$ if and only if $v \neq w$ and there exists a triangle-free path from (u, v) to w of length at most $k' + 1$. This shows that Item 2 of Claim 4.6.19 holds true for $k = k' + 1$.

This completes the proof of Claim 4.6.19. $\square$

The length of any triangle-free path in G can be at most $|V_G| - 1$. Therefore, from Items 2 and 3 of Claim 4.6.19, at the end of the while loop (lines 20-33) of Algorithm 5, P_1 and P_2 obey Items 1 and 2 of Lemma 4.6.18. This proves Lemma 4.6.18. $\square$

4.6.2 Proofs Omitted from Section 4.4

In this subsection, we provide all the missing proofs of Section 4.4.

Proof of Lemma 4.4.5. Meek [1995] gives a set of rules using which, if we have knowledge about the v-structure of an MEC, then we can construct the MEC. According to Definition 4.4.1, $\mathcal{V}(M') \subseteq \mathcal{V}(M)$. Therefore, if an edge can be directed using Meek's rule in M' , then it must be directed in M (using the same Meek's rule). This proves Lemma 4.4.5. However, we give another proof below. This proof is more detailed and has arguably better clarity. We also present this proof because this proof approach will be used multiple times in this paper.

Since M' is a projection of M , from Definition 4.4.1, $\text{skeleton}(M') = \text{skeleton}(M[V_{M'}])$. Let us assume that Lemma 4.4.5 is not true, i.e., there exists an edge $u \rightarrow v \in M'$ such that $u \rightarrow v \notin M$. Since M' is a chain graph (from Item 1 of Theorem 1.1.24), there exists a topological ordering τ of the vertices of M' such that if $u \rightarrow v \in M'$ then $\tau(u) < \tau(v)$ (see Proposition 4.6.3 for the details). Pick two vertices $u, v \in V_{M'}$ such that $u \rightarrow v \in M'$, $u \rightarrow v \notin M$, and for all edge $u' \rightarrow v' \in M'$, if $u' \rightarrow v' \notin M$ then either $\tau(v') > \tau(v)$, or $v = v'$ and $\tau(u') \leq \tau(u)$ (Chickering [1995] emphasized the utility of picking vertices in this order in the context of studying Markov equivalence). The chosen vertices u and v imply the following claim:

Claim 4.6.20. *For any edge $u' \rightarrow v' \in M'$, if $\tau(v') < \tau(v)$, or $v = v'$ and $\tau(u) < \tau(u')$ then $u' \rightarrow v' \in M$.*

Since $u \rightarrow v \in M'$, it must be strongly protected in M' (from Item 4 of Theorem 1.1.24). Then, $u \rightarrow v$ is part of one of the induced subgraphs shown in Figure 1.3. We go through each possibility and show that in each case, $u \rightarrow v \in M$, contradicting our assumption.

Case-1: Suppose $u \rightarrow v$ is a part of an induced subgraph of the form as shown in item-(a) of Figure 1.3, i.e., there exists a node w in M' such that $w \rightarrow u \rightarrow v$ is an induced subgraph in M' . Then, $\tau(w) < \tau(u) < \tau(v)$. From Claim 4.6.20, $w \rightarrow u \in M$. Then, $u \rightarrow v \notin M$, as it violates Item 3 of Theorem 1.1.24. Neither $v \rightarrow u \in M$, otherwise, $w \rightarrow u \leftarrow v$ is a v-structure in M . Since M' is a projection of M , and $u, v, w \in V_{M'}$, from Definition 4.4.1, if the v-structure $w \rightarrow u \leftarrow v \in M$ then $w \rightarrow u \leftarrow v \in M'$. But, this is a contradiction as $u \rightarrow v \in M'$. Thus the only option that remains is $u \rightarrow v \in M$.

Case-2: Suppose $u \rightarrow v$ is a part of an induced subgraph of the form as shown in item-(b) of Figure 1.3, i.e., there exists a node w in M' such that $u \rightarrow v \leftarrow w$ is an induced subgraph in M' . Since $u \rightarrow v \leftarrow w$ is a v-structure, from Definition 4.4.1, $u \rightarrow v \leftarrow w$ is also a v-structure in M , as M' is a projection of M . This implies that $u \rightarrow v \in M$.

Case-3: Suppose $u \rightarrow v$ is a part of an induced subgraph of the form as shown in item-(c) of Figure 1.3, i.e., there exists a node w in M' such that $u \rightarrow w \rightarrow v \leftarrow u$ is an induced subgraph in M' . Then, $\tau(u) < \tau(w) < \tau(v)$. From Claim 4.6.20, $u \rightarrow w \rightarrow v \in$

M . If $u - v \in M$, or $v \rightarrow u \in M$ then (u, v, w, u) is a directed cycle in M , contradicting Item 1 of Theorem 1.1.24. This implies $u \rightarrow v \in M$.

Case-4: Suppose $u \rightarrow v$ is a part of an induced subgraph of the form as shown in item-(d) of Figure 1.3, i.e., there exist two nodes w and w' in M' such that $u - w \rightarrow v \leftarrow w' - u \rightarrow v$ is an induced subgraph in M' . Since $w \rightarrow v \leftarrow w'$ is a V-structure, from Definition 4.4.1, $w \rightarrow v \leftarrow w' \in M$, as M' is a projection of M . If $v - u \in M$, or $v \rightarrow u \in M$ then $w \rightarrow u \leftarrow w' \in M$, otherwise M has a directed cycle (w, v, u, w) (if $w \rightarrow u \notin M$), or (w', v, u, w') (if $w' \rightarrow u \notin M$), which violates Item 1 of Theorem 1.1.24. Since, $w \rightarrow u \leftarrow w'$ is a V-structure of M , and $w, u, w' \in V_{M'}$, from Definition 4.4.1, $w \rightarrow u \leftarrow w' \in M'$, which is a contradiction, as $w - u - w' \in M'$. Thus the only option that remains is $u \rightarrow v \in M$.

We have shown that $u \rightarrow v \in M$ in each case. This proves Lemma 4.4.5. $\square$

4.6.3 Proofs Omitted from Section 4.5

In this subsection, we provide all the missing proofs of Section 4.5, except for the proof of Lemma 4.5.8. The proof of Lemma 4.5.8 is provided in Section 4.6.4 (page 170).

Proof of Lemma 4.5.2. Proof of Item 1: Suppose for some $a \in \{1, 2\}$, $u \rightarrow v \in O_a$. Since (O_a, P_{a1}, P_{a2}) is the shadow of M_a on $S_a \cup N(S_a, H_a)$, from Item 1 of Definition 4.3.1, $O_a = M_a[S_a \cup N(S_a, H_a)]$. This implies $u \rightarrow v \in M_a$. Since M_a is a projection of M , from Lemma 4.4.5, $u \rightarrow v \in M$.

Since $(O, P_1, P_2) = \text{shadow}(M, S_1 \cup S_2 \cup N(S_1 \cup S_2, H))$, from Item 1 of Definition 4.3.1, $O = M[S_1 \cup S_2 \cup N(S_1 \cup S_2, H)]$. This implies that $u \rightarrow v \in O$, as $u, v \in V_{O_a} = S_a \cup N(S_a, H_a) \subseteq S_1 \cup S_2 \cup N(S_1 \cup S_2, H)$. $\square$

Proof of Item 2. From our assumption, (O, P_1, P_2) is the shadow of M on $V_O = S_1 \cup S_2 \cup N(S_1 \cup S_2, H)$, and for $a \in \{1, 2\}$, (O_a, P_{a1}, P_{a2}) is the shadow of M_a on $V_{O_a} = S_a \cup N(S_a, H_a) \subseteq S_1 \cup S_2 \cup N(S_1 \cup S_2, H)$. Then, from Item 1 of Definition 4.3.1, $O = M[V_O]$ and $O_a = M_a[V_{O_a}]$. Since M_a is the projection of M on V_{M_a} , from Definition 4.4.1, $\mathcal{V}(M_a) = \mathcal{V}(M[V_{M_a}])$. This implies that $\mathcal{V}(M_a[V_{O_a}]) = \mathcal{V}(M[V_{O_a}])$, as $V_{O_a} \subseteq V_{M_a}$. Furthermore, this implies $\mathcal{V}(O_a) = \mathcal{V}(O[V_{O_a}])$, as O is an induced subgraph of M and $V_{O_a} \subseteq V_O$.

□

Proof of Item 3. We present a more robust result than Item 3 from Lemma 4.5.2.

Claim 4.6.21. *For each $a \in \{1, 2\}$, for an edge $u - v \in M_a$, $u \rightarrow v \in M$ if and only if at least one of the following holds:*

1. $u \rightarrow v$ is strongly protected in O .
2. There exists an undirected edge $x - y \in O_a$ such that $x \rightarrow y \in O$, and there exists a triangle-free path $Q = (u_1 = x, u_2 = y, \dots, u_{l-1} = u, u_l = v)$ from (x, y) to (u, v) in M_a where $l \geq 3$.
3. There exists an undirected edge $x - y \in O_a$ such that $x \rightarrow y \in O$, and there exist triangle-free paths $Q_1 = (u_1 = x, u_2 = y, \dots, u_l = v)$ from (x, y) to v in M_a , and $Q_2 = (v_1 = v, v_2 = u, \dots, v_m = x)$ from (v, u) to x in M_a where $l, m \geq 3$.

Proof. Observations 4.6.22 to 4.6.24 prove the $\rightarrow$ direction of Claim 4.6.21.

Observation 4.6.22. *If $u \rightarrow v$ is strongly protected in O , then $u \rightarrow v \in M$.*

Proof. The fact that $u \rightarrow v$ is strongly protected in O implies that $u \rightarrow v \in O$. From our assumption, (O, P_1, P_2) is the shadow of M on $V_O = S_1 \cup S_2 \cup N(S_1 \cup S_2, H)$. Therefore, from Item 1 of Definition 4.3.1, $O = M[V_O]$. Thus, $u \rightarrow v \in O$ implies that $u \rightarrow v \in M$. □

Observation 4.6.23. *For $u - v \in M_a$, if there exists $x - y \in O_a$ such that there is a triangle-free path $Q = (u_1 = x, u_2 = y, \dots, u_{l-1} = u, u_l = v)$ from (x, y) to (u, v) in M_a , and $x \rightarrow y \in O$, then for all $1 \leq i < l$, $u_i \rightarrow u_{i+1} \in M$, more specifically, $u \rightarrow v \in M$.*

Proof. Suppose $u - v, x - y \in M_a$, $x \rightarrow y \in O$, and $Q = (u_1 = x, u_2 = y, \dots, u_{l-1} = u, u_l = v)$ is a triangle-free path from (x, y) to (u, v) in M_a . Since O is an induced subgraph of M , and $x \rightarrow y \in O$, it follows that $x \rightarrow y \in M$. Then, according to Corollary 4.4.10, for all $1 \leq i < l$, $u_i \rightarrow u_{i+1} \in M$, and more specifically, $u \rightarrow v \in M$. □

Observation 4.6.24. For $u - v \in M_a$, if there exists an undirected edge $x - y \in O_a$ such that $x \rightarrow y \in O$, and in M_a , there exist triangle-free paths $Q_1 = (u_1 = x, u_2 = y, \dots, u_l = v)$ from (x, y) to v in M_a , and $Q_2 = (v_1 = v, v_2 = u, \dots, v_m = x)$ from (v, u) to x in M_a , where $l, m \geq 3$, then $u \rightarrow v \in M$.

Proof. Suppose $u - v, x - y \in O_a$, $x \rightarrow y \in O$, $Q_1 = (u_1 = x, u_2 = y, \dots, u_l = v)$ is a triangle-free path from (x, y) to v in M_a , and $Q_2 = (v_1 = v, v_2 = u, \dots, v_m = x)$ is a triangle-free path from (v, u) to x in M_a , with $l, m \geq 3$.

Since O is an induced subgraph of M and $x \rightarrow y \in O$, it follows that $x \rightarrow y \in M$. Then, according to Corollary 4.4.9, Q_1 is a triangle-free path from (x, y) to v in M . Since $x \rightarrow y \in M$, from Corollary 4.4.10, each edge of Q_1 is directed in M .

Q_2 cannot be a path in M , otherwise, we would obtain a directed cycle in M by concatenating Q_1 and Q_2 , which contradicts Item 1 of Theorem 1.1.24. But then, from Corollary 4.4.9, if $u \rightarrow v \notin M$ then Q_2 is a triangle-free path in M , a contradiction. This implies $u \rightarrow v \in M$. This completes the proof of Observation 4.6.24. $\square$

We now prove the $\leftarrow$ direction of Claim 4.6.21. Without loss of generality, let $a = 1$. Since M is a chain graph (from Item 1 of Theorem 1.1.24), $M[V_{M_1}]$ is also a chain graph. Then from Proposition 4.6.3, there must exist an ordering τ of the vertices of M_1 such that for $u, v \in V_{M_1}$, if $u \rightarrow v \in M$ then $\tau(u) < \tau(v)$. Suppose there exists an edge $u - v \in M_1$ such that $u \rightarrow v \in M$ and it does not follow Claim 4.6.21 (i.e., it follows none of Items 1 to 3 of Claim 4.6.21). We now pick such a $u - v$. Pick the least rank node v in τ such that for any edge $x - y \in M_1$, if $x \rightarrow y \in M$, and $\tau(y) < \tau(v)$ then $x \rightarrow y$ obeys Claim 4.6.21 (i.e., $x \rightarrow y$ follows one of the followings: (a) Item 1 of Claim 4.6.21, or it follows Item 2 of Claim 4.6.21, or it follows Item 3 of Claim 4.6.21). After picking such a node v , we pick u such that $u - v \in M_1$, $u \rightarrow v \in M$, and for any node $z \in V_{M_1}$ such that $\tau(u) < \tau(z) < \tau(v)$, if $z - v \in M_1$, and $z \rightarrow v \in M$ then $z \rightarrow v$ obeys Claim 4.6.21. We show that $u \rightarrow v$ also obeys Claim 4.6.21. Our selection of $u - v \in M_1$ implies the following claim:

Claim 4.6.25. For any edge $u' - v' \in M_1$ such that $u' \rightarrow v' \in M$, if $\tau(v') < \tau(v)$, or $v' = v$ and $\tau(u) < \tau(u')$ then $u' \rightarrow v'$ obeys Claim 4.6.21, i.e., either

1. $u' \rightarrow v'$ is strongly protected in O , or

2. *there exists $x - y \in O_1$ such that $x \rightarrow y \in O$ and there exists a triangle-free path from (x, y) to (u', v') in M_1 of length greater than one (i.e., the path containing more than two nodes), or*
3. *there exists $x - y \in O_1$ such that $x \rightarrow y \in O$ and there exist triangle-free paths in M_1 from (x, y) to v' , and from (v', u') to x such that each triangle-free path is of length greater than one.*

Since M is an MEC, and $u \rightarrow v \in M$, from Item 4 of Theorem 1.1.24, $u \rightarrow v$ must be strongly protected in M . This implies that $u \rightarrow v$ is part of an induced subgraph of M as shown in Figure 1.3. We show that $u \rightarrow v$ obeys Claim 4.6.21 in each such possible induced subgraph of M . We now go through all the possibilities.

Case 1: Suppose there exists $w \in V_M$ such that $u \rightarrow v$ is part of an induced subgraph $w \rightarrow u \rightarrow v$ in M (similar to Figure 1.3.a). There are two possibilities: either $w \in V_{H_1}$ or $w \notin V_{H_1}$.

If $w \notin V_{H_1}$, then $w \in V_{H_2} \setminus I$. This implies that $u \in I$, as $u \in V_{H_1}$ is a neighbor of $w \in V_{H_2} \setminus I$ in H , and I is a vertex separator of H that separates $V_{H_1} \setminus I$ and $V_{H_2} \setminus I$. This, in turn, implies that $u, v, w \in I \cup N(I, H) \subseteq V_O$, and the induced subgraph is also a part of O . Thus, $u \rightarrow v$ is strongly protected in O , as shown in Figure 1.3.a. Let us now consider the other possibility.

Suppose $w \in V_{H_1}$. Since M_1 is a projection of M on V_{H_1} , we can deduce, from Corollary 4.4.6 and Item 3 of Theorem 1.1.24, that $w - u \in M_1$. Also, as $w \rightarrow u \rightarrow v \in M$ and by the construction of τ , we have $\tau(w) < \tau(u) < \tau(v)$. According to Claim 4.6.25, $w \rightarrow u$ satisfies Claim 4.6.21, specifically, either (1) Item 1 of Claim 4.6.21, or (2) Item 2 of Claim 4.6.21, or (3) Item 3 of Claim 4.6.21. We will explore each possibility and demonstrate that in each case, $u \rightarrow v$ obeys Claim 4.6.21.

1. Suppose $w \rightarrow u$ satisfies Item 1 of Claim 4.6.21. This implies that $w \rightarrow u$ is strongly protected in O , and thus, $w \rightarrow u \in O$. In this scenario, $u \rightarrow v$ obeys Item 2, since there exists a triangle-free path $P = (w, u, v)$ from (w, u) to (u, v) in M_1 (with length greater than one), and $w \rightarrow u \in O$.
2. Suppose $w \rightarrow u$ satisfies Item 2 of Claim 4.6.21. This implies that there exists an undirected edge $x - y \in O_1$ such that $x \rightarrow y \in O$, and a triangle-free path

$Q = (u_1 = x, u_2 = y, \dots, u_{l-1} = w, u_l = u)$ from (x, y) to (w, u) in M_1 , with $l \geq 3$. We first demonstrate that Q does not contain v . According to Observation 4.6.13, Q forms a chordless path in M_1 . If Q contains v , then there must be some $i < l - 1$ where $u_i = v$ (since $u_l = u$ and $u_{l-1} = w$). This would lead to a contradiction as Q would no longer be a chordless path. Thus, Q does not contain v .

As $Q' = (w, u, v)$ is a triangle-free path from (w, u) to (u, v) in M_1 , we can use Lemma 4.2.5 to conclude that $Q'' = (u_1 = x, u_2 = y, \dots, u_{l-1} = w, u_l = u, v)$ is a triangle-free path from (x, y) to (u, v) in M_1 (by concatenating Q and Q'). Consequently, $u \rightarrow v$ satisfies Item 2 as Q'' is a triangle-free path in M_1 (of length greater than one, given that $l \geq 3$), and $x \rightarrow y \in O$.

3. Suppose $w \rightarrow u$ satisfies Item 3 of Claim 4.6.21. This implies the existence of an undirected edge $x - y \in O_1$, where $x \rightarrow y \in O$, along with triangle-free paths $Q_1 = (u_1 = x, u_2 = y, \dots, u_l = u)$ from (x, y) to u in M_1 , and $Q_2 = (v_1 = u, v_2 = w, \dots, v_m = x)$ from (u, w) to x in M_1 , with both l and m greater than two. No edge in Q_1 or Q_2 is directed in M_1 , as a directed edge in either path would imply a directed cycle in M_1 (formed by concatenating Q_1 and Q_2), contradicting Item 1 of Theorem 1.1.24.

Therefore, the nodes in Q_1 and Q_2 belong to the same undirected connected chordal component, say $\mathcal{C}$, of M_1 , and both Q_1 and Q_2 are undirected triangle-free paths in $\mathcal{C}$. By Proposition 4.6.9, both Q_1 and Q_2 are chordless paths in $\mathcal{C}$. This implies that Q_2 does not contain v , otherwise, we get a chord $u - v$ in the chordless path Q_2 , a contradiction. Also, Q_1 does not contain v , otherwise $u_{l-1} = v$, and from Corollary 4.4.10, $v \rightarrow u \in M$, a contradiction, as we have $u \rightarrow v \in M$. Since $u - v \in M_1$, v also belongs to $\mathcal{C}$.

Now, let us consider $Q_3 = (u_1 = x, u_2 = y, \dots, u_l = u, u_{l+1} = v)$. If Q_3 is a chordless path in $\mathcal{C}$, it is also a chordless path in M_1 , as $\mathcal{C}$ is an induced subgraph of M_1 . In this case, $u \rightarrow v$ satisfies Item 2, as Q_3 is a triangle-free path in M_1 .

However, if Q_3 is not a chordless path in $\mathcal{C}$, this implies that there must exist an index $i \leq l - 1$ where an edge exists between u_i and v in $\mathcal{C}$. Since u_i and v are in the same undirected connected component $\mathcal{C}$, there must exist an undirected

path from u_i to v . If there is a directed edge between u_i and v in M_1 then we get a directed cycle in M_1 by combining the undirected path from u_i to v and the directed edge, contradicting Item 1 of Theorem 1.1.24. This implies that the edge between u_i and v is an undirected edge, implying $u_i - v$ is an undirected edge in $\mathcal{C}$. Since Q_1 is a triangle-free path in M_1 and $x \rightarrow y \in M$, from Corollary 4.4.10, Q_1 forms a directed path in M . Moreover, as Q_1 is directed and $u \rightarrow v \in M$, Q_3 is also directed in M . Since Q_3 passes through u_i , there exists a directed path from u_i to v in M . This implies that $u_i \rightarrow v \in M$. Otherwise, M would have a directed cycle $(u_i, u_{i+1}, \dots, u_l = u, v, u_i)$, which contradicts Item 1 of Theorem 1.1.24, as M is an MEC.

If $i < l - 1$, then $u_i \rightarrow v \leftarrow u_l$ forms a v-structure in M (since there is no edge between u_i and u_l in M , as P_1 is a chordless path). However, this leads to a contradiction because it would imply that $\mathcal{V}(M[V_{M_1}]) \neq \mathcal{V}(M_1)$ (as $u_i \rightarrow v \leftarrow u_l \in M$ while $u_i - v - u_l \in M_1$). But, from Definition 4.4.1, since M_1 is a projection of M , both M_1 and $M[V_{M_1}]$ must have the same set of v-structures. Thus, the only possible value of i is $l - 1$. In this case, $Q_4 = (u_1 = x, u_2 = y, \dots, u_{l-1}, v)$ is a chordless path in M_1 . Additionally, since $Q_5 = (v, u, w)$ is a triangle-free path in M_1 from (v, u) to (u, w) , and Q_2 is a triangle-free path in M_1 from (u, w) to x , we can use Lemma 4.2.5 to show that $Q_6 = (v, u, w, v_3, v_4, \dots, v_m = x)$ is a triangle-free path from (v, u) to (w, x) in M_1 . Both Q_4 and Q_6 have lengths greater than one. Hence, $u \rightarrow v$ satisfies Item 3 of Claim 4.6.21 as Q_4 is a triangle-free path from (x, y) to v in M_1 , Q_6 is a triangle-free path from (v, u) to x in M_1 , and $x \rightarrow y \in O$.

This implies that in all the cases of Case 1, $u \rightarrow v$ obeys Item 3.

Case 2: Consider the scenario where $u \rightarrow v$ is strongly protected in M because it is part of an induced subgraph $u \rightarrow v \leftarrow w \in M$ (similar to Figure 1.3.b). There are two possibilities: either $w \notin V_{H_1}$ or $w \in V_{H_1}$.

If $w \notin V_{H_1}$, as shown in Case 1, the induced subgraph is also an induced subgraph of O . Therefore, $u \rightarrow v$ is strongly protected in O , and it obeys Item 1 of Claim 4.6.21. Let us now consider the other possibility.

Suppose $w \in V_{H_1}$. Then, $u \rightarrow v \leftarrow w$ is a V-structure in M such that $u, v, w \in V_{M_1}$. Since M_1 is a projection of M on V_{H_1} , from Definition 4.4.1, the set of V-structures of M_1 and the set of V-structures of $M[V_{M_1}]$ must be the same. This implies $u \rightarrow v \leftarrow w \in M_1$. This leads to a contradiction since our assumption is that $u - v \in M_1$. Therefore, this possibility cannot occur.

Case 3: Suppose $u \rightarrow v$ is strongly protected in M because it is part of an induced subgraph $u \rightarrow w \rightarrow v \leftarrow u \in M$ (same as Figure 1.3.c). There are two possibilities: either $w \notin V_{H_1}$ or $w \in V_{H_1}$.

If $w \notin V_{H_1}$, as demonstrated in Case 1, the induced subgraph is also an induced subgraph of O . Therefore, $u \rightarrow v$ is strongly protected in O , and it satisfies Item 1 of Claim 4.6.21. Let us now consider the other possibility.

Suppose $w \in V_{H_1}$. Since $u \rightarrow w$ and $w \rightarrow v \in M$, and since M_1 is a projection of M on V_{H_1} , we can deduce from Corollary 4.4.6 that either $u \rightarrow w \in M_1$ or $u - w \in M_1$, and either $w \rightarrow v \in M_1$ or $w - v \in M_1$.

Given that $u - v \in M_1$, if $u \rightarrow w \in M_1$ or $w \rightarrow v \in M_1$, it would lead to a directed cycle (u, w, v, u) in M_1 , which contradicts Item 1 of Theorem 1.1.24. Thus, we conclude that $u - w$ and $w - v \in M_1$. As $u \rightarrow w$ and $w \rightarrow v \in M$, based on the ordering in τ , it holds that $\tau(u) < \tau(w) < \tau(v)$. Referring to Claim 4.6.25, we see that both $u \rightarrow w$ and $w \rightarrow v$ follow Claim 4.6.21. This implies that either $u \rightarrow w$ obeys Item 1, or it follows Item 2, or it satisfies Item 3. Let us examine each of these possibilities.

1. Suppose $u \rightarrow w$ obeys Item 1 of Claim 4.6.21, i.e., $u \rightarrow w$ is strongly protected in O . This implies $u, w \in V_O$ and $u \rightarrow w \in O$. As shown earlier, $w \rightarrow v$ also obeys Claim 4.6.21.

This implies either $w \rightarrow v$ obeys Item 1 of Claim 4.6.21, or it obeys Item 2, or it obeys Item 3 of Claim 4.6.21. We will consider each possibility.

- (a) Suppose $w \rightarrow v$ obeys Item 1 of Claim 4.6.21, i.e., $w \rightarrow v$ is strongly protected in O . This implies $w, v \in V_O$ and $w \rightarrow v \in O$. Since $u, v, w \in V_O$, $u \rightarrow v \in M$, and O is an induced subgraph of M , we have $u \rightarrow v \in O$. This shows that $u \rightarrow v$ is also strongly protected in O , as it is part of the induced subgraph $u \rightarrow w \rightarrow v \leftarrow u \in O$. This implies $u \rightarrow v$ obeys Item 1 of Claim 4.6.21.

- (b) Suppose $w \rightarrow v$ obeys Item 2 of Claim 4.6.21, i.e., there exists an edge $x - y \in O_1$ such that there is a triangle-free path $Q = (u_1 = x, u_2 = y, \dots, u_{l-1} = w, u_l = v)$ from (x, y) to (w, v) in M_1 , with $l \geq 3$ and $x \rightarrow y \in O$.

According to Observation 4.6.13, Q is an undirected chordless path in M_1 . We first demonstrate that Q does not contain u . Suppose Q contains u , then for some $i \leq l - 2$, $u_i = u$ (since $u_l = v$ and $u_{l-1} = w$). However, this contradicts the fact that Q is a chordless path, as $u - v \in M_1$. Hence, Q does not contain u . Since $u - v \in M_1$, nodes of Q and u are in the same undirected connected chordal component of M_1 .

We now show that $u_{l-2} - u \in M_1$. From Observation 4.6.23, $u_{l-2} \rightarrow w \in M$, since Q is a triangle-free path in M_1 , and $x \rightarrow y \in M$ (because O is an induced subgraph of M and $x \rightarrow y \in O$). This implies there must be an edge between u_{l-2} and u , otherwise, $u_{l-2} \rightarrow w \leftarrow u$ would form a v-structure in M . Since M_1 is a projection of M on V_{H_1} and $u_{l-2}, w, u \in V_{H_1}$, if $u_{l-2} \rightarrow w \leftarrow u$ is a v-structure in M , then $u_{l-2} \rightarrow w \leftarrow u \in M_1$. But this is a contradiction since $w - u \in M_1$. Therefore, there must be an edge between u_{l-2} and u . Since u_{l-2} and u are in the same undirected connected component of M_1 , from Proposition 4.6.1, $u_{l-2} - u \in M_1$.

We now consider the least i such that $u_i - u \in M_1$. If $i > 1$, then $Q' = (u_1 = x, u_2 = y, \dots, u_i, u, v)$ forms a chordless path from (x, y) to (u, v) , as v cannot be a neighbor of any u_i due to Q being a chordless path and $i \leq l - 2$. This implies that $u \rightarrow v$ obeys Item 2 of Claim 4.6.21. If $i = 1$, then $Q_1 = (v, u, x)$ forms a triangle-free path from (v, u) to x in M_1 . Therefore, (u, v) obeys Item 3 of Claim 4.6.21, as Q is a triangle-free path from (x, y) to v in M_1 (of length greater than one), Q_1 is a triangle-free path from (v, u) to x in M_1 (of length greater than one), and $x \rightarrow y \in O$. This concludes the analysis of this case, showing that in each possibility $u \rightarrow v$ obeys Claim 4.6.21.

- (c) Suppose $w \rightarrow v$ obeys Item 3 of Claim 4.6.21, i.e., there exists an edge $x - y \in O_1$ such that $x \rightarrow y \in O$, and there exist triangle-free paths $Q_1 = (u_1 = x, u_2 = y, \dots, u_l = v)$ from (x, y) to v in M_1 , and $Q_2 = (v_1 = v, v_2 =$

$w, \dots, v_m = x$) from (v, w) to x in M_1 such that $l, m \geq 3$.

Q_1 and Q_2 must both be undirected paths in M_1 , otherwise combining Q_1 and Q_2 would create a directed cycle in M_1 . This implies that u and the nodes in Q_1 and Q_2 belong to the same undirected connected chordal component, say $\mathcal{C}$, of M_1 . This further implies that Q_1 and Q_2 are triangle-free paths of a chordal graph $\mathcal{C}$. Then, from Proposition 4.6.9, Q_1 and Q_2 are chordless paths of M_1 .

There are two possibilities: either u is in Q_1 , or u is not in Q_1 . If u is in Q_1 , then $u_{l-1} = u$, as Q_1 is a chordless path in $\mathcal{C}$. Then, Q_1 is a triangle-free path of M_1 from (x, y) to (u, v) . That means (u, v) obeys Item 2 of Claim 4.6.21, as Q_1 is a triangle-free path from (x, y) to (u, v) in M_1 such that $x - y \in O_1$ and $x \rightarrow y \in O$.

Now we consider the other possibility. Suppose u is not in Q_1 . There are two possibilities: either there is an edge between u_{l-1} and u in M , or there is no edge between them in M . Since Q_1 is a triangle-free path in M_1 , M_1 is a projection of M , and $x \rightarrow y \in M$ (recall $x \rightarrow y \in O$ and O is an induced subgraph of M), it follows from Corollary 4.4.10 that $u_{l-1} \rightarrow v \in M$.

If there is no edge between u_{l-1} and u in M , then $u_{l-1} \rightarrow v \leftarrow u$ forms a v-structure in M . Since M_1 is a projection of M and $u_{l-1}, v, u \in M_1$, the v-structure $u_{l-1} \rightarrow v \leftarrow u$ in M implies that $u_{l-1} \rightarrow v \leftarrow u$ is also a v-structure in M_1 . This, in turn, implies $u \rightarrow v \in M_1$. However, this leads to a contradiction, as $u - v \in M_1$.

This contradiction implies there must be an edge between u_{l-1} and u in M . From Corollaries 4.4.6 and 4.4.7, there is also an edge between u_{l-1} and u in M_1 . Since u_{l-1} and u are in the same undirected connected component of M_1 , it follows from Proposition 4.6.1 that $u_{l-1} - u \in M_1$.

We now consider a path $Q_3 := (u_1 = x, u_2 = y, \dots, u_{l-1}, u)$, obtained by replacing u_l with u in the chordless path Q_1 . There are two possibilities: either Q_3 is not a chordless path in $\mathcal{C}$, or Q_3 is a chordless path in $\mathcal{C}$. We show that in both the possibility $u \rightarrow v$ obeys Claim 4.6.21.

- i. Suppose Q_3 is not a chordless path in $\mathcal{C}$. Then there must exist a k such that $k < l - 1$ and $u_k - u \in \mathcal{C}$, because Q_1 is a chordless path and Q_3 is not a chordless path in $\mathcal{C}$. Pick the least k such that $u_k - u \in \mathcal{C}$. If $k > 1$, then this implies that $Q'_3 := (u_1 = x, u_2 = y, \dots, u_k, u, v)$ is a chordless path from (x, y) to (u, v) in $\mathcal{C}$ (since Q_1 is a chordless path and there cannot be an edge between v and any u_i s in Q_3). This further implies that $u \rightarrow v$ obeys Item 2 of Claim 4.6.21, as $x \rightarrow y \in O$ and Q'_3 is a triangle-free path from (x, y) to (u, v) in M_1 (since $\mathcal{C}$ is an undirected connected component of M_1 and every chordless path is a triangle-free path). If $k = 1$, then $Q''_3 := (v, u, x)$ is a triangle-free path from (v, u) to x in M_1 (since Q_1 is a chordless path, there cannot be an edge between x and v). This implies that $u \rightarrow v$ obeys Item 3 of Claim 4.6.21, due to the existence of triangle-free paths Q_1 , Q''_3 , and $x \rightarrow y \in O$. This shows that in each possibility of this subcase, $u \rightarrow v$ obeys Claim 4.6.21.
- ii. Suppose Q_3 is a chordless path in $\mathcal{C}$. We claim that $Q_4 := (u, v_2 = w, v_3, \dots, v_m = x)$ (obtained by replacing the first vertex v of Q_2 by u) is not a chordless path in $\mathcal{C}$.

Proof of the claim. Suppose that Q_4 is a chordless path in $\mathcal{C}$. Since Q_3 is a chordless path in $\mathcal{C}$ and $\mathcal{C}$ is an undirected connected component of M_1 , from Observation 4.6.8, Q_3 is a triangle-free path in M_1 . Since $x \rightarrow y \in O$, and Q_3 is a triangle-free path in M_1 , from Observation 4.6.23, Q_3 is a directed path in M . Also, since $u \rightarrow w \in M$ and Q_4 is a chordless path in M_1 , from Corollary 4.4.10, Q_4 is a directed path in M . This creates a directed cycle in M by concatenating Q_3 and Q_4 . However, this is a contradiction, as stated in Item 2 of Theorem 1.1.24, M cannot have a directed cycle. This proves that Q_4 is not a chordless path in M_1 . $\square$

Since Q_2 is a chordless path and Q_4 is not a chordless path, there must exist a $k > 2$ such that $u - v_k \in M_1$. Pick the highest k , i.e., for all $j > k$, $u - v_j \notin M_1$. Then $Q_5 := (u, v_k, v_{k+1}, \dots, v_m = x)$ is a chord-

less path in $\mathcal{C}$. Since Q_2 is a chordless path in M_1 , there cannot be an edge between v and v_i for $i \geq k > 2$. This further implies that $Q'_5 := (v, u, v_k, v_{k+1}, \dots, v_m = x)$ (obtained by adding v to Q_5) is a chordless path from (v, u) to x in M_1 . This implies that $u \rightarrow v$ obeys Item 3 of Claim 4.6.21, since $x \rightarrow y \in O$, Q_1 is a triangle-free path from (x, y) to v in M_1 , and Q'_5 is a triangle-free path from (v, u) to x in M_1 (every chordless path is a triangle-free path).

This concludes the analysis of all possibilities in this case, showing that in each case $u \rightarrow v$ obeys Claim 4.6.21.

2. Suppose $u \rightarrow w$ obeys Item 2 of Claim 4.6.21, i.e., there exists an edge $x - y \in O_1$ such that there exists a triangle-free path $Q := (u_1 = x, u_2 = y, \dots, u_{l-1} = u, u_l = w)$ from (x, y) to (u, w) in M_1 , and $x \rightarrow y \in O$. From Observation 4.6.13, Q is a chordless path in M_1 . This implies that Q cannot contain v , otherwise, we get a chord $w - v$ in the chordless path Q , a contradiction.

Q not containing v implies that $Q_1 := (u_1 = x, u_2 = y, \dots, u_{l-1} = u, v)$, obtained by replacing w with v in Q , is a path in M_1 . We show that Q_1 is a chordless path in M_1 . This implies that $u \rightarrow v$ obeys Item 2 of Claim 4.6.21, as $x \rightarrow y \in O$, and Q_1 is a triangle-free path from (x, y) to (u, v) in M_1 (every chordless path is a triangle-free path) of length greater than one (since $l \geq 3$). Thus, the only thing we need to complete this subcase is to show Q_1 is a chordless path in M_1 .

Suppose Q_1 is not a chordless path in M_1 . $Q' := (u_1 = x, u_2 = y, \dots, u_{l-1} = u)$, a subpath of the chordless path Q , is a chordless path from $x - y$ to u . Since Q' is a chordless path, and Q_1 is not a chordless path, there must exist an $i < l - 1$ such that $u_i - v \in M_1$. As $x \rightarrow y \in O$ and Q is a triangle-free path from (x, y) to (u, w) , from Observation 4.6.23, each edge of Q is directed in M . Since $u \rightarrow v \in M$, $Q'_1 := (u_i, u_{i+1}, \dots, u_{l-1} = u, v)$ is a directed path in M . This implies that $u_i \rightarrow v \in M$, otherwise, we get a directed cycle $C = (u_i, u_{i+1}, \dots, u_{l-1}, v, u_i)$ in M , which contradicts Item 1 of Theorem 1.1.24. Since Q is a chordless path, there is no edge between w and u_i (since $i < l - 1$). However, then $w \rightarrow v \leftarrow u_i$ forms a v-structure in M . Since M_1 is a projection of M on V_{H_1} , and $w, v, u_i \in V_{H_1}$,

from Definition 4.4.1, $w \rightarrow v \leftarrow u_i$ is also a v-structure in M_1 . But, this is a contradiction, as $w - v \in M_1$. This implies that Q_1 is a chordless path in M_1 .

As argued above this concludes the proof for the case when $u \rightarrow w$ obeys Item 2 of Claim 4.6.21. We now move to the final possibility when $u \rightarrow w$ obeys Item 3 of Claim 4.6.21.

3. Suppose $u \rightarrow w$ obeys Item 3 of Claim 4.6.21, i.e., there exists an edge $x - y \in O_1$ such that in M_1 , there exist triangle-free paths $Q_1 := (u_1 = x, u_2 = y, \dots, u_l = w)$ from (x, y) to w , and $Q_2 := (v_1 = w, v_2 = u, \dots, v_m = x)$ from (w, u) to x . Q_1 and Q_2 must both be undirected paths in M_1 , otherwise, combining Q_1 and Q_2 gives a directed cycle in M_1 , contradicting Item 1 of Theorem 1.1.24. This implies that the nodes in Q_1 , the nodes in Q_2 , and v all belong to the same undirected connected chordal component, say $\mathcal{C}$, of M_1 . This further implies that Q_1 and Q_2 are triangle-free paths of the chordal graph $\mathcal{C}$. Then, from Proposition 4.6.9, Q_1 and Q_2 are chordless paths of $\mathcal{C}$. We now show that v is neither in Q_1 nor in Q_2 . Suppose Q_1 contains v . Then, $u_{l-1} = v$, otherwise, for some $1 \leq i < l - 1$, $u_i = v$, which creates a chord $v - w$ in the chordless path Q_1 of $\mathcal{C}$, a contradiction. But, then, from Observation 4.6.23, $v \rightarrow w \in M$, as Q_1 is a triangle-free path from (x, y) to (v, w) in M_1 and $x \rightarrow y \in O$. But this is a contradiction, as from the construction $w \rightarrow v \in M$. This implies Q_1 does not contain v .

Suppose Q_2 contains v . Since Q_2 is a chordless path in M_1 , and $w - v \in M_1$, if Q_2 contains v , then $v_2 = v$. But $v_2 = u$, which is a contradiction. Therefore, Q_2 does not contain v .

We now show that either $Q_3 := (u_1 = x, u_2 = y, \dots, u_l = w, v)$ (obtained by adding v to Q_1) or $Q_4 := (u_1 = x, u_2 = y, \dots, u_{l-1}, v)$ (obtained by replacing w with v in Q_1) is a chordless path in $\mathcal{C}$, and $Q_5 := (v, v_2 = u, v_3, \dots, v_m = x)$ (obtained by replacing the node w with v in Q_2) is a chordless path in $\mathcal{C}$. This will be our main ingredient to prove that $u \rightarrow v$ obeys Claim 4.6.21.

Suppose neither Q_3 nor Q_4 is a chordless path in $\mathcal{C}$. Since Q_3 is not a chordless path in $\mathcal{C}$ and Q_1 is a chordless path in $\mathcal{C}$, for some $i \leq l - 1$, $u_i - v \in \mathcal{C}$. Pick the largest such i . If $i \neq l - 1$, then we get a chordless cycle $(u_i, u_{i+1}, \dots, u_{l-1}, u_l, v, u_i)$

in the chordal graph $\mathcal{C}$, a contradiction. This implies $u_{l-1} - v \in \mathcal{C}$. Similarly, since Q_4 is not a chordless path in $\mathcal{C}$ and Q_1 is a chordless path in $\mathcal{C}$, $u_{l-2} - v \in \mathcal{C}$. Since Q_1 is a triangle-free path in M_1 and $x \rightarrow y \in O$, from Observation 4.6.23, $u_{l-2} \rightarrow u_{l-1}, u_{l-1} \rightarrow w \in M$. Then $u_{l-2} \rightarrow v \in M$, otherwise, we get a directed cycle $C = (u_{l-2}, u_{l-1}, w, v, u_{l-2})$ in M (recall $w \rightarrow v \in M$), which is a contradiction from Item 1 of Theorem 1.1.24.

But, then, $u_{l-2} \rightarrow v \leftarrow w$ is a v-structure in M (since Q_1 is a chordless path, there is no edge between u_{l-2} and w). Since M_1 is a projection of M on V_{H_1} , and $u_{l-2}, v, w \in V_{H_1}$, from Definition 4.4.1, $u_{l-2} \rightarrow v \leftarrow w$ is also a v-structure in M_1 . But $w - v$ is an undirected edge in M_1 , a contradiction. Thus, our assumption that neither Q_3 nor Q_4 is a chordless path in $\mathcal{C}$ is wrong. This implies either Q_3 or Q_4 is a chordless path in $\mathcal{C}$. We now show that Q_5 is a chordless path in $\mathcal{C}$.

Since Q_2 is a chordless path in $\mathcal{C}$, the subpath $Q'_5 := (v_2 = u, v_3, \dots, v_m = x)$ of Q_2 is a chordless path in $\mathcal{C}$. Suppose Q_5 is not a chordless path in $\mathcal{C}$. Then, there exists a v_i such that $i > 2$, and $v - v_i \in \mathcal{C}$. Pick the highest i such that $v_i - v \in \mathcal{C}$. Then, $Q_6 := (v, v_i, v_{i+1}, \dots, v_m = x)$ is a chordless path from (v, v_i) to x in $\mathcal{C}$. Then, $v_i \rightarrow v \in M$. Otherwise, from Corollary 4.4.9, Q_6 is a triangle-free path in M , and the concatenation of Q_1 , $w \rightarrow v$, and Q_6 yield a directed cycle in M , contradicting Item 1 of Theorem 1.1.24. Since Q_2 is a chordless path in M_1 , there cannot be an edge between w and v_i (as $i > 2$). This implies that $w \rightarrow v \leftarrow v_i$ is a v-structure in M . Since M_1 is a projection of M on V_{H_1} , and $w, v, v_i \in V_{H_1}$, from Definition 4.4.1, $w \rightarrow v \leftarrow v_i$ is also a v-structure in M_1 . This gives a contradiction, as $w - v \in M_1$. This implies that Q_5 is a chordless path in M_1 .

Thus, in this case, there exists a triangle-free path from (x, y) to v , in the form of either Q_3 or Q_4 (from Observation 4.6.8, every chordless path is a triangle-free path) such that $x - y \in O_1$ and $x \rightarrow y \in O$, and there exists a triangle-free path Q_5 from (v, u) to x . Thus, $u \rightarrow v$ obeys Item 3 of Claim 4.6.21.

Case 4: Suppose $u \rightarrow v$ is strongly protected in M because it is part of an induced subgraph $u \rightarrow v \leftarrow w - u - w' \rightarrow v$ (same as Figure 1.3.d). Both w and w' cannot be in V_{H_1} , otherwise the v-structure $w \rightarrow v \leftarrow w' \in M_1$, as M_1 is a projection of M on

V_{H_1} , and $w, v, w' \in V_{H_1}$. But, then, $u - v \notin M_1$, otherwise, it creates a directed cycle (w, v, u, w) , as from Corollary 4.4.7, $u - w \in M_1$. This implies that both w and w' cannot be in V_{H_1} . And, if w or w' is not in V_{H_1} then as we have seen in Case 1, the induced subgraph is an induced subgraph of O , i.e., $u \rightarrow v$ is strongly protected in O (as it is part of an induced subgraph of O as shown in Figure 1.3.d), and $u \rightarrow v$ obeys Item 1 of Claim 4.6.21.

We show that in all the possibilities, $u \rightarrow v$ obeys Claim 4.6.21. This completes the proof of $\leftarrow$ direction of Claim 4.6.21. This further completes the proof of Claim 4.6.21. $\square$

If there exists a triangle-free path P from (u, v) to (x, y) of length more than one then $(u, v) \neq (x, y)$. Similarly, if Q is a triangle-free path from (u, v) to w of length more than one then $v \neq w$. This shows that Claim 4.6.21 implies Item 3 of Lemma 4.5.2. $\square$

Proof of Item 4. Observations 4.6.26 to 4.6.28 prove $\rightarrow$ of Item 4.

Observation 4.6.26. For $(u, v), (x, y) \in E_O$, if $(u, v) \neq (x, y)$ and there exists a triangle-free path in O from (u, v) to (x, y) , then $P_1((u, v), (x, y)) = 1$.

Proof. Since (O, P_1, P_2) is the shadow of M on V_O , from Item 1 of Definition 4.3.1, O is an induced subgraph of M , and from Item 2 of Definition 4.3.1, for $(u, v), (x, y) \in E_O$, $P_1((u, v), (x, y)) = 1$ if and only if $(u, v) \neq (x, y)$ and there exists a triangle-free path in M from (u, v) to (x, y) . For $(u, v), (x, y) \in E_O$ such that $(u, v) \neq (x, y)$, if there exists a triangle-free path in O from (u, v) to (x, y) , then the path is also a triangle-free path in M , and therefore, $P_1((u, v), (x, y)) = 1$. $\square$

Observation 4.6.27. For $(u, v), (x, y) \in E_O$, if $(u, v) \neq (x, y)$ and for some $a \in \{1, 2\}$, $P_{a1}((u, v), (x, y)) = 1$, i.e., there exists a triangle-free path in M_a from (u, v) to (x, y) , then $P_1((u, v), (x, y)) = 1$.

Proof. Since (O, P_1, P_2) is the shadow of M , as defined in Item 1 of Definition 4.3.1, O is an induced subgraph of M . Moreover, from Item 2 of Definition 4.3.1, for $(u, v), (x, y) \in E_O$, $P_1((u, v), (x, y)) = 1$ if and only if $(u, v) \neq (x, y)$ and there exists a triangle-free path in M from (u, v) to (x, y) . Let $(u, v), (x, y) \in E_O$ such that $(u, v) \neq (x, y)$ and for some $a \in \{1, 2\}$, $P_{a1}((u, v), (x, y)) = 1$, i.e., there exists a

triangle-free path P from (u, v) to (x, y) in M_a . Since $(u, v) \in E_O$, it follows that $v \rightarrow u \notin O$. This implies $v \rightarrow u \notin M$. Then, from Corollary 4.4.9, P is also a triangle-free path in M , i.e., $P_1((u, v), (x, y)) = 1$. $\square$

Observation 4.6.28. For $u, v, x, y, z_1, z_2 \in V_O$, if $(u, v) \neq (x, y)$ and $P_1((u, v), (z_1, z_2)) = P_1((z_1, z_2), (x, y)) = 1$, then $P_1((u, v), (x, y)) = 1$.

Proof. Since (O, P_1, P_2) is the shadow of M , from Item 2 of Definition 4.3.1, for $(u, v), (x, y) \in E_O$, $P_1((u, v), (x, y)) = 1$ if and only if $(u, v) \neq (x, y)$ and there exists a triangle-free path in M from (u, v) to (x, y) . Suppose $u, v, x, y, z_1, z_2 \in V_O$ such that $(u, v) \neq (x, y)$, and $P_1((u, v), (z_1, z_2)) = P_1((z_1, z_2), (x, y)) = 1$. This implies there exist triangle-free paths $Q_1 = (u_1 = u, u_2 = v, \dots, u_{l-1} = z_1, u_l = z_2)$ from (u, v) to (z_1, z_2) in M and $Q_2 = (v_1 = z_1, v_2 = z_2, \dots, v_{m-1} = x, v_m = y)$ in M , from (z_1, z_2) to (x, y) in M . Then, according to Lemma 4.2.5, $P = (u_1 = u, u_2 = v, \dots, u_{l-1} = z_1, u_l = z_2, v_1, v_2, \dots, v_{m-1} = x, v_m = y)$ is a triangle-free path from (u, v) to (x, y) in M . Thus, $P_1((u, v), (x, y)) = 1$. $\square$

We now prove $\leftarrow$ of Item 4.

Observation 4.6.29. For $((u, v), (x, y)) \in E_O \times E_O$, if $P_1((u, v), (x, y)) = 1$ then $(u, v) \neq (x, y)$ and at least one of the following occurs:

1. There exists a triangle-free path from (u, v) to (x, y) in O .
2. For $a \in \{1, 2\}$, $P_{a1}((u, v), (x, y)) = 1$ (i.e., there exists a triangle-free path from (u, v) to (x, y) in M_a).
3. There exists $(z_1, z_2) \in E_O$ such that $P_1((u, v), (z_1, z_2)) = P_1((z_1, z_2), (x, y)) = 1$ (i.e., there exist triangle-free paths in M from (u, v) to (z_1, z_2) , and from (z_1, z_2) to (x, y)).

Proof. Suppose for some $((u, v), (x, y)) \in E_O \times E_O$, $P_1((u, v), (x, y)) = 1$. Then, since (O, P_1, P_2) is the shadow of M on V_O , from Definition 4.3.1, $(u, v) \neq (x, y)$ and there exists a triangle-free path $P = (u_1 = u, u_2 = v, \dots, u_{l-1} = x, u_l = y)$ from (u, v) to (x, y) in M .

If P is completely in O (i.e., all the nodes of P belong to V_O), then P is a triangle-free path in $O = M[V_O]$ (from Definition 4.3.1, O is an induced subgraph of M). This satisfies Item 1 of Observation 4.6.29.

If all the nodes of P belong to V_{H_a} for some $a \in \{1, 2\}$, then from Corollary 4.4.8, P is a triangle-free path in M_a , i.e., $P_{a1}((u, v), (x, y)) = 1$ (from the construction, $V_O \cap V_{M_a} = V_{O_a}$, implying $(u, v), (x, y) \in E_{O_a}$). This shows that if all the nodes of P belong to V_{H_a} for some $a \in \{1, 2\}$, then Item 2 of Observation 4.6.29 is satisfied.

Suppose P is neither completely in O , nor completely in M_1 , nor completely in M_2 . There are two possibilities: either $u_3 \in V_O$ or $u_3 \notin V_O$. We deal with each possibility.

If $u_3 \in V_O$, then we have $(v, u_3) \in E_O$ such that (a) $Q_1 = (u_1 = u, u_2 = v, u_3)$, the subpath of P from u_1 to u_3 , is a triangle-free path from (u, v) to (v, u_3) in M , i.e., $P_1((u, v), (v, u_3)) = 1$, and (b) $Q_2 = (u_2 = v, u_3, \dots, u_{l-1} = x, u_l = y)$, the subpath of P from u_2 to u_l , is a triangle-free path from (v, u_3) to (x, y) in M , i.e., $P_1((v, u_3), (x, y)) = 1$ (note that $(v, u_3) \neq (x, y)$, otherwise P is completely in O , contradicting our assumption). This satisfies Item 3 of Observation 4.6.29.

Suppose $u_3 \notin V_O$. Since $(u, v) \in E_O$, either $u - v \in O$ or $u \rightarrow v \in O$. The following claim shows that either $u, v \in V_{O_1}$ or $u, v \in V_{O_2}$, or both.

Claim 4.6.30. *If $u - v \in O$ or $u \rightarrow v \in O$, then either $u, v \in V_{O_1}$ or $u, v \in V_{O_2}$.*

Proof. From the construction, $V_O = V_{O_1} \cup V_{O_2}$, and $I = V_{O_1} \cap V_{O_2}$ is a vertex separator of O . Without loss of generality, suppose $u \in V_{O_1}$. If $v \in V_{O_1}$, then we are done. Suppose, $v \notin V_{O_1}$, i.e., $v \in V_O \setminus V_{O_1} = V_{O_2} \setminus I$. Then, $u \in I$, as I is a vertex separator that separates $V_O \setminus V_{O_1}$ and $V_O \setminus V_{O_2}$. But, then, $u, v \in V_{O_2}$, as $I \subseteq V_{O_2}$. $\square$

Without loss of generality, suppose $u, v \in V_{O_1}$. Then, there must exist a node z in P such that $z \notin V_{M_1}$, otherwise, from Corollary 4.4.8, P is a triangle-free path in M_1 , contradicting our assumption. Pick the first such z in P , i.e., all the nodes before z in P are in V_{M_1} . Let z_2 be the predecessor of z in P , and z_1 be the predecessor of z_2 in P . Since z is the first node in P which is not in V_{M_1} , $z \in V_{M_2} \setminus V_{M_1}$, and $z_2 \in V_{M_1}$. Since $I = V_{H_1} \cap V_{H_2}$ is a vertex separator of H that separates $V_{H_1} = V_{M_1}$ and $V_{H_2} = V_{M_2}$, therefore, $z_2 \in I$, and $z \in V_{M_2} \setminus I$. This implies that $z, z_1, z_2 \in V_O$, as $V_O \supseteq I \cup N(I, O)$. Since u_3 is not in O , we can say that $z_1 = u_k$ for some $k > 3$. Then, $(z_1, z_2) \in E_O$, and

(a) $Q_1 = (u_1 = u, u_2 = v, \dots, u_k = z_1, u_{k+1} = z_2)$, the subpath of P from u to z_2 , is a triangle-free path from (u, v) to (z_1, z_2) in M such that $((u_1, u_2) \neq (z_1, z_2))$ (since $k > 3$), i.e., $P_1((u_1, u_2), (z_1, z_2)) = 1$, and (b) $Q_2 = (u_k = z_1, u_{k+1} = z_2, \dots, u_{l-1} = x, u_l = y)$, the subpath of P from z_1 to y , is a triangle-free path from (z_1, z_2) to (x, y) in M such that $(z_1, z_2) \neq (x, y)$ (if $(z_1, z_2) = (x, y)$, then P is completely in M_1 , contradicting our assumption), i.e., $P_1((z_1, z_2), (x, y)) = 1$. This satisfies Item 3 of Observation 4.6.29. This implies that in all the possibilities, Observation 4.6.29 is satisfied. This completes the proof. $\square$

This completes the proof of Item 4. $\square$

Proof of Item 5. Proof of Item 5 is similar to the proof of Item 4. For completeness, we give the proof below.

Observations 4.6.31 to 4.6.33 prove $\leftarrow$ of Item 5.

Observation 4.6.31. *If $v \neq w$, and there exists a triangle-free path in O from (u, v) to w , then $P_2((u, v), w) = 1$.*

Proof. Suppose $v \neq w$. If there exists a triangle-free path P in O from (u, v) to w , then P is also a triangle-free path in M . This follows from Item 1 of Definition 4.3.1, which states that O is an induced subgraph of M . Consequently, according to Item 3 of Definition 4.3.1, which specifies that $P_2((u, v), w) = 1$ if $v \neq w$ and there exists a triangle-free path from (u, v) to w , we conclude that $P_2((u, v), w) = 1$. $\square$

Observation 4.6.32. *For $((u, v), w) \in E_O \times V_O$, if $v \neq w$, and for some $a \in \{1, 2\}$, $P_{a2}((u, v), w) = 1$, i.e., there exists a triangle-free path P in M_a , from (u, v) to w , then P is also a triangle-free path in M , i.e., $P_2((u, v), w) = 1$.*

Proof. Without loss of generality, let us assume that $a = 1$. Suppose $((u, v), w) \in E_O \times V_O$, $v \neq w$, and $P_{12}((u, v), w) = 1$, i.e., there exists a triangle-free path P from (u, v) to w in M_1 . Since $(u, v) \in E_O$, therefore, $(u, v) \in E_M$. This follows from Item 1 of Definition 4.3.1, which states that O is an induced subgraph of M . This further implies that $v \rightarrow u \notin M$. Consequently, from Corollary 4.4.9, we deduce that P is also a triangle-free path in M . Thus, according to Item 3 of Definition 4.3.1, we conclude that $P_2((u, v), w) = 1$. $\square$

Observation 4.6.33. For $u, v, w, z_1, z_2 \in V_O$, if $v \neq w$, $P_1((u, v), (z_1, z_2)) = 1$, and $P_2((z_1, z_2), w) = 1$, then $P_2((u, v), w) = 1$.

Proof. Suppose $P_1((u, v), (z_1, z_2)) = P_2((z_1, z_2), w) = 1$. From Items 1 and 2 of Definition 4.3.1, there exists triangle-free paths $Q_1 = (u_1 = u, u_2 = v, \dots, u_{l-1} = z_1, u_l = z_2)$ from (u, v) to (z_1, z_2) in M (as $P_1((u, v), (z_1, z_2)) = 1$), and $Q_2 = (v_1 = z_1, v_2 = z_2, \dots, v_{m-1} = w)$ from (z_1, z_2) to w in M (as $P_2((z_1, z_2), w) = 1$). Then, from Lemma 4.2.5, $P = (u_1 = u, u_2 = v, \dots, u_{l-1} = z_1, u_l = z_2, v_3, v_4, \dots, v_{m-1}, v_m = w)$ is a triangle-free path from (u, v) to w in M . Thus, from Item 3 of Definition 4.3.1, $P_2((u, v), w) = 1$. $\square$

We now prove $\leftarrow$ of Item 5.

Observation 4.6.34. For $((u, v), w) \in E_O \times V_O$, if $P_2((u, v), w) = 1$ then $v \neq w$, and at least one of the following occurs:

1. There exists a triangle-free path from (u, v) to w in O .
2. For some $a \in \{1, 2\}$, $P_{a2}((u, v), w) = 1$ (i.e., there exists a triangle-free path from (u, v) to w).
3. There exists $(z_1, z_2) \in E_O$ such that $P_1((u, v), (z_1, z_2)) = P_2((z_1, z_2), w) = 1$ (i.e., there exist triangle-free paths in M from (u, v) to (z_1, z_2) , and from (z_1, z_2) to w).

Proof. From Item 3 of Definition 4.3.1, if $P_2((u, v), w) = 1$, then $v \neq w$, and there exists a triangle-free path $P = (u_1 = u, u_2 = v, \dots, u_l = w)$ from (u, v) to w in M .

Similar to the proof of Observation 4.6.29, either

1. P is a triangle-free path in O , or
2. P is a triangle-free path in M_1 or M_2 , i.e., for some $a \in \{1, 2\}$, $P_{a2}((u, v), w) = 1$, or
3. there exists $(z_1, z_2) \in E_O$ such that $(u, v) \neq (z_1, z_2)$, $w \neq z_2$, and there exist triangle-free paths in M from (u, v) to (z_1, z_2) and from (z_1, z_2) to w , i.e., $P_1((u, v), (z_1, z_2)) = P_2((z_1, z_2), w) = 1$.

This completes the proof of Observation 4.6.34. □

This completes the proof of $\leftarrow$ of Item 5. This further completes the proof of Item 5. □

This completes the proof of Lemma 4.5.2. □

Algorithm 6: DPF($O, P_{11}, P_{12}, P_{21}, P_{22}$)

Input : A graph O , and four functions P_{11}, P_{12}, P_{21} and P_{22} such that H, H_1 and H_2 are undirected graphs such that H_1 and H_2 are induced subgraphs of H , and $I = V_{H_1} \cap V_{H_2}$ is a vertex separator of H that separates $V_{H_1} \setminus I$ and $V_{H_2} \setminus I$, $S_1 \subseteq V_{H_1}$ and $S_2 \subseteq V_{H_2}$ such that $S_1 \cap S_2 = I$, for $a \in \{1, 2\}$, (O_a, P_{a1}, P_{a2}) is a shadow of H_a on $S_a \cup N(S_a, H_a)$ and $O \in \text{PMEC}(H[S_1 \cup S_2 \cup N(S_1 \cup S_2, H)])$ such that O obeys Items 1 to 3 of Lemma 4.5.2.

Output: $(P_1, P_2) = \text{DPF}(O, P_{11}, P_{12}, P_{21}, P_{22})$.

```

1  $(P_1, P_2) \leftarrow \text{TFP}(O)$ 
2 foreach  $a \in \{1, 2\}$  do
3   foreach  $((u, v), (x, y)) \in E_O \times E_O$  such that  $(u, v) \neq (x, y)$  do
4     if  $P_{a1}((u, v), (x, y)) = 1$  then
5        $P_1((u, v), (x, y)) = 1$ 
6     end
7   end
8   foreach  $((u, v), w) \in E_O \times V_O$  such that  $v \neq w$  do
9     if  $P_{a2}((u, v), w) = 1$  then
10        $P_2((u, v), w) = 1$ 
11     end
12   end
13 end
14 while  $\exists (u, v), (x, y) \in E_O$  such that  $(u, v) \neq (x, y)$ ,  $P_1((u, v), (x, y)) = 0$ , and
    there exists  $(z_1, z_2) \in E_O$  such that  $P_1((u, v), (z_1, z_2)) = P_1((z_1, z_2), (x, y)) = 1$  do
15    $P_1((u, v), (x, y)) = 1$ 
16 end
17 while  $\exists ((u, v), w) \in E_O \times V_O$  such that  $v \neq w$ ,  $P_1((u, v), w) = 0$ , and there exists
     $(z_1, z_2) \in E_O$  such that  $P_1((u, v), (z_1, z_2)) = P_2((z_1, z_2), w) = 1$  do
18    $P_2((u, v), w) = 1$ 
19 end
20 return  $(P_1, P_2)$ 

```

Proof of Lemma 4.5.4. We first construct Algorithm 6 to compute $\text{DPF}(O, P_{11}, P_{12}, P_{21}, P_{22})$. Line 1 of Algorithm 6 performs the initialization part (Step 1 of Definition 4.5.3). Lines 2-

13 do the first update of P_1 and P_2 (Step 2 of Definition 4.5.3). Lines 14-19 do the final update of P_1 and P_2 (Step 3 of Definition 4.5.3).

Let $\text{DPF}(O, P_{11}, P_{12}, P_{21}, P_{22}) = (P'_1, P'_2)$. We show that $(P_1, P_2) = (P'_1, P'_2)$. We first show that $P_1 = P'_1$. Since (O, P_1, P_2) is a shadow of M , therefore, from Definition 4.4.1, the domain of P_1 is $E_O \times E_O$ and the co-domain of P_1 is $\{0, 1\}$. Also, from Line 1 of Algorithm 6, since $(P'_1, P'_2) = \text{TFP}(O)$, therefore, from Algorithm 5, the domain of P'_1 is $E_O \times E_O$ and the co-domain of P'_1 is $\{0, 1\}$. This implies both P_1 and P'_1 have the same domain and co-domain. Thus, to prove $P_1 = P'_1$, it is sufficient to show that (a) $\forall ((u, v), (x, y)) \in E_O \times E_O$, if $P'_1((u, v), (x, y)) = 1$ then $P_1((u, v), (x, y)) = 1$, and (b) $\forall ((u, v), (x, y)) \in E_O \times E_O$, if $P_1((u, v), (x, y)) = 1$ then $P'_1((u, v), (x, y)) = 1$.

Claim 4.6.35. *For all $((u, v), (x, y)) \in E_O$, if $P'_1((u, v), (x, y)) = 1$, then $P_1((u, v), (x, y)) = 1$.*

Proof. From Definition 4.5.3, P'_1 reaches its final version after following steps 1, 2, and 3 of Definition 4.5.3.

We first show that at the end of step 1, for any $((u, v), (x, y)) \in E_O$, if $P'_1((u, v), (x, y)) = 1$, then $P_1((u, v), (x, y)) = 1$. From Line 1 of Algorithm 6, $(P'_1, P'_2) = \text{TFP}(O)$. Therefore, from Lemma 4.6.18, if $P'_1((u, v), (x, y)) = 1$, then $(u, v) \neq (x, y)$, and there exists a triangle-free path from (u, v) to (x, y) in O . From Item 1 of Definition 4.3.1, O is an induced subgraph of M . This implies there exists a triangle-free path from (u, v) to (x, y) in M . Then, from Item 2 of Definition 4.3.1, $P_1((u, v), (x, y)) = 1$. This implies that at the end of the initialization step (i.e., step 1) of Definition 4.5.3, if $P'_1((u, v), (x, y)) = 1$, then $P_1((u, v), (x, y)) = 1$.

We now show that at step 2 (update 1) of Definition 4.5.3, for any $((u, v), (x, y)) \in E_O \times E_O$, if P'_1 is updated with $P'_1((u, v), (x, y)) = 1$, then $P_1((u, v), (x, y)) = 1$. Suppose for some $((u, v), (x, y)) \in E_O \times E_O$, P'_1 is updated with $P'_1((u, v), (x, y)) = 1$ at step 2 of Definition 4.5.3. From Definition 4.5.3, this implies that $(u, v) \neq (x, y)$ and for some $a \in \{1, 2\}$, $P_{a1}((u, v), (x, y)) = 1$. Since P_1 obeys Item 4 of Lemma 4.5.2, from Item 4b of Lemma 4.5.2, $P_1((u, v), (x, y)) = 1$. This implies that if for some $((u, v), (x, y)) \in E_O \times E_O$, P'_1 is updated with $P'_1((u, v), (x, y)) = 1$ at step 2 of Definition 4.5.3, then $P_1((u, v), (x, y)) = 1$. This further implies that at the start of step 3, for

any $((u, v), (x, y)) \in E_O \times E_O$, if $P'_1((u, v), (x, y)) = 1$, then $P_1((u, v), (x, y)) = 1$.

Suppose while running step 3, if at some iteration, for some $((u, v), (x, y)) \in E_O \times E_O$, we update P'_1 with $P'_1((u, v), (x, y)) = 1$. Then $P_1((u, v), (x, y)) = 1$ provided that before that iteration, for all $((u', v'), (x', y')) \in E_O \times E_O$, if $P'_1((u', v'), (x', y')) = 1$, then $P_1((u', v'), (x', y')) = 1$. From Definition 4.5.3, if at step 3 of Definition 4.5.3, for any $((u, v), (x, y)) \in E_O \times E_O$, P'_1 is updated with $P'_1((u, v), (x, y)) = 1$, then $(u, v) \neq (x, y)$ and there exists a $(z_1, z_2) \in E_O$ such that $P'_1((u, v), (z_1, z_2)) = P'_1((z_1, z_2), (x, y)) = 1$. Since at the start of the iteration, $P'_1((u, v), (z_1, z_2)) = P'_1((z_1, z_2), (x, y)) = 1$, therefore, from our assumption, $P_1((u, v), (z_1, z_2)) = P_1((z_1, z_2), (x, y)) = 1$. Since P_1 follows Item 4 of Lemma 4.5.2, from Item 4c of Lemma 4.5.2, $P_1((u, v), (x, y)) = 1$. This proves that even at the end of step 3, for all $((u, v), (x, y)) \in E_O \times E_O$, if $P'_1((u, v), (x, y)) = 1$, then $P_1((u, v), (x, y)) = 1$. This proves Claim 4.6.35. $\square$

Claim 4.6.36. For all $((u, v), (x, y)) \in E_O$, if $P_1((u, v), (x, y)) = 1$, then $P'_1((u, v), (x, y)) = 1$.

Proof. Let us construct a directed acyclic graph (DAG) D such that $V_D = E_O$, and for $((u, v), (x, y)) \in E_O$, $(u, v) \rightarrow (x, y) \in E_D$ if, and only if, $P_1((u, v), (x, y)) = 1$. We first show that D is a DAG.

Since P_1 obeys Item 4 of Lemma 4.5.2, if $P_1((u, v), (x, y)) = 1$, then $(u, v) \neq (x, y)$. This implies that D does not contain any self-loops. Suppose D is not a DAG. Then, pick the shortest cycle $((u_1, v_1), (u_2, v_2), \dots, (u_n, v_n), (u_1, v_1))$ in D for some $n \geq 2$. This implies $P_1((u_1, v_1), (u_2, v_2)) = P_1((u_2, v_2), (u_3, v_3)) = \dots = P_1((u_{n-1}, v_{n-1}), (u_n, v_n)) = P_1((u_n, v_n), (u_1, v_1)) = 1$. Since P_1 obeys Item 4 of Lemma 4.5.2, it follows from Item 4c of Lemma 4.5.2 that $P_1((u_1, v_1), (u_2, v_2)) = P_1((u_2, v_2), (u_3, v_3)) = \dots = P_1((u_{n-1}, v_{n-1}), (u_n, v_n)) = 1$ implies $P_1((u_1, v_1), (u_n, v_n)) = 1$. This further implies that in M , there exist triangle-free paths from (u_1, v_1) to (u_n, v_n) , and from (u_n, v_n) to (u_1, v_1) (since $P_1((u_1, v_1), (u_n, v_n)) = P_1((u_n, v_n), (u_1, v_1)) = 1$). However, this contradicts Lemma 4.2.6. Therefore, D does not contain any cycles.

We now show that if $(u, v) \rightarrow (x, y) \in E_D$, then $P'_1((u, v), (x, y)) = 1$, which completes the proof of Claim 4.6.36 (recall from the construction of D , $(u, v) \rightarrow (x, y) \in E_D$ if and only if $P_1((u, v), (x, y)) = 1$). Let τ be an ordering on the vertices of D such

that if $(u, v) \rightarrow (x, y) \in E_D$, then $\tau((u, v)) < \tau((x, y))$ (the existence of τ follows from the fact that D is a DAG).

Suppose there exists $(u, v) \rightarrow (x, y) \in E_D$ such that $P'_1((u, v), (x, y)) = 0$. Pick such a $(u, v) \rightarrow (x, y) \in E_D$ such that for all $(u', v') \rightarrow (x', y') \in E_D$,

$$\text{if } \tau(x', y') < \tau(x, y), \text{ or } (x', y') = (x, y) \text{ and } \tau(u, v) < \tau(u', v'), \text{ then } P'_1((u', v'), (x', y')) = 1. \quad (4.6)$$

From the construction of D , since $(u, v) \rightarrow (x, y) \in E_D$, therefore, $P_1((u, v), (x, y)) = 1$. Since P_1 obeys Item 4 of Lemma 4.5.2, therefore, one of the following holds: (a) there exists a triangle-free path from (u, v) to (x, y) in O , or (b) for some $a \in \{1, 2\}$, $P_{a1}((u, v), (x, y)) = 1$, or (c) there exists a $(z_1, z_2) \in E_O$ such that $P_1((u, v), (z_1, z_2)) = P_1((z_1, z_2), (x, y)) = 1$. If $P_1((u, v), (x, y)) = 1$ due to (a) or (b), then from steps 1 and 2 of Definition 4.5.3, $P'_1((u, v), (x, y)) = 1$. If $P_1((u, v), (x, y)) = 1$ due to (c), then there exists a $(z_1, z_2) \in E_O$ such that $P_1((u, v), (z_1, z_2)) = P_1((z_1, z_2), (x, y)) = 1$. This implies $\tau((u, v)) < \tau((z_1, z_2)) < \tau((x, y))$. Then from Equation (4.6), $P'_1((u, v), (z_1, z_2)) = 1$ (because $\tau((z_1, z_2)) < \tau((x, y))$), and $P'_1((z_1, z_2), (x, y)) = 1$ (because $\tau((u, v)) < \tau((z_1, z_2))$). But, then $P'_1((u, v), (x, y)) = 1$, updated at step 3 of Definition 4.5.3. This implies that for all $(u, v), (x, y) \in E_O$, if $P_1((u, v), (x, y)) = 1$, then $P'_1((u, v), (x, y)) = 1$. This completes the proof of Claim 4.6.36. $\square$

Claims 4.6.35 and 4.6.36 proves $P_1 = P'_1$.

We now prove $P_2 = P'_2$. Since (O, P_1, P_2) is a shadow of M , therefore, according to Definition 4.4.1, the domain of P_2 is $E_O \times V_O$, and the co-domain of P_1 is $\{0, 1\}$. Also, from Line 1 of Algorithm 6, since $(P'_1, P'_2) = \text{TFP}(O)$, therefore, based on Algorithm 5, the domain of P'_2 is $E_O \times V_O$, and the co-domain of P'_1 is $\{0, 1\}$. This implies that both P_2 and P'_2 have the same domain and co-domain. Thus, to prove $P_2 = P'_2$, it is sufficient to show that (a) for all $((u, v), w) \in E_O \times V_O$, if $P_2((u, v), w) = 1$, then $P'_2((u, v), w) = 1$, and (b) for all $((u, v), w) \in E_O \times V_O$, if $P'_2((u, v), w) = 1$, then $P_2((u, v), w) = 1$. The proof of (a) is similar to Claim 4.6.35, and the proof of (b) is similar to Claim 4.6.36. For completeness, we provide the proof.

Claim 4.6.37. $\forall ((u, v), w) \in E_O \times V_O$, if $P'_2((u, v), w) = 1$ then $P_2((u, v), w) = 1$.

Proof. From Definition 4.5.3, P'_2 reaches its final version after following steps 1, 2, and 3 of Definition 4.5.3.

Firstly, we demonstrate that at the end of step 1, for any $((u, v), w) \in E_O \times V_O$, if $P'_2((u, v), w) = 1$, then $P_2((u, v), w) = 1$. From Line 1 of Algorithm 6, we have $(P'_1, P'_2) = \text{TFP}(O)$. Therefore, from Lemma 4.6.18, if $P'_2((u, v), w) = 1$, then $v \neq w$, and there exists a triangle-free path from (u, v) to w in O . Since P_2 satisfies Item 5 in Lemma 4.5.2, from Item 5a, we have $P_2((u, v), w) = 1$. This implies that at the end of the initialization step (i.e., step 1) of Definition 4.5.3, if $P'_2((u, v), w) = 1$, then $P_2((u, v), w) = 1$.

Next, we demonstrate that if at step 2 (update 1) of Definition 4.5.3, for any $((u, v), w) \in E_O \times V_O$, P'_2 is updated with $P'_2((u, v), w) = 1$, then $P_2((u, v), w) = 1$.

Suppose for some $((u, v), w) \in E_O \times V_O$, P'_2 is updated with $P'_2((u, v), w) = 1$ at step 2 of Definition 4.5.3. From Definition 4.5.3, this implies that $v \neq w$ and for some $a \in \{1, 2\}$, $P_{a2}((u, v), w) = 1$. Since P_2 satisfies Lemma 4.5.2, from Item 5b of Lemma 4.5.2, we have $P_2((u, v), w) = 1$. This implies that if for some $((u, v), w) \in E_O \times V_O$, P'_2 is updated with $P'_2((u, v), w) = 1$ at step 2, then $P_2((u, v), w) = 1$. This further implies that at the start of step 3, for any $((u, v), w) \in E_O \times V_O$, if $P'_2((u, v), w) = 1$, then $P_2((u, v), w) = 1$.

Suppose while running step 3, if at some iteration, for some $((u, v), w) \in E_O \times V_O$, we update P'_2 with $P'_2((u, v), w) = 1$. Then $P_2((u, v), w) = 1$ provided that before that iteration, for all $((u', v'), w') \in E_O \times V_O$, if $P'_2((u', v'), w') = 1$, then $P_2((u', v'), w') = 1$. From Definition 4.5.3, if at step 3 of Definition 4.5.3, for any $((u, v), w) \in E_O \times V_O$, P'_2 is updated with $P'_2((u, v), w) = 1$, then $v \neq w$, and there exists a $(z_1, z_2) \in E_O$ such that $P'_1((u, v), (z_1, z_2)) = P'_2((z_1, z_2), w) = 1$. Since we have already shown that $P_1 = P'_1$, therefore, $P_1((u, v), (z_1, z_2)) = 1$. Also, since at the start of the iteration, we have $P'_2((z_1, z_2), w) = 1$, therefore, from our assumption, $P_2((z_1, z_2), w) = 1$. Since P_2 follows Item 5 of Lemma 4.5.2, from Item 5c of Lemma 4.5.2, we have $P_2((u, v), w) = 1$. This proves that even at the end of step 3, for all $((u, v), w) \in E_O \times V_O$, if $P'_2((u, v), w) = 1$ then $P_2((u, v), w) = 1$. This completes the proof of Claim 4.6.37. $\square$

Claim 4.6.38. $\forall ((u, v), w) \in E_O \times V_O$, if $P_2((u, v), w) = 1$ then $P'_2((u, v), w) = 1$.

Proof. We show that for all $((u, v), w) \in E_O \times V_O$, if $P_2((u, v), w) = 1$ then $P'_2((u, v), w) = 1$. Let us construct a directed acyclic graph (DAG) D such that $V_D = E_O$, and for $(u, v), (x, y) \in E_O$, $(u, v) \rightarrow (x, y) \in E_D$ if, and only if, $P_1((u, v), (x, y)) = 1$. We have shown in the proof of Claim 4.6.36 that D is a DAG (see the first two paragraphs of the proof of Claim 4.6.36 for the proof that D is a DAG). Let τ be an ordering on the vertices of D such that if $(u, v) \rightarrow (x, y) \in E_D$ then $\tau((u, v)) < \tau((x, y))$ (existence of τ is because D is a DAG). If possible, suppose there exists a $((u, v), w) \in E_O \times V_O$ such that $P_2((u, v), w) = 1$ and $P'_2((u, v), w) = 0$. For w , pick such a (u, v) which is highest in τ , i.e.,

$$\forall (u', v') \in E_O \text{ such that } \tau((u, v)) < \tau((u', v')), \text{ if } P_2((u', v'), w) = 1 \text{ then } P'_2((u', v'), w) = 1. \quad (4.7)$$

Since P_2 obeys Item 5 of Lemma 4.5.2, therefore, one of the following holds: (a) there exists a triangle-free path from (u, v) to w in O , or (b) for any $a \in \{1, 2\}$, $P_{a2}((u, v), w) = 1$, and $v \rightarrow u \notin O$, or (c) there exists a $(z_1, z_2) \in E_O$ such that $P_1((u, v), (z_1, z_2)) = P_2((z_1, z_2), w) = 1$. If $P_2((u, v), w) = 1$ due to (a) or (b) then from Items 1 and 2 of Definition 4.5.3, $P'_2((u, v), w) = 1$. And, if $P_2((u, v), w) = 1$ due to (c) then there exists a $(z_1, z_2) \in E_O$ such that $P_1((u, v), (z_1, z_2)) = P_2((z_1, z_2), w) = 1$. Since, we have shown that $P_1 = P'_1$, therefore, $P'_1((u, v), (z_1, z_2)) = P_1((u, v), (z_1, z_2)) = 1$. Also, since $P_1((u, v), (z_1, z_2)) = 1$, therefore, $\tau((u, v)) < \tau((z_1, z_2))$. Then from Equation (4.7), $P'_2((z_1, z_2), w) = 1$. But, then $P'_2((u, v), w) = 1$, updated at step 3 of Definition 4.5.3. This implies that for all $((u, v), w) \in E_O \times V_O$, if $P_2((u, v), w) = 1$ then $P'_2((u, v), w) = 1$. This completes the proof of Claim 4.6.38. $\square$

Claims 4.6.37 and 4.6.38 proves $P_2 = P'_2$. This implies $(P_1, P_2) = (P'_1, P'_2) = \text{DPF}(O, P_{11}, P_{12}, P_{21}, P_{22})$. This completes the proof of Lemma 4.5.4. $\square$

4.6.4 Proof of Lemma 4.5.8

To prove Lemma 4.5.8, we need a modified version of the LBFS algorithm of Rose et al. [1976]. In the next subsection, we introduce a modified version of the LBFS algorithm. We will also see some properties of the new LBFS algorithm. All these results will be

used in the proof of Lemma 4.5.8.

4.6.4.1 Lexicographical breadth-first search (LBFS) algorithm

In this section, we present a modified version of the LBFS algorithm. Let H be an undirected graph, with H_1 and H_2 as two induced subgraphs of H such that $H = H_1 \cup H_2$, and $I = V_{H_1} \cap V_{H_2}$ is a vertex separator of H . For each $a \in \{1, 2\}$, let S_a be a subset of V_{H_a} such that $I \subseteq S_a$, M_a an MEC of H_a , and (O_a, P_{a1}, P_{a2}) the shadow of M_a on $S_a \cup N(S_a, H_a)$, i.e., $\text{shadow}(M_a, S_a \cup N(S_a, H_a)) = (O_a, P_{a1}, P_{a2})$. Let $(O, P_1, P_2) \in \text{shadow}(H[S_1 \cup S_2 \cup N(S_1 \cup S_2, H)])$ such that $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$ (i.e., (O, P_1, P_2) is an extension of $(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$; refer to Definition 4.5.6 for the definition of extension).

Algorithm 7: LBFS($\mathcal{C}, O$): A modified version of the Lexicographic BFS (Rose et al. [1976]). If the algorithm is executed with O as a null graph the algorithm performs a normal LBFS.

Input : (a) An undirected connected chordal graph $\mathcal{C}$, and (b) a chain graph O such that there exist three undirected graphs H, H_1 and H_2 such that

- (i) $H = H_1 \cup H_2$
- (ii) $V_{H_1} \cap V_{H_2} = I$ is a vertex separator of H ,
- (iii) there exist $S_1 \in V_{H_1}$ and $S_2 \in V_{H_2}$ such that $I = V_{S_1} \cap V_{S_2}$,
- (iv) M_1 and M_2 are MECs of H_1 and H_2 , respectively,
- (v) for each $a \in \{1, 2\}$, (O_a, P_{a1}, P_{a2}) is the shadow of M_a on $S_a \cup N(S_a, H_a)$,
- (vi) $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$, and
- (vii) $\mathcal{C}$ is an undirected connected component of M_1 or M_2 .

Output: an LBFS ordering τ of $\mathcal{C}$ that is consistent with O

```

1  $\mathcal{S} \leftarrow$  sequence of sets initialized with  $(V_{\mathcal{C}})$ 
2  $\tau \leftarrow$  empty list
3 while  $\mathcal{S}$  is non-empty do
4    $X \leftarrow$  first non-empty set of  $\mathcal{S}$ 
5    $v \leftarrow$  arbitrary canonical source node of  $\mathcal{U}_M(\mathcal{C}, O)[X]$ 
6   Add vertex  $v$  to the end of  $\tau$ .
7   Replace the set  $X$  in the sequence  $\mathcal{S}$  by the set  $X \setminus \{v\}$ .
8    $N(v, \mathcal{C}) \leftarrow \{x | v - x \in E_{\mathcal{C}}\}$ 
9   Denote the current  $\mathcal{S}$  by  $(S_1, \dots, S_k)$ 
10  Replace each  $S_i$  by  $S_i \cap N(v, \mathcal{C}), S_i \setminus N(v, \mathcal{C})$ .
11  Remove all empty sets from  $\mathcal{S}$ .
12 end
13 return  $\tau$ 

```

We introduce a modified LBFS algorithm, Algorithm 7. This algorithm takes as input (a) O and (b) $\mathcal{C}$, an undirected connected component of M_a for $a \in \{1, 2\}$. Algorithm 7 returns an LBFS ordering τ of $\mathcal{C}$ that obeys O , i.e., for any $u, v \in V_{\mathcal{C}}$ if $u \rightarrow v \in O$ then $\tau(u) < \tau(v)$.

Algorithm 7 is a variation of the LBFS algorithm introduced by Rose et al. [1976]. At Line 5, instead of selecting an arbitrary node v of X , Algorithm 7 chooses a canonical source node (Definition 4.6.15) in $\mathcal{U}_M(\mathcal{C}, O)[X]$, where $\mathcal{U}_M(\mathcal{C}, O)$ is the Markov union (Definition 4.6.7) of $\mathcal{C}$ and O . The crucial point to verify is the existence of a canonical source node in $\mathcal{U}_M(\mathcal{C}, O)[X]$.

Claim 4.6.39. *There always exists a canonical source vertex in $\mathcal{U}_M(\mathcal{C}, O)[X]$, where $X \subseteq V_{\mathcal{C}}$.*

Proof. Suppose $\mathcal{U}_M(\mathcal{C}, O)[X]$ does not contain any canonical source vertices. Definition 4.6.16 implies that if a node u is not a canonical source vertex in $\mathcal{U}_M(\mathcal{C}, O)[X]$ then there exists a node v that certifies that u is not a canonical source vertex. And the certification is an edge $v \rightarrow v' \in \mathcal{U}_M(\mathcal{C}, O)[X]$ and a triangle-free path from (v, v') to u in $\mathcal{U}_M(\mathcal{C}, O)[X]$. If $\mathcal{U}_M(\mathcal{C}, O)[X]$ does not contain any canonical source nodes, then, since X has a finite number of nodes, there exists a sequence $(u_1, u_2, \dots, u_l, u_{l+1} = u_1)$ such that $u_1, u_2, \dots, u_l$ are distinct, and for each $1 \leq i \leq l$, u_i is a certificate of u_{i+1} . Now, we will prove the following claim, which is an important component in demonstrating that we will not obtain any such sequence. This, in turn, implies that there always exists a canonical source node in $\mathcal{U}_M(\mathcal{C}, O)[X]$.

Claim 4.6.40. *In $\mathcal{U}_M(\mathcal{C}, O)[X]$, suppose u, v , and w are three distinct vertices such that u is a certificate of v , and v is a certificate of w . Then, either $u \rightarrow v \in O$, or u is a certificate of w .*

Proof. Suppose in $\mathcal{U}_M(\mathcal{C}, O)[X]$, u is a certificate of v , and v is a certificate of w . Then, from Definition 4.6.16, since u is a certificate of v , there must exist a node $u' \in X$ such that $u \rightarrow u' \in \mathcal{U}_M(\mathcal{C}, O)[X]$, and there exists a triangle-free path $Q_1 = (x_1 = u, x_2 = u', \dots, x_l = v)$ from (u, u') to v in $\mathcal{U}_M(\mathcal{C}, O)[X]$. Similarly, since v is a certificate of w , there exists a node $v' \in X$ such that $v \rightarrow v' \in \mathcal{U}_M(\mathcal{C}, O)[X]$, and there exists a triangle-free path $Q_2 = (y_1 = v, y_2 = v', \dots, y_m = w)$ from (v, v') to w in $\mathcal{U}_M(\mathcal{C}, O)[X]$.

We first show that for any triangle-free path P in $\mathcal{U}_M(\mathcal{C}, O)[X]$, P is a triangle-free path in $\mathcal{C}$. From Definition 4.6.7, if $u \rightarrow v \in \mathcal{U}_M(\mathcal{C}, O)$, then either $u \rightarrow v \in \mathcal{C}$ or

$u \rightarrow v \in O$. Since $\mathcal{C}$ is an undirected graph (recall that $\mathcal{C}$ is an undirected connected component of M_a), therefore, $u \rightarrow v \in O$. From the construction, for $Y = V_{\mathcal{C}} \cap V_O$, $\text{skeleton}(\mathcal{C}[Y]) = \text{skeleton}(O[Y])$. Since $X \subseteq V_{\mathcal{C}}$ ($X \in \mathcal{S}$, and in Algorithm 7, the elements of $\mathcal{S}$ are the subsets of $V_{\mathcal{C}}$), therefore, if $u \rightarrow v \in \mathcal{U}_M(\mathcal{C}, O)[X]$, then $u \rightarrow v \in O$ and $u - v \in \mathcal{C}$. This further implies that each triangle-free path in $\mathcal{U}_M(\mathcal{C}, O)[X]$ is a triangle-free path in $\mathcal{C}$.

The above discussion implies that Q_1 and Q_2 are triangle-free paths in $\mathcal{C}$ and $u \rightarrow u', v \rightarrow v' \in O$.

If $l = 2$, i.e., the length of Q_1 is one, then $u' = v$, and $u \rightarrow v \in O$, and we are done.

Suppose $l > 2$. We show that there exists a triangle-free path from (u, u') to w in $\mathcal{U}_M(\mathcal{C}, O)[X]$. As shown above, Q_1 and Q_2 are triangle-free paths in $\mathcal{C}$. Since $\mathcal{C}$ is an undirected connected component of an MEC M_a , therefore, from Item 2 of Theorem 1.1.24, $\mathcal{C}$ is chordal. Then from Observation 4.6.11, Q_1 and Q_2 are chordless paths of $\mathcal{C}$. Pick the least $i < l$ such that for some $1 \leq j \leq m$, $x_i - y_j \in \mathcal{C}$. After this, pick the highest j such that $x_i - y_j \in \mathcal{C}$. The existence of such an edge is due to $x_{l-1} - y_1 \in \mathcal{C}$ (in Q_1 , $x_{l-1} - y_1$ is an edge as $x_l = y_1$).

If $i \neq 1$, then $Q = (x_1 = u, x_2 = u', \dots, x_i, y_j, \dots, y_m = w)$ is a triangle-free path from (u, u') to w in $\mathcal{C}$. We now show that Q is also a triangle-free path in $\mathcal{U}_M(\mathcal{C}, O)[X]$.

Since Q_1 and Q_2 are triangle-free paths in $\mathcal{U}_M(\mathcal{C}, O)[X]$, to show that Q is a triangle-free path in $\mathcal{U}_M(\mathcal{C}, O)[X]$, the only thing we have to show is $y_j \rightarrow x_i \notin O$. If both x_i and y_j are not nodes of V_O , then there cannot be $y_j \rightarrow x_i \in O$. Suppose $y_j, x_i \in V_O$. Since Q is a triangle-free path in $\mathcal{C}$, the subpath $Q' = (x_1 = u, x_2 = u', \dots, x_i, y_j)$ of Q is also a triangle-free path in $\mathcal{C}$. Also, the length of Q' is at least two (since $i \neq 1$). Since $\mathcal{C}$ is an undirected connected component of M_a , a triangle-free path in $\mathcal{C}$ is a triangle-free path in M_a . This implies $P_{a1}((u, u'), (x_i, y_j)) = 1$. Since (O, P_1, P_2) is an extension of $(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$, from Item 3 of Definition 4.5.6, $x_i \rightarrow y_j \in O$ (because $P_{a1}((u, u'), (x_i, y_j)) = 1$ and $u \rightarrow u' \in O$). This shows that we get the required triangle-free path Q , from (u, u') to w , in $\mathcal{U}_M(\mathcal{C}, O)[X]$.

Thus, the only possibility that remains is $i = 1$. We show that this case cannot occur. Suppose, if possible, that $i = 1$. Recall that $l > 2$. Pick the least j such that $x_1 - y_j \in \mathcal{C}$. Since Q_1 is a chordless path in $\mathcal{C}$ and $y_1 = x_l$, j must be greater than 1. Then $Q = (y_1 =$

$v, y_2 = v', \dots, y_j, x_1 = u$) is a chordless path in $\mathcal{C}$, from (v, v') to u . As $\mathcal{C}$ is an undirected connected component of M_a , and Q_1 and Q are triangle-free paths in $\mathcal{C}$ from (u, u') to v and from (v, v') to u , respectively, therefore, $P_{a2}((u, u'), v) = P_{a2}((v, v'), u) = 1$. Since (O, P_1, P_2) is an extension of $(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$, therefore, from Item 3 of Definition 4.5.6, $v' \rightarrow v \in O$ (as $u \rightarrow u' \in O$ and $P_{a2}((u, u'), v) = P_{a2}((v, v'), u) = 1$). But, this is a contradiction, as from our assumption, $v \rightarrow v' \in O$. This shows that $i = 1$ does not occur. This completes the proof of Claim 4.6.40. $\square$

We now prove the following claim to complete the proof of Claim 4.6.39.

Claim 4.6.41. *There cannot exist a sequence $(u_1, u_2, \dots, u_l, u_{l+1} = u_1)$ such that for each $1 \leq i \leq l$, u_i is a certificate of u_{i+1} .*

We will prove Claim 4.6.41 using induction on l . From Observation 4.6.17, we know that self-certification is not possible, implying that $l \geq 2$.

Base Case: $l = 2$. Assume, for the sake of contradiction, that Claim 4.6.41 is false for $l = 2$. In other words, there exist nodes u_1 and u_2 such that u_1 is a certificate of u_2 , and u_2 is a certificate of u_1 . This implies the existence of edges $u_1 - v_1$ and $u_2 - v_2$ in $\mathcal{C}$ such that $u_1 \rightarrow v_1$ and $u_2 \rightarrow v_2$ are in O . Additionally, there exist triangle-free paths Q_1 from (u_1, v_1) to u_2 and Q_2 from (u_2, v_2) to u_1 in $\mathcal{C}$. Since $\mathcal{C}$ is an undirected connected component of the MEC M_a (for some $a \in \{1, 2\}$), $\mathcal{C}$ is an undirected chordal graph (from Item 2 of Theorem 1.1.24). Thus, Q_1 and Q_2 are triangle-free paths in M_a .

This implies that $v_1 \neq u_2$ and $v_2 \neq u_1$. If, for instance, $v_1 = u_2$, then by applying Corollary 4.6.10 to the undirected chordal graph $\mathcal{C}$, the adjacency of $v_1 = u_2$ and u_1 and the triangle-free path Q_2 from (u_2, v_2) to u_1 would imply $v_2 = u_1$ as well. Similarly, if $v_2 = u_1$, it would imply $v_1 = u_2$. Thus, we must have both $v_1 = u_2$ and $v_2 = u_1$, which leads to the existence of both $u_1 \rightarrow u_2$ and $u_2 \rightarrow u_1$ in O , a contradiction. Therefore, it follows that $v_1 \neq u_2$ and $v_2 \neq u_1$.

Given that (O, P_1, P_2) is an extension of $(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$, we can apply Item 3 of Definition 4.5.6. This reveals that $u_1 \rightarrow v_1 \in O$ together with the triangle-free paths Q_1 from (u_1, v_1) to u_2 (with $v_1 \neq u_2$) and Q_2 from (u_2, v_2) to u_1 (with $v_2 \neq u_1$) imply

$v_2 \rightarrow u_2 \in O$. This, however, contradicts the assumption that $u_2 \rightarrow v_2 \in O$. Thus, we have shown that Claim 4.6.41 holds for $l = 2$.

Induction: Assume that Claim 4.6.41 is true for $l = k - 1$, where $k \geq 3$. We will now use Claim 4.6.40 to prove Claim 4.6.41 for $l = k$.

Assume, for the sake of contradiction, that Claim 4.6.41 is not true for a sequence $(u_1, u_2, \dots, u_k, u_{k+1} = u_1)$. From Claim 4.6.40, for each $1 \leq i \leq k - 1$, either u_i is a certificate for u_{i+2} or $u_i \rightarrow u_{i+1} \in O$, and for $i = k$, either u_k is a certificate for u_2 or $u_k \rightarrow u_1 \in O$. If, for any $1 \leq i \leq k - 1$, u_i is a certificate for u_{i+2} , it would create a sequence $(u_1, u_2, \dots, u_i, u_{i+2}, \dots, u_k, u_{k+1} = u_1)$ of $k - 1$ distinct nodes that violates our induction hypothesis. Similarly, if u_k is a certificate for u_2 , it would create another sequence $(u_2, u_3, \dots, u_k, u_2)$ of $k - 1$ distinct nodes that violates our induction hypothesis. Thus, the only possibility is that for each $1 \leq i \leq k$, $u_i \rightarrow u_{i+1} \in O$. However, this leads to the presence of a directed cycle in O , which is a contradiction because O is a chain graph (as per Item 1 of Definition 4.2.1). Therefore, we have proven Claim 4.6.41 for all l by induction.

Claim 4.6.41 implies that there always exists a canonical source vertex in $\mathcal{U}_M(\mathcal{C}, O)[X]$. This completes the proof of Claim 4.6.39. $\square$

The following observation and its corollaries show that the output ordering is consistent with O .

Observation 4.6.42. *Let $\mathcal{C}$ be an undirected connected component of M_a for some $a \in \{1, 2\}$. Suppose Algorithm 7 is called for $\mathcal{C}$ and O . And, at the beginning of an iteration of the while loop (lines 3-12 of Algorithm 7), let $\tau = (a_1, a_2, \dots, a_r)$ and $\mathcal{S} = (S_1, S_2, \dots, S_l)$. Then, for any triangle-free path $P = (u_1, u_2, \dots, u_m)$ of $\mathcal{C}$ such that $u_1 \rightarrow u_2 \in O$, the following statements hold for all $1 \leq i < j \leq m$,*

1. *If $u_j \in \tau$, then $u_i \in \tau$ and u_i arrived in τ before u_j .*
2. *If $u_j \in S_x$ for some $1 \leq x \leq l$, then either $u_i \in \tau$, or $u_i \in S_y$ for $y \leq x$.*
3. *If $u_1 \in \tau$, then at most one vertex of P is in S_1 .*

Before proving Observation 4.6.42, we go through the following claim, which is a necessary ingredient for the proof of Observation 4.6.42.

Claim 4.6.43. *If $P = (u_1, u_2, \dots, u_l)$ is a triangle-free path in $\mathcal{C}$ and $u_1 \rightarrow u_2 \in O$ then P is a triangle-free path in $\mathcal{U}_M(\mathcal{C}, O)$.*

Proof. From the construction, $\text{skeleton}(\mathcal{C}[V_{\mathcal{C}} \cap V_O]) = \text{skeleton}(O[V_{\mathcal{C}} \cap V_O])$. This implies that if P is a triangle-free path in $\mathcal{C}$ then P is a triangle-free path in $\text{skeleton}(\mathcal{U}_M(\mathcal{C}, O))$ as well. Suppose P is not a triangle-free path in $\mathcal{U}_M(\mathcal{C}, O)$ then there must exist an i such that $u_i \leftarrow u_{i+1} \in \mathcal{U}_M(\mathcal{C}, O)$. Since $\mathcal{C}$ is an undirected graph, if there exists a directed edge in $\mathcal{U}_M(\mathcal{C}, O)$ then the directed edge must belong to O . This implies $u_i \leftarrow u_{i+1} \in O$. From our assumption, $u_1 \rightarrow u_2 \in O$. This implies $i > 1$. From our assumption, P is a triangle-free path in $\mathcal{C}$, an undirected connected component of M_a for some $a \in \{1, 2\}$. Also, from the construction, $V_{M_a} \cap V_O = V_{O_a}$. This implies $u_1, u_2, u_i, u_{i+1} \in V_{O_a}$. Since O_a is an induced subgraph of M_a , $(u_1, u_2), (u_i, u_{i+1}) \in E_{O_a}$. This implies $P_{a1}((u_1, u_2), (u_i, u_{i+1})) = 1$, as a subpath of P from (u_1, u_2) to (u_i, u_{i+1}) is also a triangle-free path in M_a , and $(u_1, u_2) \neq (u_i, u_{i+1})$ because $i > 1$. Since $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$, from Item 3b of Definition 4.5.6, $u_i \rightarrow u_{i+1} \in O$, a contradiction. This implies P is a triangle-free path in $\mathcal{U}_M(\mathcal{C}, O)$. $\square$

Proof of Observation 4.6.42. We will establish the proof through induction. Initially, at the start of the first iteration of the while loop in Algorithm 7, we have $\mathcal{S} = (V_{\mathcal{C}})$, and τ is an empty list. Consequently, Observation 4.6.42 is satisfied at this stage.

Now, let us assume that at the beginning of some iteration of the while loop, $\tau = (a_1, a_2, \dots, a_r)$ and $\mathcal{S} = (S_1, S_2, \dots, S_l)$, and that Observation 4.6.42 is upheld. We aim to demonstrate that, even at the end of this iteration of the while loop, Observation 4.6.42 continues to hold.

During the run of the iteration, we do the following: At Line 5, we select a canonical source node v from $\mathcal{U}_M(\mathcal{C}, O)[S_1]$. We remove vertex v from S_1 and add it to τ (lines 6-7). Subsequently, we update each set S_i as follows: S_i is replaced by $S_i \cap N(v, \mathcal{C})$ and $S_i \setminus N(v, \mathcal{C})$ (Line 10). This results in the modified $\mathcal{S} = (S_{11}, S_{12}, \dots, S_{l1}, S_{l2})$, where $S_{11} = (S_1 \setminus v) \cap N(v, \mathcal{C})$, $S_{12} = (S_1 \setminus v) \setminus N(v, \mathcal{C})$, and for $i > 1$, $S_{i1} = S_i \cap N(v, \mathcal{C})$, and $S_{i2} = S_i \setminus N(v, \mathcal{C})$. And, in the end, we remove the empty sets in $\mathcal{S}$ (Line 11).

We now demonstrate that if a triangle-free path $P = (u_1, u_2, \dots, u_m)$ satisfies $u_1 \rightarrow u_2 \in O$ and satisfies Observation 4.6.42 at the start of the iteration, it will continue to satisfy Items 1 to 3 of Observation 4.6.42 even at the end of this iteration. There are two possibilities: either P contains the picked node v (the canonical source node picked at step-5), or it does not. We go through both possibilities one by one.

1. Suppose that P contains the picked node v . Let $u_b = v$, for some $1 \leq b \leq m$.

There are two cases: either $b = 1$ (i.e., v is the first node of P), or $b > 1$.

- (a) Suppose $b = 1$. Since $u_1 = v$ has been picked from S_1 , this implies that at the start of the iteration, u_1 is in S_1 . Since, at the start of the iteration, P obeys Observation 4.6.42, Item 1 of Observation 4.6.42 implies that at the start of the iteration, none of the nodes of P are in τ .

At the end of the iteration, u_1 is in τ , and all other nodes of P remain in some set of $\mathcal{S}$. We will now demonstrate that for all possible pairs of nodes u_i and u_j , where $1 \leq i < j \leq m$, at the end of the iteration, u_i and u_j satisfy Items 1 and 2 of Observation 4.6.42. For $1 = i < j \leq m$, u_i and u_j satisfy Items 1 and 2 of Observation 4.6.42, as at the end of the iteration, u_i is in τ and u_j is in some set of $\mathcal{S}$.

Suppose $1 < i < j \leq m$. And, at the start of the iteration, u_i and u_j are in some set S_x and S_y of $\mathcal{S}$, respectively. Since at the start of the iteration, u_i and u_j satisfy Item 2 of Observation 4.6.42, we have $x \leq y$.

If $x < y$, then at the end of the iteration, the set in which u_i lies (either S_{x1} or S_{x2}) comes before the set in which u_j lies (either S_{y1} or S_{y2}). Thus, if $x < y$, then at the end of the iteration, u_i and u_j satisfy Items 1 and 2 of Observation 4.6.42.

Suppose $x = y$, i.e., at the start of the iteration, both u_i and u_j lie in the same set S_x of $\mathcal{S}$. Since P is a triangle-free path of a chordal graph $\mathcal{C}$, from Proposition 4.6.9, P is a chordless path. This implies that the only neighbor of $u_1 = v$ in P is u_2 . Therefore, if $i = 2$, then u_i is a neighbor of v and at the end of the iteration, it will move to set S_{x1} ; otherwise, u_i is not a neighbor of v and at the end of the iteration, it will move to S_{x2} . Since $1 < i < j$, j

cannot be equal to 2. This implies u_j cannot be a neighbor of v and at the end of the iteration, it will move to S_{x2} . This shows that at the end of the iteration, the set in which u_i lies does not come after the set in which u_j lies in $\mathcal{S}$. This shows that at the end of the iteration, u_i and u_j satisfy Items 1 and 2 of Observation 4.6.42.

We now show that at the end of the iteration, Item 3 of Observation 4.6.42 is also satisfied.

As shown above, at the end of the iteration, u_1 is in τ , and all the remaining nodes of P are in some set of $\mathcal{S}$. We have shown that at the end of the iteration, P obeys Items 1 and 2 of Observation 4.6.42. If at the end of the iteration, there does not exist any node of P that is in the first set of $\mathcal{S}$, then Item 3 of Observation 4.6.42 is satisfied. Suppose at the end of the iteration, there exists a node of P that lies in the first set of $\mathcal{S}$. Pick the highest i such that at the end of the iteration, u_i is in the first set of $\mathcal{S}$. If $i = 2$, then at the end of the iteration, only one node of P is in the first set of $\mathcal{S}$. This implies at the end of the iteration, Item 3 of Observation 4.6.42 is satisfied. Suppose $i > 2$, then, from Item 2 of Observation 4.6.42, at the end of the iteration, $u_2, u_3, \dots, u_i$ all are in the first set of $\mathcal{S}$. This further implies even at the start of the iteration, $u_2, u_3, \dots, u_i$ are in the same set of $\mathcal{S}$, say S_x . But, we know that among these nodes only u_2 is a neighbor of $u_1 = v$. This implies at the end of the iteration, u_2 must be in S_{x1} , and all the remaining nodes $u_3, \dots, u_i$ must be in S_{x2} . This contradicts our assumption that at the end of the iteration, $u_2, u_3, \dots, u_i$ are in the same set of $\mathcal{S}$. This implies $i > 2$ does not occur. This implies even at the end of the iteration, Item 3 of Observation 4.6.42 is obeyed. This implies that in all the possibilities of this case, Items 1 to 3 of Observation 4.6.42 are satisfied. We will now move on to the second case.

- (b) Suppose $b > 1$. Since $u_b = v$ has been picked from S_1 , this implies that at the start of the iteration, u_b is in S_1 . From Item 2 of Observation 4.6.42, it follows that at the start of the iteration, u_1 must be in either S_1 or in τ . If u_1 is in S_1 at the start of the iteration, then since u_b is also in S_1 , Item 2 of Observation 4.6.42 implies that at the start of the iteration, all the

nodes between u_1 and u_b in P (i.e., $u_1, u_2, u_3, \dots, u_{b-1}, u_b$) are in S_1 . This implies $Q = (u_1, u_2, \dots, u_b)$ is a triangle-free path in $\mathcal{C}[S_1]$. But, then, from Claim 4.6.43, u_b is not a canonical source vertex in $\mathcal{U}_M(\mathcal{C}, O)[S_1]$, due to the existence of Q and $u_1 \rightarrow u_2 \in O$. But, this is a contradiction, as we have picked u_b at Line 5 of Algorithm 7 because it is a canonical source vertex in $\mathcal{U}_M(\mathcal{C}, O)[S_1]$. Therefore, u_1 cannot be in S_1 , but must be in τ . Then, Item 3 of Observation 4.6.42 implies that at the start of the iteration, the only node of P in S_1 is $u_b = v$.

From Item 2 of Observation 4.6.42, it follows that at the start of the iteration, $u_1, u_2, \dots, u_{b-1}$ are in τ . And, at the end of the iteration, u_b also joins τ .

We will now show that, for all possible pairs of nodes u_i and u_j , where $1 \leq i < j \leq m$, at the end of the iteration, u_i and u_j obey Items 1 and 2 of Observation 4.6.42.

If at the start of the iteration, both u_i and u_j are in τ , then even at the end of the iteration, they remain in τ and obey Items 1 and 2. If at the start of the iteration, u_i is in τ and u_j is in some set of S , then at the end of the iteration, either (a) both u_i and u_j are in τ (this case occurs when $u_j = u_b = v$) and u_i joins τ before u_j , or (b) u_i is in τ , and u_j is in some set of S (this case occurs when $j \neq b$). Both possibilities obey Items 1 and 2.

Suppose at the start of the iteration, both u_i and u_j are in some sets S_x and S_y of S , respectively. Since at the start of the iteration, u_i and u_j obey Observation 4.6.42, $x \leq y$.

If $i = b$, i.e., $u_i = v$, then at the end of the iteration, the pair u_i and u_j obey Items 1 and 2, as at the end of the iteration, u_i joins τ and u_j remains in some set of S .

Suppose $i \neq b$. If $x < y$, then even at the end of the iteration, the set in which u_i lies (either S_{x1} or S_{x2}) comes before the set in which u_j lies (either S_{y1} or S_{y2}). Suppose $x = y$. Then, at the start of the iteration, both u_i and u_j are in the same set S_x . Similar to the previous case, since P is a triangle-free path of a chordal graph $\mathcal{C}$, from Proposition 4.6.9, P is a chordless path. Therefore,

there can be at most two neighbors of $v = u_b$ in P : one is u_{b-1} , and another is u_{b+1} (if $b < m$). As discussed earlier, at the start of the iteration, u_{b-1} is in τ . This, in turn, implies that either none of u_i and u_j are neighbors of $u_b = v$, or only u_i is a neighbor of v (this scenario arises when $i = b + 1$).

If neither is a neighbor of v , then at the end of the iteration, both are in S_{x2} . If u_i is a neighbor of v , then at the end of the iteration, $u_i \in S_{x1}$, and $u_j \in S_{x2}$. This implies the set in which u_i lies does not come after the set in which u_j lies in $\mathcal{S}$. This shows that, in all the possibilities of this case, Items 1 and 2 of Observation 4.6.42 are obeyed.

Similar to the previous case, we can show that at the end of the iteration, the only possible node of P which can be in the first set of P is u_{b+1} . This validates Item 3 of Observation 4.6.42. Thus, we show that in this case, Items 1 to 3 of Observation 4.6.42 are obeyed. We now move to the next possibility.

2. Suppose P does not contain v . This implies that no node of P moves to τ in this iteration. We will now demonstrate that, at the end of the iteration, for all pairs of nodes u_i and u_j in P such that $1 \leq i < j \leq m$, Observation 4.6.42 is satisfied. To prove Item 1 of Observation 4.6.42, we assume that at the end of the iteration, u_j is in τ . To prove Item 2 of Observation 4.6.42, we assume that at the end of the iteration, u_j is in some set of $\mathcal{S}$. Finally, we prove Item 3 of Observation 4.6.42.

Suppose at the end of the iteration, u_j is in τ . This implies that even at the start of the iteration, $u_j \in \tau$, as no node of P moved to τ in this iteration. Since at the start of the iteration, P obeys Observation 4.6.42, according to Item 1 of Observation 4.6.42, u_i has joined τ before u_j . Hence, at the end of the iteration, Item 1 of Observation 4.6.42 is upheld.

Suppose at the end of the iteration, u_j is in some set of $\mathcal{S}$. Then, even at the start of the iteration, u_j must be in some set, say S_y , of $\mathcal{S}$. Since at the start of the iteration, P obeys Observation 4.6.42, from Item 2 of Observation 4.6.42, at the start of the iteration, either u_i is in τ or u_i is in some set, say S_x , of $\mathcal{S}$ such that $x \leq y$.

If at the start of the iteration, u_i is in τ then even at the end of the iteration, u_i is in

τ , and u_i and u_j obey Item 2 of Observation 4.6.42. Now, suppose at the start of the iteration, u_i is in some set S_x of $\mathcal{S}$.

If $x < y$, then at the end of the iteration, u_i is in a set of $\mathcal{S}$ (either S_{x1} or S_{x2}) that comes before the set in which u_j belongs (either S_{y1} or S_{y2}), as both S_{x1} and S_{x2} come before S_{y1} and S_{y2} . This implies that Observation 4.6.42 is obeyed in this case.

If $x = y$, then either both u_i and u_j belong to the same set S_{x1} (if both are neighbors of v) or both u_i and u_j belong to the same set S_{x2} (if neither of them is a neighbor of v) or $u_i \in S_{x1}$ and $u_j \in S_{x2}$ (if u_i is a neighbor of v , and u_j is not a neighbor of v) or $u_j \in S_{x1}$ and $u_i \in S_{x2}$ (if u_j is a neighbor of v , and u_i is not a neighbor of v). The first three possibilities obey Observation 4.6.42, as in the first and second cases, u_i and u_j both are in the same set of $\mathcal{S}$, and in the third case, u_i is in S_{x1} , which comes before S_{x2} in which u_j belongs. We now show that the last possibility cannot occur.

Suppose u_j is a neighbor of v , and u_i is not a neighbor of v . Pick the least k such that u_k comes after u_i in P , and u_k is a neighbor of v . Such a u_k exists due to the presence of u_j .

Then, $Q = (u_1, u_2, \dots, u_k, v)$ is a triangle-free path in $\mathcal{C}$ such that $u_1 \rightarrow u_2 \in O$.

Now, from our induction hypothesis, at the start of the iteration, all the triangle-free paths of $\mathcal{C}$ obey Observation 4.6.42. This implies Q also obeys Observation 4.6.42 at the start of the iteration.

v has been picked in this iteration implying that at the start of the iteration, $v \in S_1$. From Item 2 of Observation 4.6.42, at the start of the iteration, all other nodes of Q are either in τ or in S_1 . Since u_i is not in τ at the start of the iteration, this implies that at the start of the iteration, u_i is in S_1 along with v . Then, Item 3 of Observation 4.6.42 implies that at the start of the iteration, all the nodes of Q are in S_1 . But then, v cannot be a canonical source node in $\mathcal{U}_M(\mathcal{C}, O)[S_1]$ due to the existence of a triangle-free path Q in $\mathcal{U}_M(\mathcal{C}, O)[S_1]$ and $u_1 \rightarrow u_2 \in O$. The fact that Q is a triangle-free path in $\mathcal{U}_M(\mathcal{C}, O)$ follows from Claim 4.6.43. This implies this possibility cannot occur. This shows that in this case, Items 1 and 2 of

Observation 4.6.42 are obeyed.

We now show that at the end of the iteration, P also obeys Item 3 of Observation 4.6.42. Since in this iteration, the picked node v is not a node of P , this implies that the set of nodes of P in τ at the start of the iteration is the same as the nodes of P in τ at the end of the iteration. If none of the nodes of P are in τ , then Item 3 is satisfied.

Suppose a node of P is in τ . Since at the end of the iteration, P obeys Items 1 and 2 of Observation 4.6.42, there exists some $i \geq 1$ such that at the end of the iteration, $u_1, u_2, \dots, u_i$ are in τ and each of the remaining nodes of P is in some set of $\mathcal{S}$. If at the end of the iteration, no node of P is in the first set of $\mathcal{S}$, then Item 3 is obeyed.

Suppose at the end of the iteration, there exists a node of P in the first set of $\mathcal{S}$. Pick the highest j such that at the end of the iteration, u_j is in the first set of $\mathcal{S}$. Since at the end of the iteration, P obeys Items 1 and 2 of Observation 4.6.42, at the end of the iteration, there must be $u_{i+1}, u_{i+2}, \dots, u_j$ all are in the first set of $\mathcal{S}$. If $j = i + 1$, then Item 3 is obeyed, as in that case there exists only a single node of P in the first set of $\mathcal{S}$. We now show that $j = i + 1$.

Suppose $j > i + 1$. Then, at the start of the iteration when u_i has been picked and has been moved to τ , $u_{i+1}, u_{i+2}, \dots, u_j$ must be in the same set, say S_x , of $\mathcal{S}$ (because from Algorithm 7, once two nodes of $\mathcal{C}$ gets separated and move to two different set of $\mathcal{S}$, they never unite). Since P is a triangle-free path of a chordal graph $\mathcal{C}$, from Proposition 4.6.9, P is a chordless path. This implies that only u_{i+1} is a neighbor of u_i and rest of the nodes $u_{i+2}, u_{i+3}, \dots, u_j$ are not the neighbors of u_i in P . This further implies that at the end of the iteration in which u_i has been picked at step 5 of Algorithm 7, u_{i+1} moves to the set S_{x1} of $\mathcal{S}$ and u_j moves to S_{x2} of $\mathcal{S}$. This further implies that even at the end of the current iteration, u_{i+1} and u_j are in different sets of $\mathcal{S}$, a contradiction. This implies $j > i + 1$ cannot occur. This shows that at the end of the iteration, Item 3 of Observation 4.6.42 is obeyed.

This demonstrates that in all scenarios, at the end of the iteration, Items 1 to 3 of Observation 4.6.42 are true. Thus, Observation 4.6.42 is validated. $\square$

Observation 4.6.42 implies the following corollaries:

Corollary 4.6.44. *For input $\mathcal{C}$ and O , let τ be an LBFS ordering returned by Algorithm 7. If $(u_1, u_2, \dots, u_l)$ is a chordless path in $\mathcal{C}$ such that $u_1 \rightarrow u_2 \in O$, then $\tau(u_1) < \tau(u_2) < \dots < \tau(u_l)$.*

Proof. From Observation 4.6.42, for $1 \leq i < j \leq l$, u_i enters τ before u_j . Suppose, for contradiction, that u_j enters τ before u_i . Then, at the start of the next iteration, Item 2 would not hold, which leads to a contradiction. $\square$

Corollary 4.6.45. *For input $\mathcal{C}$ and O , let τ be an LBFS ordering returned by Algorithm 7. For $a - b \in \mathcal{C}$, if $a \rightarrow b \in O$, then $\tau(a) < \tau(b)$.*

Proof. Corollary 4.6.45 is a direct consequence of Corollary 4.6.44 when $l = 2$. $\square$

As discussed in Chapter 1, an LBFS ordering τ (Definition 1.1.13) of a chordal graph G is a perfect elimination ordering (Definition 1.1.12) of G , i.e., for $u - v, w - v \in E_G$, if $\tau(u) < \tau(v)$ and $\tau(w) < \tau(v)$ (both u and w come before v in τ), then $u - w \in E_G$. For the completeness, we show that the ordering τ returned by Algorithm 7 is a perfect elimination ordering of $\mathcal{C}$, where $\mathcal{C}$ is the input undirected connected chordal graph. The following two claims will help us in establishing this.

Claim 4.6.46. *At the end of any iteration of the while loop (lines 3-12) of Algorithm 7, the induced subgraph of $\mathcal{C}$ on the nodes in τ is a connected graph.*

Proof. We prove this by induction. At the end of the first iteration of the while loop, τ consists of a single node. Thus, Claim 4.6.46 holds true at the end of the first iteration.

At each iteration of the while loop, we add one node in τ (at Line 6). For some $i > 1$, assume that at the start of the i th iteration, $\tau = (a_1, a_2, \dots, a_{i-1})$ and $\mathcal{S} = (S_1, S_2, \dots, S_l)$. Let $v \in S_1$ be the node picked at Line 5 of Algorithm 7, and added to τ at Line 6. We show that v is adjacent to at least one node in τ .

Suppose, for contradiction, that v is not adjacent to any node in τ . From the construction of S_1 , any node x in τ is either adjacent to all nodes in S_1 or to none of them (as per Line 10 of Algorithm 7). This implies that no node in S_1 is adjacent to any node in τ . We now demonstrate that $l = 1$, i.e., S_1 is the only set in $\mathcal{S}$.

Assume $l > 1$. Pick a node $y \in S_2$. Initially, at the start of the first iteration, both v and y were in the same set of $\mathcal{S}$ (from Line 1 of Algorithm 7). Nodes in the same set

of $\mathcal{S}$ are split into different sets only at Line 10. From the construction of Algorithm 7, once a node a is moved to a set ahead of the set containing node b , a will remain in a set ahead of b 's set until a is added to τ . Therefore, there must exist an iteration in which a node, say x' , is added to τ , causing v and y to be separated, with v moving to a set ahead of y 's set. From Line 10 of Algorithm 7, v must be a neighbor of x' while y is not, contradicting our assumption that v has no neighbors in τ . Hence, $l = 1$.

If $l = 1$, the nodes in $\mathcal{C}$ are partitioned into two disjoint subsets: τ and S_1 . Since no node in S_1 is adjacent to any node in τ , $\mathcal{C}$ is not connected, a contradiction. Therefore, v must have a neighbor in τ . This ensures that Claim 4.6.46 remains true at the end of the i th iteration.

By induction, Claim 4.6.46 is valid at the end of every iteration. This completes the proof. $\square$

Claim 4.6.47. *Let, at the start of some iteration of the while loop (lines 3-12) of Algorithm 7, $\tau = (a_1, a_2, \dots, a_r)$ and $\mathcal{S} = (S_1, S_2, \dots, S_l)$. For all $u, v, w \in V_{\mathcal{C}}$ (where $\mathcal{C}$ is the input undirected connected chordal graph of Algorithm 7), such that $u - v, u - w \in E_{\mathcal{C}}$, if u is in some S_x , v belongs to either τ or some S_y such that $y < x$, and w belongs to either τ or some S_z such that $z < x$ (where y and z may or may not be equal), then $v - w \in E_{\mathcal{C}}$.*

Proof. We prove this by induction. At the start of the first iteration, τ is empty, and $\mathcal{S}$ contains only one set $V_{\mathcal{C}}$ (from lines 1-2). This implies that at the start of the first iteration, Claim 4.6.47 holds true.

For some $i \geq 1$, assume that at the start of the i th iteration, $\tau = (a_1, a_2, \dots, a_r)$, $\mathcal{S} = (S_1, S_2, \dots, S_l)$, and Claim 4.6.47 holds true. We show that even at the end of the i th iteration, Claim 4.6.47 holds true. Suppose a node $p \in S_1$ is added to τ in the i th iteration. Then, at the end of the iteration, $\tau = (a_1, a_2, \dots, a_r, p)$. Let $\mathcal{S} = (S'_1, S'_2, \dots, S'_m)$ at the end of the iteration, and suppose Claim 4.6.47 does not hold. This implies that there exist $u, v, w \in V_{\mathcal{C}}$ such that $u - v, u - w \in E_{\mathcal{C}}$, u is in some S'_x , v belongs to either τ or some S'_y such that $y < x$, and w belongs to either τ or some S'_z such that $z < x$, and $v - w \notin E_{\mathcal{C}}$.

Let u be in a set S_a of $\mathcal{S}$ at the start of the iteration. From the construction of Algorithm 7, for two distinct nodes x and y , once x is moved to a set ahead of the set

containing the node y in $\mathcal{S}$, x will remain in a set ahead of y 's set in $\mathcal{S}$ until x is added to τ . This implies that at the start of the iteration, v was either already in τ or in some set S_b of $\mathcal{S}$ such that $b \leq a$. Similarly, at the start of the iteration, w was either already in τ or in some set S_c of $\mathcal{S}$ such that $c \leq a$.

If, at the start of the iteration, v was in τ or in some set S_b such that $b < a$, and w was either in τ or in some set S_c such that $c < a$, then even at the start of the iteration, Claim 4.6.47 does not hold, contradicting our assumption. This implies that at the start of the iteration, at least one among v and w was with u in S_a . Without loss of generality, assume that at the start of the iteration, u and w were in S_a .

Due to Line 10 of Algorithm 7, at any instance while running the algorithm, for any node q in τ and a set S in $\mathcal{S}$, in the graph $\mathcal{C}$, either q is a neighbor of all the nodes in S , or it is a neighbor of no nodes in S . This implies that at the start of the iteration, v cannot be in τ , otherwise, we get a node v in τ and a set S_a in $\mathcal{S}$ such that S_a contains two nodes u and w such that u is a neighbor of v and w is not a neighbor of v , a contradiction. This implies that at the start of the iteration, v is in some set S_b such that $b \leq a$. Additionally, v is not added to τ in this iteration, i.e., $v \neq p$, otherwise, at the end of the iteration, due to Line 10, S'_x would be ahead of S'_y , contradicting our assumption. Therefore, there are two possibilities: either $a = b$ (i.e., at the start of the iteration, u , v , and w are in the same set S_a) or $b < a$ (i.e., at the start of the iteration, u and v are not in the same set).

Suppose $a = b$. Since at the end of the iteration, v and w are in a set of $\mathcal{S}$ that comes before u 's set (due to Line 10), this implies v and w are neighbors of p , and u is not a neighbor of p . But this implies there exists a chordless cycle (p, v, u, w, p) in the chordal graph $\mathcal{C}$, a contradiction. Therefore, this possibility cannot occur.

Suppose $b < a$. Since at the end of the iteration, w is in a set of $\mathcal{S}$ that comes before u 's set, this implies w is a neighbor of p , and u is not a neighbor of p . If v were a neighbor of p , then we would have a chordless cycle (p, v, u, w, p) in the chordal graph $\mathcal{C}$, a contradiction. Since at the start of the first iteration, u and v were in the same set of $\mathcal{S}$ (Line 1), and at the start of the i th iteration, v 's set in $\mathcal{S}$ comes before u 's set, there must exist some q in τ such that v is a neighbor of q and u is not. Since at the start of the iteration, u and w were in the same set, w must not be a neighbor of q . From Claim 4.6.46, there exists a chordless path $(u_1 = q, u_2, \dots, u_t = p)$ in $\mathcal{C}$, from q to p ,

such that each node of the path is in τ .

Let i be such that $u_i - v \in E_{\mathcal{C}}$ and for all $j > i$, $u_j - v \notin E_{\mathcal{C}}$. For all $i < k < t$, u must not be a neighbor of u_k ; otherwise, even at the start of the iteration, we would have a triplet (u, v, u_k) that disobeys Claim 4.6.47, contradicting our assumption. As discussed earlier, at the start of the iteration, for any node x in τ , either both u and w are neighbors of x , or neither of them is a neighbor of x . This implies that for all $i < k < t$, neither u nor w is a neighbor of u_k . This implies that $(v, u_i, u_{i+1}, \dots, u_t, w, u, v)$ is a chordless cycle in the chordal graph $\mathcal{C}$, a contradiction.

Therefore, even at the end of the i th iteration, we do not have u , v , and w that disobey Claim 4.6.47. This implies that even at the end of the i th iteration, Claim 4.6.47 holds. This completes the proof of Claim 4.6.47. $\square$

Claim 4.6.47 shows that, at the end of the while loop in Algorithm 7, the ordering τ returned by the algorithm is a perfect elimination ordering of the input chordal graph $\mathcal{C}$.

As discussed earlier, Algorithm 7 is a modified LBFS ordering of the LBFS ordering given by Rose et al. (Rose et al. [1976]). Instead of picking a node arbitrarily at Line 5 of Algorithm 7, we choose an arbitrary canonical source node. Since there always exists a canonical source node (as shown in Claim 4.6.39), Algorithm 7 outputs an LBFS ordering (verified by Claim 4.6.47). Corollary 4.6.45 demonstrates that the ordering returned by Algorithm 7 is consistent with O . Lemma 4.6.48 summarizes this.

Lemma 4.6.48. *Let H be an undirected graph, H_1 and H_2 be two induced subgraphs of H such that $H = H_1 \cup H_2$, and $I = V_{H_1} \cap V_{H_2}$ is a vertex separator of H . For each $i \in \{1, 2\}$, let S_i be a subset of V_{H_i} such that $I \subseteq S_i$, M_i be an MEC of H_i , and the shadow of M_i on $S_i \cup N(S_i, H_i)$ be (O_i, P_{i1}, P_{i1}) . Let $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, P_{12}, O_2, P_{21}, P_{22})$. For $i \in \{1, 2\}$, let $\mathcal{C}$ be an undirected connected component of M_i . For inputs $\mathcal{C}$ and O , Algorithm 7 returns an LBFS ordering τ that satisfies O , i.e., for $u \rightarrow v \in O$ such that $u, v \in V_{\mathcal{C}}$, $\tau(u) < \tau(v)$.*

4.6.4.2 Proof of Lemma 4.5.8

We now prove Lemma 4.5.8.

Algorithm 8: Construct MEC

Input : Three graphs M_1, M_2 , and O such that

- (i) H is an undirected graph and
- (ii) H_1 and H_2 are two induced subgraphs of H such that $H = H_1 \cup H_2$, and $I = V_{H_1} \cap V_{H_2}$ is a vertex separator of H .
- (iii) S_1 and S_2 are subsets of V_{H_1} and V_{H_2} , respectively, such that $S_1 \cap S_2 = I$.
- (iv) M_1 and M_2 are MECs of H_1 and H_2 , respectively.
- (v) $\forall a \in \{1, 2\}$, the shadow of M_a on $S_a \cup N(S_a, H_a)$ is (O_a, P_{a1}, P_{a2}) .
- (vi) $(O, P_1, P_2) \in \text{shadow}(H[S_1 \cup S_2 \cup N(S_1 \cup S_2, H)])$, and
- (vii) $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$.

Output: M , an MEC of H such that $M \in \text{MEC}(H, O, P_1, P_2)$ and $\mathcal{P}(M, V_{H_1}, V_{H_2}) = (M_1, M_2)$.

```
1  $M \leftarrow \mathcal{U}_M(M_1, M_2, O)$ 
2  $I \leftarrow V_{M_1} \cap V_{M_2}$ 
3 foreach  $i \in \{1, 2\}$  do
4   foreach undirected connected component  $\mathcal{C}$  of  $M_i$  do
5      $\tau \leftarrow \text{LBFS}(\mathcal{C}, O)$ .
6     Denote a vertex  $v$  of  $\mathcal{C}$  by  $u_i$  if  $\tau(v) = i$ .
7      $a \leftarrow 2$ 
8     while  $a \leq |V_{\mathcal{C}}|$  do
9        $b \leftarrow a - 1$ 
10      while  $b \geq 1$  do
11        if  $\exists c$  such that  $c < b$ , and  $u_c - u_b - u_a$  is an induced subgraph of  $\mathcal{C}$ ,
          and  $u_c \rightarrow u_b - u_a$  is an induced subgraph of  $M$ , or  $\exists c$  such that
           $b < c < a$ , and  $u_b - u_c - u_a - u_b \in \mathcal{C}$ , and  $u_b \rightarrow u_c \rightarrow u_a - u_b$  is
          an induced subgraph of  $M$  then
12          | Replace the edge  $u_b - u_a$  in  $M$  with  $u_b \rightarrow u_a$ .
13          end
14           $b = b - 1$ 
15        end
16       $a = a + 1$ 
17    end
18  end
19 end
20 return  $M$ 
```

Proof of Lemma 4.5.8. Let M_1 and M_2 be the MECs of H_1 and H_2 , respectively, such that $\text{shadow}(M_1, S_1 \cup N(S_1, H_1)) = (O_1, P_{11}, P_{12})$ and $\text{shadow}(M_2, S_2 \cup N(S_2, H_2)) = (O_2, P_{21}, P_{22})$. Furthermore, let $(O, P_1, P_2) \in \text{shadow}(H[S_1 \cup S_2 \cup N(S_1 \cup S_2, H)])$ such that $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$. We show that there exists a unique MEC M such that $\mathcal{P}(M, V_{H_1}, V_{H_2}) = (M_1, M_2)$ and the shadow of M on V_O is (O, P_1, P_2) . We first construct such an MEC M using Algorithm 8. Algorithm 8 follows the following steps to construct the MEC:

1. Initialize $M = \mathcal{U}_M(M_1, M_2, O)$ (Line 1 of Algorithm 8), a Markov union (Definition 4.6.7) of M_1, M_2 and O .
2. For each $i \in \{1, 2\}$ (lines 3-19 of Algorithm 8), for each undirected connected component $\mathcal{C}$ of M_i (lines 4-18 of Algorithm 8), we do the following:
 - (a) Construct an LBFS ordering τ of $\mathcal{C}$ using Algorithm 7 for input $\mathcal{C}$ and O (Line 5 of Algorithm 8).
 - (b) For $u - v \in M$ such that $\tau(u) < \tau(v)$, replace $u - v$ with $u \rightarrow v$ (Line 12 of Algorithm 8) if either
 - i. there exists a vertex $x \in \mathcal{C}$ such that $\tau(x) < \tau(u)$, and there exists an induced subgraph $x - u - v$ of $\mathcal{C}$ such that $x \rightarrow u \in M$ (Line 11), or
 - ii. there exists a vertex $x \in \mathcal{C}$ such that $\tau(u) < \tau(x) < \tau(v)$, and $u \rightarrow x, x \rightarrow v \in M$ (Line 11).

We begin by demonstrating that Algorithm 8 correctly executes steps 1 and 2. The implementation of step 1 can be found at Line 1 of Algorithm 8. Next, we verify that for each $i \in \{1, 2\}$ and for every undirected connected component $\mathcal{C}$ of M_i , steps 2a and 2b are executed correctly.

Step 2a is implemented at Line 5 of Algorithm 8, where we construct an LBFS ordering τ for $\mathcal{C}$. To show that step 2b is also executed correctly, we use a corollary (Corollary 4.6.56) that is stated and proved later.

Let us suppose that step 2b is not accurately executed by Algorithm 8. This would imply the existence of an x such that either (a) $x - u - v$ forms an induced subgraph of $\mathcal{C}$, with $\tau(x) < \tau(u) < \tau(v)$, and $x \rightarrow u - v$ is an induced subgraph in M , or

(b) $u - x - v - u$ forms an induced subgraph of $\mathcal{C}$, with $\tau(u) < \tau(x) < \tau(v)$, and $u \rightarrow x \rightarrow v - u$ is an induced subgraph in M .

Let us address both cases individually:

Case (a): Assuming there exists an x such that $x - u - v$ forms an induced subgraph of $\mathcal{C}$, $\tau(x) < \tau(u) < \tau(v)$, and $x \rightarrow u - v$ is an induced subgraph in M . From Corollary 4.6.56, $x \rightarrow u$ was added to M either during initialization (Line 1 of Algorithm 8) or in the second foreach loop (lines 4-18 of Algorithm 8) when $\mathcal{C}$ is considered, and when $a = \tau(v)$ and $b = \tau(u)$. This implies that when we have $a = \tau(v)$ and $b = \tau(u)$, $x \rightarrow u$ is in M . However, during the execution of lines 11-13 (when we have $a = \tau(v)$ and $b = \tau(u)$), $u \rightarrow v$ is added to M , leading to a contradiction. Therefore, this case cannot occur.

Case (b): Assuming there exists an x such that $u - x - v - u$ forms an induced subgraph of $\mathcal{C}$, with $\tau(u) < \tau(x) < \tau(v)$, and $u \rightarrow x, x \rightarrow v - u \in M$. According to Corollary 4.6.56, $u \rightarrow x$ was added to M either during initialization or in the second foreach loop (lines 4-18 of Algorithm 8) when $\mathcal{C}$ is considered, and when $a = \tau(x)$ and $b = \tau(u)$, and $x \rightarrow v$ was added to M either during initialization or when $a = \tau(v)$ and $b = \tau(x)$. This implies that when we have $a = \tau(v)$ and $b = \tau(u)$, both $u \rightarrow x$ and $x \rightarrow u$ are in M . However, during the execution of lines 11-13 (when we have $a = \tau(v)$ and $b = \tau(u)$), $u \rightarrow v$ is added to M , again leading to a contradiction. Therefore, this case also cannot occur.

In conclusion, we have demonstrated that step 2b is executed correctly by Algorithm 8. This further affirms that Algorithm 8 accurately implements steps 1 and 2.

Lemmas 4.6.49 to 4.6.53 validate M .

Lemma 4.6.49. $M[V_O] = O$.

Lemma 4.6.50. M is an MEC of H .

Lemma 4.6.51. $\mathcal{P}(M, V_{M_1}, V_{M_2}) = (M_1, M_2)$.

Lemma 4.6.52. If there exists an MEC M' of H such that $M'[V_O] = O$, and $\mathcal{P}(M', V_{M_1}, V_{M_2}) = (M_1, M_2)$ then $M' = M$.

Lemma 4.6.53. $\text{shadow}(M, V_O) = (O, P_1, P_2)$.

Lemma 4.6.50 shows that the graph returned by Algorithm 8 is an MEC of H . Lemma 4.6.49 shows that O is an induced subgraph of M , i.e., $M[V_O] = O$, and Lemma 4.6.51 shows that M_1 and M_2 are projections of M . Lemma 4.6.52 shows that there exists a unique MEC with O as its induced subgraph, and M_1 and M_2 as its projections. Additionally, Lemma 4.6.53 shows that $\text{shadow}(M, V_O) = (O, P_1, P_2)$ (recall that we have assumed in the first paragraph of the proof that $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$).

Proofs of Lemmas 4.6.49 to 4.6.53 are given below. Given these lemmas, Lemma 4.5.8 follows immediately. □

We now prove Lemma 4.6.49. The proofs of Lemma 4.6.50, Lemma 4.6.51, Lemma 4.6.52, and Lemma 4.6.53 are at pages 206, 231, 231, and 232, respectively.

In preparation for proving Lemma 4.6.49, we first prove the following observation. The proof of Lemma 4.6.49 appears on page 203.

Observation 4.6.54. *For $i \in \{1, 2\}$, for any undirected connected component $\mathcal{C}$ of M_i , at Line 5 of Algorithm 8, let τ be the LBFS ordering returned by Algorithm 7. Then for $u - v \in \mathcal{C}$, at the start (and also at the end) of any iteration of the while loop (lines 8-17) of Algorithm 8, if $u \rightarrow v \in M$ then at least one of the following occurs:*

1. $u \rightarrow v \in O$
2. $\exists a \rightarrow b \in O$ such that there exists a chordless path from (a, b) to (u, v) in $\mathcal{C}$ such that $(a, b) \neq (u, v)$.
3. $\exists a \rightarrow b \in O$ such that there exist chordless paths P and Q such that P is a chordless path from (a, b) to v in $\mathcal{C}$, and Q is a chordless path from (v, u) to a in $\mathcal{C}$, such that $b \neq v$ and $u \neq a$.

Proof. We start with the following observation:

Observation 4.6.55. *For $i \in \{1, 2\}$, if $u - v \in M_i$ and $u \rightarrow v \in \mathcal{U}_M(M_1, M_2, O)$ then $u \rightarrow v \in O$.*

Proof. From Definition 4.6.7, if $u - v \in M_1$ and $u \rightarrow v \in \mathcal{U}_M(M_1, M_2, O)$ then either $u \rightarrow v \in M_2$ or $u \rightarrow v \in O$. $u \rightarrow v \in M_2$ implies that $u, v \in V_{M_1} \cap V_{M_2} = I$. Since (O_2, P_{21}, P_{22}) is the shadow of M_2 on V_{O_2} , therefore, from Item 1 of Definition 4.3.1, $O_2 = M_2[V_{O_2}]$. This implies $u \rightarrow v \in O_2$ as from the construction, $I \subseteq V_{O_2}$. Since $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$, therefore, from Item 1 of Definition 4.5.6, we have $u \rightarrow v \in O$. Thus, we can say that if $u - v \in M_1$ and $u \rightarrow v \in \mathcal{U}_M(M_1, M_2, O)$ then $u \rightarrow v \in O$. Similarly, if $u - v \in M_2$ and $u \rightarrow v \in \mathcal{U}_M(M_1, M_2, O)$ then $u \rightarrow v \in O$. This completes the proof of Observation 4.6.55. $\square$

Let us number the undirected connected components of M_1 and M_2 in the sequence in which they are considered on Line 4 as $\mathcal{C}_1, \mathcal{C}_2, \dots$. For some $\alpha \geq 1$, suppose that Observation 4.6.54 is true for all $\mathcal{C}_\beta$ with $1 \leq \beta < \alpha$. We now prove by induction on the iteration of the second while loop (lines 10-15) of Algorithm 8 corresponding to the connected component $\mathcal{C}_\alpha$ that the Observation 4.6.54 is true for $\mathcal{C} = \mathcal{C}_\alpha$ as well.

Base case: We first show that at the start of the first run of the second while loop (lines 10-15) corresponding to $\mathcal{C} = \mathcal{C}_\alpha$ (i.e., when $a = 2$ and $b = 1$), it must be the case that for any edge $u - v \in \mathcal{C}$, if $u \rightarrow v \in M$ then $u \rightarrow v \in O$. This will establish that Observation 4.6.54 holds at the start of the first run of the while loop corresponding to the undirected connected chordal component $\mathcal{C} = \mathcal{C}_\alpha$.

Suppose, if possible, at the start of the first run of the while loop, there exists $u - v \in \mathcal{C}$ such that $u \rightarrow v \in M$. There are two possibilities: either $u \rightarrow v$ is added at the initialization step, i.e., at Line 1 of Algorithm 8, or $u \rightarrow v$ is added to M during the processing of some $\mathcal{C}_\gamma$ with $\gamma < \alpha$ at lines 3-19 of Algorithm 8. At the initialization step, $M = \mathcal{U}_M(M_1, M_2, O)$. Since $\mathcal{C}$ is an undirected connected component of M_i for some $i \in \{1, 2\}$ and $u - v \in \mathcal{C}$, $u - v \in M_i$. Then from Observation 4.6.55, if $u \rightarrow v \in \mathcal{U}_M(M_1, M_2, O)$ then $u \rightarrow v \in O$. This validates Observation 4.6.54. We now go through the other possibility that $u \rightarrow v$ has been added to M during the processing of some $\mathcal{C}_\gamma$ with $\gamma < \alpha$.

Without loss of generality, let us assume that $\mathcal{C}_\alpha$ is an undirected connected component of M_1 . Then $\mathcal{C}_\gamma$ cannot be an undirected connected component of M_1 (two distinct undirected connected components of a graph do not share any vertex). This implies

$\mathcal{C}_\gamma$ is an undirected connected component of M_2 . This further implies $u - v$ is in both M_1 and M_2 . That means $u, v \in V_{M_1} \cap V_{M_2} = I \subseteq V_O$. Since $u - v \in \mathcal{C}_\gamma$, an undirected connected component of M_2 , therefore, $u - v \in M_2$. Also, since $u, v \in I \subseteq V_{O_2}$, and O_2 is an induced subgraph of M_2 (from Item 1 of Definition 4.3.1 as (O_2, P_{21}, P_{22}) is a shadow of M_2), therefore, $u - v \in O_2$. Since we assumed that Observation 4.6.54 is true for such a $\mathcal{C}_\gamma$, one of the three conditions (Items 1 to 3) of Observation 4.6.54 must be true for $u - v$, as $u - v$ is considered as an edge in $\mathcal{C}_\gamma$.

If Item 1 of Observation 4.6.54 holds then we get $u \rightarrow v \in O$. If one of Item 2 or Item 3 of Observation 4.6.54 holds, then in the undirected connected chordal component $\mathcal{C}_\gamma$ of M_2 , there exists an edge $x - y \in \mathcal{C}_\gamma$ such that $x \rightarrow y \in O$, and either (a) $(x, y) \neq (u, v)$ and there exists a chordless path from (x, y) to (u, v) in $\mathcal{C}_\gamma$ (due to Item 2 of Observation 4.6.54), or (b) $y \neq v, u \neq x$, and there exist chordless paths Q_1 and Q_2 in $\mathcal{C}_\gamma$ such that Q_1 is a chordless path from (x, y) to v , and Q_2 is a chordless path from (v, u) to x (due to Item 3 of Observation 4.6.54). Since $\mathcal{C}_\gamma$ is an undirected connected component of M_2 , therefore, a chordless path in $\mathcal{C}_\gamma$ is a chordless path of M_2 . And, from Observation 4.6.8, every chordless path is a triangle-free path. This implies chordless paths in $\mathcal{C}_\gamma$ are triangle-free paths of M_2 . Also, since $(x, y) \in \mathcal{C}_\gamma$, an undirected connected component of M_2 , and $x \rightarrow y \in O$, this implies $x, y \in V_{M_2} \cap V_O = V_{O_2}$. Since (O_2, P_{21}, P_{22}) is the shadow of M_2 on V_{O_2} , from Definition 4.3.1, O_2 is an induced subgraph of M_2 . This implies $(x, y) \in E_{O_2}$, and if there exists a chordless path in $\mathcal{C}_\gamma$, from (x, y) to (u, v) , then $P_{21}((x, y), (u, v)) = 1$, and if there exist chordless paths in $\mathcal{C}_\gamma$, from (x, y) to v and from (v, u) to x then $P_{22}((x, y), v) = P_{22}((v, u), x) = 1$. Since $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$, therefore, in both the possibilities, from Item 3 of Definition 4.5.6, we again have $u \rightarrow v \in O$. This shows that for any $u - v \in \mathcal{C}_\alpha$, when $\mathcal{C}_\alpha$ is processed at lines 4-18 of Algorithm 8, at the start of the first run of the while loop (at lines 10-15), if $u \rightarrow v \in M$ then $u \rightarrow v \in O$. This establishes the base case of the induction.

Induction step: Suppose now that at the start of an iteration of the second while loop (at lines 10-15 of Algorithm 8) corresponding to $\mathcal{C} = \mathcal{C}_\alpha$, Observation 4.6.54 is valid. This means that at the start of the iteration, for any edge $u' - v' \in \mathcal{C}$, if $u' \rightarrow v' \in M$,

then $u' \rightarrow v'$ follows Observation 4.6.54. We show that even at the end of the iteration, Observation 4.6.54 is valid, i.e., for any $u - v \in \mathcal{C}$, during the iteration, if $u \rightarrow v$ is added to M , then the new directed edge $u \rightarrow v$ obeys Observation 4.6.54.

During the iteration, let there exist an edge $u - v \in \mathcal{C}$ such that $u \rightarrow v$ is added to M in the while loop while processing $\mathcal{C} = \mathcal{C}_\alpha$. We show that $u \rightarrow v$ obeys Observation 4.6.54.

Let τ be the LBFS ordering of $\mathcal{C}$ returned by Algorithm 7 at Line 5 of Algorithm 8 for the input $\mathcal{C}$ and O . From lines 11-13 of Algorithm 8, if $u \rightarrow v$ is added to M , then at the start of the iteration, either of the following has occurred: (a) there exists an induced subgraph $x - u - v$ of $\mathcal{C}$ such that $\tau(x) < \tau(u) < \tau(v)$, and $x \rightarrow u - v$ is an induced subgraph of M , or (b) there exists an induced subgraph $u - x - v - u$ of $\mathcal{C}$ such that $\tau(u) < \tau(x) < \tau(v)$, and $u \rightarrow x \rightarrow v - u$ is an induced subgraph of M .

We go through each case and show that in each case $u \rightarrow v$ obeys Observation 4.6.54.

1. Suppose that during the iteration, $u - v \in M$ is replaced with $u \rightarrow v$ because there exists an induced subgraph $x - u - v$ of $\mathcal{C}$ such that $\tau(x) < \tau(u) < \tau(v)$, and at the start of the iteration, $x \rightarrow u - v$ is an induced subgraph of M . By the induction hypothesis, we have that $x \rightarrow u \in M$ because one of the following holds: (i) $x \rightarrow u \in O$, or (ii) $\exists a \rightarrow b \in O$ such that there exists a chordless path $(p_1 = a, p_2 = b, \dots, p_{l-1} = x, p_l = u)$ from $a - b$ to $x - u$ in $\mathcal{C}$ (with $(a, b) \neq (x, u)$), or (iii) $\exists a \rightarrow b \in O$ such that there exists a chordless path from $a - b$ to u , and a chordless path from $u - x$ to a in $\mathcal{C}$ (with $b \neq u$ and $x \neq a$). We start with the first possibility.

(a) Suppose that $x \rightarrow u \in M$ because $x \rightarrow u \in O$. Then the new directed edge $u \rightarrow v$ obeys Item 2 of Observation 4.6.54, due to the existence of the chordless path (x, u, v) in $\mathcal{C}$ and $x \rightarrow u \in O$. We now go through the second possibility.

(b) Suppose that $x \rightarrow u \in M$ because $\exists a \rightarrow b \in O$ such that there exists a chordless path $Q = (p_1 = a, p_2 = b, \dots, p_{l-1} = x, p_l = u)$ from $a - b$ to $x - u$ in $\mathcal{C}$ (with $(a, b) \neq (x, u)$). By the hypothesis of this case, $Q' = (x, u, v)$ is a chordless path in $\mathcal{C}$ (recall that in this case, $x - u - v$ is an induced

subgraph of $\mathcal{C}$). Then, from Lemma 4.6.14, $P = (p_1 = a, p_2 = b, \dots, p_{l-1} = x, p_l = u, v)$, concatenation of Q and Q' , is a chordless path in $\mathcal{C}$.

Thus, we have a chordless path P in $\mathcal{C}$ from (a, b) to (u, v) with $(a, b) \neq (u, v)$ (Q being a path precludes $a = u$) and $a \rightarrow b \in O$. This further implies that the new directed edge $u \rightarrow v$ obeys Item 2 of Observation 4.6.54. We now move to the third possibility.

- (c) Suppose that $x \rightarrow u \in M$ because $\exists a \rightarrow b \in O$ such that there exists a chordless path $Q_1 = (p_1 = a, p_2 = b, \dots, p_l = u)$ from (a, b) to u in $\mathcal{C}$, and a chordless path $Q_2 = (q_1 = u, q_2 = x, \dots, q_m = a)$ from (u, x) to a in $\mathcal{C}$ (with $b \neq u$ and $x \neq a$).

Since Q_1 is a chordless path and $a \rightarrow b \in O$, according to Corollary 4.6.44, we have $\tau(a) = \tau(p_1) < \tau(p_2) < \tau(p_3) < \dots < \tau(p_l) = \tau(u)$.

Now, we claim that

$$\text{For } 1 \leq i \leq l-2, v - p_i \notin \mathcal{C}. \quad (4.8)$$

To see this, suppose, if possible, that for some $i \leq l-2$, $v - p_i \in \mathcal{C}$. As argued above, $\tau(p_i) < \tau(u)$, and by the hypothesis for this case, we have $\tau(u) < \tau(v)$. Then, from Observation 4.6.4, we must have $p_i - u \in \mathcal{C}$. Since $i \leq l-2$, this constitutes a chord of Q_1 and contradicts the fact that Q_1 is a chordless path. This establishes Equation (4.8).

Due to Equation (4.8), only the following two options are now allowed: either (i) for each $1 \leq i \leq l-1$, $v - p_i \notin \mathcal{C}$, or (ii) $v - p_{l-1} \in \mathcal{C}$ and for each $1 \leq i \leq l-2$, $v - p_i \notin \mathcal{C}$.

If the first option holds, then $(p_1 = a, p_2 = b, \dots, p_{l-1}, p_l = u, v)$ is a chordless path (because Q_1 is a chordless path) from (a, b) to (u, v) with $(a, b) \neq (u, v)$ ($a = u$ is precluded by Q_1 being a path), and this implies that $u \rightarrow v$ obeys Item 2 of Observation 4.6.54.

In case the second option holds, $Q_3 = (p_1 = a, p_2 = b, \dots, p_{l-1}, v)$ is a chordless path from (a, b) to v . Since $\tau(p_2) = \tau(b) < \tau(u)$ (as argued above) and $\tau(u) < \tau(v)$ (by the hypothesis of this case), we also have $b \neq v$.

Now, we claim that

$$Q_4 = (v, q_1 = u, q_2 = x, \dots, q_m = a) \text{ is a chordless path in } \mathcal{C} \text{ from } (v, u) \text{ to } a. \quad (4.9)$$

By the hypothesis of this case, $Q'_2 = (v, u, x)$ is a chordless path in $\mathcal{C}$ (recall $x - u - v$ is an induced subgraph of $\mathcal{C}$ in this case). From Lemma 4.6.14, $Q_4 = (v, q_1 = u, q_2 = x, q_3, \dots, q_m = a)$, the concatenation of Q'_2 and Q_2 , is a chordless path in $\mathcal{C}$. Also, Q_2 being a path precludes $u = a$.

We thus obtain a chordless path Q_4 from (v, u) to a (with $u \neq a$). Along with the chordless path Q_3 from (a, b) to v (with $b \neq v$) and $a \rightarrow b \in O$, this implies that $u \rightarrow v$ obeys Item 3 of Observation 4.6.54.

We now move to the second case.

2. Suppose that during the iteration, $u - v \in M$ is replaced by $u \rightarrow v$ because there exists an induced subgraph $u - x - v - u$ of $\mathcal{C}$ such that $\tau(u) < \tau(x) < \tau(v)$, $u \rightarrow x, x \rightarrow v \in M$ and $u - v \in M$. Again, by the induction hypothesis, we have that $u \rightarrow x \in M$ because one of the following holds: (i) $u \rightarrow x \in O$, or (ii) $\exists a \rightarrow b \in O$ such that there exists a chordless path $(p_1 = a, p_2 = b, \dots, p_{l-1} = u, p_l = x)$ from (a, b) to (u, x) in $\mathcal{C}$ (with $(a, b) \neq (u, x)$), or (iii) $\exists a \rightarrow b \in O$ such that there exists a chordless path from (a, b) to v , and a chordless path from (v, x) to a in $\mathcal{C}$ (with $b \neq v$ and $x \neq a$). We again go through each possibility.

- (a) Suppose $u \rightarrow x \in M$ because $u \rightarrow x \in O$. Again, by the induction hypothesis, we have that $x \rightarrow v \in M$ because one of the following holds: (i) $x \rightarrow v \in O$, or (ii) $\exists c \rightarrow d \in O$ such that there exists a chordless path $(q_1 = c, q_2 = d, \dots, q_{m-1} = x, q_m = v)$ from (c, d) to (x, v) in $\mathcal{C}$ (with $(c, d) \neq (x, v)$), or (iii) $\exists c \rightarrow d \in O$ such that there exists a chordless path from (c, d) to v , and a chordless path from (v, x) to c in $\mathcal{C}$ (with $d \neq v$ and $x \neq c$). We go through each sub-case.

- i. Let us assume that $x \rightarrow v \in M$ due to the fact that $x \rightarrow v \in O$. The presence of both $u \rightarrow x$ and $x \rightarrow v$ in O implies that u, v , and x are all elements of V_O .

At the start of the iteration, we have $u - v \in M$. This implies even after the initialization step (Line 1), we have $u - v \in M$. From the construction of M , $\text{skeleton}(M[V_O]) = \text{skeleton}(O)$. That means $u - v \in \text{skeleton}(O)$. Since (O, P_1, P_2) is an extension of $(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$, by Definition 4.5.6, O is a partial MEC. Then, from Item 1 of Definition 4.2.1, O is a chain graph. This implies $u \rightarrow v \in O$ (or else O would contain a directed cycle (u, x, v, u)).

But then, from the initialization of M (recall M is initialized with $\mathcal{U}_M(M_1, M_2, O)$), $u \rightarrow v \in M$, a contradiction. This contradiction implies that this particular sub-case cannot occur. We will now proceed to examine the remaining sub-cases.

- ii. Let us assume that $x \rightarrow v \in M$ because there exists $c \rightarrow d \in O$ such that there is a chordless path $Q = (p_1 = c, p_2 = d, \dots, p_{l-2}, p_{l-1} = x, p_l = v)$ from (c, d) to (x, v) in $\mathcal{C}$ (with $(c, d) \neq (x, v)$). Since $(c, d) \neq (x, v)$, $l \geq 3$. Also, for all $1 \leq i \leq l - 2$, $u \neq p_i$, otherwise, there would be a chord $u - v$ in the chordless path Q , leading to a contradiction. In particular, $u \neq c$.

According to Corollary 4.6.44, we have $\tau(c) = \tau(p_1) < \tau(p_2) < \dots < \tau(p_{l-1}) = \tau(x) < \tau(p_l) = \tau(v)$. Since $\tau(p_{l-2}) < \tau(x)$, and $\tau(u) < \tau(x)$ (as per the hypothesis of this case), Observation 4.6.4 implies that $u - p_{l-2} \in \mathcal{C}$ (since x is adjacent to both u and p_{l-2} in $\mathcal{C}$).

Let $1 \leq i \leq l - 2$ be the smallest number such that $u - p_i \in \mathcal{C}$. If $i > 1$, then, as Q is a chordless path, $Q' = (p_1 = c, p_2 = d, \dots, p_i, u, v)$ is a chordless path in $\mathcal{C}$ from (c, d) to (u, v) with $(c, d) \neq (u, v)$ (since $c \neq u$). Since $c \rightarrow d \in O$, this implies that $u \rightarrow v$ satisfies Item 2 of Observation 4.6.54.

If $i = 1$, then $Q_1 = (v, u, c)$ is a chordless path from (v, u) to c (c is not adjacent to v in $\mathcal{C}$ because Q is a chordless path). Since $Q = (p_1 = c, p_2 = d, \dots, p_{l-1} = x, p_l = v)$ is a chordless path from (c, d) to v (with $d \neq v$) and Q_1 is a chordless path from (v, u) to c (with $c \neq u$), $u \rightarrow v$ satisfies Item 3 of Observation 4.6.54.

Therefore, in this subcase, $u \rightarrow v$ complies with Observation 4.6.54. We will now consider the remaining sub-case.

- iii. Suppose $x \rightarrow v \in M$ because $\exists c \rightarrow d \in O$ such that there exists a chordless path $Q_1 = (q_1 = c, q_2 = d, \dots, q_m = v)$ from (c, d) to v in $\mathcal{C}$ (in which $d \neq v$), and a chordless path $Q_2 = (p_1 = v, p_2 = x, \dots, p_l = c)$ from (v, x) to c in $\mathcal{C}$ (in which $c \neq x$). Note that c is not adjacent to v (since Q_1 is a chordless path), and u is adjacent to v (by the hypothesis of this case). We thus also have $u \neq c$.

We now claim that there exists a chordless path Q from (v, u) to c in $\mathcal{C}$. This further implies that $u \rightarrow v$ obeys Item 3 of Observation 4.6.54 due to the existence of chordless paths Q_1 from (c, d) to v (with $d \neq v$) and Q from (v, u) to c (with $u \neq c$) such that $c \rightarrow d \in O$.

Proof of the claim. There are two possibilities: either $\tau(p_3) < \tau(p_2)$, or $\tau(p_3) > \tau(p_2)$. We go through both possibilities and show that in each possibility there exists a chordless path from (v, u) to c .

Suppose $\tau(p_3) < \tau(p_2 = x)$. Since we also have $\tau(u) < \tau(x)$ (by the hypothesis of this case), and since x is adjacent to both u and p_3 , Observation 4.6.4 implies that $u - p_3 \in \mathcal{C}$. Pick the highest i such that $u - p_i \in \mathcal{C}$. Since $u - p_3 \in \mathcal{C}$, i must be greater than 2. Then, $Q = (v, u, p_i, \dots, p_l = c)$ is a chordless path from (v, u) to c (since v is adjacent to u and Q_1 is a chordless path). We now move to the other possibility.

Suppose $\tau(p_3) > \tau(p_2 = x)$. Since Q_2 is a chordless path in $\mathcal{C}$, Observation 4.6.5 implies that $\tau(p_2 = x) < \tau(p_3) < \dots < \tau(p_l = c)$, i.e., $\tau(x) < \tau(c)$. From the hypothesis of this case, $\tau(u) < \tau(x)$. This implies that $\tau(u) < \tau(c)$. We now claim that u must be adjacent to every vertex in the chordless path Q_1 , and in particular to c . Since Q_1 is a chordless path and $c \rightarrow d \in O$, therefore, from Corollary 4.6.44, $\tau(q_1) = \tau(c) < \tau(q_2) = \tau(d) < \dots < \tau(q_{m-1}) < \tau(q_m) = \tau(v)$. Since $\tau(u) < \tau(c)$, we have $\tau(u) < \tau(q_i) < \tau(q_{i+1})$ for every $1 \leq i \leq m - 1$.

Suppose if possible that there exists an i such that $q_i - u \notin \mathcal{C}$, then pick the highest such i . Since u is adjacent to $v = q_m$ in $\mathcal{C}$ (by the hypothesis of this case), we must have $i \leq m - 1$. Further, by the choice of i , we must have that u is adjacent to q_{i+1} . But, then, since $\tau(q_i) < \tau(q_{i+1})$ and $\tau(u) < \tau(q_{i+1})$, Observation 4.6.4 implies that $q_i - u \in \mathcal{C}$ (since q_i and u both are adjacent to q_{i+1}), which is a contradiction to the choice of i . Thus, u is adjacent to every vertex in Q_1 , and in particular to c . This implies that $Q = (v, u, c)$ is a chordless path in $\mathcal{C}$ from (v, u) to c (as argued above v and c are not adjacent and $u \neq c$).

Thus, we show that in both possibilities there exists a chordless path Q from (v, u) to c . $\square$

We now move to the next possibility.

- (b) Let us assume that $u \rightarrow x \in M$ because there exists an edge $a - b \in \mathcal{C}$ such that $a \rightarrow b \in O$, and there exists a chordless path $Q = (p_1 = a, p_2 = b, \dots, p_{l-1} = u, p_l = x)$ from (a, b) to (u, x) in $\mathcal{C}$ (with $(a, b) \neq (u, x)$). Note that we trivially also have $a \neq u$, since Q is a path.

Now, by Corollary 4.6.44 applied to Q , we also have $\tau(p_1) = \tau(a) < \tau(p_2) = \tau(b) < \dots < \tau(p_{l-1}) < \tau(p_l)$.

We now claim that for each $1 \leq i \leq l - 2$, $v - p_i \notin \mathcal{C}$. To see this, suppose, if possible, that v is adjacent to p_i for some $1 \leq i \leq l - 2$. Then, since $\tau(p_i) < \tau(x)$, $\tau(x) < \tau(v)$ (by the hypothesis of this case), and since v is adjacent to both x and p_i , Observation 4.6.4 would imply that $x - p_i \in \mathcal{C}$. But this would be a chord in the chordless path Q , thereby leading to a contradiction. We conclude that for each $1 \leq i \leq l - 2$, $v - p_i \notin \mathcal{C}$. This implies that $(p_1 = a, p_2 = b, \dots, p_{l-1} = u, v)$ is a chordless path from (a, b) to (u, v) with $(a, b) \neq (u, v)$ (since $a \neq u$).

This further implies that in this case, $u \rightarrow v$ obeys Item 2 of Observation 4.6.54.

- (c) Suppose that $u \rightarrow x \in M$ because there exists an edge $a - b \in \mathcal{C}$ such that $a \rightarrow b \in O$, and there exists a chordless path $Q_1 = (p_1 = a, p_2 = b, \dots, p_l =$

x) from (a, b) to x in $\mathcal{C}$ (with $b \neq x$), and a chordless path $Q_2 = (q_1 = x, q_2 = u, \dots, q_m = a)$ from (x, u) to a in $\mathcal{C}$ (with $u \neq a$). By Corollary 4.6.44 applied to Q_1 , we have $\tau(p_1) = \tau(a) < \tau(p_2) < \dots < \tau(p_{l-1}) < \tau(p_l) = \tau(x)$. Since $\tau(q_m) = \tau(a) < \tau(x) = \tau(q_1)$, Observation 4.6.5 applied to the LBFS ordering τ and the chordless path Q_2 of the undirected chordal graph $\mathcal{C}$ then also implies that we must have $\tau(q_i) < \tau(x)$ for every $2 \leq i \leq m$. By the hypothesis of this case, $\tau(x) < \tau(v)$. Therefore, for all i , $\tau(q_i) < \tau(v)$. We now claim that

$$\text{For all } 3 \leq i \leq m, v - q_i \notin \mathcal{C}. \quad (4.10)$$

To see this, suppose, if possible, that for some $3 \leq i \leq m$, $v - q_i \in \mathcal{C}$. As argued above, we have $\tau(q_i) < \tau(v)$ and $\tau(q_1) < \tau(v)$. Therefore, from Observation 4.6.4, $q_1 - q_i \in \mathcal{C}$ (as both q_1 and q_i are adjacent to v). Since $3 \leq i \leq m$, this constitutes a chord of Q_2 and thus contradicts that Q_2 is a chordless path. This establishes Equation (4.10). Due to Equation (4.10), we have that $Q_3 = (v, q_2 = u, q_3, \dots, q_m = a)$ (which we get by replacing q_1 with v in Q_2) is a chordless path in $\mathcal{C}$.

We now show the existence of a chordless path from (a, b) to v in $\mathcal{C}$. Since Q_1 is a chordless path, and $a \rightarrow b \in O$, from Corollary 4.6.44, $\tau(p_1 = a) < \tau(p_2 = b) < \dots < \tau(p_l = x)$. Suppose for some $i \leq l - 2$, $v - p_i \in \mathcal{C}$. Then, from Observation 4.6.4, $p_i - p_l \in \mathcal{C}$, as $\tau(p_i) < \tau(p_l = x) < \tau(v)$ (from our hypothesis of this case, $\tau(x) < \tau(v)$), which further implies that Q_1 is not a chordless path, which is a contradiction. This implies that for all $i \leq l - 2$, $v - p_i \notin \mathcal{C}$. Thus, we get that either $Q_4 := (p_1 = a, p_2 = b, \dots, p_{l-1}, v)$ is a chordless path from (a, b) to v (if $p_{l-1} - v \in \mathcal{C}$), or else $Q_4 = (p_1 = a, p_2 = b, \dots, p_l, v)$ is a chordless path from (a, b) to v (if $p_{l-1} - v \notin \mathcal{C}$). Note that in either case, we have $b \neq v$ since $\tau(b) = \tau(p_2) < \tau(x)$ and also $\tau(x) < \tau(v)$ (by the hypothesis of this case).

Existence of the chordless paths Q_3 and Q_4 implies that $u \rightarrow v$ obeys Item 3 of Observation 4.6.54.

This concludes the second case.

We have thus shown that in all cases $u \rightarrow v$ obeys Observation 4.6.54. This completes the induction, and hence also the proof of Observation 4.6.54. $\square$

Observation 4.6.54 implies the following corollary.

Corollary 4.6.56. *For $i \in \{1, 2\}$, let $\mathcal{C}$ be an undirected connected component of M_i . Let τ be the LBFS ordering returned at Line 5 of Algorithm 8 in the iteration of the second foreach loop (lines 4-18 of Algorithm 8) when $\mathcal{C}$ is considered. For an undirected edge $u - v$ in $\mathcal{C}$, if $u \rightarrow v \in M$ then either $u \rightarrow v$ is added to M in the initialization step (Line 1 of Algorithm 8) or in the second foreach loop (lines 4-18 of Algorithm 8) when $\mathcal{C}$ is considered, in the iteration of the first while loop (lines 8-17 of Algorithm 8) when $a = \tau(v)$ and in the iteration of the second while loop (lines 10-15 of Algorithm 8) when $b = \tau(u)$.*

Proof. Without loss of generality, let us assume that $i = 1$, i.e., $\mathcal{C}$ is an undirected connected component of M_1 . From the construction of M , it's evident that if $u \rightarrow v \in M$, then it's either added at the initialization step (Line 1 of Algorithm 8), or $u - v$ is in some undirected connected component $\mathcal{C}'$ of M_1 or M_2 , and it gets directed at Line 12 when $\mathcal{C}'$ is considered in the second foreach loop (lines 4-18 of Algorithm 8).

If $u \rightarrow v$ is added at the initialization step, then we are done. Suppose $u \rightarrow v$ is not added at the initialization step; we'll now show that $\mathcal{C} = \mathcal{C}'$.

Suppose $\mathcal{C} \neq \mathcal{C}'$. Then, $\mathcal{C}'$ must be an undirected connected component of M_2 , as two different undirected connected component of a graph cannot share a node, and both $\mathcal{C}$ and $\mathcal{C}'$ contain $u - v$. This implies $u, v \in V_{M_1} \cap V_{M_2} = I \subseteq V_O$.

Since $u - v \in \mathcal{C}'$ and $u \rightarrow v \in M$, $u \rightarrow v$ must obey Observation 4.6.54. We now show that this implies $u \rightarrow v \in O$.

If Item 1 of Observation 4.6.54 holds, then we have $u \rightarrow v \in O$. If one of Item 2 or Item 3 of Observation 4.6.54 holds, then in the undirected connected chordal component $\mathcal{C}'$ of M_2 , there exists an edge $x - y \in \mathcal{C}'$ such that $x \rightarrow y \in O$, and either (a) $(x, y) \neq (u, v)$ and there exists a chordless path from (x, y) to (u, v) in $\mathcal{C}'$ (due to Item 2 of Observation 4.6.54), or (b) $y \neq v, u \neq x$, and there exist chordless paths Q_1 and Q_2 in $\mathcal{C}'$ such that Q_1 is a chordless path from (x, y) to v , and Q_2 is a chordless path from (v, u) to x (due to Item 3 of Observation 4.6.54).

Since $\mathcal{C}'$ is an undirected connected chordal component of M_2 (from Item 2 of Theorem 1.1.24), a chordless path in $\mathcal{C}'$ is a chordless path of M_2 . And, from Observation 4.6.8, every chordless path in $\mathcal{C}'$ is a triangle-free path. This implies chordless paths in $\mathcal{C}'$ are triangle-free paths of M_2 . Also, since $(x, y) \in \mathcal{C}'$, an undirected connected component of M_2 , and $x \rightarrow y \in O$, this implies $x, y \in V_{M_2} \cap V_O = V_{O_2}$.

Since (O_2, P_{21}, P_{22}) is the shadow of M_2 on V_{O_2} , from Definition 4.3.1, O_2 is an induced subgraph of M_2 . This implies $(x, y) \in E_{O_2}$, and if there exists a chordless path in $\mathcal{C}'$, from (x, y) to (u, v) , then $P_{21}((x, y), (u, v)) = 1$, and if there exist chordless paths in $\mathcal{C}'$, from (x, y) to v and from (v, u) to x then $P_{22}((x, y), v) = P_{22}((v, u), x) = 1$.

Since $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$, therefore, in both possibilities, from Item 3 of Definition 4.5.6, we again have $u \rightarrow v \in O$.

But, if $u \rightarrow v \in O$, then $u \rightarrow v$ has been added to M at the initialization step, contradicting our assumption. This implies $\mathcal{C} = \mathcal{C}'$. This further implies $u \rightarrow v$ is added to M at the iteration of the second foreach loop (lines 3-19) when $\mathcal{C}$ is considered. It is obvious from the algorithm that in that iteration, $u \rightarrow v$ is added to M when $a = \tau(v)$ and $b = \tau(u)$. This completes the proof of Corollary 4.6.56. $\square$

The following corollary is implied by Corollaries 4.6.44 and 4.6.45 when applied to Observation 4.6.54.

Corollary 4.6.57. *For $i \in \{1, 2\}$, and any undirected connected component $\mathcal{C}$ of M_i at Line 5 of Algorithm 8, let τ be the LBFS ordering returned by Algorithm 7. Then, for $u - v \in \mathcal{C}$, at any iteration of the while loop (lines 8-17) of Algorithm 8, if $u \rightarrow v \in M$, then at least one of the following occurs:*

1. $u \rightarrow v$ is strongly protected in O , and $\tau(u) < \tau(v)$, or
2. $\exists a \rightarrow b \in O$ such that there exists a chordless path $Q = (p_1 = a, p_2 = b, \dots, p_{l-1} = u, p_l = v)$ from (a, b) to (u, v) in $\mathcal{C}$ such that $(a, b) \neq (u, v)$ and $\tau(p_1) < \tau(p_2) < \dots < \tau(p_{l-1}) < \tau(p_l)$. In particular, $\tau(u) < \tau(v)$.
3. $\exists a \rightarrow b \in O$ such that there exist chordless paths Q_1 and Q_2 such that $Q_1 = (p_1 = a, p_2 = b, \dots, p_{l-1}, p_l = v)$ is a chordless path from (a, b) to v in $\mathcal{C}$ and Q_2 is a chordless path from (v, u) to a in $\mathcal{C}$ such that $b \neq v$ and $u \neq a$, and $\tau(p_1 = a) < \tau(p_2 = b) <$

$\dots < \tau(p_{l-1}) < \tau(p_l = v)$, and $\tau(z) < \tau(v)$ for every node $z \neq v$ in Q_2 (in particular, $\tau(u) < \tau(v)$).

Proof. We prove the relative ordering of nodes (mentioned in Items 1 to 3 of Corollary 4.6.57) at the end of the proof. We first prove the remaining parts of the corollary. Let for $i \in \{1, 2\}$, $\mathcal{C}$ be an undirected connected component of M_i , and $u - v \in \mathcal{C}$. From Observation 4.6.54, if $u \rightarrow v \in M$ then either (a) $u \rightarrow v \in O$ (Item 1 of Observation 4.6.54), or (b) there exists an edge $a - b \in \mathcal{C}$ such that there exists a chordless path from (a, b) to (u, v) in $\mathcal{C}$ (Item 2 of Observation 4.6.54), or (c) there exists an edge $a - b \in \mathcal{C}$ such that there exist chordless paths in $\mathcal{C}$ from (a, b) to v and from (v, u) to a (Item 3 of Observation 4.6.54).

Suppose $u - v \in \mathcal{C}$, and $u \rightarrow v \in M$ because it obeys Item 1 of Observation 4.6.54, i.e., $u \rightarrow v \in O$. This implies $u, v \in V_{M_i} \cap V_O = I \subseteq V_{O_i}$. As (O_i, P_{i1}, P_{i2}) is the shadow of M_i on V_{O_i} , therefore, from Item 1 of Definition 4.3.1, O_i is an induced subgraph of M_i . Since $u, v \in V_{O_i}$ and $u - v \in \mathcal{C}$, an undirected connected component of M_i , therefore, $u - v \in O_i$. Since $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$, therefore, O obeys Definition 4.5.6. Item 3 of Definition 4.5.6 implies that if $u \rightarrow v \in O$ then either (i) $u \rightarrow v$ is strongly protected in O , or (ii) there exists an edge $a - b \in O_i$ such that $a \rightarrow b \in O$ and $P_{i1}((a, b), (u, v)) = 1$, or (iii) there exists an edge $a - b \in O_i$ such that $a \rightarrow b \in O$ and $P_{i2}((a, b), v) = P_{i2}((v, u), a) = 1$.

If $u \rightarrow v$ is strongly protected in O then it obeys Item 1 of Corollary 4.6.57.

Suppose $u \rightarrow v \in O$ and there exists an edge $a - b \in O_i$ such that $P_{i1}((a, b), (u, v)) = 1$. We first show that $a - b$ and $u - v$ are in the same undirected connected component $\mathcal{C}$ of M_i . Since $P_{i1}((a, b), (u, v)) = 1$ this implies $(a, b) \neq (u, v)$ and there exists a triangle-free path $Q = (q_1 = a, q_2 = b, \dots, q_{l-1} = u, q_l = v)$ from (a, b) to (u, v) in M_i . We claim that all the edges of P are undirected. If this is not true then pick a directed edge $u_j \rightarrow u_{j+1}$ of P such that j is maximum. Since $u - v$ is an undirected edge in M_i , therefore, $j < l - 1$. But, then $u_j \rightarrow u_{j+1} - u_{j+2}$ is an induced subgraph of M_i , contradicting Item 3 of Theorem 1.1.24 (recall M_i is an MEC). This validates our claim that all the edges of Q are undirected. This implies Q is a triangle-free path in the undirected connected component $\mathcal{C}$ of M_i . From Item 2 of Theorem 1.1.24, $\mathcal{C}$ is a chordal graph. Therefore, from Proposition 4.6.9, Q is a chordless path in $\mathcal{C}$. This

implies $u \rightarrow v$ obeys Item 2 of Corollary 4.6.57.

Suppose $u \rightarrow v \in O$ because there exists an edge $a - b \in O_i$ such that $P_{i2}((a, b), v) = P_{i2}((v, u), a) = 1$. This implies there exist triangle-free paths Q_1 and Q_2 in M_i such that Q_1 is a triangle-free path from (a, b) to v , and Q_2 is a triangle-free path from (v, u) to a . All the edges in Q_1 and Q_2 must be undirected, otherwise, by concatenating Q_1 and Q_2 we get a directed cycle in M_i , contradicting Item 1 of Theorem 1.1.24, as M_i is an MEC. This implies Q_1 and Q_2 are triangle-free paths in $\mathcal{C}$, an undirected connected component of M_i . From Item 2 of Theorem 1.1.24, $\mathcal{C}$ is a chordal graph. Therefore, from Proposition 4.6.9, Q_1 and Q_2 are chordless paths in $\mathcal{C}$. This implies $u \rightarrow v$ obeys Item 3 of Corollary 4.6.57.

Suppose $u \rightarrow v \in M$ because it obeys Item 2 of Observation 4.6.54. Then, $u \rightarrow v$ obeys Item 2 of Corollary 4.6.57. Similarly, if $u \rightarrow v \in M$ because it obeys Item 3 of Observation 4.6.54. Then, $u \rightarrow v$ obeys Item 3 of Corollary 4.6.57.

The ranking of the nodes comes from Observation 4.6.5 and Corollary 4.6.45. If $u \rightarrow v \in O$ then from Corollary 4.6.45, $\tau(u) < \tau(v)$. In the chordless path of Item 2 of Corollary 4.6.57, $a \rightarrow b \in O$. From Corollary 4.6.45, $\tau(a) < \tau(b)$. Then, from Observation 4.6.5, the nodes of the chordless path follows the order: $\tau(p_1 = a) < \tau(p_2 = b) < \dots < \tau(p_{l-1}) < \tau(p_l)$. Similarly, the nodes in the chordless path P_1 of Item 3 of Corollary 4.6.57 follow the order: $\tau(p_1 = a) < \tau(p_2 = b) < \dots < \tau(p_{l-1}) < \tau(p_l = v)$. In particular, this also implies $\tau(a) < \tau(v)$. Thus, Observation 4.6.5 implies that for all the nodes $z \neq v$ of the chordless path P_2 of Item 3 of Corollary 4.6.57, we have $\tau(z) < \tau(v)$. This completes the proof of Corollary 4.6.57. $\square$

We are now ready to prove Lemma 4.6.49.

Proof of Lemma 4.6.49. We initialize M as $\mathcal{U}_M(M_1, M_2, O)$ (Line 1 of Algorithm 8). This implies that after the initialization step, if $u \rightarrow v \in O$ then $u \rightarrow v \in M$. Throughout the algorithm, we only convert an undirected edge into a directed one (lines 11-13). This implies that if $u \rightarrow v \in O$ then $u \rightarrow v$ belongs to the graph M returned by the algorithm. To complete the proof, we have to show that even for any undirected edge $u - v \in O$, M returned by Algorithm 8 has $u - v$.

Suppose, if possible, that there is an edge $u - v$ of O that is directed as $u \rightarrow v$ in

M . If $u - v \in O$ then $u - v \in \text{skeleton}(O)$. From the construction, $\text{skeleton}(O) = \text{skeleton}(O_1) \cup \text{skeleton}(O_2)$. This implies either $u - v \in \text{skeleton}(O_1)$ or $u - v \in \text{skeleton}(O_2)$ or both.

Without loss of generality, let us assume that $u - v \in \text{skeleton}(O_1)$. Since $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$ and $u - v \in O$, from Item 1 of Definition 4.5.6, $u - v \in O_1$. This implies $u - v \in M_1$ as O_1 is an induced subgraph of M_1 (from Item 1 of Definition 4.3.1, as (O_1, P_{11}, P_{12}) is the shadow of M_1 on V_{O_1}).

This further implies that there exists an undirected connected component $\mathcal{C}$ of M_1 that contains $u - v$. Since from our assumption, $u \rightarrow v \in M$ and $u \rightarrow v \notin O$, therefore, from Observation 4.6.54, either (a) there exists $a - b \in \mathcal{C}$ such that $a \rightarrow b \in O$ and there exists a chordless path from (a, b) to (u, v) in $\mathcal{C}$, or (b) there exists $a - b \in \mathcal{C}$ such that there exist chordless paths in $\mathcal{C}$ from (a, b) to v and from (v, u) to a .

In both of the possibilities, $a \rightarrow b \in O$ and $a - b \in M_1$ (recall $\mathcal{C}$ is an undirected connected component of M_1). This implies $a, b \in V_{M_1} \cap V_O = I \subseteq V_{O_1}$. This further implies $a - b \in O_1$ as O_1 is an induced subgraph of M_1 . In the first possibility, since there is a chordless path from (a, b) to (u, v) in $\mathcal{C}$, therefore, there is a chordless path from (a, b) to (u, v) in M_1 . From Observation 4.6.8, this further implies that there is a triangle-free path from (a, b) to (u, v) in M_1 . That means $P_{11}((a, b), (u, v)) = 1$. But, then from Item 3 of Definition 4.5.6, $u \rightarrow v \in O$, a contradiction.

Similarly, in the second possibility, we have $a - b \in O_1$ and there exist triangle-free paths from (a, b) to v and from (v, u) to a in M_1 . This implies $P_{22}((a, b), v) = P_{22}((v, u), a) = 1$. But, then from Item 3 of Definition 4.5.6, $u \rightarrow v \in O$, a contradiction. This implies that for any undirected edge $u - v \in O$, M returned by Algorithm 8 has $u - v$. The above discussion implies that $M[V_O] = O$. $\square$

The following observations can be easily derived from Corollary 4.6.57. These observations will be used later, and, if preferred, the reader may skip ahead to page 206, where we prove that M is an MEC of H (Lemma 4.6.50).

Observation 4.6.58. *For $a \in \{1, 2\}$, and for any undirected connected component $\mathcal{C}$ of M_a , let τ be the LBFS ordering obtained from Algorithm 7 when it is called at Line 5 of Algorithm 8. Then, for $u - v \in \mathcal{C}$, if $u \rightarrow v \in O$ then $u \rightarrow v \in M$, and $\tau(u) < \tau(v)$.*

Proof. At the initialization step, Line 1 of Algorithm 8, if $u \rightarrow v \in O$, then $u \rightarrow v \in M$. From Corollary 4.6.45, it follows that $\tau(u) < \tau(v)$. $\square$

Observation 4.6.59. For $a \in \{1, 2\}$, and for any undirected connected component $\mathcal{C}$ of M_a , let τ be the LBFS ordering returned by Algorithm 7 when it is called at Line 5 of Algorithm 8. Then, for $u - v \in \mathcal{C}$, if $u \rightarrow v \in M$, then $\tau(u) < \tau(v)$.

Proof. Without loss of generality, let us assume that $a = 1$. For $u - v \in \mathcal{C}$, if $u \rightarrow v \in M$, then either it got directed at the initialization step (Line 1 of Algorithm 8), or it gets directed at lines 11-13 of Algorithm 8.

At the initialization step, $M = \mathcal{U}_M(M_1, M_2, O)$. Therefore, if $u \rightarrow v \in M$ and $u \rightarrow v \notin M_1$, then either $u \rightarrow v \in M_2$ or $u \rightarrow v \in O$. If $u \rightarrow v \in M_2$, then $u, v \in V_{M_1} \cap V_{M_2} = I \cup V_{O_2}$. Since O_2 is an induced subgraph of M_2 , therefore $u \rightarrow v \in O_2$.

Since (O, P_1, P_2) is an extension of $(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$, from Definition 4.5.6, $u \rightarrow v \in O$. This further implies that if $u \rightarrow v \in M$, then either $u \rightarrow v \in O$ or it gets directed during the run of lines 11-13 of Algorithm 8.

If $u \rightarrow v \in O$, then from Observation 4.6.58, $\tau(u) < \tau(v)$. And, if $u \rightarrow v$ is directed during the run of lines 11-13 of Algorithm 8, then from Items 2 and 3 of Corollary 4.6.57, $\tau(u) < \tau(v)$. $\square$

Observation 4.6.60. For $a \in \{1, 2\}$, and for any undirected connected component $\mathcal{C}$ of M_a , let τ be the LBFS ordering obtained from Algorithm 7. Then, for $u - v \in \mathcal{C}$, if there exists $a - b \in \mathcal{C}$ such that either $a \rightarrow b \in M$ or $a \rightarrow b \in O$, and there exists a chordless path $Q = (u_1 = a, u_2 = b, \dots, u_{l-1} = u, u_l = v)$ from (a, b) to (u, v) in $\mathcal{C}$ then $\tau(u_1) < \tau(u_2) < \dots < \tau(u_l)$, and for all $1 \leq i \leq l-1$, $u_i \rightarrow u_{i+1} \in M$. In particular, $u \rightarrow v \in M$.

Proof. Note that when $a \rightarrow b \in O$, from the initialization step (Line 1) of Algorithm 7, we have $a \rightarrow b \in M$. Thus, we can assume that $a \rightarrow b \in M$.

Now, since $Q = (u_1 = a, u_2 = b, \dots, u_{l-1} = u, u_l = v)$ is a chordless path from (a, b) to (u, v) in $\mathcal{C}$, and $a \rightarrow b \in M$, Observation 4.6.59 implies that $\tau(u_1 = a) < \tau(u_2 = b)$. Since Q is a chordless path in the undirected chordal graph $\mathcal{C}$ and τ is an LBFS ordering of $\mathcal{C}$, Observation 4.6.5 implies that we must then have $\tau(u_1) < \tau(u_2) < \dots < \tau(u_{l-1}) < \tau(u_l)$.

Since $u_1 = a$ and $u_2 = b$, we have $u_1 \rightarrow u_2 \in M$. Lines 8-17 of Algorithm 8 show that for $1 < i < l$, in the iteration of the while loop at Line 8 when $a = i + 1$, before the end of the iteration of the while loop at Line 10 when $b = i$, we get $u_i \rightarrow u_{i+1} \in M$. Because, if at the initialization step, we do not have $u_i \rightarrow u_{i+1}$ in M then due to the existence of the induced subgraph $u_{i-1} \rightarrow u_i \rightarrow u_{i+1}$ in M , the if condition at Line 11 is satisfied and we get $u_i \rightarrow u_{i+1} \in M$ at Line 12. Thus, we conclude that for all $1 \leq i \leq l - 1$, $u_i \rightarrow u_{i+1} \in M$. $\square$

This completes the proof of all the claims and observations used in the proof of Lemma 4.6.49. To recap, we first argue that each directed edge in O is a directed edge in M (this follows from the initialization step in the construction of M , Line 1 of Algorithm 8). Then, we show that each undirected edge in O is an undirected edge in M . This validates that O is an induced subgraph of M . Lemma 4.6.49 states the same.

We now proceed to prove Lemma 4.6.50.

Proof of Lemma 4.6.50. We first show that M is an MEC. Then we show that $\text{skeleton}(M) = \text{skeleton}(H)$. Together these two imply that M is an MEC of H . Theorem 1.1.24 provides necessary and sufficient conditions for an MEC. Lemmas 4.6.61 to 4.6.64 validate that M obeys the required conditions mentioned in Theorem 1.1.24 to be an MEC. This further implies that M is an MEC.

Lemma 4.6.61. *M is a chain graph.*

Lemma 4.6.62. *Every undirected connected component of M is chordal.*

Lemma 4.6.63. *M does not have an induced subgraph of the form $a \rightarrow b - c$.*

Lemma 4.6.64. *Every directed edge $u \rightarrow v \in M$ is strongly protected in M .*

Proofs of Lemmas 4.6.61 to 4.6.64 are given below.

We now prove that $\text{skeleton}(M) = H$. From the initialization step (Line 1 of Algorithm 8), $M = \mathcal{U}_M(M_1, M_2, O)$. The remaining lines of Algorithm 8 orient the undirected edges of M and do not change the skeleton of M . Therefore, $\text{skeleton}(M) = \text{skeleton}(M_1) \cup \text{skeleton}(M_2) \cup \text{skeleton}(O)$.

From the construction, $\text{skeleton}(M_1) = \text{skeleton}(H_1)$, $\text{skeleton}(M_2) = \text{skeleton}(H_2)$, and $\text{skeleton}(H) = \text{skeleton}(H_1) \cup \text{skeleton}(H_2) \cup \text{skeleton}(O)$. This implies $\text{skeleton}(M) = \text{skeleton}(H)$. Thus, M is an MEC of H , as it is an MEC with H as its skeleton. $\square$

We will now prove Lemmas 4.6.61 to 4.6.64. If preferred, the reader may choose to bypass the proofs and proceed directly to the proof of Lemma 4.6.51 (page 231), where we demonstrate that M_1 and M_2 are projections of M . The proofs of Lemmas 4.6.61, 4.6.62, 4.6.63, and 4.6.64 can be found on pages 207, 227, 227, and 228, respectively.

Proof of Lemma 4.6.61. The following observations, to be proved later, imply that M is a chain graph.

Observation 4.6.65. For $a \in \{1, 2\}$, $M[V_{H_a}]$ is a chain graph.

Observation 4.6.66. M does not have any chordless cycle.

Suppose M is not a chain graph. Then M has a directed cycle. Choose a cycle $C = (u_1, u_2, \dots, u_l, u_{l+1} = u_1)$ of the smallest length. l must be greater than two.

Suppose $l = 3$. We claim that either all the nodes of the cycle are in M_1 (i.e., $u_1, u_2, u_3 \in V_{M_1}$) or all the nodes of the cycle are in M_2 (i.e., $u_1, u_2, u_3 \in V_{M_2}$).

Suppose the claim is not true. Then there exists a node in the cycle that is in M_1 but not in M_2 , and there exists a node in the cycle that is in M_2 but not in M_1 . Without loss of generality, let us assume that u_1 is in M_1 but not in M_2 (i.e., $u_1 \in V_{M_1} \setminus V_{M_2} = V_{M_1} \setminus I$, where $I = V_{H_1} \cap V_{H_2}$ is a vertex separator of H separating $V_{H_1} \setminus I$ and $V_{H_2} \setminus I$). Then u_2 and u_3 , the neighbors of u_1 in C , must be in M_1 (neither u_2 nor u_3 can be in $V_{M_2} \setminus I$ since there is no edge between $V_{M_1} \setminus I$ and $V_{M_2} \setminus I$, as they are separated by I). This validates our claim.

From Observation 4.6.65, there cannot be a directed cycle in $M[V_{M_1}]$ or $M[V_{M_2}]$. Therefore, C must have a length greater than three.

Suppose $l > 3$. Since we have chosen a cycle of the smallest length, C must be a chordless cycle. If C had a chord (an edge between two non-adjacent nodes of the cycle), we would obtain a directed cycle of smaller length in M , regardless of whether the chord is directed or undirected. But from Observation 4.6.66, M cannot have any chordless cycle. This implies this case cannot occur.

Thus, we have shown that M does not have any directed cycle, i.e., M is a chain graph. This completes the proof of Lemma 4.6.61. $\square$

We now prove Observations 4.6.65 and 4.6.66. The proof of Observation 4.6.65 is given below, and the proof of Observation 4.6.66 is given at page 224.

Proof of Observation 4.6.65. If $M[V_{H_a}]$ is not a chain graph, then there exists a directed cycle $C = (u_1, u_2, \dots, u_l, u_{l+1} = u_1)$ in M such that each node of the cycle is in V_{H_a} .

From the construction of M , for $p, q \in V_{H_a}$, if there is an edge $p \rightarrow q$ in M , then either $p \rightarrow q \in M_a$ or $p - q \in M_a$, and if $p - q \in M$, then $p - q \in M_a$. This implies C is also a cycle in M_a . This further implies that none of the edges in C are directed in M_a , otherwise M_a contains a directed cycle, contradicting Item 1 of Theorem 1.1.24.

This implies all the edges of the cycle are undirected edges in M_a , and all the nodes of the cycle are part of the same undirected connected component $\mathcal{C}$ of M_a . This further implies that for some undirected connected component $\mathcal{C}$ of M_a , $M[V_{\mathcal{C}}]$ contains a directed cycle. From Observation 4.6.67, this is not possible.

Observation 4.6.67. For $a \in \{1, 2\}$, for any undirected connected component $\mathcal{C}$ of M_a , there does not exist a directed cycle in $M[V_{\mathcal{C}}]$.

Proof of Observation 4.6.67 is given below. This completes the proof of Observation 4.6.65. $\square$

Proof of Observation 4.6.67. Suppose for some $a \in \{1, 2\}$, for some undirected connected component $\mathcal{C}$ of M_a , there exist a directed cycle in $M[V_{\mathcal{C}}]$. Pick a directed cycle $C = (u_1, u_2, \dots, u_l, u_{l+1} = u_1)$ in $M[V_{\mathcal{C}}]$ of the smallest length. We first show that l cannot be greater than three.

Suppose $l > 3$. Then, C must be a chordless cycle, because if C has a chord then whatever the direction of the chord, we get a directed cycle of smaller length. From the construction of M , $\text{skeleton}(M[V_{\mathcal{C}}]) = \text{skeleton}(M_a[V_{\mathcal{C}}]) = \mathcal{C}$. This further implies C is a chordless cycle in $\mathcal{C}$. But, since $\mathcal{C}$ is a chordal graph (from Item 2 of Theorem 1.1.24, as $\mathcal{C}$ is an undirected connected component of the MEC M_a), it cannot contain any chordless cycle, a contradiction. Therefore, if there exists any directed cycle in $M[V_{\mathcal{C}}]$ then the shortest length directed cycle must contain only three vertices.

There are three possibilities: either all the edges of the cycle are directed, or at most two edges of the cycle are directed, or only one edge of the cycle is directed.

Suppose there exist vertices $x, y, z \in \mathcal{C}$ such that $x - y - z - x \in \mathcal{C}$ and $x \rightarrow y \rightarrow z \rightarrow x \in M$, i.e., all edges of the cycle are directed. Let τ be the LBFS ordering of $\mathcal{C}$ we get at Line 5 of Algorithm 8. From Observation 4.6.59, for $u - v \in V_{\mathcal{C}}$, if $u \rightarrow v \in M$ then $\tau(u) < \tau(v)$. This implies that $x \rightarrow y \rightarrow z \rightarrow x$ cannot occur in M , otherwise, we have $\tau(x) < \tau(y) < \tau(z) < \tau(x)$, a contradiction.

Suppose there exist vertices $x, y, z \in \mathcal{C}$ such that $x - y - z - x \in \mathcal{C}$ and $x \rightarrow y \rightarrow z - x \in M$ (two edges of the cycle are directed). Then from Observation 4.6.59, $\tau(x) < \tau(y) < \tau(z)$. And, while processing the edge $x - z$ on lines 11-13 of Algorithm 8, we replace $x - z$ with $x \rightarrow z$. This implies that $x \rightarrow y \rightarrow z - x$ cannot occur in M . We thus have the following: if vertices x, y, z in $\mathcal{C}$ form a directed cycle $\mathcal{C}$ in M , then $\mathcal{C}$ has exactly one directed edge. The only possibility that remains therefore is of a cycle of the form $x \rightarrow y - z - x$. Observation 4.6.68, stated below, shows that even this possibility cannot occur.

Observation 4.6.68. *For some $a \in \{1, 2\}$, let $\mathcal{C}$ be an undirected connected component of M_a , and M be the graph returned by Algorithm 8 for input M_1, M_2 and O . There does not exist vertices $x, y, z \in \mathcal{C}$ such that $x - y - z - x \in \mathcal{C}$ and $x \rightarrow y - z - x \in M$.*

The proof of Observation 4.6.68 is presented below. This completes the proof of Observation 4.6.67. $\square$

Proof of Observation 4.6.68. Let τ be the LBFS ordering returned by Algorithm 7 at Line 5 of Algorithm 8, when $\mathcal{C}$ is being processed. From Observation 4.6.59, for $p - q \in V_{\mathcal{C}}$, if $p \rightarrow q \in M$, then $\tau(p) < \tau(q)$. We proceed by structuring the analysis based on the relative ordering of x, y , and z in τ . Claim 4.6.69 lists all such potential orderings and shows that none of them are allowed, thus completing the proof of Observation 4.6.68.

Claim 4.6.69. *There do not exist $u, v, w \in \mathcal{C}$ such that $\tau(u) < \tau(v) < \tau(w)$, $u - v - w - u \in \mathcal{C}$ and one of the following holds: (a) $u \rightarrow v - w - u \in M$, (b) $v \rightarrow w - u - v \in M$, or (c) $u \rightarrow w - v - u \in M$.*

We establish the validity of Claim 4.6.69 using induction on $\tau(w)$.

Base Case ($\tau(w) = 2$): This case is vacuously true, as any node in τ must have a rank greater than or equal to 1. Therefore, there are no nodes u and v satisfying $1 \leq \tau(u) < \tau(v) < \tau(w) = 2$.

Inductive Hypothesis: Suppose there exists some $k \geq 2$ such that for all $2 \leq j \leq k$, if $\tau(w) = j$ then Claim 4.6.69 holds. That means, for all $2 \leq j \leq k$, there do not exist $u, v, w \in \mathcal{C}$ such that $\tau(u) < \tau(v) < \tau(w) = j$, $u - v - w - u \in \mathcal{C}$ and one of the following holds: (a) $u \rightarrow v - w - u \in M$, (b) $v \rightarrow w - u - v \in M$, or (c) $u \rightarrow w - v - u \in M$.

Inductive Step: We now demonstrate that Claim 4.6.69 holds for $\tau(w) = k + 1$.

Suppose, for the sake of argument, that there exist $u, v, w \in \mathcal{C}$ such that $\tau(u) < \tau(v) < \tau(w) = k + 1$, and either $u \rightarrow v - w - u \in M$, or $u - v \rightarrow w - u \in M$, or $u - v - w \leftarrow u \in M$.

We select u to be the node with the maximum possible rank, i.e., for any u' with $\tau(u) < \tau(u')$, there does not exist any v satisfying $\tau(u') < \tau(v) < \tau(w) = k + 1$ and either $u' \rightarrow v - w - u \in M$, or $u' - v \rightarrow w - u \in M$, or $u' - v - w \leftarrow u \in M$ hold. We then select v to be the node with the minimum possible rank, i.e., for any v' with $\tau(u) < \tau(v') < \tau(v) < \tau(w) = k + 1$, neither $u \rightarrow v' - w - u \in M$, nor $u - v' \rightarrow w - u \in M$, nor $u - v' - w \leftarrow u \in M$ hold.

The following claims show that neither $u \rightarrow v - w - u \in M$, nor $u - v \rightarrow w - u \in M$, nor $u - v - w \leftarrow u \in M$ is satisfied:

Claim 4.6.70. $u \rightarrow v - w - u \notin M$.

Claim 4.6.71. $u - v \rightarrow w - u \notin M$.

Claim 4.6.72. $u - v - w \leftarrow u \notin M$.

Proofs of Claims 4.6.70 to 4.6.72 are given below. These claims validate Claim 4.6.69. □

Proof of Claim 4.6.70. Suppose $u \rightarrow v - w - u \in M$. From Corollary 4.6.57, at least one of the following has occurred:

1. $u \rightarrow v$ is strongly protected in O .

2. There exists $a \rightarrow b \in O$ such that $(a, b) \neq (u, v)$ and there exists a chordless path from (a, b) to (u, v) in $\mathcal{C}$.
3. There exists $a \rightarrow b \in O$ such that $b \neq v, u \neq a$ and there exist chordless paths in $\mathcal{C}$ from (a, b) to v and from (v, u) to a .

We go through each possibility.

1. Assume that $u \rightarrow v$ is strongly protected in O . Consequently, $u \rightarrow v$ is part of one of the induced subgraphs of O , as depicted in Figure 1.3.

(a) Consider the scenario where $u \rightarrow v$ is strongly protected due to its presence in an induced subgraph $p \rightarrow u \rightarrow v \in O$ (similar to Figure 1.3.a). Keep in mind that u, v , and w are members of $\mathcal{C}$, an undirected connected component of M_a for some $a \in \{1, 2\}$. We need to examine whether p belongs to M_a or not.

Suppose $p \notin M_a$. Without loss of generality, let us assume that $a = 1$. From the construction, if $p \notin V_{M_1}$, then u , a neighbor of p in M , belongs to $I = V_{M_1} \cap V_{M_2}$, as $u \in V_{M_1}$ (recall that u belongs to $\mathcal{C}$, an undirected connected component of M_a , and I is a vertex separator of H that separates $V_{M_1} \setminus I$ and $V_{M_2} \setminus I$). However, this implies $u, v, w \in I \cup N(I, H) \subseteq V_O$. Consequently, $u \rightarrow v - w - u \in O$, as from Lemma 4.6.49, $M[V_O] = O$. Nevertheless, this contradicts Item 1 of Definition 4.2.1, as O is a partial MEC. Thus, the only remaining possibility is $p \in M_a$.

Suppose $p \in M_a$. Since $u, p \in V_{M_a}$ and $p \rightarrow u \in M$, from the construction of M , either $p \rightarrow u \in M_a$ or $p - u \in M_a$. If $p \rightarrow u \in M_a$, then $p \rightarrow u - v$ is an induced subgraph in M_a , a contradiction from Item 2 of Theorem 1.1.24. This implies $p - u \in M_a$, which means $p \in \mathcal{C}$.

From Observation 4.6.58, $\tau(p) < \tau(u)$. This further implies $\tau(p) < \tau(u) < \tau(v) < \tau(w)$ (from our hypothesis $\tau(u) < \tau(v) < \tau(w)$). Since $p, v \in \mathcal{C}$, and there is no edge between p and v in M , from the construction of M , there must be no edge between p and v in $\mathcal{C}$. This also implies $p - w \notin \mathcal{C}$; otherwise, since $\tau(v) < \tau(w)$, $\tau(p) < \tau(w)$, and $v - w \in \mathcal{C}$, from Observation 4.6.4,

$p - v \in \mathcal{C}$, leading to a contradiction. This implies that $p \rightarrow u - w \in M$. However, this is not possible as while running lines 11-13 of Algorithm 8, we get $u \rightarrow w \in M$. Hence, this case does not occur.

Considering the above analysis, it is evident that neither $p \in M_a$ nor $p \notin M_a$ holds true. Consequently, this case cannot occur. We will now proceed to explore other possibilities.

- (b) Suppose $u \rightarrow v$ is strongly protected because it is part of an induced subgraph $u \rightarrow v \leftarrow p \in O$ (same as Figure 1.3.b). Similar to the above case, the possibility that $p \notin M_a$, and the possibility that $p \in M_a$ and $v \rightarrow p \in M_a$, would not occur. This further implies that the only possibility we need to consider is when $p \in M_a$ and $v - p \in M_a$, which means $p \in \mathcal{C}$. Since $u, v, p \in V_O$ as well as $\mathcal{C}$, an undirected connected component of M_a , we have $u, v, p \in O_a = V_O \cap V_{M_a}$. As $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$, from Item 2 of Definition 4.5.6, the v-structure $u \rightarrow v \leftarrow p$ also belongs to O_a . Since (O_a, P_{a1}, P_{a2}) is a shadow of M_a on V_{O_a} , from Item 1 of Definition 4.3.1, O_a is an induced subgraph of M_a . This further implies $u \rightarrow v \leftarrow p \in M_a$. However, this leads to a contradiction, as $u - v \in M_a$ (since $\mathcal{C}$ is an undirected connected component of M_a , if $u - v \in \mathcal{C}$, then $u - v \in M_a$). This contradiction implies that this case also does not occur.

- (c) Suppose $u \rightarrow v$ is strongly protected in O because it is part of an induced subgraph $u \rightarrow p \rightarrow v \leftarrow u \in O$ (same as Figure 1.3.c). Similar to the above Case 1.a, the only possibility we need to consider is when $p \in \mathcal{C}$. Then from Observation 4.6.59, $\tau(u) < \tau(p) < \tau(v)$. Combining this with our induction hypothesis, we get $\tau(u) < \tau(p) < \tau(v) < \tau(w)$. There are two possibilities: either $p - w \notin \mathcal{C}$, or $p - w \in \mathcal{C}$.

If $p - w \notin \mathcal{C}$ then while running lines 11-13 of Algorithm 8, we get $v \rightarrow w \in M$ (because $\tau(p) < \tau(v) < \tau(w)$ and $p \rightarrow v - w$ is an induced subgraph in M). This means that this case does not occur as $v - w \in M$.

Suppose then that $p - w \in \mathcal{C}$. Then there are three possibilities: either $p - w \in M$, or $p \leftarrow w \in M$, or $p \rightarrow w \in M$. We will show that none of the

three can occur.

- i. If $p - w \in M$, then we would have a triplet (p, v, w) that satisfies the conditions $\tau(u) < \tau(p) < \tau(v) < \tau(w)$, and $p \rightarrow v - w - p \in M$. However, this directly contradicts our choice of u as the vertex with the maximal $\tau(u)$ (recall that we have selected u as a node with maximum rank in τ such that there exists a v satisfying $\tau(u) < \tau(v) < \tau(w) = k + 1$ and $u \rightarrow v - w - u \in M$). Hence, it is concluded that $p - w \notin M$.
- ii. Since $\tau(p) < \tau(w)$, therefore, according to Observation 4.6.59, $w \rightarrow p \notin M$.
- iii. If $p \rightarrow w \in M$ then from lines 11-13 of Algorithm 8, $u \rightarrow w \in M$ (due to $\tau(u) < \tau(p) < \tau(w)$, and $u \rightarrow p \rightarrow w - u \in M$). However, this leads to a contradiction as $u - w \in M$.

This analysis demonstrates that none of the three possibilities can occur, thereby ruling out the possibility of case $p - w \in \mathcal{C}$. With this, we conclude that this case also does not occur.

- (d) Suppose $u \rightarrow v$ is strongly protected in O because it is part of an induced subgraph $u \rightarrow v \leftarrow p - u - p' \rightarrow v$ (same as Figure 1.3.d). Similar to above case 1.a, the only possibility we need to consider is when $p, p' \in \mathcal{C}$. Similar to Case 1.b, the v-structure $p \rightarrow v \leftarrow p'$ in O implies that the v-structure also belongs to M_a , i.e., $p \rightarrow v \leftarrow p' \in M_a$. But, since $\mathcal{C}$ is an undirected connected component of M_a , and $p, v \in \mathcal{C}$, therefore, from Proposition 4.6.1, $p \rightarrow v$ cannot be in M_a . This leads to a contradiction. Hence, it can be deduced that this particular case also does not come to pass.

We have thus shown that $u \rightarrow v$ is a part of none of the induced subgraphs shown in Figure 1.3. This implies that this possibility does not occur, i.e., $u \rightarrow v$ is not strongly protected in O . We now move to the other possibilities.

2. Suppose $u \rightarrow v \in M$ because of the existence of $x - y \in \mathcal{C}$ such that $x \rightarrow y \in O$, and there exists a chordless path $Q = (p_1 = x, p_2 = y, \dots, p_{l-1} = u, p_l = v)$ from (x, y) to (u, v) in $\mathcal{C}$. From Observation 4.6.60, $p_{l-2} \rightarrow p_{l-1} = p_{l-2} \rightarrow u \in M$. We

thus have the induced sub-graph $p \rightarrow u \rightarrow v$ in M with $p = p_{l-2}$. This case is now exactly the same as the first case of the first possibility (the possibility when $p \in M_a$; note that there we only use the fact that u and v are in M_a , not the fact that u and v are in O), and hence as proved there, cannot occur.

3. Suppose there exists $x - y \in O$ such that $x \rightarrow y \in O$, and there exist chordless paths in $\mathcal{C}$: $Q_1 = (p_1 = x, p_2 = y, \dots, p_l = v)$ from (x, y) to v and a chordless path $Q_2 = (q_1 = v, q_2 = u, \dots, q_m = x)$ from (v, u) to x in $\mathcal{C}$, such that $y \neq v$ and $u \neq x$.

Note that by Observation 4.6.60, we then have

$$\tau(p_i) < \tau(v) \text{ and } p_i \rightarrow p_{i+1} \in M \text{ for every } 1 \leq i \leq l-1. \quad (4.11)$$

There are two possibilities: either $Q'_1 = (p_1 = x, p_2 = y, \dots, p_l = v, w)$ is a chordless path in $\mathcal{C}$, or Q'_1 is not a chordless path in $\mathcal{C}$. If Q'_1 were a chordless path, then from Observation 4.6.60, $v \rightarrow w \in M$. This is a contradiction, since by hypothesis, we have $v - w \in M$. Thus, Q'_1 cannot be a chordless path.

We now show that $Q := (p_1 = x, p_2 = y, \dots, p_{l-1}, w)$ (obtained by replacing the node p_l with w in Q_1) is a chordless path.

Since Q_1 is a chordless path in $\mathcal{C}$, and Q'_1 is not a chordless path in $\mathcal{C}$, there must exist an edge $p_i - w \in \mathcal{C}$ for some $i < l$. As observed in Equation (4.11), for every $i \leq l-1$, $\tau(p_i) < \tau(v) < \tau(w)$ (the second inequality comes from the hypothesis of Claim 4.6.69). Thus, for any $i \leq l-2$, if $p_i - w \in \mathcal{C}$, then from Observation 4.6.4, $p_i - v \in \mathcal{C}$. But, then Q_1 is not a chordless path, a contradiction. This implies that the neighbor of w in Q_1 , responsible for ensuring that Q'_1 is not a chordless path, must be p_{l-1} . This further implies that Q is a chordless path.

Note now that since $\tau(p_{l-1}) < \tau(v) < \tau(w)$ and $\tau(u) < \tau(w)$, Observation 4.6.4 implies that $p_{l-1} - u \in \mathcal{C}$ (since w is adjacent to both p_{l-1} and u). We will now show that

$$p_{l-1} \rightarrow w \in M \text{ and } u \rightarrow p_{l-1} \notin M. \quad (4.12)$$

The first follows from Observation 4.6.60 since Q as constructed above is a chordless path in $\mathcal{C}$ from (x, y) to (p_{l-1}, w) where $x \rightarrow y \in M$. Suppose, if possible, that $u \rightarrow p_{l-1} \in M$. Then, by Observation 4.6.59 and the observation preceding Equation (4.12), $\tau(u) < \tau(p_{l-1}) < \tau(w)$. From Corollary 4.6.56, this implies that before the edge $u - w$ is processed at lines 11-13 of Algorithm 8, $u \rightarrow p_{l-1}$ is present in M . Thus, while running lines 11-13 of Algorithm 8, we get $u \rightarrow w \in M$, as $\tau(u) < \tau(p_{l-1}) < \tau(w)$ and $u \rightarrow p_{l-1}, p_{l-1} \rightarrow w \in M$. This is a contradiction, as $u - w \in M$. Therefore, $u \rightarrow p_{l-1} \notin M$.

We now show that there exists some $i \geq 2$ for which $q_{i+1} \rightarrow q_i \in M$. Recall that $Q_2 = (q_1 = v, q_2 = u, \dots, q_m = x)$ is a chordless path from (v, u) to x in $\mathcal{C}$, an undirected connected component of M_a , as guaranteed by the hypothesis for this case.

From the construction, $\text{skeleton}(M[V_{\mathcal{C}}]) = \mathcal{C}$, a chordal graph. Since $u - p_{l-1} \in \mathcal{C}$, therefore, $u - p_{l-1} \in \text{skeleton}(M)$. This implies from Equation (4.12), either $p_{l-1} \rightarrow u \in M$ or $p_{l-1} - u \in M$. Thus, from Equation (4.11), if we do not have $q_{i+1} \rightarrow q_i \in M$ for any $i \geq 2$, then $M[V_{\mathcal{C}}]$ has a directed cycle contained in the sequence $C_1 := (p_1 = x, p_2, \dots, p_{l-1}, q_2 = u, q_3, \dots, q_m = x)$. From Proposition 4.6.2, there must thus exist a directed cycle of the form $C = (a, b, c)$ such that a, b, c are nodes of the cycle C_1 .

From Equation (4.11), $\tau(p_1 = q_m) < \tau(v)$, therefore, from Observation 4.6.5, for path Q_2 , for $i > 1$, $\tau(q_i) < \tau(v)$. Also, from Equation (4.11), for $1 \leq i < l$, $\tau(p_i) < \tau(v)$. Therefore, for $1 \leq i < l$, $\tau(p_i) < \tau(w)$, and for $i > 1$, $\tau(q_i) < \tau(w)$ (since $\tau(v) < \tau(w)$). This implies each node of the cycle C_1 has a rank lower than the rank of w in τ . Therefore, the rank of the highest rank node of the cycle C in τ is less than $k + 1$, the rank of w .

As proved in the paragraphs preceding Observation 4.6.68, only certain kinds of directed three-length cycles (which have only one directed edge) can possibly occur in $M[V_{\mathcal{C}}]$. By our induction hypothesis, none of these cycles can have the highest-ranked node with a rank less than $k + 1$. Therefore, the existence of the directed cycle C leads to a contradiction. This confirms that for some $i \geq 2$,

$$q_{i+1} \rightarrow q_i \in M.$$

Since Q_2 is an undirected chordless path in $\mathcal{C}$, therefore, $\bar{Q}_2 = (q_m, q_{m-1}, \dots, q_2, q_1)$, the reverse of the path Q_2 , is also a chordless path. Consequently, $Q_3 = (q_{i+1}, q_i, \dots, q_2, q_1)$, a subpath of $\bar{Q}_2$, is also a chordless path in $\mathcal{C}$. Since $q_{i+1} \rightarrow q_i \in M$, according to Observation 4.6.60, $\tau(q_{i+1}) < \tau(q_i) < \dots, \tau(q_2) < \tau(q_1)$, and for each $i + 1 \leq j \leq 2$, $q_j \rightarrow q_{j-1} \in M$.

We thus have the induced sub-graph $p \rightarrow u \rightarrow v$ in M with $p = q_3$ (recall $q_2 = u$ and $q_1 = v$). This case is now exactly the same as the first case of the first possibility, and hence as proved there, this case cannot occur.

The above discussion shows that it is not possible that $u \rightarrow v - w - u \in M$. $\square$

Proof of Claim 4.6.71. Suppose that $u - v \rightarrow w - u \in M$. From Corollary 4.6.57, at least one of the following has occurred:

1. $v \rightarrow w$ is strongly protected in O .
2. There exists $a \rightarrow b \in O$ such that there exists a chordless path $Q = (p_1 = a, p_2 = b, \dots, p_{l-1} = v, p_l = w)$ from (a, b) to (v, w) in $\mathcal{C}$ such that $(a, b) \neq (v, w)$ and $\tau(p_1) < \tau(p_2) < \dots < \tau(p_{l-1}) < \tau(p_l)$.
3. There exists $a \rightarrow b \in O$ such that there exist chordless paths Q_1 and Q_2 such that $Q_1 = (p_1 = a, p_2 = b, \dots, p_{l-1}, p_l = w)$ is a chordless path from (a, b) to w in $\mathcal{C}$ and Q_2 is a chordless path from (w, v) to a in $\mathcal{C}$ such that $b \neq w$ and $v \neq a$, and $\tau(p_1 = a) < \tau(p_2 = b) < \dots < \tau(p_{l-1}) < \tau(p_l = w)$, and $\tau(z) < \tau(w)$ for every node $z \neq w$ in Q_2 .

Let us examine each possibility.

1. Suppose $v \rightarrow w$ is strongly protected in O . This implies that $v \rightarrow w$ is part of one of the induced subgraphs of O , as illustrated in Figure 1.3.
 - (a) Assume that $v \rightarrow w$ is strongly protected due to its presence in an induced subgraph $p \rightarrow v \rightarrow w \in O$ (similar to Figure 1.3.a). Similar to the first case

of the first possibility of the proof of Claim 4.6.70, the only relevant scenario is when p belongs to $\mathcal{C}$.

From the construction, $\text{skeleton}(M[V_{\mathcal{C}}]) = \mathcal{C}$. This implies $p - v \in \mathcal{C}$ and $p - w \notin \mathcal{C}$. Given that $p - v \in \mathcal{C}$ and $p \rightarrow v \in M$, we can infer from Observation 4.6.59 that $\tau(p) < \tau(v)$. Given that $u - v \in \mathcal{C}$ and $\tau(u) < \tau(v)$, we can deduce from Observation 4.6.4 that $p - u \in \mathcal{C}$. Since $\text{skeleton}(M[V_{\mathcal{C}}]) = \mathcal{C}$, therefore, either (1) $p - u \in M$ or (2) $p \rightarrow u \in M$ or (3) $u \rightarrow p \in M$. We will show that none of the three can occur.

- i. If $p - u \in M$, then M would possess a directed cycle (p, v, u, p) with the highest rank node v having a rank less than $\tau(w)$, thereby contradicting our induction hypothesis.
- ii. If $p \rightarrow u \in M$ then using Observation 4.6.59, we have $\tau(p) < \tau(u)$. However, since $p - w \notin \mathcal{C}$ and $\tau(p) < \tau(u) < \tau(w)$, applying lines 11-13 of Algorithm 8 would lead to $u \rightarrow w \in M$, which contradicts $u - w \in M$.
- iii. And, if $u \rightarrow p \in M$ then using Observation 4.6.59, we have $\tau(u) < \tau(p)$. Therefore, since $\tau(u) < \tau(p) < \tau(v)$ and $u \rightarrow p \rightarrow v - u \in M$, applying lines 11-13 of Algorithm 8 would lead to $u \rightarrow v \in M$, which contradicts $u - v \in M$.

Hence, based on the analysis above, it becomes evident that none of these scenarios are feasible. Therefore, we conclude that this case cannot occur.

- (b) Assume that $v \rightarrow w$ is strongly protected due to the presence in an induced subgraph $v \rightarrow w \leftarrow p \in O$ (as shown in Figure 1.3.b). Similar to the second case of the first possibility of the proof of Claim 4.6.70, this case does not occur.
- (c) Suppose $v \rightarrow w$ is strongly protected in O because it is part of an induced subgraph $v \rightarrow p \rightarrow w \leftarrow v \in O$ (same as Figure 1.3.c). Similar to the third case of the first possibility of the proof of Claim 4.6.70, $p \in \mathcal{C}$. From Lemma 4.6.49, $v \rightarrow p \rightarrow w \leftarrow v \in M$. From the construction, $\text{skeleton}(M[V_{\mathcal{C}}]) = \mathcal{C}$. This implies $p - w \in \mathcal{C}$ (as $p \rightarrow w \in M$ and $p, w \in \mathcal{C}$). From Observation 4.6.59,

$\tau(v) < \tau(p) < \tau(w)$. From Observation 4.6.4, this implies $u - p \in \mathcal{C}$ (since $p - w, u - w \in \mathcal{C}, \tau(p) < \tau(w)$ and $\tau(u) < \tau(w)$). Since $u - p \in \mathcal{C}$ and $\text{skeleton}(M[V_{\mathcal{C}}]) = \mathcal{C}$, therefore, either (1) $u - p \in M$, or (2) $u \rightarrow p \in M$, or (3) $p \rightarrow u \in M$. We show that none of these cases occur.

- i. If $u - p \in M$ then M has a directed cycle $u - v \rightarrow p - u$ such that the rank of the highest rank node of the cycle is less than $\tau(w)$, which contradicts our induction hypothesis. This implies $u - p \notin M$.
- ii. Suppose $u \rightarrow p \in M$. Then, since $u \rightarrow p, p \rightarrow w \in M$, while running lines 11-13 of Algorithm 8, we get $u \rightarrow w \in M$, a contradiction, as $u - w \in M$. This implies $u \rightarrow p \notin M$ does not occur.
- iii. Since $\tau(u) < \tau(p)$, from Observation 4.6.59, $p \rightarrow u \notin M$.

The above analysis shows that neither $u - p \in M$ nor $u \rightarrow p \in M$ nor $p \rightarrow u \in M$. This implies that this case does not occur.

- (d) Suppose $v \rightarrow w$ is strongly protected in O because it is part of an induced subgraph $v \rightarrow w \leftarrow p - v - p' \rightarrow w$ (same as Figure 1.3.d). Similar to the fourth case of the first possibility of the proof of Claim 4.6.70, this case does not occur.

2. Suppose $v \rightarrow w \in M$ because of the existence of some $x - y \in \mathcal{C}$ such that $x \rightarrow y \in O$, and there exists a chordless path $P = (p_1 = x, p_2 = y, \dots, p_{l-1} = v, p_l = w)$ from (x, y) to (v, w) in $\mathcal{C}$. From Observation 4.6.60, $p_{l-2} \rightarrow p_{l-1} \in M$. We thus have the induced sub-graph $p \rightarrow v \rightarrow w$ in M with $p = p_{l-2}$. This case is now exactly the same as the first case of the first possibility (note that there we only use the fact that p is a node in $\mathcal{C}$, not the fact that $p \rightarrow v \rightarrow w \in O$), and hence as proved there, cannot occur.
3. Suppose there exists $x - y \in O$ such that $x \rightarrow y \in O$, and there exists a chordless path in $\mathcal{C}$: $Q_1 = (p_1 = x, p_2 = y, \dots, p_l = w)$ from (x, y) to w in $\mathcal{C}$, and $Q_2 = (q_1 = w, q_2 = v, \dots, q_m = x)$ from (w, v) to x in $\mathcal{C}$ such that $y \neq w$ and $v \neq x$, and $\tau(p_1 = x) < \tau(p_2 = y) < \dots < \tau(p_{l-1}) < \tau(p_l = w)$, and for all $2 \leq i \leq m$, $\tau(q_i) < \tau(w)$. According to Observation 4.6.60, we have $p_{l-1} \rightarrow w \in M$.

Considering that $\tau(p_{l-1}) < \tau(w)$, $\tau(v) < \tau(w)$, and $\tau(u) < \tau(w)$, it follows from Observation 4.6.4 that $p_{l-1} - u$ and $p_{l-1} - v$ are in $\mathcal{C}$.

There are two possibilities: either $v \rightarrow p_{l-1} \in M$, or $v \rightarrow p_{l-1} \notin M$. Let us examine both possibilities:

- (a) Suppose $v \rightarrow p_{l-1} \in M$. From Observation 4.6.59, $\tau(v) < \tau(p_{l-1})$. Since $u - p_{l-1} \in \mathcal{C}$, we have either $u - p_{l-1} \in M$, or $u \rightarrow p_{l-1} \in M$, or $p_{l-1} \rightarrow u \in M$.
 - i. Suppose $u - p_{l-1} \in M$. This would result in a directed cycle (u, v, p_{l-1}, u) in M where the node with the highest rank, p_{l-1} , has a rank lower than $\tau(w)$, contradicting our induction hypothesis.
 - ii. Suppose $u \rightarrow p_{l-1} \in M$. Since $p_{l-1} \rightarrow w \in M$, running lines 11-13 of Algorithm 8 would yield $u \rightarrow w \in M$. This contradicts the assumption $u - w \in M$. Thus, $u \rightarrow p_{l-1} \notin M$.
 - iii. From the hypothesis of Claim 4.6.69, $\tau(u) < \tau(v)$. Since from our assumption, $v \rightarrow p_{l-1} \in M$, therefore, from Observation 4.6.59, $\tau(v) < \tau(p_{l-1})$. This implies $\tau(u) < \tau(p_{l-1})$. Then, from Observation 4.6.59, $p_{l-1} \rightarrow u \notin M$.

The above analysis shows that neither $u - p_{l-1} \in M$ nor $u \rightarrow p_{l-1} \in M$ nor $p_{l-1} \rightarrow u \in M$. Thus, we conclude that $v \rightarrow p_{l-1}$ is not in M .

- (b) Suppose $v \rightarrow p_{l-1} \notin M$. Then for some $i \geq 2$, $q_{i+1} \rightarrow q_i$; otherwise, we get a directed cycle $(p_1 = x, p_2, \dots, p_{l-1}, q_2 = v, q_3, \dots, q_m = x)$ in M such that the rank of each node in the cycle is less than the rank of w . From Proposition 4.6.2, this would lead to a directed cycle in $M[V_{\mathcal{C}}](= \mathcal{C}$, a chordal graph) of length 3 such that the rank of the highest rank node in the cycle is less than $\tau(w)$, contradicting our induction hypothesis. Hence, for some $i \geq 2$, $q_{i+1} \rightarrow q_i \in M$. Since Q_2 is an undirected chordless path in $\mathcal{C}$, therefore, $\bar{Q}_2 = (q_m, q_{m-1}, \dots, q_2, q_1)$, the reverse of the path Q_2 , is also a chordless path. Consequently, $Q_3 = (q_{i+1}, q_i, \dots, q_2, q_1)$, a subpath of $\bar{Q}_2$, is also a chordless path in $\mathcal{C}$. Since $q_{i+1} \rightarrow q_i \in M$, according to Observation 4.6.60, $\tau(q_{i+1}) < \tau(q_i) < \dots, \tau(q_2) < \tau(q_1)$, and for each

$i + 1 \leq j \leq 2$, $q_j \rightarrow q_{j-1} \in M$. We thus have the induced sub-graph $p \rightarrow v \rightarrow w$ in M with $p = q_3$ (recall Q_2 is a chordless path, $q_1 = w$ and $q_2 = v$). This case is now exactly the same as the first case of the first possibility (note that there we only use the fact that p is a node in $\mathcal{C}$, not the fact that $p \rightarrow v \rightarrow w \in O$), and hence as proved there, cannot occur.

The above discussion shows a contradiction in each possibility, thereby demonstrating that $u - v \rightarrow w - u \notin M$. This establishes the truth of Claim 4.6.71.

□

Proof of Claim 4.6.72. Suppose $u - v - w \leftarrow u \in M$. From Corollary 4.6.57, at least one of the following occurred:

1. $u \rightarrow w$ is strongly protected in O .
2. There exists $a \rightarrow b \in O$ such that there exists a chordless path $Q = (p_1 = a, p_2 = b, \dots, p_{l-1} = u, p_l = w)$ from (a, b) to (u, w) in $\mathcal{C}$ such that $(a, b) \neq (u, w)$ and $\tau(p_1) < \tau(p_2) < \dots < \tau(p_{l-1}) < \tau(p_l)$.
3. There exists $a \rightarrow b \in O$ such that there exist chordless paths Q_1 and Q_2 such that $Q_1 = (p_1 = a, p_2 = b, \dots, p_{l-1}, p_l = w)$ is a chordless path from (a, b) to w in $\mathcal{C}$ and Q_2 is a chordless path from (w, u) to a in $\mathcal{C}$ such that $b \neq w$ and $u \neq a$, and $\tau(p_1 = a) < \tau(p_2 = b) < \dots < \tau(p_{l-1}) < \tau(p_l = w)$, and $\tau(z) < \tau(w)$ for every node $z \neq w$ in Q_2 .

We will consider each possibility.

1. $u \rightarrow w$ is strongly protected in O . Then, $u \rightarrow w$ is part of one of the induced subgraphs of O in Figure 1.3.
 - (a) Suppose $u \rightarrow w$ is strongly protected because it is part of an induced sub-graph $p \rightarrow u \rightarrow w \in O$ (same as Figure 1.3.a). Similar to the first case of the first possibility of the proof of Claim 4.6.70, the only possibility we need to consider is when $p \in \mathcal{C}$. From Observation 4.6.59, $\tau(p) < \tau(u)$. If $p - v \notin \mathcal{C}$, then while running lines 11-13 of Algorithm 8, we get $u \rightarrow v \in M$, a contradiction, as $p \rightarrow u - v$ is an induced subgraph in M and $\tau(p) < \tau(u) < \tau(v)$

(second inequality comes from the hypothesis of Claim 4.6.69). However, this is a contradiction as $u - v \in M$. This implies $p - v \in \mathcal{C}$. There are three possibilities: either $v - p \in M$, or $v \rightarrow p \in M$, or $p \rightarrow v \in M$. Let us consider each one.

- i. If $v - p \in M$ then we have a directed cycle (p, u, v, p) in M such that the rank of the highest rank node of the cycle (which is v) is less than the rank of w , this contradicts the induction hypothesis of Claim 4.6.69. Hence implies $v - p \notin M$.
- ii. From Observation 4.6.59, $v \rightarrow p \notin M$ as $\tau(p) < \tau(v)$.
- iii. Suppose $p \rightarrow v \in M$. But, then while running lines 11-13 of Algorithm 8, we get $v \rightarrow w \in M$, a contradiction, as $p \rightarrow v - w$ is an induced subgraph in M and $\tau(p) < \tau(v) < \tau(w)$.

We have shown that in all the possibilities of this subcase, we get a contradiction. Therefore, this subcase cannot occur.

- (b) Suppose that $u \rightarrow w$ is strongly protected because it is part of an induced subgraph $u \rightarrow w \leftarrow p \in O$ (same as Figure 1.3.b). Similar to the second case of the first possibility of the proof of Claim 4.6.70, this case does not occur.
- (c) Suppose $u \rightarrow w$ is strongly protected in O because it is part of an induced subgraph $u \rightarrow p \rightarrow w \leftarrow u \in O$ (same as Figure 1.3.c). Similar to the third case of the first possibility of the proof of Claim 4.6.70, the only possibility we need to consider is when $p \in \mathcal{C}$. From Observation 4.6.59, $\tau(u) < \tau(p) < \tau(w)$. From Observation 4.6.4, $v - p \in \mathcal{C}$ (since $p - w, v - w \in \mathcal{C}, \tau(p) < \tau(w)$ and $\tau(v) < \tau(w)$). From the construction, $\text{skeleton}(M[V_{\mathcal{C}}]) = \mathcal{C}$. Therefore, $v - p \in \text{skeleton}(M)$. This implies either $v - p \in M$ or $v \rightarrow p \in M$ or $p \rightarrow v \in M$. We go through each possibility.
 - i. Suppose $v - p \in M$. Then we get a directed cycle (u, p, v, u) in M such that the highest rank node (which could be either v or p) in the cycle is less than the rank of w in τ , this contradicts the induction hypothesis of Claim 4.6.69. This implies $v - p \notin M$.
 - ii. Suppose $v \rightarrow p \in M$. Then from Observation 4.6.59, $\tau(v) < \tau(p)$.

Then while running lines 11-13 of Algorithm 8, we get $v \rightarrow w \in M$ (as $\tau(v) < \tau(p) < \tau(w)$ and $v \rightarrow p, p \rightarrow w \in M$). But, this is a contradiction as $v - w \in M$. This implies $v \rightarrow p \notin M$.

- iii. Suppose $p \rightarrow v \in M$. Then from Observation 4.6.59, $\tau(p) < \tau(v)$. Then while running lines 11-13 of Algorithm 8, we get $u \rightarrow v \in M$ (as $\tau(u) < \tau(p) < \tau(v)$ and $u \rightarrow p, p \rightarrow v \in M$). But, this is a contradiction as $u - v \in M$. This implies $p \rightarrow v \notin M$.

The above analysis shows that none of the possibilities can occur. This implies that this case does not occur.

- (d) Suppose $u \rightarrow w$ is strongly protected in O because it is part of an induced subgraph $u \rightarrow w \leftarrow p - u - p' \rightarrow w$ (same as Figure 1.3.d). Similar to the fourth case of the first possibility of the proof of Claim 4.6.70, this case does not occur.

2. Suppose $u \rightarrow w \in M$ because of the existence of some $x - y \in \mathcal{C}$ such that $x \rightarrow y \in O$, and there exists a chordless path $P = (p_1 = x, p_2 = y, \dots, p_{l-1} = u, p_l = w)$ from (x, y) to (u, w) in $\mathcal{C}$. From Observation 4.6.60, $p_{l-2} \rightarrow p_{l-1} \in M$. We thus have the induced sub-graph $p \rightarrow u \rightarrow w$ in M with $p = p_{l-2}$. This case is now exactly the same as the first case of the first possibility (note that there we only use the fact that p is a node in $\mathcal{C}$, not the fact that $p \rightarrow u \rightarrow w \in O$), and hence as proved there, cannot occur.
3. Suppose there exists $x - y \in O$ such that $x \rightarrow y \in O$, and there exist chordless paths in $\mathcal{C}$: $Q_1 = (p_1 = x, p_2 = y, \dots, p_l = w)$ from (x, y) to w in $\mathcal{C}$, and $Q_2 = (q_1 = w, q_2 = u, \dots, q_m = x)$ from (w, u) to x such that $y \neq w, u \neq x, \tau(p_1 = x) < \tau(p_2 = y) < \dots < \tau(p_{l-1}) < \tau(p_l = w)$, and for all $2 \leq i \leq m, \tau(q_i) < \tau(w)$. From Observation 4.6.60, $p_{l-1} \rightarrow w \in M$. Since $\tau(p_{l-1}) < \tau(w), \tau(v) < \tau(w)$, and $\tau(u) < \tau(w)$ (the last two inequality comes from the hypothesis of Claim 4.6.69), therefore, from Observation 4.6.4, $u - p_{l-1}, v - p_{l-1} \in \mathcal{C}$. From the construction, $\text{skeleton}(M[V_{\mathcal{C}}]) = \mathcal{C}$. This implies $u - p_{l-1}, v - p_{l-1} \in \text{skeleton}(M)$. There are two possibilities: either $u \rightarrow p_{l-1} \in M$, or $u \rightarrow p_{l-1} \notin M$ (i.e., either $u - p_{l-1} \in M$ or $p_{l-1} \rightarrow u \in M$). We go through each possibility.

(a) Suppose $u \rightarrow p_{l-1} \in M$. Since $v - p_{l-1} \in \text{skeleton}(M)$, either we have $v - p_{l-1} \in M$ or $v \rightarrow p_{l-1} \in M$ or $v \leftarrow p_{l-1} \in M$.

i. Suppose $v - p_{l-1} \in M$. Then we have a directed cycle (u, p_{l-1}, v, u) in M such that the rank of the highest rank node in the cycle is less than $\tau(w)$ in τ . This contradicts the induction hypothesis of Claim 4.6.69. This implies $v - p_{l-1} \notin M$.

ii. Suppose $v \rightarrow p_{l-1} \in M$. Then, since $p_{l-1} \rightarrow w \in M$, therefore, while running lines 11-13 of Algorithm 8, we get $v \rightarrow w \in M$ (as $v \rightarrow p_{l-1}, p_{l-1} \rightarrow w \in M$). But, from our assumption, $v - w \in M$. This implies that $v \rightarrow p_{l-1} \notin M$.

iii. Suppose $v \leftarrow p_{l-1} \in M$. Then, while running lines 11-13 of Algorithm 8, we get $u \rightarrow v \in M$ (as $u \rightarrow p_{l-1} \in M, p_{l-1} \rightarrow v \in M$). But, from our assumption $u - v \in M$. This implies that $v \leftarrow p_{l-1} \notin M$.

The above analysis shows that when $u \rightarrow p_{l-1} \in M$, neither $v - p_{l-1} \in M$ nor $v \rightarrow p_{l-1} \in M$ nor $v \leftarrow p_{l-1} \in M$. This implies $u \rightarrow p_{l-1} \in M$ does not occur.

(b) Suppose $u \rightarrow p_{l-1} \notin M$. Then for some $i \geq 2$, $q_{i+1} \rightarrow q_i$; otherwise, we get a directed cycle $(p_1 = x, p_2, \dots, p_{l-1}, q_2 = u, q_3, \dots, q_m = x)$ in M such that the rank of each node in the cycle is less than the rank of w . From Proposition 4.6.2, this would lead to a directed cycle in $M[V_{\mathcal{C}}](= \mathcal{C}$, a chordal graph) of length 3 such that the rank of the highest rank node in the cycle is less than $\tau(w)$, contradicting our induction hypothesis. Hence, for some $i \geq 2$, $q_{i+1} \rightarrow q_i \in M$. Since Q_2 is an undirected chordless path, $\bar{Q}_2 = (q_m, q_{m-1}, \dots, q_2, q_1)$, the reverse of the path Q_2 , is also a chordless path. Consequently, $Q_3 = (q_{i+1}, q_i, \dots, q_2, q_1)$, a subpath of $\bar{Q}_2$, is also a chordless path in $\mathcal{C}$. Since $q_{i+1} \rightarrow q_i \in M$, according to Observation 4.6.60, $\tau(q_{i+1}) < \tau(q_i) < \dots, \tau(q_2) < \tau(q_1)$, and for each $i+1 \leq j \leq 2$, $q_j \rightarrow q_{j-1} \in M$. We thus have the induced sub-graph $p \rightarrow u \rightarrow w$ in M with $p = q_3$ (recall Q_2 is a chordless path, $q_1 = w$ and $q_2 = u$). This case is now exactly the same as the first case of the first possibility (note that there we only use

the fact that p is a node in $\mathcal{C}$, not the fact that $p \rightarrow v \rightarrow w \in O$), and hence as proved there, cannot occur.

This indicates a contradiction in all possibilities, thereby demonstrating that $u - v - w \leftarrow u \notin M$. This establishes the truth of Claim 4.6.72.

□

This completes the proofs of all the claims and observations used in the proof of Observation 4.6.65. To recap, we first showed that if, for some $a \in \{1, 2\}$, there exists a directed cycle in $M[V_{M_a}]$, then all the nodes of the cycle belong to the same undirected connected component of M_a . We then used the chordal property of the undirected connected component of the MECs to establish that if there exists a directed cycle in $M[V_{M_a}]$, the shortest such directed cycle must be of length three. Using Claims 4.6.70 to 4.6.72, we show that no three-length directed cycle exists in M with all its nodes in the same undirected connected component of M_a . This validates Observation 4.6.65 that no directed cycle exists in $M[V_{M_a}]$.

We now proceed to prove Observation 4.6.66.

Proof of Observation 4.6.66. Let $C = (u_1, u_2, \dots, u_l, u_{l+1} = u_1)$ be a chordless cycle in M . If C lies entirely within M_1 (i.e., each node of C belongs to V_{M_1}), then by Observation 4.6.65, C is undirected. However, from the construction of M , (a) for any $u, v \in V_{M_1}$, if $u - v \in M$, then $u - v \in M_1$ (due to the initialization step, Line 1 of Algorithm 8, a directed edge in M_1 is a directed edge in M), and (b) if there is no chord in C in M then there is no chord in C in M_1 as well (because all the nodes of C are in M_1 and $\text{skeleton}(M[V_{M_1}]) = \text{skeleton}(M_1)$). This implies that C is an undirected chordless cycle in M_1 , which contradicts Item 2 of Theorem 1.1.24. Hence, the nodes in C cannot be exclusively within M_1 or M_2 . Thus, there must exist a node u_i in C that is in $V_{M_1} \setminus I$, a node u_j in C that is in $V_{M_2} \setminus I$, and a node u_k in C that is in I , as $I = V_{M_1} \cap V_{M_2}$ is a vertex separator of $H = \text{skeleton}(M)$ that separates $V_{M_1} \setminus I$ and $V_{M_2} \setminus I$.

Without loss of generality, let us assume that $u_1 \in I$. Therefore, $u_1, u_2, u_l \in V_O$ since u_2 and u_l are neighbors of u_1 , and $I \cup N(I, H) \subseteq V_O$. Since O is an induced subgraph of M (from Lemma 4.6.49), and C is a cycle in M , for any edge (u_i, u_{i+1}) in C , if $u_i, u_{i+1} \in V_O$, then $(u_i, u_{i+1}) \in O$. More specifically, $(u_1, u_2), (u_l, u_1) \in E_O$.

We enumerate the edges in C present in O in the order they appear, starting from (u_1, u_2) and ending with (u_l, u_1) . This enumeration is represented as $L = ((x_1, y_1), (x_2, y_2), \dots, (x_m, y_m))$, where $(x_1, y_1) = (u_1, u_2)$ and $(x_m, y_m) = (u_l, u_1)$. Notably, $m \geq 2$ since the list contains the edges (u_1, u_2) and (u_l, u_1) . Additionally, let $(x_{m+1}, y_{m+1}) = (x_1, y_1)$. From Item 4 of Definition 4.5.6, $(P_1, P_2) = \text{DPF}(O, P_{11}, P_{12}, P_{21}, P_{22})$ (recall from our assumption that $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$). We present the following claims for the edges in this enumeration:

1. For $1 \leq i \leq m$, if $y_i = x_{i+1}$, then (x_i, y_i, y_{i+1}) is a triangle-free path in O , and $P_1((x_i, y_i), (x_{i+1}, y_{i+1})) = 1$.
2. For $1 \leq i \leq m$, if $y_i \neq x_{i+1}$, then either there exists a triangle-free path from (x_i, y_i) to (x_{i+1}, y_{i+1}) in M_1 , or there exists a triangle-free path from (x_i, y_i) to (x_{i+1}, y_{i+1}) in M_2 , and $P_1((x_i, y_i), (x_{i+1}, y_{i+1})) = 1$.

Proof of the claim. For any $1 \leq i \leq m$ where $y_i = x_{i+1}$, due to the chordless nature of C and the presence of (x_i, y_i) and (x_{i+1}, y_{i+1}) as edges in O , it follows that (x_i, y_i, y_{i+1}) is a triangle-free path in O . Consequently, considering that $(P_1, P_2) = \text{DPF}(O, P_{11}, P_{12}, P_{21}, P_{22})$, by Item 1 of Definition 4.5.3, $P_1((x_i, y_i), (x_{i+1}, y_{i+1}))$ is initialized to 1 due to the presence of a triangle-free path from (x_i, y_i) to (x_{i+1}, y_{i+1}) in O . This validates the first part of the claim.

Suppose for some $1 \leq i \leq m$, $y_i \neq x_{i+1}$. Then, y_i cannot be part of I . Otherwise, the neighbors of y_i would be within O (since $I \cup N(I, H) \subseteq V_O$). That means, for $1 \leq j \leq l$, if $y_i = u_j$ then the next node u_{j+1} in C is also in V_O . This implies (u_j, u_{j+1}) is in O , as C is a cycle in M , $u_j, u_{j+1} \in V_O$, and O is an induced subgraph of M . This implies the next edge after (x_i, y_i) in L is $(u_j, u_{j+1}) = (x_{i+1}, y_{i+1})$. But, then $y_i = x_{i+1}$, a contradiction. This implies that y_i must either be in $V_{M_1} \setminus I$ or in $V_{M_2} \setminus I$.

Without loss of generality, assume that $y_i \in V_{M_1} \setminus I$. Consequently, x_{i+1} must not be in $V_{M_2} \setminus I$, or else the path between y_i and x_{i+1} in C would contain an I node, as I serves as a vertex separator between $V_{H_1} \setminus I$ and $V_{H_2} \setminus I$. This implies that within the path from y_i to x_{i+1} in C , there exists an edge from C in O due to the containment of $I \cup N(I, H) \subseteq V_O$ and the fact that C is a cycle in M , and O is an induced subgraph of M . Similarly, x_{i+1} must not be in I ; otherwise, the edge right before (x_{i+1}, y_{i+1}) in C

would be in O as well (since $I \cup N(I, H) \subseteq V_O$), but the edge prior to (x_{i+1}, y_{i+1}) in O is (x_i, y_i) . This implies that (x_i, y_i) is right before (x_{i+1}, y_{i+1}) in C , leading to $y_i = x_{i+1}$, contradicting the assumption. Hence, $x_{i+1} \in V_{M_1} \setminus I$, which further implies $y_{i+1} \in V_{M_1}$ since a neighbor of a node in $V_{M_1} \setminus I$ within V_O must belong to V_{M_1} , and y_{i+1} is the neighbor of x_{i+1} and lies in V_O (due to $(x_{i+1}, y_{i+1}) \in O$). Thus, x_i, y_i, x_{i+1} , and y_{i+1} all belong to M_1 . Furthermore, the path from (x_i, y_i) to (x_{i+1}, y_{i+1}) must exclusively lie within $M[V_{M_1}]$, otherwise there would be an edge in O between (x_i, y_i) and (x_{i+1}, y_{i+1}) in C , a contradiction. With our construction of M , for any $u, v \in V_{M_1}$, if $(u, v) \in M$, then $(u, v) \in M_1$. This implies that the chordless path from (x_i, y_i) to (x_{i+1}, y_{i+1}) in C is also a chordless path in M_1 , meaning $P_{a1}((x_i, y_i), (x_{i+1}, y_{i+1})) = 1$ (since every chordless path is a triangle-free path). Also, $y_1 \rightarrow x_i \notin M$, otherwise C wouldn't possess the edge (x_i, y_i) . Therefore, by Item 2 of Definition 4.5.3, $P_1((x_i, y_i), (x_{i+1}, y_{i+1})) = 1$. $\square$

It follows that for all $1 \leq i \leq m$, $P_1((x_i, y_i), (x_{i+1}, y_{i+1})) = 1$. Now, we claim that $P_1((x_1, y_1), (x_m, y_m)) = 1$. Suppose this is not true and select the smallest $j \leq m$ for which $P_1((x_1, y_1), (x_j, y_j)) = 0$. Given the previous discussion, since $P_1((x_i, y_i), (x_{i+1}, y_{i+1})) = 1$ for all $1 \leq i \leq m$, it implies that $j > 2$. This further leads to $P_1((x_1, y_1), (x_{j-1}, y_{j-1})) = P_1((x_{j-1}, y_{j-1}), (x_j, y_j)) = 1$. But, from Item 3 of Definition 4.5.3, we get $P_1((x_1, y_1), (x_i, y_i)) = 1$, a contradiction. Hence, $P_1((x_1, y_1), (x_m, y_m)) = 1$. Additionally, from our earlier claim, $P_1((x_m, y_m), (x_{m+1}, y_{m+1})) = P_1((x_1, y_1), (x_m, y_m)) = 1$.

According to the assumption of Lemma 4.5.8, (O, P_1, P_2) is an extension of $(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$. Thus, by Definition 4.5.6, $\text{DPF}(O, P_{11}, P_{12}, P_{21}, P_{22})$ is valid (as per Definition 4.5.5). However, the fact that $P_1((x_1, y_1), (x_m, y_m)) = P_1((x_m, y_m), (x_1, y_1)) = 1$ demonstrates that $\text{DPF}(O, P_{11}, P_{12}, P_{21}, P_{22})$ is invalid, leading to a contradiction. Consequently, the existence of a chordless cycle in M is not possible. $\square$

This completes the proof of all the claims and observations used in the proof of Lemma 4.6.61. To recap, Lemma 4.6.61 states that M is a chain graph, i.e., M does not have a directed cycle. We show that M does not contain any chordless cycle. This implies that if a directed cycle exists in M , then the shortest such cycle has length three. This further implies that all the nodes of the shortest cycle belong to either V_{M_1} , or V_{M_2} , or V_O . We show that $M[V_{M_1}]$ and $M[V_{M_2}]$ are chain graphs. Since O is a chain graph, O

also cannot contain a directed cycle, implying that M is a chain graph.

We now proceed to the proof of Lemma 4.6.62.

Proof of Lemma 4.6.62. Assume, for the sake of contradiction, that Lemma 4.6.62 is false, implying the existence of an undirected connected component $\mathcal{C}$ in M that is not chordal. Since $\mathcal{C}$ is not chordal, there must exist an undirected chordless cycle $C = (u_1, u_2, \dots, u_l, u_{l+1} = u_1)$, where l is greater than or equal to 4. This implies that no edges exist between non-adjacent nodes of C within $\mathcal{C}$. We consider two possibilities: (a) there is an edge between non-adjacent nodes of C in M , but it is directed; or (b) no edge, either directed or undirected, exists between non-adjacent nodes of C in M .

Let us analyze these cases. Firstly, suppose there is a directed edge between two non-adjacent nodes of C in M . This would create a directed cycle in M , contradicting Lemma 4.6.61 which states that M is a chain graph.

Next, consider the second possibility, where there are no edges, neither directed nor undirected, between non-adjacent nodes in M . As a result, C would be a chordless cycle in M . However, this contradicts Observation 4.6.66, which establishes that M does not contain any chordless cycles.

Consequently, neither of these cases is feasible. Hence, we conclude that Lemma 4.6.62 holds true, and there are no undirected connected components in M that are not chordal. $\square$

Proof of Lemma 4.6.63. We will first demonstrate that there are no induced subgraphs of the form $a \rightarrow b - c$ within $M[V_{M_1}]$ and $M[V_{M_2}]$.

Let us consider the scenario where there is an induced subgraph of the form $a \rightarrow b - c$ within $M[V_{M_i}]$ for some $i \in \{1, 2\}$. Based on the construction of M , either $a \rightarrow b - c$ or $a - b - c$ is an induced subgraph of M_i . According to Item 3 of Theorem 1.1.24, $a \rightarrow b - c$ cannot be an induced subgraph of M_i .

If $a - b - c \in M_i$, then a, b , and c all belong to an undirected connected component $\mathcal{C}$ of M_i , and (a, b, c) forms a chordless path from (a, b) to (b, c) within $\mathcal{C}$. Now, considering Observation 4.6.60, since (a, b, c) is a chordless path and $a \rightarrow b \in M$, it follows that $b \rightarrow c \in M$. This leads to a contradiction since we assumed that $b - c \in M$. Thus, we can conclude that there cannot exist an induced subgraph of the form $a \rightarrow b - c$ within

$M[V_{M_i}]$.

The above analysis implies that if there exists an induced subgraph $a \rightarrow b - c$ in M , then all the nodes of the induced subgraph cannot belong exclusively to V_{M_1} or V_{M_2} . This implies that at least one node must be in $V_{M_1} \setminus I$ (where $I = V_{H_1} \cap V_{H_2}$), and one node must be in $V_{M_2} \setminus I$. Since I acts as a separator between $V_{H_1} \setminus I$ and $V_{H_2} \setminus I$, the induced subgraph must contain at least one node from I . Therefore, we have two possibilities: (i) $a \in V_{M_1} \setminus I$, $b \in I$, and $c \in V_{M_2} \setminus I$, or (ii) $a \in V_{M_2} \setminus I$, $b \in I$, and $c \in V_{M_1} \setminus I$. In both cases, b belongs to I .

Consequently, we have $a, b, c \in I \cup N(I, H) \subseteq V_O$. This implies that the induced subgraph is in O as O is an induced subgraph of M (Lemma 4.6.49). However, this contradicts the fact that O is a partial MEC, as stated in Item 3 of Definition 4.2.1. Nonetheless, (O, P_1, P_2) is an extension of $(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$, and Definition 4.5.6 implies that O is a partial MEC. This creates a contradiction. Hence, there cannot be any such induced subgraph in M . Thus, Lemma 4.6.63 is validated. $\square$

Proof of Lemma 4.6.64. Suppose $u \rightarrow v \in M$. Since, from the construction, $\text{skeleton}(M) = \text{skeleton}(M_1) \cup \text{skeleton}(M_2) = H_1 \cup H_2$, there are two possibilities: either $u - v \in E_{H_1}$ or $u - v \in E_{H_2}$. Without loss of generality, we assume $u - v \in E_{H_1}$. From the construction of M , either $u \rightarrow v \in M_1$ or $u - v \in M_1$. Because if $v \rightarrow u \in M_1$ then due to the initialization step, Line 1 of Algorithm 8, we have $v \rightarrow u \in M$, which contradicts our assumption that $u \rightarrow v \in M$. We will show that in both cases, $u \rightarrow v$ is strongly protected in M . We will consider each possibility in turn:

1. Suppose $u \rightarrow v \in M_1$. Then $u \rightarrow v$ is strongly protected in M_1 according to Item 4 of Theorem 1.1.24. As per the construction of M , all the directed edges in M_1 are also directed in M (due to Line 1 of Algorithm 8). Now, let us consider the induced subgraphs shown in (a), (b), (c), and (d) of Figure 1.3.

If $u \rightarrow v$ is strongly protected in M_1 due to its inclusion in an induced subgraph shown in (a), (b), or (c) of Figure 1.3, then the fact that all the edges in those induced subgraphs are directed implies that the induced subgraph is also an induced subgraph of M (due to Line 1 of Algorithm 8). This further implies $u \rightarrow v$ is also strongly protected in M .

Now, let us focus on the case where $u \rightarrow v$ is strongly protected in M_1 because it is part of the induced subgraph shown in Figure 1.3-(d). In this induced subgraph, the edges $w \rightarrow v$, $w' \rightarrow v$, and $u \rightarrow v$ are directed, indicating that $w \rightarrow v$, $w' \rightarrow v$, and $u \rightarrow v$ are also present in M due to the construction of M . To establish the strong protection of $u \rightarrow v$ in M , we consider different cases. If $w - u$ and $w' - u$ are present in M , then $u \rightarrow v$ is part of the induced subgraph shown in Figure 1.3-(d), making it strongly protected in M . If $u \rightarrow w$ or $u \rightarrow w'$ exists in M , then again $u \rightarrow v$ is part of an induced subgraph, either as shown in Figure 1.3-(c) with $u \rightarrow w \rightarrow v \leftarrow u$, or as shown in Figure 1.3-(c) with $u \rightarrow w' \rightarrow v \leftarrow u$. In both cases, $u \rightarrow v$ is strongly protected in M .

The remaining possibility is that $w \rightarrow u$ or $w' \rightarrow u$ are present in M . Since $w - u - w'$ is present in M_1 , it implies that w , u , and w' belong to the same undirected connected component $\mathcal{C}$ of M_1 . Now, considering the fact that $w - u - w'$ forms a chordless path in $\mathcal{C}$, according to Observation 4.6.60, if $w \rightarrow u$ exists in M , then $u \rightarrow w'$ must also be present in M . Similarly, if $w' \rightarrow u \in M$, then $u \rightarrow w \in M$. And, as shown above, if $u \rightarrow w$ or $u \rightarrow w'$ exists in M , then $u \rightarrow v$ is strongly protected.

In conclusion, we have shown that in all possible cases, if $u \rightarrow v$ is strongly protected in M_1 , then it is also strongly protected in M . Hence, the claim is verified.

2. Suppose $u - v \in M_1$ and $u \rightarrow v \in M$. From Observation 4.6.54, $u \rightarrow v \in M$ either because (a) $u \rightarrow v$ is strongly protected in O , (b) there exists $x \rightarrow y \in O$ with $(u, v) \neq (x, y)$ and a chordless path from (x, y) to (u, v) in M_1 , or (c) there exist $x \rightarrow y \in O$ and chordless paths in M_1 from (x, y) to v and from (v, u) to x , with $y \neq v$ and $u \neq x$. We will consider each possibility in detail.

(a) If $u \rightarrow v$ is strongly protected in O , then since O is an induced subgraph of M (from Lemma 4.6.49), if $u \rightarrow v$ is strongly protected in O , then it is strongly protected in M .

(b) Suppose there exists an edge $x \rightarrow y \in O$ with a chordless path $Q = (u_1 = x, u_2 = y, \dots, u_{l-1} = u, u_l = v)$ from (x, y) to (u, v) in $\mathcal{C}$ where $(x, y) \neq (u, v)$.

According to Observation 4.6.60, $u_{l-2} \rightarrow u_{l-1} \rightarrow u_l \in M$. This implies that $u \rightarrow v$ is strongly protected in M as it is part of an induced subgraph shown in Figure 1.3-(a).

- (c) Suppose there are chordless paths $Q_1 = (u_1 = x, u_2 = y, \dots, u_l = u)$ in M_1 from (x, y) to u , and $Q_2 = (v_1 = v, v_2 = u, \dots, v_m = x)$ from (v, u) to x , where $y \neq u$ and $u \neq x$. All the edges of Q_1 and Q_2 must be undirected in M_1 , otherwise, concatenating Q_1 and Q_2 would create a directed cycle in M_1 , contradicting Item 1 of Theorem 1.1.24. This implies that all nodes in Q_1 and Q_2 belong to an undirected connected component $\mathcal{C}$ of M_1 .

Let τ be the LBFS ordering returned by Algorithm 7 when called at Line 5 of Algorithm 8 for the undirected connected component $\mathcal{C}$ of M_1 . According to Observation 4.6.60, $\tau(u_1) < \tau(u_2) < \dots < \tau(u_l)$, and for all $1 \leq i < l-1$, $u_i \rightarrow u_{i+1} \in M$. According to Observation 4.6.59, $\tau(u) < \tau(v)$. Based on Observation 4.6.4, $u_{l-1} - u \in \mathcal{C}$.

If $u \rightarrow u_{l-1} \in M$, then $u \rightarrow v$ is strongly protected in M due to the induced subgraph $u \rightarrow u_{l-1} \rightarrow v \leftarrow u$ (as shown in Figure 1.3-(c)).

Suppose $u \rightarrow u_{l-1} \notin M$. Then there must exist $j \geq 2$ such that $v_{j+1} \rightarrow v_j \in M$, otherwise, $C = (u_1 = x, u_2, \dots, u_{l-1}, v_2, \dots, v_m = x)$ is a directed cycle in M , contradicting Lemma 4.6.61. Thus, $v_{j+1} \rightarrow v_j \in M$ for some $j \geq 2$. Then $Q'_2 = (v_{j+1}, v_j, \dots, v_2, v_1)$ is a chordless path and $v_{j+1} \rightarrow v_j \in M$. Then, according to Observation 4.6.60, $v_3 \rightarrow v_2 \rightarrow v_1 \in M$. This implies that $u \rightarrow v$ is strongly protected in M as it is part of an induced subgraph $v_3 \rightarrow v_2 = u \rightarrow v_1 = v$ (as shown in Figure 1.3-(a)).

Thus, we have shown that in all possible cases, $u \rightarrow v$ is strongly protected in M . This verifies the claim.

This completes the proof of Lemma 4.6.64. □

This completes the proof of all the claims and observations used in the proof of Lemma 4.6.50. To recap, Lemma 4.6.50 states that M is an MEC. To prove this, we showed that it follows Items 1 to 4 of Theorem 1.1.24.

We now proceed to prove Lemma 4.6.51.

Proof of Lemma 4.6.51. Without loss of generality, we prove that $\mathcal{P}(M, V_{H_1}) = M_1$. Similarly, we can claim that $\mathcal{P}(M, V_{H_2}) = M_2$, which completes the proof of Lemma 4.6.51.

Since M_1 is an MEC of H_1 , it suffices to show that $\mathcal{V}(M_1) = \mathcal{V}(M[V_{H_1}])$ (from Definition 4.4.1). From the initialization step, all directed edges of M_1 are also directed edges of M . This implies $\mathcal{V}(M_1) \subseteq \mathcal{V}(M[V_{H_1}])$ as from the construction of M , $\text{skeleton}(M[V_{M_1}]) = \text{skeleton}(M_1)$. Thus, to establish $\mathcal{V}(M_1) = \mathcal{V}(M[V_{H_1}])$, we only need to demonstrate $\mathcal{V}(M[V_{H_1}]) \subseteq \mathcal{V}(M_1)$.

Suppose $\mathcal{V}(M[V_{H_1}]) \not\subseteq \mathcal{V}(M_1)$. This implies the existence of $u, v, w \in V_{H_1}$ such that $u \rightarrow v \leftarrow w \in M$ and $u \rightarrow v \leftarrow w \notin M_1$. From the construction of M , if $u \rightarrow v \in M$, then either $u \rightarrow v \in M_1$ or $u - v \in M_1$ (because if $v \rightarrow u \in M_1$ from Line 1 of Algorithm 8, $v \rightarrow u \in M$). Similarly, if $w \rightarrow v \in M$, then either $w \rightarrow v \in M_1$ or $w - v \in M_1$. Since $u \rightarrow v \leftarrow w \notin M_1$, either $u \rightarrow v - w \in M_1$ or $u - v \leftarrow w \in M_1$ or $u - v - w \in M_1$. From Item 3 of Theorem 1.1.24, the first two possibilities are not possible. Thus, the only remaining possibility is that $u - v - w$ is an induced subgraph of M . However, this contradicts Observation 4.6.60, as if $u \rightarrow v \in M$, then $v \rightarrow w \in M$, which contradicts the presence of $w \rightarrow v \in M$. This confirms that there are no such $u, v, w \in V_{H_1}$ for which $\mathcal{V}(M[V_{H_1}]) \not\subseteq \mathcal{V}(M_1)$.

Thus, we have established $\mathcal{V}(M[V_{H_1}]) \subseteq \mathcal{V}(M_1)$. Combined with the earlier inclusion that $\mathcal{V}(M_1) \subseteq \mathcal{V}(M[V_{H_1}])$, we conclude that $\mathcal{V}(M_1) = \mathcal{V}(M[V_{H_1}])$.

Hence, we have successfully shown that $\mathcal{P}(M, V_{H_1}) = M_1$, completing the proof of Lemma 4.6.51. $\square$

Proof of Lemma 4.6.52. Suppose there exists an MEC M' such that $M'[V_O] = O$, and $\mathcal{P}(M', V_{M_1}, V_{M_2}) = (M_1, M_2)$. We will show that $M' = M$.

We know that each MEC of a graph has a unique set of v-structures. Our aim is to demonstrate that both M and M' contain the same set of v-structures. This implies that $M = M'$.

From Lemma 4.6.49, we have $M[V_O] = O$. Also, from Lemma 4.6.51, we know that $\mathcal{P}(M, V_{M_1}, V_{M_2}) = (M_1, M_2)$. Since both M and M' are MECs of H , and $\text{skeleton}(H) = \text{skeleton}(H_1) \cup \text{skeleton}(H_2)$, it follows that for any v-structure $u \rightarrow v \leftarrow w$ that

belongs to either M or M' , one of the following cases holds: (a) $u, v, w \in V_{M_1}$, (b) $u, v, w \in V_{M_2}$, (c) $v \in I = V_{M_1} \cap V_{M_2}$, $u \in V_{M_1} \setminus I$, and $w \in V_{M_2} \setminus I$, (d) $u \in V_{M_2} \setminus I$, $v \in I$, and $w \in V_{M_1} \setminus I$.

We now address each of these cases.

For case (a), if $u \rightarrow v \leftarrow w$ is a v-structure in M with $u, v, w \in V_{M_1}$, since M_1 is a projection of M on V_{M_1} , the v-structure $u \rightarrow v \leftarrow w$ also belongs to M_1 . Similarly, as M_1 is a projection of M' on V_{M_1} , the same v-structure belongs to M' . A similar argument can be made for v-structures in M' . The same argument holds for v-structures when $u, v, w \in V_{M_2}$ (i.e., for case (b)).

For case (c), if $u \rightarrow v \leftarrow w$ is a v-structure in M with $u \in V_{M_1} \setminus I$, $v \in I$, and $w \in V_{M_2} \setminus I$, this v-structure is in O as $I \cup N(I, H) \subseteq V_O$, and O is an induced subgraph of M . Since O is also an induced subgraph of M' , the v-structure also belongs to M' . A similar argument can be made for v-structures in M' with $u \in V_{M_1} \setminus I$, $v \in I$, and $w \in V_{M_2} \setminus I$. The same argument holds for v-structures when $u \in V_{M_2} \setminus I$, $v \in I$, and $w \in V_{M_1} \setminus I$ (i.e., for case (d)).

The above analysis demonstrates that a v-structure $u \rightarrow v \leftarrow w$ is present in M if and only if it is also present in M' . This implies that both M and M' have the same set of v-structures, and therefore, $M = M'$. This completes the proof of Lemma 4.6.52. $\square$

Proof of Lemma 4.6.53. Let (O', P'_1, P'_2) be the shadow of M on V_O . We aim to show that $(O, P_1, P_2) = (O', P'_1, P'_2)$. From Item 1 of Definition 4.3.1, $O' = M[V_O]$. Furthermore, from Lemma 4.6.49, $O = M[V_O]$. Thus, we conclude that $O = O'$.

We are given that the shadow of M_1 on $S_1 \cup N(S_1, H_1)$ is (O_1, P_{11}, P_{12}) , and the shadow of M_2 on $S_2 \cup N(S_2, H_2)$ is (O_2, P_{21}, P_{22}) . Furthermore, from Lemma 4.6.51, we have $\mathcal{P}(M, V_{H_1}, V_{H_2}) = (M_1, M_2)$. Therefore, from Lemma 4.5.7, $(O', P'_1, P'_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$. Also, we have given that $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$. Therefore, from Item 4 of Definition 4.5.6 and Item 1 of Definition 4.5.5, $(P'_1, P'_2) = (P_1, P_2) = \text{DPF}(O' = O, P_{11}, P_{12}, P_{21}, P_{22})$. This shows that (O, P_1, P_2) is the shadow of M on V_O , i.e., $M \in \text{MEC}(H, O, P_1, P_2)$. $\square$

This completes the proof of all the claims and observations used in the proof of

Lemma 4.5.8. To recap, to prove Lemma 4.5.8, we first construct an MEC M of H such that $\mathcal{P}(M, V_{H_1}, V_{H_2}) = (M_1, M_2)$ and $\text{shadow}(M, V_O) = (O, P_1, P_2)$. Additionally, we show that M is unique.

4.7 Algorithm for Counting MECs of a Graph

In this section, based on the discussion in Section 4.5, we construct Algorithm 9 to count the MECs of an undirected connected graph. Algorithm 9 takes an undirected graph G as its input, and returns $|\text{MEC}(G)|$ as the output. The “Count” function used at Line 3 of Algorithm 9 comes from Algorithm 10. Lemma 4.7.1 validates this.

Algorithm 9: Count-MEC(G)

Input : An undirected connected graph G

Output: $|\text{MEC}(G)|$

```

1  $(T, \beta) \leftarrow$  a tree decomposition of  $G$ .
2  $X_1 \leftarrow$  an arbitrarily picked root node of  $T$ 
3  $F \leftarrow \text{Count}(G, (T, \beta), X_1)$ ;
4  $sum \leftarrow 0$ 
5  $S_1 \leftarrow \beta(X_1)$ 
6 foreach  $(O, P_1, P_2) \in \text{shadow}(G[S_1 \cup N(S_1, G)])$  do
7    $sum = sum + F(O, P_1, P_2)$ 
8 end
9 return  $sum$ 
```

Lemma 4.7.1. *Let G be an undirected connected graph. Given input G , Algorithm 9 returns $|\text{MEC}(G)|$.*

Proof. Algorithm 9 constructs a tree decomposition (T, β) of G (at line 1 of Algorithm 9). At Line 2, it arbitrarily picks a node X_1 of T as the root node of T . At Line 3, it calls Algorithm 10 with inputs G , (T, β) , and X_1 . Let $S_1 = \beta(X_1)$. From Lemma 4.7.2, stated and proved below, Algorithm 10 returns a function F such that $\text{shadow}(G[S_1 \cup N(S_1, G)])$ is the domain of F , and for each triplet $(O, P_1, P_2) \in \text{shadow}(G[S_1 \cup N(S_1, G)])$, $F(O, P_1, P_2) = |\text{MEC}(G, O, P_1, P_2)|$. Further, lines 6–8 of Algorithm 9 implement Lemma 4.3.7 to obtain $|\text{MEC}(G)|$, which is returned at line 9. $\square$

It remains to prove the following lemma to complete the proof of Lemma 4.7.1.

Algorithm 10: Count($G, (T, \beta), X_1$)

Input : An undirected graph G , a tree decomposition (T, β) of G , and the root node X_1 of T

Output: A function $F : \text{shadow}(G[\beta(X_1) \cup N(\beta(X_1), G)]) \rightarrow \mathbb{Z}$ such that $F(O, P_1, P_2) = |\text{MEC}(G, O, P_1, P_2)|$

```
1 if  $V_T = \{X_1\}$  then
2   | return BRUTEFORCECOUNT( $G$ )
3 else
4   |  $X_2 \leftarrow$  a child of  $X_1$  in  $T$ .
5   | Cut the edge  $X_1 - X_2$  of  $T$ .
6   |  $T_1 \leftarrow$  induced subtree of  $T$  containing  $X_1$ 
7   |  $T_2 \leftarrow$  induced subtree of  $T$  containing  $X_2$ 
8   |  $\beta_1 \leftarrow$  a function with domain  $V_{T_1}$  such that  $\forall X \in V_{T_1}, \beta_1(X) = \beta(X)$ .
9   |  $\beta_2 \leftarrow$  a function with domain  $V_{T_2}$  such that  $\forall X \in V_{T_2}, \beta_2(X) = \beta(X)$ .
10  |  $G_1 \leftarrow$  induced subgraph of  $G$  such that  $(T_1, \beta_1)$  is its tree decomposition
11  |  $G_2 \leftarrow$  induced subgraph of  $G$  such that  $(T_2, \beta_2)$  is its tree decomposition
12  |  $S_1 \leftarrow \beta(X_1)$ 
13  |  $S_2 \leftarrow \beta(X_2)$ 
14  |  $F_1 \leftarrow \text{Count}(G_1, (T_1, \beta_1), X_1)$ 
15  |  $F_2 \leftarrow \text{Count}(G_2, (T_2, \beta_2), X_2)$ 
16  |  $F : \text{shadow}(G[S_1 \cup N(S_1, G)]) \rightarrow \mathbb{Z}$ 
17  | foreach  $(O', P'_1, P'_2) \in \text{shadow}(G[S_1 \cup N(S_1, G)])$  do
18  |   |  $F(O', P'_1, P'_2) \leftarrow 0$ 
19  | end
20  | foreach  $(O, P_1, P_2) \in \text{shadow}(G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)])$  do
21  |   | foreach  $(O_1, P_{11}, P_{12}) \in \text{shadow}(G_1[S_1 \cup N(S_1, G_1)])$  do
22  |   |   | foreach  $(O_2, P_{21}, P_{22}) \in \text{shadow}(G_2[S_2 \cup N(S_2, G_2)])$  do
23  |   |   |   | if  $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$  then
24  |   |   |   |   |  $(O', P'_1, P'_2) \leftarrow \mathcal{P}(O, P_1, P_2, S_1 \cup N(S_1, G))$ 
25  |   |   |   |   |  $F(O', P'_1, P'_2) = F(O', P'_1, P'_2) + F_1(O_1, P_{11}, P_{12}) \times F_2(O_2, P_{21}, P_{22})$ 
26  |   |   |   | end
27  |   |   | end
28  |   | end
29  | end
30  | return  $F$ 
31 end
```

Lemma 4.7.2. *Let G be an undirected graph, (T, β) be a tree decomposition of G , $X_1 \in V_T$ be the root node of T , and $S_1 = \beta(X_1)$. For input G , (T, β) , and X_1 , Algorithm 10 returns a function F such that $\text{shadow}(G[S_1 \cup N(S_1, G)])$ is the domain of F , and for each $(O, P_1, P_2) \in \text{shadow}(G[S_1 \cup N(S_1, G)])$, $F(O, P_1, P_2) = |\text{MEC}(G, O, P_1, P_2)|$.*

Proof. If the tree T has only one node, the root node X_1 , i.e., $V_T = \{X_1\}$, then Algorithm 10 calls Algorithm 11 (see lines 1–3 of Algorithm 10). In this case, $\beta(X_1) = V_G$, implying $G[\beta(X_1) \cup N(\beta(X_1), G)] = G$. Lemma 4.7.3 shows that Algorithm 11 returns the required function F such that the domain of F is $\text{shadow}(G)$, and for all $(O, P_1, P_2) \in \text{shadow}(G)$, $F(O, P_1, P_2) = |\text{MEC}(G, O, P_1, P_2)|$.

If T has more than one node, we pick a child, X_2 , of X_1 in T (line 4 of Algorithm 10) and cut the edge $X_1 - X_2$ of T (line 5 of Algorithm 10). This operation results in two induced subgraphs, T_1 and T_2 , of T , where T_1 contains X_1 , and T_2 contains X_2 (lines 6–7). Let β_1 be a function with domain V_{T_1} such that for all $X \in V_{T_1}$, $\beta_1(X) = \beta(X)$, and β_2 a function with domain V_{T_2} such that for all $X \in V_{T_2}$, $\beta_2(X) = \beta(X)$. This results in two induced subgraphs, G_1 and G_2 , of G such that (T_1, β_1) is a tree decomposition of G_1 , and (T_2, β_2) is a tree decomposition of G_2 (lines 10–11 of Algorithm 10).

From Observation 4.5.1, $G = G_1 \cup G_2$, and $\beta(X_1) \cap \beta(X_2)$ is a vertex separator of G . This scenario matches the discussion following Observation 4.5.1. Specifically, G corresponds to H , G_1 and G_2 correspond to H_1 and H_2 , respectively, and $\beta(X_1)$ and $\beta(X_2)$ correspond to S_1 and S_2 , respectively. And, therefore, to get the required function F , we implement Lemma 4.5.18.

For simplification, we define $S_1 = \beta(X_1)$ and $S_2 = \beta(X_2)$ (lines 12–13 of Algorithm 10). To implement Lemma 4.5.18, we require functions F_1 and F_2 such that for any triplet $(O_1, P_{11}, P_{12}) \in \text{shadow}(G_1[S_1 \cup N(S_1, G_1)])$, $F_1(O_1, P_{11}, P_{12}) = |\text{MEC}(G_1, O_1, P_{11}, P_{12})|$, and for any triplet $(O_2, P_{21}, P_{22}) \in \text{shadow}(G_2[S_2 \cup N(S_2, G_2)])$, $F_2(O_2, P_{21}, P_{22}) = |\text{MEC}(G_2, O_2, P_{21}, P_{22})|$. Recursively, we compute the functions F_1 and F_2 (lines 14–15 of Algorithm 10). After obtaining F_1 and F_2 , we implement Lemma 4.5.18 (lines 20–29) to compute the required function F . Here the function $\mathcal{P}$ used at Line 24 is the projection of shadows as defined in Definition 4.5.14.

□

Algorithm 11: BRUTEFORCECOUNT(G)

Input : An undirected connected graph G

Output: A function $F : \text{shadow}(G) \rightarrow \mathbb{Z}$ such that for each $(O, P_1, P_2) \in \text{shadow}(G)$, $F(O, P_1, P_2) = |\text{MEC}(G, O, P_1, P_2)|$

```
1  $F : \text{shadow}(G) \rightarrow \mathbb{Z}$ 
2 foreach  $(O, P_1, P_2) \in \text{shadow}(G)$  do
3   if  $O \in \text{MEC}(G)$  and  $(P_1, P_2) = \text{TFP}(O)$  then
4      $F(O, P_1, P_2) \leftarrow 1$ 
5   else
6      $F(O, P_1, P_2) \leftarrow 0$ 
7   end
8 end
9 return  $F$ 
```

Lemma 4.7.3. *Let G be an undirected graph. For input G , Algorithm 11 returns a function F such that for any triplet $(O, P_1, P_2) \in \text{shadow}(G)$, $F(O, P_1, P_2) = |\text{MEC}(G, O, P_1, P_2)|$.*

Proof. Recall from Lemma 4.6.18 that $\text{TFP}(O)$ (as defined in Algorithm 5) provides a pair of functions (P_1, P_2) such that:

1. $P_1 : E_O \times E_O \rightarrow \{0, 1\}$ is a function such that $P_1((u, v), (x, y)) = 1$ if and only if $(u, v) \neq (x, y)$ and there exists a triangle-free path from (u, v) to (x, y) in O , and
2. $P_2 : E_O \times V_O \rightarrow \{0, 1\}$ is a function such that $P_2((u, v), w) = 1$ if and only if $v \neq w$ and a triangle-free path exists from (u, v) to w in O .

For an $M \in \text{MEC}(G)$ and $(O, P_1, P_2) \in \text{shadow}(G)$, if $M \in \text{MEC}(G, O, P_1, P_2)$, then from Definition 4.3.1, $O = M[V_O] = M[V_G] = M$, i.e., O is M itself, and $(P_1, P_2) = \text{TFP}(O)$. This implies that for a triplet $(O, P_1, P_2) \in \text{shadow}(G)$, $|\text{MEC}(G, O, P_1, P_2)| \leq 1$. Moreover, for any $(O, P_1, P_2) \in \text{shadow}(G)$, if O is an MEC of G and $(P_1, P_2) = \text{TFP}(O)$, then $|\text{MEC}(G, O, P_1, P_2)| = 1$; otherwise, $|\text{MEC}(G, O, P_1, P_2)| = 0$. The function F returned by Algorithm 11 implements the same (see lines 2–8 of Algorithm 11).

This completes the proof of Lemma 4.7.3. □

4.8 Time Complexity Analysis

In this section, we analyze the time complexity of the algorithms discussed in this chapter. Throughout this section, we assume that G is a connected graph.

Lemma 4.8.1. *For an input chain graph G with chordal undirected connected components, the time complexity of Algorithm 5 is $O(|E_G|^2 \cdot |V_G|)$.*

Proof. The running time of lines 3–5 in Algorithm 5 is $O(|E_G|^2)$, as the first foreach loop iterates $|E_G|^2$ times, and each iteration takes $O(1)$ time. Similarly, the running time of lines 6–8 is $O(|V_G| \cdot |E_G|)$, since the second foreach loop iterates $|V_G| \times |E_G|$ times, and each iteration takes $O(1)$ time. Furthermore, the running time of lines 12–18 is $O(|V_G|^3)$, as the fourth foreach loop iterates $|V_G|^3$ times, and each iteration takes $O(1)$ time.

From Item 1 of Claim 4.6.19, each $((u, v), (x, y)) \in E_G \times E_G$ can be in at most one Q_i . This implies that the cumulative size of all the Q_i s is $O(|E_G|^2)$. Since the neighbors of a node can be at most $|V_G| - 1$, the number of distinct $x - y$ edges, at Line 22, can be at most $|V_G| - 1$. This implies that the cumulative running time of lines 21–27 is $O(|E_G|^2 \cdot |V_G|)$. Similarly, the cumulative running time of lines 28–31 is $O(|E_G|^2)$.

Combining the above, the total running time complexity of the algorithm is $O(|E_G|^2 \cdot |V_G|)$. □

Lemma 4.8.2. *For the input elements $O, P_{11}, P_{12}, P_{21}$, and P_{22} , the time complexity of Algorithm 6 is $O(|E_O|^5)$.*

Proof. Line 1 of Algorithm 6 takes $O(|E_O|^5)$ time as it calls Algorithm 5. Lines 3–7 run for $O(|E_O|^2)$ time, where the foreach loop at line 3 iterates $O(|E_O|^2)$ times, with each iteration taking $O(1)$ time. Similarly, lines 8–11 run for $O(|V_O| \cdot |E_O|)$ time. The foreach loop at line 8 iterates for $O(|V_O| \cdot |E_O|)$ time, and each iteration takes $O(1)$ time. Thus, the overall running time of lines Line 2–13 is $O(|E_O|^2)$.

Similar to Algorithm 5, the running time complexity of lines 14–16 and lines 17–19 is $O(|E_O|^5)$ and $O(|V_O|^2 \cdot |E_O|^3)$, respectively. Therefore, the time complexity of Algorithm 6 is $O(|E_O|^5)$. □

Lemma 4.8.3. *For an input undirected graph G , the time complexity of Algorithm 11 is $2^{O(|E_G|^2)}$.*

Proof. For any edge $u - v \in E_G$, in a partial MEC O of G , there are three possibilities: either $u - v \in O$, or $u \rightarrow v \in O$, or $v \rightarrow u \in O$. Therefore, the number of partial MECs

of G is at most $3^{|E_G|}$. For any partial MEC O of G , the number of different functions $P_1 : E_O \times E_O \rightarrow \{0,1\}$ is $2^{|E_O|^2} = 2^{O(|E_G|^2)}$. Similarly, for any partial MEC O of G , the number of different functions $P_2 : E_O \times V_O \rightarrow \{0,1\}$ is $2^{|E_O| \cdot |V_O|} = 2^{O(|E_G| \cdot |V_G|)}$. This implies the number of shadows of G is $2^{O(|E_G|^2)}$. Therefore, the foreach loop (lines 2–8 of Algorithm 11) runs for $2^{O(|E_G|^2)}$ iterations.

Verifying whether O is an MEC or not, at line 3, will take $O(|V_G|^2 \cdot |E_G|)$ time, as verifying each condition of Items 1 to 4 will take at most $O(|V_G|^2 \cdot |E_G|)$ time. From Lemma 4.8.1, computation of $\text{TFP}(O)$, at line 3, takes $O(|E_G|^2 \cdot |V_G|)$ time. And, verification of $(P_1, P_2) = \text{TFP}(O)$ will take $O(|E_G|^2)$ time, as the size of (P_1, P_2) is $O(|E_G|^2)$. The initialization of $F(O, P_1, P_2)$, at lines 4 and 6, takes $O(1)$ time. Therefore, the overall runtime of the foreach loop (lines 2–8) takes $2^{O(|E_G|^2)}$ time. Thus, the time complexity of Algorithm 11 is $2^{O(|E_G|^2)}$. $\square$

Lemma 4.8.4. *For the input elements G , (T, β) , and X_1 , the time complexity of Algorithm 10 is $O(n(2^{O(d^4 \cdot \delta^4)} + n^2))$, where d is the width of the tree decomposition (T, β) of G , and δ is the maximum degree of a node in G .*

Proof. If $V_T = \{X_1\}$, the tree T consists of a single node X_1 , and $V_G = \beta(X_1)$. If the width of T is d , then $|V_G| = d + 1$ and $|E_G| = O(d^2)$. By Lemma 4.8.3, the time complexity to compute $\text{BRUTEFORCECOUNT}(G)$ is $2^{O(d^4)}$. Thus, the runtime of lines 1–3 is $2^{O(d^4)}$.

The time complexity of lines 4–13 is $O(|V_G|^2)$. If the width of T is d , then the sizes of S_1 and S_2 are at most $d + 1$. And, if the maximum degree of a node in G is δ , then the sizes of $S_1 \cup N(S_1, G_1)$, $S_2 \cup N(S_2, G_2)$, $S_1 \cup N(S_1, G)$, and $S_1 \cup S_2 \cup N(S_1 \cup S_2, G)$ are $O(d\delta)$. This implies that the number of edges in the induced subgraphs $G_1[S_1 \cup N(S_1, G_1)]$, $G_2[S_2 \cup N(S_2, G_2)]$, $G[S_1 \cup N(S_1, G)]$, and $G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)]$ is $O(d^2\delta^2)$. Therefore, as explained in Theorem 4.8.5, the sizes of $\text{shadow}(G_1[S_1 \cup N(S_1, G_1)])$, $\text{shadow}(G_2[S_2 \cup N(S_2, G_2)])$, $\text{shadow}(G[S_1 \cup N(S_1, G)])$, and $\text{shadow}(G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)])$ are $2^{O(d^4\delta^4)}$.

The initialization of F (lines 17–19) requires $2^{O(d^4\delta^4)}$, as $\text{shadow}(G[S_1 \cup N(S_1, G)])$ has size $2^{O(d^4\delta^4)}$. Each foreach loop (lines 20, 21, and 22) runs for $2^{O(d^4\delta^4)}$ iterations.

Verifying whether $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$ (line 23) involves:

- Checking Items 1 to 3, which takes $O(|E_O|^2)$ time.
- Verifying Item 4, which takes $O(d^{10}\delta^{10})$ time. By Lemma 4.8.2, $\text{DPF}(O, P_{11}, P_{12}, P_{21}, P_{22})$ can be computed in $O(|E_O|^5) = O(d^{10}\delta^{10})$ time (since $|E_O| = O(d^2\delta^2)$).

The projection (O', P'_1, P'_2) (line 24) is computed following Items 1 to 3 of Definition 4.5.14, which takes $O(|E_O|^2) = O(d^4\delta^4)$ time.

At lines 14–15, Algorithm 10 calls itself. Each recursive call splits T into two disjoint parts T_1 and T_2 , reducing the number of edges in T by one. Since T has $O(|V_G|)$ edges (Bodlaender and Kloks [1996]), the number of recursive calls is $O(|V_G|)$.

Combining these results, the total runtime of Algorithm 10 is $O(|V_G| \cdot (2^{O(d^4\delta^4)} + |V_G|^2)) = O(n(2^{O(d^4\delta^4)} + n^2))$. This completes the proof. $\square$

Theorem 4.8.5. *For an input undirected graph G , the time complexity of Algorithm 9 is $O(n(2^{O(k^4\delta^4)} + n^2))$, where δ is the maximum degree of a node in G , and k denotes the treewidth of G .*

Proof. For a graph of treewidth k , we use Korhonen’s algorithm (Korhonen [2022]) to construct a tree decomposition (T, β) of width d , where $k \leq d \leq 2k + 1$, as executed at line 1 of Algorithm 9. Korhonen’s algorithm outputs a tree decomposition for an n -vertex graph G in time $2^{O(k)} \cdot n$.

At line 3, we compute the required function F using Algorithm 10. From Lemma 4.8.4, this step takes $O(n(2^{O(d^4\delta^4)} + n^2))$ time.

Since the width of the tree decomposition is d , the size of $S_1 = \beta(X_1)$ is $d + 1$. Additionally, as the maximum degree of a node in G is δ , the number of nodes in $S_1 \cup N(S_1, G)$ is at most $(1 + \delta)(1 + d)$. Therefore, the size of $G[S_1 \cup N(S_1, G)]$ is $O(d^2\delta^2)$, and the number of partial MECs of $G[S_1 \cup N(S_1, G)]$ is $3^{O(d^2\delta^2)}$, as each edge $u - v \in G[S_1 \cup N(S_1, G)]$ has three possibilities in a partial MEC $O \in \text{PMEC}(G[S_1 \cup N(S_1, G)])$: either $u \rightarrow v \in O, v \rightarrow u \in O$, or $u - v \in O$.

For a partial MEC O of $G[S_1 \cup N(S_1, G)]$:

1. The number of distinct functions $P_1 : E_O \times E_O \rightarrow \{0, 1\}$ is at most $2^{|E_O|^2} = 2^{O(d^4\delta^4)}$.
2. The number of distinct functions $P_2 : E_O \times V_O \rightarrow \{0, 1\}$ is at most $2^{|E_O| \cdot |V_O|} = 2^{O(d^3\delta^3)}$.

Thus, the number of shadows $(O, P_1, P_2) \in \text{shadow}(G[S_1 \cup N(S_1, G)])$ is $2^{O(d^4\delta^4)}$. This implies that the foreach loop at line 6 executes $2^{O(d^4\delta^4)}$ iterations. Each iteration of the foreach loop (line 7) takes $O(1)$ time. Consequently, the total running time of lines 6–8 is $2^{O(d^4\delta^4)}$. Since $k \leq d \leq 2k + 1$, the overall running time of the algorithm is $O(n(2^{O(k^4\delta^4)} + n^2))$. $\square$

Theorem 4.8.5 and Lemma 4.7.1 prove Theorem 4.1.1.

In this chapter, we construct an FPT algorithm (Algorithm 9) that counts the MECs of an undirected graph. This is the first non-trivial algorithm for counting MECs in general graphs. The notion of a “shadow” is a key component of the algorithm. A shadow of an MEC has three parts: the first is an induced subgraph of the MEC, while the second and third parts provide information about the triangle-free paths in the MEC. For chordal graphs, the information about triangle-free paths can be derived directly from the induced subgraph. Therefore, we do not need to store additional information about the triangle-free paths. In the next chapter, we first prove how the triangle-free paths can be extracted from the induced subgraph of an MEC of a chordal graph. We then use this result to construct a more efficient algorithm for counting MECs of chordal graphs.

Chapter 5

Counting MECs with chordal graph skeleton

The objective of this chapter is to solve Problem 1.2 when the input graph is a chordal graph. We formalize the problem as follows:

Problem 5.1. *INPUT: A chordal graph G .*

OUTPUT: $|\text{MEC}(G)|$, the number of MECs of G , i.e., the number of MECs with G as their skeleton.

5.1 Overview

Theorem 5.1.1 summarizes our result.

Theorem 5.1.1 (Restating Theorem 1.3.2). *For a chordal graph G , there exists a fixed-parameter tractable algorithm that counts the MECs with skeleton G in $O(n(2^{O(\delta^2 k^2)} + n^2))$ time, where n is the number of nodes in G , and the parameters δ and k are, respectively, the maximum degree of a node in G and the treewidth of G .*

In the previous chapter (Chapter 4), we solved the problem of counting MECs for general graphs. There, we defined a triplet (O, P_1, P_2) as a unique identifier for an MEC (referred to as the “shadow of the MEC”) and used it to partition the MECs of a graph. In this triplet, P_1 provides information about the existence of a triangle-free path in the MEC, of length three or more, between two edges of O , and P_2 provides information

about the existence of a triangle-free path in the MEC, of length three or more, between an edge and a node of O . For chordal graphs, we can compute P_1 and P_2 directly from O (Proposition 5.2.1), which implies that P_1 and P_2 do not provide any additional information when the input graph is chordal. Consequently, we update the algorithm presented in Chapter 4 for counting the MECs of a graph, reducing its time complexity when the input graph is a chordal graph.

Here is a detailed technical overview of this chapter.

Technical Overview: In this chapter, we provide a fixed-parameter-tractable (FPT) algorithm to count the MECs of a chordal graph such that the runtime of our algorithm significantly improves the runtime of the algorithm (Algorithm 9) presented in Chapter 4.

For any MEC M and $S \subseteq V_M$, the induced subgraph of M on S is unique. We use this to partition the MECs of a chordal graph G . From Observation 4.2.3, $M[S]$ is a partial MEC. For any $O \in \text{PMEC}(G[S])$, let $\text{MEC}(G, O)$ be the set of MECs M of G such that $O = M[S]$. Then, for any $S \subseteq V_G$, the summation of $|\text{MEC}(G, O)|$, over all $O \in \text{PMEC}(G[S])$, equals $|\text{MEC}(G)|$ (Equation (5.1)). Therefore, to compute $|\text{MEC}(G)|$, we fix an $S \subseteq V_G$ and construct a recursive algorithm to compute $|\text{MEC}(G, O)|$ for all $O \in \text{PMEC}(G[S])$. We use a clique tree decomposition (Definition 1.1.16) of G for this purpose.

For a chordal graph G , we construct a rooted clique tree decomposition (T, β, X_1) of G and develop a recursive algorithm to compute $|\text{MEC}(G, O)|$ for each $O \in \text{PMEC}(G[\beta(X_1) \cup N(\beta(X_1), G)])$. To achieve this, we cut an edge $X_1 - X_2$ in the clique tree, resulting in two subtrees: T_1 (containing X_1) and T_2 (containing X_2). These correspond to two induced subgraphs of G : $G_1 = G[\cup_{X_i \in V_{T_1}} \beta(X_i)]$ and $G_2 = G[\cup_{X_i \in V_{T_2}} \beta(X_i)]$. Let β_1 and β_2 be the restrictions of β to V_{T_1} and V_{T_2} , respectively, i.e., $\beta_1 : V_{T_1} \rightarrow 2^{V_{G_1}}$ such that for any $u \in V_{T_1}$, $\beta_1(u) = \beta(u)$, and $\beta_2 : V_{T_2} \rightarrow 2^{V_{G_2}}$ such that for any $u \in V_{T_2}$, $\beta_2(u) = \beta(u)$. Then (T_1, β_1) and (T_2, β_2) are clique tree decompositions of G_1 and G_2 (same as Chapter 4). Let $\beta(X_1) = S_1$ and $\beta(X_2) = S_2$. We show that, given $|\text{MEC}(G_1, O_1)|$ and $|\text{MEC}(G_2, O_2)|$ for all $O_1 \in \text{PMEC}(G_1[S_1 \cup N(S_1, G_1)])$ and $O_2 \in \text{PMEC}(G_2[S_2 \cup N(S_2, G_2)])$, we can compute $|\text{MEC}(G, O)|$ for all $O \in$

$\text{PMEC}(G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)])$.

Let $O \in \text{PMEC}(G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)])$, $O_1 \in \text{PMEC}(G_1[S_1 \cup N(S_1, G_1)])$, and $O_2 \in \text{PMEC}(G_2[S_2 \cup N(S_2, G_2)])$. For MECs $M \in \text{MEC}(G, O)$, $M_1 \in \text{MEC}(G_1, O_1)$, and $M_2 \in \text{MEC}(G_2, O_2)$ such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$ (Definition 4.4.1), we find a necessary condition obeyed by O , O_1 , and O_2 (Lemma 5.2.6). We define “extension of partial MECs” to relate triplets (O, O_1, O_2) that satisfy the necessary condition (Definition 5.2.5). We also find that for any $O \in \text{PMEC}(G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)])$, $O_1 \in \text{PMEC}(G_1[S_1 \cup N(S_1, G_1)])$, and $O_2 \in \text{PMEC}(G_2[S_2 \cup N(S_2, G_2)])$ satisfying the necessary condition, for each pair $(M_1, M_2) \in \text{MEC}(G_1, O_1) \times \text{MEC}(G_2, O_2)$, there exists a unique MEC $M \in \text{MEC}(G, O)$ such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$ (Lemma 5.2.7).

Lemmas 5.2.6 and 5.2.7 provide us a method to compute $|\text{MEC}(G, O)|$ for each $O \in \text{PMEC}(G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)])$. For recursion, we require $|\text{MEC}(G, O)|$ for each $O \in \text{PMEC}(G[S_1 \cup N(S_1, G)])$. Since $M[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)]$ provides more information than $M[S_1 \cup N(S_1, G)]$, we can compute $|\text{MEC}(G, O')|$ for each $O' \in \text{PMEC}(G[S_1 \cup N(S_1, G)])$ given $|\text{MEC}(G, O)|$ for each $O \in \text{PMEC}(G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)])$ (Lemma 5.2.12). This completes our solution.

Organization of the chapter: Section 5.2 presents the construction of the algorithm to count the MECs of a chordal graph. We first show that for a chordal graph G , for any clique X of G , in the shadow (O, P_1, P_2) of G on X (Definition 4.3.1), O can provide all the information stored by P_1 and P_2 (Proposition 5.2.1). In Section 5.2.1, we partition the set of MECs of a chordal graph (Equation (5.1)) based on the induced subgraphs of the MECs. In this subsection, we also present Lemma 5.2.2, which provides a necessary condition that is satisfied by the induced subgraphs of the MECs of G , G_1 , and G_2 (recall from the technical overview that G_1 and G_2 are induced subgraphs of G). Let $\beta(X_1) = S_1$ and $\beta(X_2) = S_2$. Definition 5.2.5 defines the extension of partial MECs O_1 and O_2 of $G_1[S_1 \cup N(S_1, G_1)]$ and $G_2[S_2 \cup N(S_2, G_2)]$, respectively. Lemma 5.2.7 shows that the necessary condition is also sufficient. Lemma 5.2.8 uses the necessary and sufficient condition to construct a recursive relation to compute $|\text{MEC}(G, O)|$ for all partial MECs O of $G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)]$. Lemma 5.2.12 provides an approach with which, for all partial MECs O of $G[S_1 \cup N(S_1, G)]$, we can compute $|\text{MEC}(G, O)|$ when

given $|\text{MEC}(G, O')|$ for all partial MECs O' of $G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)]$. Algorithm 13 implements our idea and outputs $|\text{MEC}(G)|$ for any connected chordal graph G . Section 5.3 provides the time complexity of the algorithms discussed in this chapter. As the proof of Lemma 5.2.7 is lengthy, we present it in a separate section, Section 5.4.

5.2 Counting MECs of a Chordal Graph

In this section, we solve Problem 5.1. We start with the following result:

Proposition 5.2.1. *Let M be an MEC of a chordal graph G , S be a clique of G , and $O = M[S \cup N(S, G)]$. For all distinct nodes $u, v, w \in V_O$, if M has a triangle-free path from (u, v) to w in M , then each node of the triangle-free path belongs to V_O . In particular, P is a triangle-free path in O .*

Proof. Suppose M has a triangle-free path $P = (u_1 = u, u_2 = v, \dots, u_l = w)$ from (u, v) to w . We show that each node in P belongs to $S \cup N(S, G)$, implying P is also a triangle-free path in O , since $O = M[S \cup N(S, G)]$.

Since M is an MEC of G , $\text{skeleton}(M) = G$, implying P is also a triangle-free path in G . From Proposition 4.6.9, P is a chordless path in G . There are three possibilities: either both endpoints of P , i.e., u and w , are in S , or exactly one of them is in S , or neither is in S .

The first possibility cannot occur; otherwise, $u - w \in E_G$ (since S is a clique in G), and P would not be a chordless path in G , leading to a contradiction.

Suppose exactly one of u and w is in S . In this case, we will only consider P as a chordless path from u to w in the undirected chordal graph G . Without loss of generality, let $u \in S$ and $w \notin S$. This implies $w \in N(S, G) \setminus S$, as $w \in V_O = S \cup N(S, G)$. Consequently, there exists a node $x \in S$ such that $w - x \in E_G$. Note that $x \neq u$, otherwise P would not be a chordless path in G , leading to a contradiction. Then, $x - u \in E_G$ (since both u and x belong to S , a clique in G). We claim that each node in P is a neighbor of x . Suppose there exists a node u_i in P such that $u_i - x \notin E_G$. Then, $1 < i < l$ (since $x - u, x - w \in E_G$). Pick the smallest such i . This implies that $x - u_{i-1} \in E_G$. Now pick the first node u_j after u_i in P that is a neighbor of x

in G (such a u_j exists because w is a neighbor of x in G). Then, $(x, u_{i-1}, u_i, \dots, u_j, x)$ forms a chordless cycle in the chordal graph G , which is a contradiction. This implies that each node in P is a neighbor of x in G . Since $x \in S$, each node in P belongs to $S \cup N(S, G) = V_O$, implying that P is also a triangle-free path in O .

Suppose now that neither u nor w belongs to S , i.e., $u, w \in N(S, G) \setminus S$. This implies that either there exists an $x \in S$ such that $x - u, x - w \in E_G$, or there exist $x, y \in S$ such that $x - u, y - w \in E_G$ and $x - w, y - u \notin E_G$. Let us consider the first possibility. Suppose there exists an $x \in S$ such that $x - u, x - w \in E_G$. Similar to the previous case, all the nodes of P are neighbors of x . Since $x \in S$, each node of P belongs to $S \cup N(S, G) = V_O$, implying P is also a triangle-free path in O .

Now consider the second possibility, where there exist $x, y \in S$ such that $x - u, y - w \in E_G$ and $x - w, y - u \notin E_G$. Since $x, y \in S$, a clique in G , we have $x - y \in E_G$. We show that each node of P is either a neighbor of x or a neighbor of y . Suppose there exists a node u_i in P that is not a neighbor of x , i.e., $x - u_i \notin E_G$. Since $u_1 = u$ is a neighbor of x , $1 < i \leq l$. Pick the smallest such i . We first show that for all $j > i$, $x - u_j \notin E_G$. Suppose there exists a $j > i$ such that $x - u_j \in E_G$. Pick the smallest such j . Then, $(x, u_{i-1}, u_i, \dots, u_j, x)$ forms a chordless cycle in the chordal graph G , which is a contradiction. This implies that for all $j \geq i$, $x - u_j \notin E_G$. We next show that $y - u_{i-1} \in E_G$. Suppose $y - u_{i-1} \notin E_G$. Pick the smallest $j \geq i$ such that $y - u_j \in E_G$ (such a j must exist since $w - y \in E_G$). Then, $(x, u_{i-1}, u_i, \dots, u_j, y, x)$ forms a chordless cycle in the chordal graph G , which is a contradiction. This implies that $y - u_{i-1} \in E_G$. We now show that for all $j \geq i$, $y - u_j \in E_G$. Suppose there exists a $j \geq i$ such that $y - u_j \notin E_G$. Pick the smallest such j . Now pick the smallest $k > j$ such that $u_k - y \in E_G$ (such a k must exist as $w - y \in E_G$). Then, $(y, u_{j-1}, u_j, \dots, u_k, y)$ is a chordless cycle in the chordal graph G , which is a contradiction. This implies that for all $j \geq i - 1$, $y - u_j \in E_G$. Consequently, each node in P is either a neighbor of x or a neighbor of y . Since $x, y \in S$, nodes in P belong to $V_O = S \cup N(S, G)$. This implies P is a triangle-free path in O .

We have shown that in all possibilities, P is a triangle-free path in O . This completes the proof of Proposition 5.2.1. $\square$

Let S be a clique of a chordal graph G . Proposition 5.2.1 shows that for any MEC M

of G , in the triplet $(O, P_1, P_2) = \text{shadow}(M, S \cup N(S, G))$ (see Definition 4.3.1 for the definition of shadow), P_1 and P_2 do not contain any extra information that cannot be extracted from O . This makes P_1 and P_2 irrelevant in $\text{shadow}(M, S \cup N(S, G))$. For a chordal graph, we can construct a clique tree decomposition (T, β) such that for all $X \in V_T$, $\beta(X)$ is a clique. Therefore, we modify the algorithm for counting MECs for a general graph (Algorithm 9) to construct an algorithm for counting MECs of a chordal graph by just using the induced subgraph O in the triplet (O, P_1, P_2) of the shadow of an MEC. We show that this approach significantly improves the time complexity.

5.2.1 Construction of the Algorithm to Count the MECs of a Chordal Graph

We now construct an algorithm that for a chordal graph G , counts the MECs of G . We partition the set of MECs M of G based on the induced subgraph of the MECs. We denote by $\text{MEC}(G, O)$ the set of MECs M of G for which O is an induced subgraph of M . More formally, $\text{MEC}(G, O) = \{M : M \in \text{MEC}(G) \text{ and } M[V_O] = O\}$.

For any MEC M of G , and for any $Y \subseteq V_G$, the induced subgraph of M on Y is uniquely determined by M and Y . Additionally, from Observation 4.2.3, $M[Y]$ is a partial MEC (Definition 4.2.1) of $G[Y]$. This implies that for each MEC M of G , there exists a unique partial MEC O of $G[Y]$ such that $M \in \text{MEC}(G, O)$. Therefore, the sum of $|\text{MEC}(G, O)|$ over all partial MECs O of $G[Y]$ is $|\text{MEC}(G)|$, i.e., for any $Y \subseteq V_G$,

$$|\text{MEC}(G)| = \sum_{O \in \text{PMEC}(G[Y])} |\text{MEC}(G, O)| \quad (5.1)$$

Equation (5.1) reduces the problem of counting MECs of G to the following problem:

Problem 5.2. *Input:* An undirected connected chordal graph G , and $Y \subseteq V_G$. *Output:* A function $F : \text{PMEC}(G[Y]) \rightarrow \mathbb{N}$ such that for $O \in \text{PMEC}(G[Y])$, $F(O) = |\text{MEC}(G, O)|$.

To compute $|\text{MEC}(G)|$, we first fix a Y and then provide a recursive method to solve Problem 5.2 for inputs G and Y . Using Equation (5.1), we are able to compute $|\text{MEC}(G)|$.

To compute Y , we construct a clique tree decomposition (Definition 1.1.16) (T, β) of the chordal graph G . We arbitrarily select a node X_1 as the root node of T . We then apply a recursive method to solve Problem 5.2 for $Y = \beta(X_1) \cup N(\beta(X_1), G)$. Our approach to compute $|\text{MEC}(G, O)|$ for all $O \in \text{PMEC}(G[Y])$ is similar to Chapter 4 where we compute $|\text{MEC}(G, O, P_1, P_2)|$ for all $(O, P_1, P_2) \in \text{shadow}(G[Y])$.

We arbitrarily pick a neighbor X_2 of X_1 in T and cut the edge $X_1 - X_2$ in T . This gives us two induced subgraphs of T : T_1 (containing node X_1), and T_2 (containing node X_2), and two induced subgraphs of G : $G_1 = G[\cup_{X \in V_{T_1}} \beta(X)]$ and $G_2 = G[\cup_{X \in V_{T_2}} \beta(X)]$. For $a \in \{1, 2\}$, let β_a be a function with domain set V_{T_a} such that for all $X \in V_{T_a}$, $\beta_a(X) = \beta(X)$. Then, (T_1, β_1) and (T_2, β_2) are tree decompositions of G_1 and G_2 , respectively. The induced subgraphs G_1 and G_2 of G have the following properties: (a) $G_1 \cup G_2 = G$, and (b) $I = V_{G_1} \cap V_{G_2} = \beta(X_1) \cap \beta(X_2)$ is a vertex separator of G that separates $V_{G_1} \setminus I$ and $V_{G_2} \setminus I$ [Blair and Peyton, 1993]. We show that if for each $a \in \{1, 2\}$ and for each $O_a \in \text{PMEC}(G_a[\beta(X_a) \cup N(\beta(X_a), G_a)])$, we know $|\text{MEC}(G_a, O_a)|$, then for each $O \in \text{PMEC}(G[\beta(X_1) \cup N(\beta(X_1), G)])$, we can compute $|\text{MEC}(G, O)|$.

We first show the relation between $\text{MEC}(G)$, $\text{MEC}(G_1)$, and $\text{MEC}(G_2)$, using which we construct a recursive algorithm to count the MECs of G . To understand this relation, we abstract out the following basic setup underlying it. The technical results below will all be under this abstract setup.

Setup for technical results Let G , G_1 , and G_2 be three undirected chordal graphs, and S_1 and S_2 be cliques of V_{G_1} and V_{G_2} , respectively, such that

1. G_1 and G_2 are induced subgraphs of G such that $G = G_1 \cup G_2$,
2. $I = V_{G_1} \cap V_{G_2}$ is a vertex separator of G that separates $V_{G_1} \setminus I$ and $V_{G_2} \setminus I$, and
3. both S_1 and S_2 contain the nodes of I , i.e., $S_1 \cap S_2 = I$.

In this setup, S_1 resembles $\beta(X_1)$, and S_2 resembles $\beta(X_2)$. Note that the above setup is analogous to the setup for technical results in Chapter 4 (where G , G_1 , and G_2 are referred to as H , H_1 , and H_2 , respectively; this is the only difference). Therefore, when needed, we can leverage the results from Chapter 4.

We start by establishing a relation between the MECs of G, G_1 , and G_2 . The following result is a chordal graph version of Lemma 4.5.2. Lemma 5.2.2 (along with its reformulation in Lemma 5.2.6) and Lemma 5.2.7 are the main results of this section, and serve as the foundations of our counting principle.

Lemma 5.2.2. *Let M, M_1 , and M_2 be MECs of, respectively, G, G_1 , and G_2 such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. Let $O = M[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)]$, $O_1 = M_1[S_1 \cup N(S_1, G_1)]$, and $O_2 = M_2[S_2 \cup N(S_2, G_2)]$. Then, for each $a \in \{1, 2\}$:*

1. *If $u \rightarrow v \in O_a$, then $u \rightarrow v \in O$.*
2. *$\mathcal{V}(O_a) = \mathcal{V}(O[V_{O_a}])$.*
3. *If $u - v \in O_a$ and $u \rightarrow v \in O$, then $u \rightarrow v$ is strongly protected in O .*

Proof. From Definition 4.3.1, the first component of the shadow of M on $S_1 \cup S_2 \cup N(S_1 \cup S_2, G)$ is O , the first component of the shadow of M_1 on $S_1 \cup N(S_1, G_1)$ is O_1 , and the first component of the shadow of M_2 on $S_2 \cup N(S_2, G_2)$ is O_2 . Let $(O, P_1, P_2) = \text{shadow}(M, S_1 \cup S_2 \cup N(S_1 \cup S_2, G))$, $(O_1, P_{11}, P_{12}) = \text{shadow}(M_1, S_1 \cup N(S_1, G_1))$, and $(O_2, P_{21}, P_{22}) = \text{shadow}(M_2, S_2 \cup N(S_2, G_2))$. Then, from Item 1 of Lemma 4.5.2, for $a \in \{1, 2\}$, if $u \rightarrow v \in O_a$, then $u \rightarrow v \in O$. Furthermore, from Item 2 of Lemma 4.5.2, for $a \in \{1, 2\}$, $\mathcal{V}(O_a) = \mathcal{V}(O[V_{O_a}])$. This proves Items 1 and 2 of Lemma 5.2.2. We now prove Item 3 of Lemma 5.2.2.

From Item 3 of Lemma 4.5.2, for $a \in \{1, 2\}$, if $u - v \in O_a$, then $u \rightarrow v \in O$ if and only if one of the following occurs:

1. $u \rightarrow v$ is strongly protected in O .
2. There exists $x - y \in O_a$ such that $x \rightarrow y \in O$, and $P_{a1}((x, y), (u, v)) = 1$ (i.e., there exists a triangle-free path from (x, y) to (u, v) in M_a with $(x, y) \neq (u, v)$).
3. There exists $x - y \in O_a$ such that $x \rightarrow y \in O$, and $P_{a2}((x, y), v) = P_{a2}((v, u), x) = 1$ (i.e., $y \neq v, u \neq x$, and there exist triangle-free paths in M_a from (x, y) to v , and from (v, u) to x).

We show that Items 2 and 3 above imply Item 1. The following two claims prove this.

Claim 5.2.3. *If there exists $x - y \in O_a$ such that $x \rightarrow y \in O$, and $P_{a1}((x, y), (u, v)) = 1$ (i.e., there exists a triangle-free path from (x, y) to (u, v) in M_a with $(x, y) \neq (u, v)$), then $u \rightarrow v$ is strongly protected in O .*

Proof. Let $x - y \in O_a$, $x \rightarrow y \in O$, and $P = (u_1 = x, u_2 = y, \dots, u_{l-1} = u, u_l = v)$ be a triangle-free path from (x, y) to (u, v) in M_a . Since O is an induced subgraph of M , $x \rightarrow y \in M$. Since M_1 is a projection of M and P is a triangle-free path from (x, y) to (u, v) in M_a such that $x \rightarrow y \in M$, from Corollary 4.4.10, for all $1 \leq i < l$, $u_i \rightarrow u_{i+1} \in M$. From Definition 4.3.1, the domain of the function P_{a1} is $E_{O_a} \times E_{O_a}$, implying $u, v, x, y \in V_{O_a}$. Since M_a is an MEC of the chordal graph G_a , $x, y, v \in V_{O_a}$, and P is a triangle-free path in M_a from (x, y) to v , from Proposition 5.2.1, each node of the path belongs to V_{O_a} . From the construction, $V_{O_a} = S_a \cup N(S_a, G_a) \subseteq S_1 \cup S_2 \cup N(S_1 \cup S_2, G) = V_O$. This implies that all the nodes of P are in V_O . This further implies $u \rightarrow v$ is strongly protected in O as it is part of an induced subgraph $u_{l-2} \rightarrow (u_{l-1} = u) \rightarrow (u_l = v)$ in O . This completes the proof of Claim 5.2.3. $\square$

Claim 5.2.4. *If there exists $x - y \in O_a$ such that $x \rightarrow y \in O$, and $P_{a2}((x, y), v) = P_{a2}((v, u), x) = 1$ (i.e., $y \neq v, u \neq x$, and there exist triangle-free paths in M_a from (x, y) to v , and from (v, u) to x), then $u \rightarrow v$ is strongly protected in O .*

Proof. Let $x - y \in O_a$, $x \rightarrow y \in O$, $P = (u_1 = x, u_2 = y, \dots, u_l = v)$ be a triangle-free path from (x, y) to v in M_a , and $Q = (v_1 = v, v_2 = u, \dots, v_m = x)$ be a triangle-free path from (v, u) to x in M_a . Since M_a is an MEC of G_a , P and Q are triangle-free paths in G_a as well. Since G_a is a chordal graph, from Proposition 4.6.9, P and Q are chordless paths in G_a . Since G_a is an induced subgraph of G , P and Q are also chordless paths in G . From Definition 4.3.1, the domain of P_{a2} is $E_{O_a} \times V_{O_a}$, implying $u, v, x, y \in V_{O_a}$. Then, similar to the previous claim, all the nodes of P and Q belong to V_{O_a} .

All the edges of P and Q must be undirected in M_a , as otherwise the concatenation of P and Q forms a directed cycle in M_a , contradicting Item 1 of Theorem 1.1.24. We now show that $u_{l-1} - v_2 \in E_G$.

Suppose $u_{l-1} - v_2 \notin E_G$. There are two possibilities: either there exists a $j < l - 1$ such that $u_j - v_2 \in E_G$, or no such j exists. If there exists a $j < l - 1$ such that $u_j - v_2 \in E_G$, pick the largest such j . Then, $(u_j, u_{j+1}, \dots, u_l = v = v_1, v_2, u_j)$ forms a

chordless cycle in the chordal graph G , a contradiction. Suppose no such $j < l - 1$ exists. Then, find the smallest $i > 2$ such that there exists a $j > 1$ with $v_i - u_j \in E_G$ (such an i exists as $(v_m = u_1) - u_2 \in E_G$). Now, find the largest j such that $v_i - u_j \in E_G$ (such a j exists by the choice of i). $j < l$, because $u_l = v_1$, and Q is a chordless path. Then, $(v_i, u_j, u_{j+1}, \dots, u_l = v = v_1, v_2, \dots, v_i)$ forms a chordless cycle in the chordal graph G , a contradiction. This implies $u_{l-1} - v_2 \in E_G$.

Since $x \rightarrow y \in O$, similar to the previous claim, for all $1 \leq i < l$, $u_i \rightarrow u_{i+1} \in O$. There are two possibilities: either $v_2 \rightarrow u_{l-1} \in O$, or $v_2 \rightarrow u_{l-1} \notin O$ (which means either $v_2 \leftarrow u_{l-1} \in O$ or $v_2 - u_{l-1} \in O$). If $v_2 \rightarrow u_{l-1} \in O$, then $u \rightarrow v$ is strongly protected in O as it is part of an induced subgraph $u \rightarrow u_{l-1} \rightarrow (u_l = v) \leftarrow u$.

Suppose $v_2 \rightarrow u_{l-1} \notin O$. Then, there must exist an $i > 2$ such that $v_i \rightarrow v_{i-1} \in O$, as otherwise, $(x = u_1, u_2, \dots, u_{l-1}, v_2 = u, v_3, \dots, v_m = x)$ yields a directed cycle in O . Since O is an induced subgraph of the MEC M , this directed cycle must also be in M , contradicting Item 1 of Theorem 1.1.24. This implies there exists an $i > 2$ such that $v_i \rightarrow v_{i-1} \in O$. Then for all $1 < j \leq i$, $v_j \rightarrow v_{j-1} \in O$, as otherwise, for some $j \leq i$, either $v_j \rightarrow v_{j-1} - v_{j-2} \in M$, contradicting Item 3 of Theorem 1.1.24, or we have a V-structure $v_j \rightarrow v_{j-1} \leftarrow v_{j-2}$ in $O[V_{O_a}]$ but not in O_a , contradicting Item 2 of Lemma 5.2.2. Therefore, $u \rightarrow v$ is strongly protected in O as it is part of an induced subgraph $v_3 \rightarrow (v_2 = u) \rightarrow (v_1 = v)$. $\square$

Claims 5.2.3 and 5.2.4 along with Item 3 of Lemma 4.5.2 prove Item 3 of Lemma 5.2.2. This completes the proof of Lemma 5.2.2. $\square$

We define the following to summarize the necessary conditions provided by Lemma 5.2.2. (The definition of “Extension” below is the chordal graph version of the term “Extension” defined in Chapter 4.)

Definition 5.2.5 (Extension). Let $O \in \text{PMEC}(G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)])$, $O_1 \in \text{PMEC}(G_1[S_1 \cup N(S_1, G_1)])$, and $O_2 \in \text{PMEC}(G_2[S_2 \cup N(S_2, G_2)])$. We call O an extension of O_1 and O_2 , denoted by $O \in \mathcal{E}(O_1, O_2)$, if for each $a \in \{1, 2\}$,

1. If $u \rightarrow v \in O_a$, then $u \rightarrow v \in O$.

2. $\mathcal{V}(O_a) = \mathcal{V}(O[V_{O_a}])$.

3. For $u - v \in O_a$, if $u \rightarrow v \in O$, then $u \rightarrow v$ is strongly protected in O .

Since O, O_1 and O_2 used in Lemma 5.2.2 are partial MECs of, respectively, $G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)]$, $G_1[S_1 \cup N(S_1, G_1)]$, and $G_2[S_2 \cup N(S_2, G_2)]$ (from Observation 4.2.3), we summarize Lemma 5.2.2 as follows:

Lemma 5.2.6. *Let M, M_1 and M_2 be MECs of, respectively, G, G_1 , and G_2 such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. Let $O = M[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)]$, $O_1 = M_1[S_1 \cup N(S_1, G_1)]$, and $O_2 = M_2[S_2 \cup N(S_2, G_2)]$. Then, $O \in \mathcal{E}(O_1, O_2)$.*

We now show that the necessary conditions that are obeyed by the partial MECs O, O_1 , and O_2 in Lemma 5.2.6 are sufficient to answer whether there exists an MEC $M \in \text{MEC}(G, O)$ for any pair of MECs $(M_1, M_2) \in \text{MEC}(G_1, O_1) \times \text{MEC}(G_2, O_2)$ such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.

Lemma 5.2.7. *Let $O \in \text{PMEC}(G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)])$, $O_1 \in \text{PMEC}(G_1[S_1 \cup N(S_1, G_1)])$, and $O_2 \in \text{PMEC}(G_2[S_2 \cup N(S_2, G_2)])$ such that $O \in \mathcal{E}(O_1, O_2)$. Then, for each pair of MECs $(M_1, M_2) \in \text{MEC}(G_1, O_1) \times \text{MEC}(G_2, O_2)$, there exists a unique MEC $M \in \text{MEC}(G, O)$ such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.*

The proof of Lemma 5.2.7 is given in Section 5.4 (page 260).

We use Lemmas 5.2.6 and 5.2.7 to construct a formula that computes $|\text{MEC}(G, O)|$ for any partial MEC $O \in \text{PMEC}(G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)])$. Given Lemmas 5.2.6 and 5.2.7, the proof of the formula itself is routine, but we provide all the details for completeness. The reader can skip the proof of Lemma 5.2.8 and move ahead to page 253.

Lemma 5.2.8. *For any $O \in \text{PMEC}(G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)])$, $|\text{MEC}(G, O)|$ equals*

$$\sum_{\substack{O_1 \in \text{PMEC}(G_1[S_1 \cup N(S_1, G_1)]) \\ O_2 \in \text{PMEC}(G_2[S_2 \cup N(S_2, G_2)]) \\ \text{such that } O \in \mathcal{E}(O_1, O_2)}} |\text{MEC}(G_1, O_1)| \times |\text{MEC}(G_2, O_2)|. \quad (5.2)$$

Proof. We define a bijective function f_O . This function is used to compute $|\text{MEC}(G, O)|$.

Definition 5.2.9 (Function f_O). Let $O \in \text{PMEC}(G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)])$, $T_1 = \text{MEC}(G, O)$, and $T_2 = \{(M_1, M_2) : \exists (O_1, O_2) \in \text{PMEC}(G_1[S_1 \cup N(S_1, G_1)]) \times \text{PMEC}(G_2[S_2 \cup N(S_2, G_2)]) \text{ such that } O \in \mathcal{E}(O_1, O_2), M_1 \in \text{MEC}(G_1, O_1), \text{ and } M_2 \in \text{MEC}(G_2, O_2)\}$. The function $f_O : T_1 \rightarrow T_2$ is defined as $f_O(M) = (M_1, M_2)$ if $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$.

We first show that f_O is a bijection between T_1 and T_2 . The following claims establish this.

Claim 5.2.10. For each $M \in \text{MEC}(G, O)$, $f_O(M) = (M_1, M_2)$ belongs to T_2 .

Proof. Let $M \in \text{MEC}(G, O)$. This implies that M is an MEC of G , and $M[V_O] = O$. Let $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. From Corollary 4.4.3, there exists a unique (M_1, M_2) such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. Thus, we only need to show that $(M_1, M_2) \in T_2$.

From the construction, $G_1 = G[V_{G_1}]$, and $G_2 = G[V_{G_2}]$. Therefore, from Definition 4.4.1, $M_1 \in \text{MEC}(G_1)$, and $M_2 \in \text{MEC}(G_2)$. Let $O_1 = M_1[S_1 \cup N(S_1, G_1)]$ and $O_2 = M_2[S_2 \cup N(S_2, G_2)]$. From Observation 4.2.3, $O_1 \in \text{PMEC}(G_1[S_1 \cup N(S_1, G_1)])$ and $O_2 \in \text{PMEC}(G_2[S_2 \cup N(S_2, G_2)])$. This implies that $M_1 \in \text{MEC}(G_1, O_1)$ and $M_2 \in \text{MEC}(G_2, O_2)$. From Lemma 5.2.6, $O \in \mathcal{E}(O_1, O_2)$. Thus, by the definition of T_2 , $(M_1, M_2) \in T_2$. Therefore, $f_O(M) = (M_1, M_2)$ belongs to T_2 . $\square$

Claim 5.2.11. Let O_1 be a partial MEC of $G_1[S_1 \cup N(S_1, G_1)]$ and O_2 a partial MEC of $G_2[S_2 \cup N(S_2, G_2)]$ such that O is an extension of (O_1, O_2) . For each $M_1 \in \text{MEC}(G_1, O_1)$ and $M_2 \in \text{MEC}(G_2, O_2)$, there exists a unique $M \in \text{MEC}(G, O)$ such that $f_O(M) = (M_1, M_2)$.

Proof. Lemma 5.2.7 and Definition 5.2.9 imply this claim. $\square$

Claims 5.2.10 and 5.2.11 imply that f_O is a bijection between T_1 and T_2 . This further implies that $|T_1| = |T_2|$. Since $T_1 = \text{MEC}(G, O)$, therefore, $|T_1| = |\text{MEC}(G, O)|$. Since T_2 contains tuples (M_1, M_2) for some $(O_1, O_2) \in \text{PMEC}(G_1[S_1 \cup N(S_1, G_1)]) \times \text{PMEC}(G_2[S_2 \cup N(S_2, G_2)])$ such that $O \in \mathcal{E}(O_1, O_2)$, $M_1 \in \text{MEC}(G_1, O_1)$, and $M_2 \in \text{MEC}(G_2, O_2)$, this implies

$$T_2 = \bigcup_{\substack{O_1 \in \text{PMEC}(G_1[S_1 \cup N(S_1, G_1)]) \\ O_2 \in \text{PMEC}(G_2[S_2 \cup N(S_2, G_2)]) \\ \text{such that } O \in \mathcal{E}(O_1, O_2)}} \text{MEC}(G_1, O_1) \times \text{MEC}(G_2, O_2). \quad (5.3)$$

We now show that the sets $\text{MEC}(G_1, O_1) \times \text{MEC}(G_2, O_2)$ in Equation (5.3) are disjoint. Suppose they are not. Then there exists (M_1, M_2) such that $(M_1, M_2) \in \text{MEC}(G_1, O_1) \times \text{MEC}(G_2, O_2)$ and $(M_1, M_2) \in \text{MEC}(G_1, O'_1) \times \text{MEC}(G_2, O'_2)$ for some $O_1, O'_1 \in \text{PMEC}(G_1[S_1 \cup N(S_1, G_1)])$ and $O_2, O'_2 \in \text{PMEC}(G_2[S_2 \cup N(S_2, G_2)])$.

This implies $M_1 \in \text{MEC}(G_1, O_1)$ and $M_1 \in \text{MEC}(G_1, O'_1)$. Since $V_{O_1} = V_{O'_1} = S_1 \cup N(S_1, G_1)$, $O_1 = O'_1 = M[S_1 \cup N(S_1, G_1)]$ (recall: from our definition of $\text{MEC}(G, O)$, if $M \in \text{MEC}(G, O)$ then $O = M[V_O]$). Similarly, $O_2 = O'_2$. This shows $(O_1, O_2) = (O'_1, O'_2)$, proving the sets are disjoint.

Therefore,

$$|T_2| = \sum_{\substack{O_1 \in \text{PMEC}(G_1[S_1 \cup N(S_1, G_1)]) \\ O_2 \in \text{PMEC}(G_2[S_2 \cup N(S_2, G_2)]) \\ \text{such that } O \in \mathcal{E}(O_1, O_2)}} |\text{MEC}(G_1, O_1)| \times |\text{MEC}(G_2, O_2)|. \quad (5.4)$$

This completes the proof of Lemma 5.2.8. $\square$

Using Lemma 5.2.8, knowing $|\text{MEC}(G_1, O_1)|$ for each $O_1 \in \text{PMEC}(G[S_1 \cup N(S_1, G_1)])$, and $|\text{MEC}(G_2, O_2)|$ for each $O_2 \in \text{PMEC}(G[S_2 \cup N(S_2, G_2)])$, we can compute $|\text{MEC}(G, O)|$ for each $O \in \text{PMEC}(G[S = S_1 \cup S_2 \cup N(S_1 \cup S_2, G)])$. But, for the recursion, we need to compute $|\text{MEC}(G, O)|$ for each $O \in \text{PMEC}(G[S' = S_1 \cup N(S_1, G)])$. For any MEC M of G , $M[S']$ is an induced subgraph of $M[S]$. We use this to evaluate $|\text{MEC}(G, O')|$ for all $O' \in \text{PMEC}(G[S'])$, when we have the knowledge of $|\text{MEC}(G, O)|$ for all $O \in \text{PMEC}(G[S])$.

Lemma 5.2.12. *Let G be a chordal graph, and $S' \subseteq S \subseteq V_G$. Then, for any $O' \in \text{PMEC}(G[S'])$,*

$$|\text{MEC}(G, O')| = \sum_{\substack{O \in \text{PMEC}(G[S]) \\ \text{such that } O' = O[S']}} |\text{MEC}(G, O)|. \quad (5.5)$$

Proof. We first prove the following:

$$\text{MEC}(G, O') = \bigcup_{\substack{O \in \text{PMEC}(G[S]) \\ \text{such that } O' = O[S']}} \text{MEC}(G, O). \quad (5.6)$$

Claims 5.2.13 and 5.2.14 prove the correctness of Equation (5.6).

Claim 5.2.13. *Let $M \in \text{MEC}(G, O')$. Then, there exists a partial MEC $O \in \text{PMEC}(G[S])$ such that $O' = O[S']$ and $M \in \text{MEC}(G, O)$.*

Proof. Since $M \in \text{MEC}(G, O')$, $M[V_{O'} = S'] = O'$. Let $O = M[S]$. This implies that $M \in \text{MEC}(G, O)$. From Observation 4.2.3, $O \in \text{PMEC}(G[S])$. Also, since $S' \subseteq S$, and both O and O' are induced subgraph of M , $O[S'] = M[S'] = O'$. This proves Claim 5.2.13. $\square$

Claim 5.2.14. *Let $O \in \text{PMEC}(G[S])$ such that $O' = O[S']$. If $M \in \text{MEC}(G, O)$ then $M \in \text{MEC}(G, O')$.*

Proof. Let $M \in \text{MEC}(G, O)$. This implies $M[V_O = S] = O$. Since $S' \subseteq S$, $M[S'] = O[S'] = O'$. This implies O' is an induced subgraph of M , i.e., $M \in \text{MEC}(G, O')$. This proves Claim 5.2.14. $\square$

From Claim 5.2.13, If M is in the L.H.S. of Equation (5.6), then M is also in the R.H.S. of Equation (5.6). And, from Claim 5.2.14, if M is in the R.H.S. of Equation (5.6), then M is also in the L.H.S. of Equation (5.6). This implies L.H.S. and R.H.S. are equal in Equation (5.6).

Each MEC has a unique induced subgraph on S . This implies the R.H.S. of Equation (5.6) is a disjoint union of sets $\text{MEC}(G, O)$ for $O \in \text{PMEC}(G[S])$ and $O[S'] = O'$. This further implies Equation (5.5). This completes the proof of Lemma 5.2.12. $\square$

Lemmas 5.2.8 and 5.2.12 imply the following:

Lemma 5.2.15. *For any $O' \in \text{PMEC}(G[S_1 \cup N(S_1, G)])$, $|\text{MEC}(G, O')|$ equals*

$$\sum_{\substack{O_1 \in \text{PMEC}(G_1[S_1 \cup N(S_1, G_1)]) \\ O_2 \in \text{PMEC}(G_2[S_2 \cup N(S_2, G_2)]) \\ O \in \text{PMEC}(G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)]) \\ \text{such that } O \in \mathcal{E}(O_1, O_2) \\ \text{and } O' = O[S_1 \cup N(S_1, G)]}} |\text{MEC}(G_1, O_1)| \times |\text{MEC}(G_2, O_2)|. \quad (5.7)$$

Proof. From Lemma 5.2.12, for any $O' \in \text{PMEC}(G[S_1 \cup N(S_1, G)])$,

$$|\text{MEC}(G, O')| = \sum_{\substack{O \in \text{PMEC}(G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)]) \\ \text{such that } O' = O[S_1 \cup N(S_1, G)]}} |\text{MEC}(G, O)|. \quad (5.8)$$

From Lemma 5.2.8, for any $O \in \text{PMEC}(G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)])$, $|\text{MEC}(G, O)|$ equals

$$\sum_{\substack{O_1 \in \text{PMEC}(G_1[S_1 \cup N(S_1, G_1)]) \\ O_2 \in \text{PMEC}(G_2[S_2 \cup N(S_2, G_2)]) \\ \text{such that } O \in \mathcal{E}(O_1, O_2)}} |\text{MEC}(G_1, O_1)| \times |\text{MEC}(G_2, O_2)|. \quad (5.9)$$

Combining Equations (5.8) and (5.9), we get Equation (5.7). $\square$

We now use Lemma 5.2.15 to construct a recursive algorithm (Algorithm 12) that for inputs G , a clique tree representation (T, β) of G , and a node X_1 of T , outputs a function F with domain $\text{PMEC}(G[\beta(X_1) \cup N(\beta(X_1), G)])$ such that for $O \in \text{PMEC}(G[\beta(X_1) \cup N(\beta(X_1), G)])$, $F(O) = |\text{MEC}(G, O)|$. Lemma 5.2.16 proves the correctness of Algorithm 12.

Lemma 5.2.16. *Let G be a chordal graph, (T, β) be a clique tree representation of G , and X_1 be the root node of T . For inputs G , (T, β) , and X_1 , Algorithm 12 outputs a function $F : \text{PMEC}(G[\beta(X_1) \cup N(\beta(X_1), G)]) \rightarrow \mathbb{N}$ such that for $O \in \text{PMEC}(G[\beta(X_1) \cup N(\beta(X_1), G)])$, $F(O) = |\text{MEC}(G, O)|$.*

Proof. At Line 1, the algorithm constructs the function F . For each partial MEC $O \in \text{PMEC}(G[\beta(X_1) \cup N(\beta(X_1), G)])$, it initializes $F(O) = 0$ (at lines 2–4).

Lines 5–8 handle the base case, when the degree of X_1 in T is zero. In this case, $V_G = \beta(X_1)$, and G is a clique. For a clique graph G , there exists only one MEC, which is G itself.

To verify, suppose M is an MEC of the clique graph G , and there exists a directed edge in M . Choose $u \rightarrow v \in M$ such that no edges are directed towards u , and v has no directed edges coming in from any directed children of u , i.e., for all $w \in V_M$, if $u \rightarrow w \in M$ then $w \rightarrow v \notin M$. Since M is a chain graph (from Item 1 of Theorem 1.1.24), such a $u \rightarrow v$ must exist. However, since M is an MEC of a clique graph, $u \rightarrow v$ cannot belong to any induced subgraph as shown in Figure 1.3.a, Figure 1.3.b, or Figure 1.3.d. Furthermore, due to the choice of u and v , $u \rightarrow v$ cannot belong to any induced subgraph as shown in Figure 1.3.c. Thus, $u \rightarrow v$ is not strongly protected, contradicting Item 4 of Theorem 1.1.24. Hence, M cannot have any directed edges. This implies all edges of M are undirected, meaning M is G itself. Since G is chordal, it satisfies the

Algorithm 12: Count($G, (T, \beta), X_1$)

Input : (a) A connected chordal graph G , (b) a clique tree decomposition (T, β) of G , (c) the root node $X_1 \in V_T$ of T .

Output: A function $F : \text{PMEC}(G[\beta(X_1) \cup N(\beta(X_1), G)]) \rightarrow \mathbb{N}$ such that for $O \in \text{PMEC}(G[\beta(X_1) \cup N(\beta(X_1), G)])$, $f(O) = |\text{MEC}(G, O)|$.

```
1 Create a function  $F : \text{PMEC}(G[\beta(X_1) \cup N(\beta(X_1), G)]) \rightarrow \mathbb{N}$ 
2 foreach  $O \in \text{PMEC}(G[\beta(X_1) \cup N(\beta(X_1), G)])$  do
3   |  $F(O) \leftarrow 0$ 
4 end
5 if degree of  $X_1$  in  $T = 0$  then
6   |  $F(G) = 1$ 
7   | return  $F$ 
8 end
9 Pick an arbitrary edge  $X_1 - X_2 \in E_T$ 
10 Cut the edge  $X_1 - X_2$  of  $T$ 
11  $T_1 \leftarrow$  induced subtree of  $T$  containing  $X_1$ 
12  $T_2 \leftarrow$  induced subtree of  $T$  containing  $X_2$ 
13  $G_1 \leftarrow G[\cup_{X \in V_{T_1}} \beta(X)]$ 
14  $G_2 \leftarrow G[\cup_{X \in V_{T_2}} \beta(X)]$ 
15  $\beta_1 \leftarrow \beta|_{V_{T_1}}$  (i.e.,  $\beta_1 : V_{T_1} \rightarrow 2^{V_{G_1}}$  such that  $\beta_1(X) = \beta(X)$ )
16  $\beta_2 \leftarrow \beta|_{V_{T_2}}$  (i.e.,  $\beta_2 : V_{T_2} \rightarrow 2^{V_{G_2}}$  such that  $\beta_2(X) = \beta(X)$ )
17  $F_1 \leftarrow \text{Count}(G_1, (T_1, \beta_1), X_1)$ 
18  $F_2 \leftarrow \text{Count}(G_2, (T_2, \beta_2), X_2)$ 
19 foreach  $O \in \text{PMEC}(G[\beta(X_1) \cup \beta(X_2) \cup N(\beta(X_1) \cup \beta(X_2), G)])$  do
20   | foreach  $O_1 \in \text{PMEC}(G_1[\beta(X_1) \cup N(\beta(X_1), G_1)])$  do
21     | foreach  $O_2 \in \text{PMEC}(G_2[\beta(X_2) \cup N(\beta(X_2), G_2)])$  do
22       | if  $O \in \mathcal{E}(O_1, O_2)$  then
23         | |  $O' \leftarrow O[\beta(X_1) \cup N(\beta(X_1), G)]$ 
24         | |  $F(O') = F(O') + F_1(O_1) \times F_2(O_2)$ 
25       | end
26     | end
27   | end
28 end
29 return  $F$ 
```

conditions of Items 1 to 4 of Theorem 3.2.1. Thus, $|\text{MEC}(G)| = 1$. This also implies that the induced subgraph of the only MEC M of G on V_G is G , i.e., $|\text{MEC}(G, G)| = 1$. Consequently, the algorithm updates F at Line 6 with $F(G) = 1$. Therefore, when the degree of X_1 is zero, the algorithm outputs the correct function.

If the degree of X_1 is not zero, then the algorithm does the same as we discussed at the beginning of this subsection. The algorithm selects an edge $X_1 - X_2$ in T and cuts this edge, resulting in two subtrees: T_1 (containing X_1) and T_2 (containing X_2). Corresponding to T_1 and T_2 , the algorithm constructs two induced subgraphs G_1 and G_2 of G . It then recursively calls Algorithm 12 to obtain functions $F_1 : \text{PMEC}(G_1[\beta(X_1) \cup N(\beta(X_1), G_1)]) \rightarrow \mathbb{N}$ and $F_2 : \text{PMEC}(G_2[\beta(X_2) \cup N(\beta(X_2), G_2)]) \rightarrow \mathbb{N}$. For all $O_1 \in \text{PMEC}(G_1[\beta(X_1) \cup N(\beta(X_1), G_1)])$, $F_1(O_1) = |\text{MEC}(G_1, O_1)|$, and for all $O_2 \in \text{PMEC}(G_2[\beta(X_2) \cup N(\beta(X_2), G_2)])$, $F_2(O_2) = |\text{MEC}(G_2, O_2)|$.

The algorithm implements Lemma 5.2.15 between lines 19–28 to compute the required function F . Lemma 5.2.15 validates the correctness of F , which is returned at Line 29. This completes the proof of Lemma 5.2.16. $\square$

Algorithm 13: Count-MEC(G)

Input : A connected chordal graph G

Output: $|\text{MEC}(G)|$

- 1 Construct a clique tree decomposition (T, β) of G .
 - 2 Pick an arbitrary root node X_1 of T .
 - 3 $F \leftarrow \text{Count}(G, T, X_1)$
 - 4 $\text{sum} \leftarrow 0$
 - 5 **foreach** $O \in \text{PMEC}(G[\beta(X_1) \cup N(\beta(X_1), G)])$ **do**
 - 6 $\text{sum} = \text{sum} + F[O]$
 - 7 **end**
 - 8 **return** sum ;
-

We use Algorithm 12 and Equation (5.1) to construct Algorithm 13 that for an input chordal graph G , outputs $|\text{MEC}(G)|$. Lemma 5.2.17 provides the correctness of Algorithm 13.

Lemma 5.2.17. *Let G be a chordal graph. For input G , Algorithm 13 outputs $|\text{MEC}(G)|$.*

Proof. At Line 1, the algorithm constructs a clique tree decomposition (T, β) of G . Then, it selects an arbitrary root node X_1 of T (Line 2). At Line 3, the algorithm calls Algorithm 12, which returns a function $F : \text{PMEC}(G[\beta(X_1) \cup N(\beta(X_1), G)]) \rightarrow \mathbb{N}$ such that for any $O \in \text{PMEC}(G[\beta(X_1) \cup N(\beta(X_1), G)])$, $F(O) = |\text{MEC}(G, O)|$ (from Lemma 5.2.16). Subsequently, lines 5–7 implement Equation (5.1) to compute $|\text{MEC}(G)|$. Finally, Line 8 returns the computed value $|\text{MEC}(G)|$. This completes the proof of Lemma 5.2.17. $\square$

5.3 Time Complexity Analysis

In this section, we analyze the time complexity of the algorithms discussed in this chapter. In addition to this, we also prove Theorem 5.1.1.

Lemma 5.3.1. *Let G be a chordal graph, (T, β) be a tree decomposition of G , and X_1 be the root node of T . For input G , (T, β) , and X_1 , the time complexity of Algorithm 12 is $O(n(2^{O(\delta^2 k^2)} + n^2))$, where δ is the maximum degree of a node in the graph, k is the treewidth of G , and n is the number of nodes in G .*

Proof. At Line 10 of Algorithm 12, we cut an edge of T to recursively call Algorithm 12 (at lines 17 and 18). For a clique tree decomposition (T, β) of a chordal graph G , the number of edges in T is $O(n)$ [Blair and Peyton, 1993]. Thus, the number of recursive calls made by Algorithm 12 is $O(n)$. We will therefore account for all the time taken in each recursive call, ignoring the recursive calls themselves, and then we will multiply this by $O(n)$.

If the treewidth of G is k , then the size of $\beta(X_1)$ is $k + 1$. If the maximum degree of a node in G is δ , then the number of nodes in $\beta(X_1) \cup N(\beta(X_1), G)$ is at most $(\delta + 1)(k + 1)$, i.e., $O(\delta k)$. Furthermore, the number of edges in $G[\beta(X_1) \cup N(\beta(X_1), G)]$ is $O(\delta^2 k^2)$. For any partial MEC O of $G[\beta(X_1) \cup N(\beta(X_1), G)]$, for each $u - v \in E_{G[\beta(X_1) \cup N(\beta(X_1), G)]}$, we have $u - v \in O$, $u \rightarrow v \in O$, or $v \rightarrow u \in O$. Hence, the number of partial MECs is at most $3^{O(\delta^2 k^2)}$, i.e., $O(2^{O(\delta^2 k^2)})$. Therefore, the initialization of F (lines 1–4) requires $O(2^{O(\delta^2 k^2)})$ time.

The execution of lines 5–8 takes $O(1)$ time. Lines 9–14 require $O(|V_G| + |E_G|)$ time.

Similar to the partial MECs of $G[\beta(X_1) \cup N(\beta(X_1), G)]$, the number of partial MECs of $G_1[\beta(X_1) \cup N(\beta(X_1), G_1)]$, $G_2[\beta(X_2) \cup N(\beta(X_2), G_2)]$, and $G[\beta(X_1) \cup \beta(X_2) \cup N(\beta(X_1) \cup \beta(X_2), G)]$ is $O(2^{O(\delta^2 k^2)})$. Hence, each of the for loops at lines 20, 21, and 19 runs for $O(2^{O(\delta^2 k^2)})$ iterations.

Using Definition 5.2.5, we now show that we can verify whether $O \in \mathcal{E}(O_1, O_2)$ in $O(\delta^3 k^2)$ time. The number of edges in O , O_1 , and O_2 is $O(\delta^2 k^2)$. Verification of Item 1 of Definition 5.2.5 takes $O(\delta^2 k^2)$ time. For each edge $u \rightarrow v \in O_a$, the number of v-structures it is part of is at most δ (as if $u \rightarrow v \leftarrow w$ is a v-structure in O_a , then w is a neighbor of v in G_a). Therefore, the total number of v-structures in O_a is $O(\delta^3 k^2)$. For a v-structure $u \rightarrow v \leftarrow w \in O_a$, it takes $O(1)$ time to verify whether the v-structure is in $O[V_{O_a}]$ or not. Thus, the verification of Item 2 of Definition 5.2.5 takes $O(\delta^3 k^2)$ time. For an edge $u \rightarrow v \in O$, verifying whether it is strongly protected in O involves checking neighbors of u and v , which can be at most 2δ . Therefore, the verification of Item 3 of Definition 5.2.5 takes $O(\delta^3 k^2)$ time. Consequently, we can verify whether $O \in \mathcal{E}(O_1, O_2)$ at Line 22 in $O(\delta^3 k^2)$ time.

Constructing O' at Line 23 takes $O(\delta^2 k^2)$ time, and updating F at Line 24 requires $O(1)$ time.

Combining these results, the total running time of Algorithm 12 is $O(n(2^{O(\delta^2 k^2)} + n^2))$. The n^2 term in the time complexity comes from the running time of lines 9–14, which is $O(|V_G| + |E_G|)$, as shown above. This completes the proof of Lemma 5.3.1. $\square$

Lemma 5.3.2. *For an input chordal graph G , the time complexity to run Algorithm 13 is $O(n(2^{O(\delta^2 k^2)} + n^2))$, where δ is the maximum degree of a node in G , k is the treewidth of G , and n is the number of nodes in G .*

Proof. A clique tree representation of a chordal graph G (computed at Line 1 of Algorithm 13) can be obtained in $O(|V_G| + |E_G|)$ time [Blair and Peyton, 1993]. From Lemma 5.3.1, the running time of Line 3 is $O(n(2^{O(\delta^2 k^2)} + n^2))$.

As discussed in the proof of Lemma 5.3.1, the number of partial MECs in $G[\beta(X_1) \cup N(\beta(X_1), G)]$ is $O(2^{O(\delta^2 k^2)})$. Therefore, the running time of lines 5–7 is $O(2^{O(\delta^2 k^2)})$.

Combining these, the overall running time of Algorithm 13 is $O(n(2^{O(\delta^2 k^2)} + n^2))$. This completes the proof of Lemma 5.3.2. $\square$

Proof of Theorem 5.1.1. Lemmas 5.2.17 and 5.3.2 validates Theorem 5.1.1. $\square$

To recap, we show that the MECs of a chordal graph can be counted in $O(n(2^{O(\delta^2 k^2)} + n^2))$ time (Theorem 5.1.1). We provide the proofs of all auxiliary lemmas, claims, and observations stated to prove Theorem 5.1.1, except for the proof of Lemma 5.2.7. This proof is provided in the next section.

5.4 Proof of Lemma 5.2.7

From the hypotheses of Lemma 5.2.7, we have $O \in \text{PMEC}(G[S_1 \cup S_2 \cup N(S_1 \cup S_2, G)])$, $O_1 \in \text{PMEC}(G_1[S_1 \cup N(S_1, G_1)])$, and $O_2 \in \text{PMEC}(G_2[S_2 \cup N(S_2, G_2)])$ such that $O \in \mathcal{E}(O_1, O_2)$. Let $M_1 \in \text{MEC}(G_1, O_1)$ and $M_2 \in \text{MEC}(G_2, O_2)$. We use Lemma 4.5.8 to prove that there exists a unique MEC $M \in \text{MEC}(G, O)$ such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. This will prove Lemma 5.2.7.

We first go through the following observations:

Observation 5.4.1. *For some $a \in \{1, 2\}$, and for distinct nodes $u, v, w \in V_{O_a}$, let $P = (u_1 = u, u_2 = v, \dots, u_l = w)$ be a triangle-free path in M_a . If $u \rightarrow v \in O$, then for all $1 \leq i < l$, $u_i \rightarrow u_{i+1} \in O$. In particular, P is a directed triangle-free path in O .*

Proof. From Proposition 5.2.1, P is a triangle-free path in O_a . By construction, $V_{O_a} \subseteq V_O$. This implies that each node of P is in V_O . For $i = 1$, we have $u_i \rightarrow u_{i+1} \in O$ since $u \rightarrow v \in O$. Suppose for some $1 < i < l$, $u_i \rightarrow u_{i+1} \notin O$. Pick the smallest such i . This implies $u_{i-1} \rightarrow u_i \in O$. By construction, $\text{skeleton}(O_a) = \text{skeleton}(O[V_{O_a}])$. This implies that either $u_i - u_{i+1} \in O$ or $u_i \leftarrow u_{i+1} \in O$. Suppose $u_i - u_{i+1} \in O$. Then, since $\text{skeleton}(O_a) = \text{skeleton}(O[V_{O_a}])$ and Q is a triangle-free path, $u_{i-1} \rightarrow u_i - u_{i+1}$ is an induced subgraph of O , contradicting Item 3 of Definition 4.2.1 (recall that O is a partial MEC). If $u_i \leftarrow u_{i+1} \in O$, then $u_{i-1} \rightarrow u_i \leftarrow u_{i+1}$ is a v-structure in O , but not in O_a (since P is a path in O_a , $u_i \leftarrow u_{i+1}$ cannot be in O_a), contradicting Item 2 of Definition 5.2.5 (recall that $O \in \mathcal{E}(O_1, O_2)$). This implies that for all $1 \leq i < l$, $u_i \rightarrow u_{i+1} \in O$. Since P is a triangle-free path in O_a and $\text{skeleton}(O_a) = \text{skeleton}(O[V_{O_a}])$, this further implies that P is a triangle-free path in O . $\square$

Observation 5.4.2. For some $a \in \{1, 2\}$, if $Q = (u_1 = u, u_2 = v, \dots, u_l)$ is a triangle-free path in O_a and $v \rightarrow u \notin O$, then Q is a triangle-free path in O as well.

Proof. Suppose $v \rightarrow u \notin O$. We show that for all $2 \leq i \leq l$, $u_i \rightarrow u_{i-1} \notin O$. From the construction, $\text{skeleton}(O_a) = \text{skeleton}(O[V_{O_a}])$. Suppose for some i , $u_i \rightarrow u_{i-1} \in O$. Pick the smallest such i . Since $v \rightarrow u \notin O$, we must have $i > 2$. This implies that $u_{i-1} \rightarrow u_{i-2} \notin O$. Since $\text{skeleton}(O_a) = \text{skeleton}(O[V_{O_a}])$, this further implies that either $u_{i-1} - u_{i-2} \in O$ or $u_{i-1} \leftarrow u_{i-2} \in O$. If $u_{i-1} - u_{i-2} \in O$, then $u_i \rightarrow u_{i-1} - u_{i-2}$ is an induced subgraph of O , contradicting Item 3 of Definition 4.2.1 (recall that O is a partial MEC). If $u_{i-1} \leftarrow u_{i-2} \in O$, then $u_i \rightarrow u_{i-1} \leftarrow u_{i-2}$ is a v-structure in O but not in O_a , contradicting Item 2 of Definition 5.2.5 (recall that $O \in \mathcal{E}(O_1, O_2)$). This implies that for all $2 \leq i \leq l$, $u_i \rightarrow u_{i-1} \notin O$. This further implies that Q is a triangle-free path in O . $\square$

We now begin the proof of Lemma 5.2.7.

Proof of Lemma 5.2.7. Since $M_1 \in \text{MEC}(G_1, O_1)$, O_1 is an induced subgraph of M_1 , i.e., $M_1[V_{O_1} = S_1 \cup N(S_1, G_1)] = O_1$. Let $(O_1, P_1, P_2) = \text{shadow}(M_1, S_1 \cup N(S_1, G_1))$ (from Definition 4.3.1, O_1 is the first term in the shadow of M_1 on $S_1 \cup N(S_1, G_1)$). Similarly, let $(O_2, P_{21}, P_{22}) = \text{shadow}(M_2, S_2 \cup N(S_2, G_2))$. Let $(P_1, P_2) = \text{TFP}(O)$ (recall from Algorithm 5, $P_1 : E_O \times E_O \rightarrow \{0, 1\}$ is a function such that for $(u, v), (x, y) \in E_O$, $P_1((u, v), (x, y)) = 1$ if $(u, v) \neq (x, y)$ and there exists a triangle-free path from (u, v) to (x, y) in O , else $P_1((u, v), (x, y)) = 0$, and $P_2 : E_O \times V_O \rightarrow \{0, 1\}$ is a function such that for $(u, v) \in E_O$ and $w \in V_O$, $P_2((u, v), w) = 1$ if $v \neq w$ and there exists a triangle-free path from (u, v) to w in O , else $P_2((u, v), w) = 0$).

The following claim shows that (O, P_1, P_2) is an extension of (O_1, P_{11}, P_{12}) and (O_2, P_{21}, P_{22}) , i.e., $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$ (note that here we are using the extension of shadows, as defined in Definition 4.5.6; this should not be confused with the extension of partial MECs defined in Definition 5.2.5).

Claim 5.4.3. $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$.

Proof. To prove Claim 5.4.3, we have to show that (O, P_1, P_2) obeys Items 1 to 4 of Definition 4.5.6.

We are given that $O \in \mathcal{E}(O_1, O_2)$ (Definition 5.2.5). Items 1 and 2 of Definition 5.2.5 show that (O, P_1, P_2) obeys Items 1 and 2 of Definition 4.5.6.

Item 3 of Definition 5.2.5 shows that (O, P_1, P_2) satisfies $\leftarrow$ of Item 3 of Definition 4.5.6. We now show that (O, P_1, P_2) also satisfies $\rightarrow$ of Item 3 of Definition 4.5.6, i.e., we show the following: for some $a \in \{1, 2\}$, for $u - v \in O_a$, (a) if $u \rightarrow v$ is strongly protected in O then $u \rightarrow v \in O$, (b) for $x - y \in O_a$, if $P_{a1}((x, y), (u, v)) = 1$ and $x \rightarrow y \in O$ then $u \rightarrow v \in O$, and (c) for $x - y \in O_a$, if $P_{a2}((x, y), v) = P_{a2}((v, u), x) = 1$ and $x \rightarrow y \in O$ then $u \rightarrow v \in O$.

It is obvious that if $u \rightarrow v$ is strongly protected in O then $u \rightarrow v \in O$.

For $a \in \{1, 2\}$, since $(O_a, P_{a1}, P_{a2}) = \text{shadow}(M_a, S_a \cup N(S_a, G_a))$, from Definition 4.3.1, for $x - y, u - v \in O_a$, if $P_{a1}((x, y), (u, v)) = 1$ then $(x, y) \neq (u, v)$ and there exists a triangle-free path $Q = (u_1 = x, u_2 = y, \dots, u_{l-1} = u, u_l = v)$ from (x, y) to (u, v) in M_1 . From Observation 5.4.1, if $x \rightarrow y \in O$ then $u \rightarrow v \in O$.

For $a \in \{1, 2\}$, since $(O_a, P_{a1}, P_{a2}) = \text{shadow}(M_a, S_a \cup N(S_a, G_a))$, from Definition 4.3.1, for $x - y, u - v \in O_a$, if $P_{a2}((x, y), v) = P_{a2}((v, u), x) = 1$ then $y \neq v, u \neq x$, and there exist two triangle-free paths $Q_1 = (u_1 = x, u_2 = y, \dots, u_{l-1}, u_l = v)$ from (x, y) to v in M_a , and $Q_2 = (v_1 = v, v_2 = u, \dots, v_m = x)$ from (v, u) to x in M_a . Suppose $x \rightarrow y \in O$. We show that $u \rightarrow v \in O$. From Observation 5.4.1, Q_1 is a directed path in O . From Proposition 5.2.1, Q_2 is a triangle-free path in O_a . Suppose $u \rightarrow v \notin O$. Then, from Observation 5.4.2, Q_2 is a triangle-free path in O , and the concatenation of Q_1 and Q_2 provides a directed cycle in O , contradicting Item 1 of Definition 4.2.1 (recall that O is a partial MEC). This implies $u \rightarrow v \in O$.

The above discussion implies that (O, P_1, P_2) satisfies $\rightarrow$ of Item 3 of Definition 4.5.6. This further implies that (O, P_1, P_2) satisfies Item 3 of Definition 4.5.6.

We now show that (P_1, P_2) is the valid derived path function (Definitions 4.5.3 and 4.5.5) of $(O, P_{11}, P_{12}, P_{21}, P_{22})$, so that (O, P_1, P_2) also satisfies Item 4 of Definition 4.5.6. We first show that (P_1, P_2) is a derived path function of $(O, P_{11}, P_{12}, P_{21}, P_{22})$.

Claim 5.4.4. (P_1, P_2) is the derived path function of $(O, P_{11}, P_{12}, P_{21}, P_{22})$.

Proof. Let (P'_1, P'_2) be the derived path function of $(O, P_{11}, P_{12}, P_{21}, P_{22})$. We show that $(P_1, P_2) = (P'_1, P'_2)$. From the construction of (P_1, P_2) and Definition 4.5.3, the domain of

both P_1 and P'_1 is $E_O \times E_O$, and the codomain is $\{0, 1\}$; the domain of both P_2 and P'_2 is $E_O \times V_O$, and the codomain is $\{0, 1\}$. From Definition 4.5.3, we construct (P'_1, P'_2) by following steps 1-3 of Definition 4.5.3. We show that at the end of each step, we have $(P_1, P_2) = (P'_1, P'_2)$.

At the end of step 1, for $(u, v), (x, y) \in E_O$, $P'_1((u, v), (x, y)) = 1$ if $(u, v) \neq (x, y)$ and there exists a triangle-free path from (u, v) to (x, y) in O , otherwise $P'_1((u, v), (x, y)) = 0$, and $P'_2((u, v), w) = 1$ if $v \neq w$ and there exists a triangle-free path from (u, v) to w in O , otherwise $P'_2((u, v), w) = 0$. From our construction of (P_1, P_2) , $(P_1, P_2) = \text{TFP}(O)$. Therefore, from Lemma 4.6.18, at the end of step 1, $(P_1, P_2) = (P'_1, P'_2)$.

We now show that there do not exist $(u, v), (x, y) \in E_O$ such that at the end of step 1, $P'_1((u, v), (x, y)) = 0$ and at the end of step 2, $P'_1((u, v), (x, y)) = 1$. Suppose there exist $(u, v), (x, y) \in E_O$ such that at the end of step 1, $P'_1((u, v), (x, y)) = 0$, and at the end of step 2, $P'_1((u, v), (x, y)) = 1$. Since at the end of step 1, $P_1 = P'_1$, this implies there does not exist a triangle-free path from (u, v) to (x, y) in O . Since at the end of step 2, $P'_1((u, v), (x, y)) = 1$, from Definition 4.5.3, for some $a \in \{1, 2\}$, we have $P_{a1}((u, v), (x, y)) = 1$. This implies that there exists a triangle-free path Q from (u, v) to (x, y) in M_a . From Proposition 5.2.1, Q is a triangle-free path in O_a . Since $(u, v) \in E_O$, $v \rightarrow u \notin O$. Therefore, from Observation 5.4.2, Q is a triangle-free path in O , implying that at the end of step 1, $P'_1((u, v), (x, y)) = 1$, contradicting our assumption. This implies that there do not exist $(u, v), (x, y) \in E_O$ such that at the end of step 1, $P'_1((u, v), (x, y)) = 0$ and at the end of step 2, $P'_1((u, v), (x, y)) = 1$. For any $(u, v), (x, y) \in E_O$ such that at the end of step 1, $P'_1((u, v), (x, y)) = 1$, step 2 does not change $P'_1((u, v), (x, y))$ to 0. This implies that at step 2, P'_1 does not change. This further implies that at the end of step 2, $P_1 = P'_1$. Using a similar argument, we can show that at the end of step 2, we have $P_2 = P'_2$.

We now show that there do not exist $(u, v), (x, y) \in E_O$ such that at the end of step 2, $P'_1((u, v), (x, y)) = 0$ and at the end of step 3, $P'_1((u, v), (x, y)) = 1$. Suppose there exist $(u, v), (x, y) \in E_O$ such that at the end of step 2, $P'_1((u, v), (x, y)) = 0$, and at the end of step 3, $P'_1((u, v), (x, y)) = 1$. Pick the first such $(u, v), (x, y) \in E_O \times E_O$ that got updated at step 3. Then, from step 3, there exists $(z_1, z_2) \in E_O$ such that $P'_1((u, v), (z_1, z_2)) = P'_1((z_1, z_2), (x, y)) = 1$. Since at the end of step 2, $P_1 = P'_1$, this

implies there exist triangle-free paths Q_1 from (u, v) to (z_1, z_2) in O , and Q_2 from (z_1, z_2) to (x, y) . Then, by applying Lemma 4.2.5 to the chain graph O with chordal undirected components, the concatenation of Q_1 and Q_2 yields a triangle-free path from (u, v) to (x, y) . This implies that at the end of step 2, we have $P'_1((u, v), (x, y)) = 1$, leading to a contradiction. This implies that there do not exist $(u, v), (x, y) \in E_O$ such that at the end of step 2, $P'_1((u, v), (x, y)) = 0$ and at the end of step 3, $P'_1((u, v), (x, y)) = 1$. For any $(u, v), (x, y) \in E_O$ such that at the end of step 2, $P'_1((u, v), (x, y)) = 1$, step 3 does not change $P'_1((u, v), (x, y))$ to 0. This implies that at step 3, P'_1 does not change. This further implies that at the end of step 3, $P_1 = P'_1$. Using a similar argument, we can show that at the end of step 3, we have $P_2 = P'_2$. This implies that (P_1, P_2) is the derived path function of $(O, P_{11}, P_{12}, P_{21}, P_{22})$. $\square$

We now show that (P_1, P_2) is the valid derived path function (Definition 4.5.5) of $(O, P_{11}, P_{12}, P_{21}, P_{22})$. Since we have already shown that (P_1, P_2) is a derived path function of $(O, P_{11}, P_{12}, P_{21}, P_{22})$, we only need to prove Item 2 of Definition 4.5.5. The following claim proves this.

Claim 5.4.5. *For all $(u, v), (x, y) \in E_O$, if $P_1((u, v), (x, y)) = 1$ then $P_1((x, y), (u, v)) = 0$.*

Proof. Suppose, if possible, that there exist $(u, v), (x, y) \in E_O$, such that $P_1((u, v), (x, y)) = P_1((x, y), (u, v)) = 1$. This implies that there exist triangle-free paths P_1 from (u, v) to (x, y) and P_2 from (x, y) to u (P_2 is a prefix of the hypothesized triangle-free path from (x, y) to (u, v) due to $P_1((x, y), (u, v)) = 1$). Then, by applying Lemma 4.2.5 to the chain graph O with chordal undirected components, the concatenation of P_1 and P_2 yields a triangle-free path from (u, v) to u . But this is a contradiction, since by the definition of path, Definition 1.1.2, no vertex can appear twice in a path. This implies there cannot exist a triangle-free path from (x, y) to u . Thus, there cannot exist a triangle-free path from (x, y) to (u, v) , implying $P_1((x, y), (u, v)) = 0$. $\square$

Claim 5.4.5 completes our argument that (P_1, P_2) is the valid derived path function of $(O, P_{11}, P_{12}, P_{21}, P_{22})$, implying (O, P_1, P_2) obeys Item 4 of Definition 4.5.6. As (O, P_1, P_2) obeys Items 1 to 4 of Definition 4.5.6, $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$. This completes the proof of Claim 5.4.3. $\square$

Since $M_1 \in \text{MEC}(G_1, O_1, P_{11}, P_{12})$, $M_2 \in \text{MEC}(G_2, O_2, P_{21}, P_{22})$, and $(O, P_1, P_2) \in \mathcal{E}(O_1, P_{11}, P_{12}, O_2, P_{21}, P_{22})$, from Lemma 4.5.8, there exists a unique MEC $M \in \text{MEC}(G, O, P_1, P_2)$ such that $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. From Definition 4.3.1, $M[V_O] = O$, implying $M \in \text{MEC}(G, O)$. From Definition 4.5.3, there cannot be two distinct pairs (P_1, P_2) and (P'_1, P'_2) such that both (P_1, P_2) and (P'_1, P'_2) are derived path functions of $(O, P_{11}, P_{12}, P_{21}, P_{22})$. This implies that for the pair of MECs (M_1, M_2) discussed above, there exists a unique MEC M such that $M \in \text{MEC}(G, O)$ and $\mathcal{P}(M, V_{G_1}, V_{G_2}) = (M_1, M_2)$. This completes the proof of Lemma 5.2.7. $\square$

Chapter 6

Conclusion and Open Problems

In this thesis, we study two problems related to graphical models: (a) counting DAGs in a Markov equivalence class (MEC), and (b) counting the number of MECs that share the same skeleton.

We address the more general problem of counting background knowledge consistent DAGs within an MEC (Chapter 2), extending the first problem. This generalization goes back to the classic paper of Meek [1995]. For the special case where no background knowledge is present, a breakthrough result of Wienöbst et al. [2021] (see also Wienöbst et al. [2023]) gave a polynomial time algorithm to count the DAGs of an MEC. We build upon their approach to solve the general problem. This constitutes the first non-trivial result for counting background knowledge consistent DAGs of an MEC.

Our main result shows that the “max-clique-knowledge” parameter we introduce is pivotal in determining the algorithmic complexity of counting Markov equivalent DAGs under background knowledge constraints. Specifically, it allows for a polynomial-time algorithm in the special case of graphs with bounded maximum clique size. However, a polynomial-time algorithm for the general case is precluded by the #P-hardness result of Wienöbst et al. [2021], under standard complexity theoretic assumptions. Determining the optimal dependence of the runtime on the max-clique-knowledge parameter remains an open question. Interestingly, as we discussed in Chapter 2, the parameter “max-clique-knowledge” can be much smaller than the size of background knowledge.

For the second problem, counting MECs that share the same skeleton (Problem 1.2),

no non-trivial solution was previously known for general graphs to the best of our knowledge, and only very special graph skeletons had been studied in the literature (Radhakrishnan et al. [2018]). We first solve the problem for skeletons that are tree graphs (Chapter 3), providing a polynomial-time solution for this special case. Building on this result, we develop an algorithm to count MECs for general graphs (Chapter 4). The algorithm is fixed-parameter tractable with runtime $O(n(2^{O(k^4\delta^4)} + n^2))$, where n is the number of nodes, and k and δ are the treewidth and maximum degree of the input graph, respectively.

Our analysis suggests that for certain special cases, such as chordal graphs, the runtime dependence on treewidth can be significantly improved. In Chapter 5, we address Problem 1.2 for chordal graphs, modifying the algorithm for general graphs to solve the special case where the skeleton is chordal. We provide a fixed-parameter tractable algorithm with runtime $O(n(2^{O(k^2\delta^2)} + n^2))$ for this case.

The computational complexity of the second problem, counting MECs with the same skeleton, remains open. The primary open question is whether a fully polynomial-time algorithm exists for this problem or whether it is computationally hard in general. Improving the runtime of the fixed-parameter tractable algorithm, possibly using different parameters, represents an intermediate step toward fully understanding the computational complexity.

We regard the results presented in this thesis as a foundational step toward understanding the computational complexity of counting MECs with the same skeleton. We hope that the notions (such as shadows of MECs, extensions of shadows, etc.) and associated techniques introduced here will facilitate further study of this challenging problem.

List of Notations

Symbol	Meaning
V_G	Vertex set of the graph G
E_G	Edge set of the graph G
$\Pi(G)$	The set of maximal cliques of a graph G
$\mathcal{V}(G)$	The set of v-structures (Definition 1.1.8) in a graph G
$\text{AMO}(G)$	The set of acyclic moral orientations (AMOs, Definition 2.2.1) of G
$\text{AMO}(G, \mathcal{K})$	The set of $\mathcal{K}$ -consistent AMOs of G
$\text{MEC}(G)$	The set of MECs having skeleton G
$\text{PMEC}(G)$	The set of partial MECs having skeleton G
$\mathcal{P}(M, X)$	The projection (Definition 4.4.1) of an MEC M on $X \subseteq V_M$
$\text{shadow}(M, Y)$	The shadow (Definition 4.3.1) of an MEC M on $Y \subseteq V_M$
$\text{shadow}(G)$	The set of shadows (Definition 4.3.4) of an undirected graph G
$\text{MEC}(G, O, P_1, P_2)$	The set of MECs of G with shadow (O, P_1, P_2)

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